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Numerical and Experimental Modeling of Shotcrete and Application to Finite Element Analyses of Deep Tunnel Advance

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Dissertation

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Kurzfassung

Die vorliegende Dissertation umfasst die Beschreibung des Materialverhaltens von Spritzbeton, mit Schwerpunkt auf Anwendungen in numerischen Simulationen von Tunnelvortrieben. Zu diesem Zweck wird ein neues Materialmodell vorgestellt, welches auf drei Kontinuumsmodellen für Normalbeton basiert. Diese drei Modelle umfassen das Schädigungs-Plastizitätsmodell nach Grassl und Jirásek, die Solidification Theory nach Bažant und Prasannan sowie das semiempirische Modell nach Bažant und Panula zur Beschreibung des Schwindverhaltens.

In diesem neuen Modell für Spritzbeton, im Folgenden als das SCDP Modell bezeichnet, werden die zeitliche Entwicklung der Materialeigenschaften infolge der Hydratation, ver- und entfestigendes Materialverhalten infolge mechanischer Belastung, plastische Verformungen, die Abnahme der Materialsteifigkeit infolge von Schädigung, Schwinden sowie nichtlineares Kriechen beschrieben. Zur Bewertung des neuen Modells wird eine ausführliche Studie vorgestellt, welche einen Vergleich mit zwei weiteren Spritzbetonmodellen aus der Literatur, nämlich den Modellen nach Meschke und nach Schädlich und Schweiger, beinhaltet. Dieser Vergleich basiert sowohl auf einer Validierung anhand von Daten aus Laborversuchen, welche in der Literatur verfügbar sind, als auch auf einem Benchmark Finite Elemente Modell des Vortriebs eines tiefliegenden Tunnels zur Bewertung der Materialmodelle auf Strukturebene.

Das vorgestellte Finite Elemente Modell für eine Benchmark Studie basiert auf einem Abschnitt des Brenner Basistunnels, für welchen ein Messprogramm zur Bestimmung der Verformungen des umgebenden Gebirges während des Vortriebs durchgeführt wurde. Der Vergleich der Messdaten mit den numerischen Ergebnissen zeigt die realistische Prognose basierend auf dem neuen SCDP Modell.

Weiters wird ein neues Versuchsprogramm für Spritzbeton präsentiert, welches die Bestimmung der zeitlichen Entwicklung des E-Moduls und der einaxialen Druckfestigkeit sowie des Schwind- und Kriechverhaltens von jungem Spritzbeton umfasst. Es wird gezeigt, dass sich die Materialeigenschaften der untersuchten Spritzbetonrezeptur deutlich von den Eigenschaften der in der Literatur behandelten älteren Rezepturen unterscheiden. Auf Basis der neuen Versuchsdaten wird die Kalibrierung der untersuchten Spritzbetonmodelle durchgeführt.

Abstract

The present thesis deals with the description of the material behavior of shotcrete, with special focus on applications in numerical simulations of deep tunnel advance. To this end, a new material model for shotcrete is proposed, which is based on three continuum models for concrete. They comprise the damage plasticity model for concrete by Grassl and Jirásek, the solidification theory for concrete creep by Bažant and Prasannan, and the semi-empirical model by Bažant and Panula for describing shrinkage.

The new model for shotcrete, denoted as the SCDP model, represents the time-dependent evolution of material properties due to hydration, hardening and softening material behavior under mechanical loading, inelastic deformations, stiffness degradation due to damage, as well as shrinkage and nonlinear creep. For an assessment of the new model an extensive study is presented, which includes a comparison with two other models for shotcrete selected from the literature, which are the model by Meschke and the model by Schädlich and Schweiger. This comparison consists of numerical simulations of experimental tests on shotcrete available in the literature, as well as structural analyses based on a benchmark finite element model of deep tunnel advance for evaluating the models at structural level.

The presented finite element model for a benchmark study is based on a stretch of the Brenner Base Tunnel, for which an in-situ measurement program was carried out. It is demonstrated that based on the new SCDP model the realistic prediction of the mechanical response during tunnel advance can be obtained.

Furthermore, a new experimental program on shotcrete is presented, in which the evolution of the Young's modulus, the uniaxial compressive strength, as well as the shrinkage and creep behavior of young shotcrete is investigated. It is shown that the material properties of the investigated shotcrete composition are substantially different from those of older shotcrete compositions for which test data sets are available in the literature. Based on the test data of the new experimental program, the three investigated material models are calibrated.

Contents

1 Introduction

The *New Austrian Tunneling Method* (NATM) is a modern method for designing and constructing tunnels, taking advantage of the inherent geological strength of the surrounding rock mass or soil to stabilize the tunnel. The name of the method goes back to Rabcewicz (1964), one of the pioneers who developed the method in the 1950s. A clear definition is constituted in (ÖIAV, 1980) in terms of 22 elementary laws originally formulated by L. Müller and Fecker (1978).

Basically, the NATM can be described as follows: During excavation, the bearing strength of the surrounding soil or rock mass is tried to be sustained, and it is supported by a layer of shotcrete and optional reinforcements like rock anchors and steel arches. The process is accompanied by continuous exploration and monitoring to control the stabilization of the structure, and to optimize the construction process.

1.1 Motivation – Shotcrete Shells as Part of the NATM

The use of shotcrete is an essential part of the NATM: One aspect of tunnel construction is the transmission of stresses from the surrounding rock mass to the tunnel support structure. In this context, shotcrete is considered as the *ideal* securing measure (Maidl, Heimbecher, et al., 1997), since it allows for creating a void free bond between the excavation surface and the tunnel support structure. Shotcrete is conveyed under pressure through a pneumatic hose or pipe, and projected into place at high velocity with simultaneous compaction, forming a supporting shell (DIN 18851, 2005). Due to an added chemical accelerator, a fast setting of the fresh shotcrete is achieved, resulting in a good adhesion on excavation surfaces and making it possible to apply thick layers of shotcrete even at overhead areas.

Shotcrete can be applied immediately after an excavation step, requires no form work, allows to form complex geometries like those present at junctions, and the application in partial areas is possible (Maidl, 1992; Maidl, Thewes, et al., 2013). Due to the spraying process, the thickness of the shotcrete shell can be adjusted to meet specific local requirements, and shotcrete shells can be combined with additional securing measures like rock anchors or steel arches, as depicted in Figure 1.1. In contrast to normal concrete structures which are loaded at matured material stages, shotcrete shells are loaded already at a very early age of the material, i.e., usually a couple of hours after application due to further tunnel advance. In this context, one import property of shotcrete is its ability to sustain large deformations imposed by movements of surrounding rock mass due to stress rearrangement especially at early ages of the material (Maidl, Heimbecher, et al., 1997). This very ductile behavior is commonly denoted as the *good-natured* behavior of shotcrete (Pöttler, 1990a).

Figure 1.1: Application of shotcrete as securing measure during tunnel advance, in combination with additional rock anchors and steel arches.

Basically, two different techniques are employed for creating shotcrete compositions (Thomas, 2008): *Dry-mix* shotcrete and *wet-mix* shotcrete. For the former technique, a dry composition, consisting of sand, cement and aggregate is conveyed pneumatically to the spraying nozzle, and water and a chemical accelerator are added directly at the nozzle. For the latter technique, a wet ready-mixed composition is conveyed to the nozzle, where solely the accelerator is added. Both techniques are widely used, but in general wet-mix processes became more popular in the past 20 years for large scale applications like tunnel construction, since the wet-mix process allows for a better quality control, produces less rebound, i.e., blown back non-adhering shotcrete, and the spraying process associated with a lower dust pollution. Compared to normal concrete, modern wet-mix shotcrete compositions are characterized by a higher water to cement ratio, a smaller maximum aggregate size, the chemical accelerator which is added for an increased hydration speed, and additional admixtures for improving properties like adhesion or workability of shotcrete (Thomas, 2008). Optional reinforcements consisting of steel meshes or short steel or synthetic fibres are used to increase the tensile strength of shotcrete.

For non-mechanized tunneling, the conventional construction methods of tunnel linings are *primary and secondary linings* (Thomas, 2008): The shotcrete shell applied as a securing measure during the excavation process is employed exclusively as a temporary securing measure, denoted as the *outer* shell, whereas a later on installed second shell, the *inner* shell, made either from cast in place concrete or shotcrete is used as part of the permanent support structure. Usually, the inner shell and the outer shell are separated by waterproofing measures like sheet membranes or spray-on membranes. Following this design approach, for dimensioning of the inner shell it is assumed that the outer shotcrete shell deteriorates completely with ongoing time and accordingly, the complete load is carried exclusively by the inner shell. Clearly, the assumption of a complete deterioration of the inner shell is in contrast to results of long-term observations (Galler and Lorenz, 2017).

Alternatively, the *single shell lining* approach aims at permanent usage of the shotcrete shell as part of the tunnel support structure (Maidl, 1992; Golser and Kienberger, 1997; Thomas, 2008). This permanent shotcrete shell can consist of several layers of shotcrete, which can be applied at different times. Compared to primary and secondary linings, considerable cost reductions can be achieved by this approach. A comprehensive summary of tunnels with single shell linings can be found in (Franzen et al., 2001).

For numerical analyses of tunnel constructions and tunnel advance, constitutive models for representing the behavior of the involved materials are necessary. Compared to normal concrete, material modeling of shotcrete is represented considerably less in the literature. While this is clearly related to the special field of application, to a certain extent this might be attributed also to the complex nature of numerical simulations of tunnel advance: Although often simplified by means of two dimensional models, tunnel advance is a three dimensional mechanical problem, which is time-dependent due to the sequential construction process, and due to the time-dependent behavior of the involved materials, in particular that of shotcrete. Since material properties of shotcrete are continuously developing during the hydration process, constitutive modeling of shotcrete can be performed only in a reasonable way by considering its material properties as time-dependent. In addition to the complex construction process and the nonlinear, time-dependent material behavior, the actual loads imposed on the tunnel support structure can be estimated only roughly due to unknown in-situ stresses prevailing in the surrounding rock mass, due to the complex and often anisotropic material behavior of rock, due to possible joints and fault zones within the surrounding rock mass, and due to limited exploration techniques for estimating those joints and fault zones.

The present work addresses the complex material behavior of shotcrete with a focus on shotcrete as a securing measure in tunneling, and methods to account for the time-dependent, nonlinear material behavior of shotcrete in numerical simulations. Reliable predictions of the short-term and long-term behavior of shotcrete shells are of great interest for estimating structural safety on the one hand, but also for economic reasons on the other hand, with regard to increasingly more important single shell linings.

1.2 Material Behavior of Shotcrete

Similar to concrete, the description of the material behavior of shotcrete is a complex issue since the material exhibits a number of nonlinear and time-dependent phenomena. These phenomena are summarized according to Meschke (1996) and Thomas (2008) as:

- The material properties are dependent on the degree of hydration, and thus dependent on the material age. For instance, stiffness and strength are increasing during hydration, while ductility is decreasing. Compared to normal concrete, due to the added accelerator material properties are evolving considerably faster at early material ages, i.e., in the first hours after application of the shotcrete shell.
- The theory of linear elasticity for describing the stress-strain relation for an instant of time is admissible only for comparatively low loads: In uniaxial compression, approximately linear-elastic material behavior is observed for ratios of the applied stress to the uniaxial compressive strength of about up to 30 %. For higher load levels, inelastic (plastic) strains are observed after unloading. Furthermore, shotcrete shows hardening material behavior in compression, and behaves increasingly more ductile with increasing confining pressure (Sezaki et al., 1989). After attaining material strength, shotcrete shows softening material behavior, accompanied by stiffness degradation. Material failure in compression is observed in the form of cracking or crushing, dependent on the level of confinement.
- By contrast, in uniaxial tension shotcrete shows approximately linear-elastic behavior nearly up to peak stress, followed by brittle failure due to cracking. Compared to its uniaxial compressive strength, shotcrete exhibits a considerably lower uniaxial tensile strength. Experimental tests indicate a ratio of the uniaxial tensile strength to the uniaxial compressive strength of 10 % to 20 %. As demonstrated by means of experimental tests reported by Hammer (2018), this ratio is dependent on the degree of hydration, with higher ratios observed at younger material ages.
- Shotcrete exhibits shrinkage, which denotes the phenomenon of a time-dependent decrease in volume, independent of any applied load. According to the ACI Committee 209 (1992), shrinkage can be subdivided into autogenous shrinkage, which is the part of shrinkage which is exclusively related to the hydration in absence of moisture exchange, and drying shrinkage, which is related to moisture exchange.
- Under sustained loading, shotcrete tends to creep, which denotes the phenomenon of a load-dependent, time-dependent increase of deformations. Creep is subdivided into basic creep, which is the part of creep which is independent of any moisture exchange, and drying creep, which is the additional part of creep which is related to simultaneous drying. Similar to the evolution of stiffness and strength, the creep behavior of shotcrete is dependent on the degree of hydration, and thus dependent on the material age.

- Compared to normal concrete, nonlinear creep is an important issue in the construction of shotcrete shells: Linear creep is characterized by validity of the principle of superposition, i.e., an assumed linear relation between the creep strain accumulated in a certain time period and an applied constant stress. Experimental results on concrete, e.g. by Mazzotti and Savoia (2002), indicate that for uniaxial compression the assumption of linear creep is valid up to ratios of the applied stress to the compressive strength of approximately 30 % to 40 %, while for higher load levels a disproportionally higher increase of the creep strain rate is observed. However, while the assumption of linear creep is usually admissible for normal concrete structures, shotcrete shells may undergo considerably higher degrees of loading, which makes the principle of superposition invalid.
- The coefficient of thermal expansion is dependent on the degree of hydration, and thus dependent on the age of the material.

Advanced constitutive models aim at a realistic representation of those nonlinear, time-dependent material phenomena.

For calibrating such advanced material models, experimental results on the material behavior are necessary. However, the conduction of experimental tests on shotcrete is considerably more complex compared to normal concrete: While common, casted concrete specimens usually can be sampled under controlled laboratory conditions, shotcrete specimens must be sampled employing the same spraying process as used on the construction site since the spraying process has a substantial influence on the resulting properties of shotcrete. An attempt to simulate the spraying process under controlled laboratory conditions is reported in (Thewes and Vollmann, 2007; Nakashima et al., 2015), employing a test rig equipped with a robot arm from the automotive industry for the application of shotcrete.

1.3 State of the Art – Constitutive Models for Shotcrete

Only few constitutive models explicitly developed for shotcrete have been proposed in the literature so far.

The most simple approach for representing the material behavior of shotcrete in numerical simulations is the assumption of linear-elastic material behavior (Thomas, 2008). Commonly, a time-dependent Young's modulus is assumed to account for the evolution of material stiffness. Linear-elastic models are still widely used in engineering practice to describe the material behavior of shotcrete in tunneling simulations, and reports of application can be found in, e.g., (Schuller and Schweiger, 2002; Ng et al., 2004; Yeo et al., 2009; Kivi et al., 2012).

In this context, an early attempt to account for the time-dependent material behaviour of shotcrete in an approximate manner was proposed by Pöttler (1990b). In this relatively simple approach, a reduced Young's modulus for shotcrete is used to consider effects like time-dependent stiffness, creep, shrinkage and even the time-dependent loading during the excavation process. By combining these effects in a hypothetical modulus of elasticity (HME), approximate solutions for problems with axisymmetric boundary conditions can be obtained. Problematic is the lack of calibration parameters for the model in the literature, and thus, the HME can be based more or less only on experience.

A model for describing the evolution of stiffness and creep of shotcrete was published as the rate of flow method. The model was originally proposed by Schubert (1988) for shotcrete, and extended by Aldrian (1991) to account for nonlinear creep. In (M. Müller, 2001), the calibration of the material model based on a comprehensive experimental program on a dry-mix shotcrete composition is presented. A brief summary of the model and a critical discussion are presented in (Thomas, 2008).

A viscoplastic constitutive model for shotcrete was presented by Meschke (1996). The model is based on a hardening Drucker-Prager yield criterion for describing the material behaviour for compressive and mixed stress states, and a softening Rankine criterion is used for describing the material behaviour in tension. Associated flow rules are employed to describe the evolution of viscoplastic strains. The model takes into account time-dependent stiffness due to hydration based on hyperelastic relations, and the evolution of material strength and ductility is considered by means of empirical time functions. An application of the model is reported in (Meschke et al., 1996).

Shotcrete models based on the theory of thermo-chemo-plasticity consider a dependency of material properties on the degree of hydration. In (Ulm and Coussy, 1995), a general strategy to develop such constitutive models is documented, based on the theory of porous media proposed by Coussy (1995). A constitutive model based on thermo-chemo-plasticity theory was proposed by Hellmich, Ulm, et al. (1999a) and was further developed by Sercombe et al. (2001) and Lackner, Hellmich, et al. (2002). In (Hellmich, Ulm, et al., 1999b), a report of the application of the thermo-chemo-plastic model can be found. To consider early age cracking of shotcrete, the model was further extended by Lackner and Mang (2003, 2004).

Approaches to represent the evolution of stiffness and strength of shotcrete by means of multi-scale models were presented by Hellmich and Mang (2005) and Pichler et al. (2008).

A constitutive model based on the theories of viscoelasticity and plasticity representing nonlinear strain hardening and softening, creep and shrinkage was presented by Schütz et al. (2011). Partially based on this model, recently a similar approach was presented by Schädlich and Schweiger (2014). The model by Schädlich and Schweiger includes strain hardening and strain softening material behavior, time-dependent evolution of stiffness and strength, and nonlinear creep within the framework of viscoelasticity. To delimit the elastic domain, the Mohr-Coulomb and the Rankine yield criteria are employed, and plastic flow is modeled by means of non-associated flow rules. The model was validated and applied in (Schädlich, Schweiger, Marcher, et al., 2014a) and (Schädlich, Schweiger, Marcher, et al., 2014b).

All of the mentioned models belong to the class of continuum models, i.e., they describe the constitutive relations in terms of stress-strain relations for an infinitesimal material point. To the author's best knowledge, other approaches like discrete models, or continuum models with discontinuities, such as proposed for normal concrete, have not been presented for shotcrete in the literature so far.

1.4 Objectives of the Thesis

The scope of this thesis is to present a contribution to constitutive modeling of shotcrete in terms of a new shotcrete model, a comparison of the new model with existing models, calibration and validation of the model by means of results from a new experimental program, as well as the application of the new model in a numerical simulation of deep tunnel advance. To this end, following objectives are constituted:

- **(i)** Review on the state of the art of constitutive models for shotcrete.
- **(ii)** Proposal of a new material model for shotcrete.
- **(iii)** Implementation of the new material model and two selected models from the literature into a finite element framework.
- **(iv)** Design and performance of a new experimental program on a state-of-the-art wet-mix shotcrete composition.
- **(v)** Calibration of the three investigated material models based on the experimental results from the new experimental program.
- **(vi)** Application and comparison of the new model with the two models selected from the literature.
- **(vii)** Assessment of an advanced regularization technique for the new shotcrete model.

In particular, these objectives comprise:

(i) *A review on the state of the art and the conduction of a literature survey*. In this context, the fundamental aspects of different constitutive models and approaches for representing the material behavior of shotcrete are presented and explained. The perceptions gained within this objective form the basis for the subsequent objectives.

(ii) *The proposal of a novel damage plasticity model for shotcrete*. The new model is based on three constitutive models for concrete, which are the damage plasticity model by Grassl and Jirásek (2006a), the solidification theory for concrete creep by Bažant and Prasannan (1989a), and the semi-empirical shrinkage model by Bažant and Panula (1978). The new material model, henceforth denoted as the Shotcrete Damage Plasticity (SCDP) model, incorporates these constituents into a common framework. It represents time-dependent evolution of material properties, hardening and softening material behavior, inelastic strains, stiffness degradation due to damage, as well as nonlinear creep and shrinkage. The representation of thermal strains is neglected in the current version of the model. The model by Grassl and Jirásek is chosen as a point of departure, since its good performance was assessed and proven by Valentini (2011). For describing the evolution of stiffness and nonlinear creep, the solidification theory is employed. It is adapted in order to meet the requirements for representing the material behavior of shotcrete. Furthermore, empirical time functions for describing the evolution of strength and ductility are employed.

(iii) *The implementation of the SCDP model into a finite element framework*, ensuring a robust and efficient stress update. In addition, two shotcrete models available in the literature are selected for a comparison with the new SCDP model. The model by Meschke (1996) is chosen as a representative for a viscoplastic material model formulated in terms of associated plastic flow rules. The second model is the viscoelastic-plastic model by Schädlich and Schweiger (2014), which is the most recently proposed model in the literature prior to the SCDP model. Similar to the SCDP model, both models are implemented into a finite element framework.

(iv) *The conduction of a new experimental program*, which focuses on a modern wet-mix shotcrete composition used within the NATM. To this end, specimens are sampled directly at a construction site of the Brenner Base Tunnel. Suitable methods for sampling shotcrete specimens at tunnel construction sites are investigated, and experimental results for calibrating the shotcrete models are presented. Based on the new experimental program and based on a comparison with experimental results from the literature, the performance of the modern shotcrete composition is discussed.

(v) *The calibration of the three investigated material models* based on the test data from the new experimental program. As a result, material parameter sets for the three models for representing the material behavior of a modern wet-mix shotcrete composition are available, and the performance of the constitutive models in representing that shotcrete composition is discussed.

(vi) *The application and comparison of the investigated models*, and the assessment of the capabilities of the material models. This comparison is based on two extensive studies, which comprise

- a validation by means of a comparison with experimental results, in which the ability of the models to reproduce experimentally observed material behavior is assessed, and
- a benchmark study based on a finite element model of deep tunnel advance, which allows to evaluate the influence of the shotcrete model on the predicted mechanical response of the shotcrete shell. The finite element model is derived from a stretch of the Brenner Base Tunnel, for which in-situ measurement results on the deformations in the surrounding rock mass during the excavation process are available. A comparison with simpler approaches such as commonly used in engineering practice and with geodetic and in-situ measurement records is presented.

Within both studies, experimental results from the literature as well as the new experimental program are used for validating the material models.

(vii) *An advanced regularization scheme for the SCDP model is discussed*, as an outlook for future research activities. Since softening material behavior described by continuum models leads to a localization of deformations in finite element simulations, a proper regularization technique is required to avoid mesh-sensitive results. For the model by Meschke, the model by Schädlich and Schweiger, and the SCDP model, the crack band approach is used, as proposed in the respective publications. However, due to several issues related to the crack band approach, an alternative regularization technique based on the over-nonlocal implicit gradient-enhanced formulation is adopted for the SCDP model. This formulation was proposed and discussed by Poh and Swaddiwudhipong (2009b) for the concrete model by Grassl and Jirásek. The gradient-enhnced formulation is implemented into a finite element framework and applied to examples of mode I and mode II failure to assess its performance, and advantages and shortcomings of this approach are discussed.

Finally, the obtained results are summarized and discussed, and proposals for future research activities are given.

1.5 Outline of the Thesis

This thesis has a cumulative character, and the presented scientific publications are considered as the main part of the thesis. These publications are presented in the subsequent chapters, which are building up on each other.

This work is structured as follows: In Chapter 2 to Chapter 7, the motivation and a short summary for each publication are presented, along with the respective publication. Furthermore, the scientific contribution by the author is declared. Chapter 8 presents a summary of the numerical implementation of the investigated models into a finite element framework, since details on the numerical implementation were not part of any of the publications. Finally, Chapter 9 concludes with a summary and an outlook on recommended future research activities related to constitutive modeling of shotcrete.

1.6 Publications within the Thesis

Following publications are part of this thesis:

Publication I

Neuner M., P. Gamnitzer, and G. Hofstetter (2017). "An Extended Damage Plasticity model for Shotcrete: Formulation and Comparison with Other Shotcrete Models". *Materials* **10** (1)

Publication II

Neuner M., M. Schreter, D. Unteregger, and G. Hofstetter (2017). "Influence of the Constitutive Model for Shotcrete on the Predicted Structural Behavior of the Shotcrete Shell of a Deep Tunnel". *Materials* **10** (6)

Publication III

Theiner Y., M. Drexel, M. Neuner, and G. Hofstetter (2017). "Comprehensive study of concrete creep, shrinkage, and water content evolution under sealed and drying conditions". *Strain* **53** (2)

Publication IV

Neuner M., T. Cordes, M. Drexel, and G. Hofstetter (2017). "Time-Dependent Material Properties of Shotcrete: Experimental and Numerical Study". *Materials* **10** (9)

Publication V (accepted for publication)

Neuner M., M. Schreter, and G. Hofstetter (2018). "Comparative Investigation of Constitutive Models for Shotcrete Based on Numerical Simulations of Deep Tunnel Advance". *Numerical Methods in Geotechnical Engineering: 9th European Conference on Numerical Methods in Geotechnical Engineering*. CRC Press Taylor & Francis Group

Publication VI

Neuner M., M. Schreter, and G. Hofstetter (2017). "Evaluation of a Gradient Enhanced Damage Plasticity Model for Shotcrete". *7th GACM Colloquium on Computational Mechanics for Young Scientists from Academia and Industry*. Ed. by M. v. Scheven, M.-A. Keip, and N. Karajan. University of Stuttgart, Stuttgart, Germany

Publication VII

Schreter M., M. Neuner, and G. Hofstetter (2017). "Study of the Regularization Scheme of an Advanced Rock Model". *7th GACM Colloquium on Computational Mechanics for Young Scientists from Academia and Industry*. Ed. by M. v. Scheven, M.-A. Keip, and N. Karajan. University of Stuttgart, Stuttgart, Germany

2 The SCDP Model – A Damage Plasticity Model for Shotcrete

In Publication I, *An Extended Damage Plasticity Model for Shotcrete: Formulation and Comparison with Other Shotcrete Models*, a brief summary of material models for shotcrete is presented, and the new SCDP model is proposed.

For a comparison with the SCDP model, two models from the literature are selected, which are the model by Meschke (1996), henceforth denoted simply as the Meschke model, and the model by Schädlich and Schweiger (2014), henceforth denoted as the Schädlich model. Since the SCDP model, the Meschke model and the Schädlich model are representatives for single-field models, a multi-field model for concrete by Gawin et al. (2006a,b), calibrated to represent the material behavior of shotcrete, is included additionally in the comparison. The models are compared by means of numerical simulations of experimental tests on shotcrete specimens which were presented in the literature, in particular those by Huber (1991) and M. Müller (2001).

It is demonstrated that the SCDP model is in good agreement with the experimentally observed material behavior of shotcrete. However, it is remarked that only few experimental results are available in the literature, and that several shortcomings which complicate the calibration procedure are associated with the available test results.

The scientific contribution by the author comprises

- the literature survey on the state of the art of constitutive modeling of shotcrete, as well as a literature survey on experimental data on shotcrete,
- the development of the SCDP model, and its implementation into a finite element framework,
- the implementation of the constitutive models by Meschke and by Schädlich and Schweiger into a finite element framework,
- the calibration of the three models based on test data from the literature,
- the performance of the numerical simulations employing the three single-field models (SCDP model, Meschke model, Schädlich model),
- the evaluation of the comparison of the four models,
- the preparation of the manuscript, jointly with Peter Gamnitzer and Günter Hofstetter.

materials

MDPI

Article

An Extended Damage Plasticity Model for Shotcrete: Formulation and Comparison with Other Shotcrete Models

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Abstract: The aims of the present paper are (i) to briefly review single-field and multi-field shotcrete models proposed in the literature; (ii) to propose the extension of a damage-plasticity model for concrete to shotcrete; and (iii) to evaluate the capabilities of the proposed extended damage-plasticity model for shotcrete by comparing the predicted response with experimental data for shotcrete and with the response predicted by shotcrete models, available in the literature. The results of the evaluation will be used for recommendations concerning the application and further improvements of the investigated shotcrete models and they will serve as a basis for the design of a new lab test program, complementing the existing ones.

Keywords: shotcrete; constitutive model; experimental data; tunnel advance; numerical simulation

1. Introduction

The use of shotcrete for supporting the surrounding rock or soil of a tunnel is an essential part of the New Austrian Tunneling method (NATM). Hence, for numerical simulations of tunnel advance based on the NATM, in addition to a constitutive model for the rock mass, a constitutive model for shotcrete is required for representing the mechanical behavior of the shotcrete shell. Shotcrete is characterized by continuously developing material properties during hydration, nonlinear stress–strain relations, time-dependent material behavior due to creep and shrinkage and fracture. In contrast to concrete, shotcrete applied for securing excavations in tunneling is already loaded at very early ages. Compared to the large body of literature on experimental investigations and constitutive modeling of concrete, only a relatively small number of experiments, focusing on the characterization of the complex material behavior of shotcrete, and only a few constitutive models for shotcrete have been reported in the literature up to now.

An early attempt to account for the time-dependent shotcrete behavior in numerical simulations of tunnel advance was proposed in [1]. In this practical approach, a reduced Young's modulus for shotcrete is used to consider effects such as evolving stiffness and strength during hydration, creep, shrinkage and even effects due to time-dependent loading caused by consecutive excavation steps. By combining these effects in a hypothetical modulus of elasticity (HME), approximate axisymmetric solutions for tunneling can be obtained. Problematic is the lack of calibration parameters for the model in the literature, and thus, the magnitude of the HME is commonly based only on experience.

Constitutive models for shotcrete, formulated within the framework of continuum mechanics, can be broadly subdivided into single-field and multi-field models. The former only focus on the mechanical behavior by providing time-dependent stress–strain relations, whereas the latter account for interactions between the mechanical behavior and chemical, thermal and hygral processes, related to hydration and influenced by the ambient conditions.

Representatives of the single-field shotcrete models are the shotcrete models, proposed by Meschke [2], by Schütz et al. [3] and, partially based on the latter, the recently proposed shotcrete model by Schädlich and Schweiger [4]. The single-field shotcrete models have in common the description of the evolution of material stiffness and strength in terms of the shotcrete age, the application of a yield surface for delimiting the domain of elastic material behavior and consideration of hardening and/or softening material behavior, the latter being regularized within the framework of the Finite Element Method (FEM) by means of the specific fracture energy and a characteristic element length. Main differences of the models proposed in [2,4], are related to the applied type of the flow rule and to the employed theory for modeling creep.

In the single-field shotcrete models, several phenomena, such as the evolution of mechanical properties, shrinkage and creep are described separately by empirical relations, ignoring interactions among them. Distinctions between autogenous shrinkage and drying shrinkage and between basic creep and drying creep are only made indirectly by adjusting the ultimate shrinkage strain or the creep coefficient. Furthermore, hydration and aging of shotcrete do not only depend on the shotcrete age but also on the water content of shotcrete and on temperature. Temperature, in turn, is increased by hydration whereas the water content is decreased. Both temperature and water content also depend on the ambient conditions. Autogenous shrinkage depends on hydration. Drying shrinkage is caused by the increase of capillary stress, acting in the unsaturated pores of the porous material concrete, due to the decrease of the degree of water saturation in the pores of the porous material concrete. Creep depends on hydration and moisture content, in addition to the acting sustained stress. The increase of capillary stress due to the decrease in moisture content in the pores is related to the Pickett effect [5] observed in drying creep. Hence, it follows that contrary to the simplifying assumptions inherent in the single-field models, these phenomena are actually coupled and because of the interactions and the dependency on the ambient conditions, the material properties are not uniformly distributed in a shotcrete shell.

For the above reasons, an alternative to single-field shotcrete models are multi-field models. In the latter, concrete is described as a porous material, consisting of several interacting phases, namely the solid phase with an embedded pore structure, containing one or several fluid phases (water phase, dry air phase and vapor phase). The first multi-field model for shotcrete was proposed by Hellmich et al. [6]. Considering the dependency of shotcrete properties on the degree of hydration, i.e., on the time- and temperature-dependent chemical reaction between cement and water, it is based on thermo–chemo-plasticity theory along the lines of the thermo–chemo-mechanical framework, developed by Ulm and Coussy [7]. This model was further developed in [8] to consider early age cracking of shotcrete. An even broader framework for concrete was proposed by Gawin et al. [9,10] by including hygral effects, resulting in a hygro–thermo–chemo-mechanical concrete model. The latter was modified by several authors, e.g., recently by Sciume et al. [11] for modeling repair work.

Similar to the available literature on shotcrete models, also the number of experimental studies on the early age behavior of shotcrete is limited up to the present day. Hence, it is not surprising that most of the mentioned shotcrete models were calibrated mainly on the basis of the same sets of results of laboratory tests and, due to the limited experimental data for shotcrete, frequently experimental results for normal concrete were used for calibration. Results of experimental programs, specifically devoted to shotcrete, are documented in the following publications:

- Sezaki et al. [12] published test results on the evolution of Young's modulus and uniaxial compressive strength up to the age of 28 days and stress–strain relations from short-term uniaxial and triaxial compression tests on specimens of different age.
- Aldrian [13] and Golser et al. [14] presented experimental results on the evolution of the uniaxial compressive strength up to the age of 28 days, results of uniaxial compression tests and of the evolution of the total strain in creep and shrinkage tests.

- Huber [15] investigated the evolution of temperature due to hydration, of the Young's modulus and of the uniaxial compressive strength up to the age of 7 days as well as the evolution of the total strain in shrinkage and creep tests.
- Fischnaller [16] presented test results on the evolution of Young's modulus and uniaxial compressive strength and the results of relaxation and shrinkage tests up to the age of 7 days.
- Müller [17] published test results on the evolution of stiffness and strength, results of short-term uniaxial compression tests and the evolution of the total strain in shrinkage and creep tests.

In the mentioned laboratory tests on the shrinkage behavior of shotcrete, conducted on sealed and unsealed specimens, only the evolution of the total strain is reported. Thus, in the case of sealed specimens, the measured strain consists of the autogenous shrinkage strain and the thermally induced strain due to the hydration heat for very early shotcrete ages. In the case of unsealed specimens, furthermore the drying shrinkage strain is included in the measured total strain. Similarly, the reported measured total strain of creep tests includes a combination of strain components induced by different phenomena. In the case of creep tests on sealed specimens, they consist of the basic creep strain in addition to the strain components measured in the shrinkage tests on sealed specimens. In the case of creep tests on unsealed specimens, they include the basic creep strain and drying creep strain in addition to the strain components measured in the shrinkage tests on unsealed specimens.

The paper is organized as follows. In the next section, the damage-plasticity model for concrete, proposed by Grassl and Jirásek [18], is extended to meet the specific requirements of a constitutive model for shotcrete concerning the time-dependent evolution of material properties and of the strain due to creep and shrinkage. In Section 3, the performance of the proposed extended damage plasticity model for shotcrete is evaluated by comparing the predicted response with experimental data and with the response of the shotcrete models proposed by Meschke [2] and by Schädlich et al. [4] and with the response of the multi-field concrete model, proposed by Gawin et al. [9,10]. In the interests of brevity, equations for the latter models will not be provided as they are available in the respective papers for the interested reader. The comparison of experimental and numerical results will be used for recommendations concerning the application and further improvement of the investigated shotcrete models and it will serve as a basis for the design of a new lab test program, complementing the existing ones.

2. An Extended Damage Plasticity Model for Shotcrete

The damage plasticity model serves as the starting point for the formulation of a constitutive model for shotcrete, proposed by Grassl and Jirásek [18] for describing the time-independent material behavior of concrete. The model is here denoted as the Concrete Damage Plastic (CDP) model. In the latter, the nonlinear mechanical behavior of concrete is modeled by means of a combination of plasticity theory and the theory of damage mechanics. The smooth yield surface is expanding during hardening until at peak stress it attains the triaxial strength envelope for concrete, proposed by Menétrey and Willam [19]. The plastic strain rate is determined by means of a non-associated flow rule. Beyond peak strength, stiffness degradation and softening material behavior are described by means of an isotropic damage model.

The CDP model is extended by considering the aging of shotcrete and creep by means of the solidification theory [20] and shrinkage on the basis of the Bažant–Panula model [21]. To improve the representation of the evolution of the material properties of shotcrete, especially up to the age of 24 h, several modifications are proposed. The extended model is denoted as Shotcrete Damage Plasticity (SCDP) model.

The solidification theory is incorporated into the damage-plasticity framework by modeling creep in the effective stress space. The nonlinear stress–strain relation is given in total form as

$$\boldsymbol{\sigma} = (1-\omega)\,\mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}} - \boldsymbol{\varepsilon}^{\mathrm{ve}} - \boldsymbol{\varepsilon}^{\mathrm{f}} - \boldsymbol{\varepsilon}^{\mathrm{shr}}). \tag{1}$$

The total strain $\boldsymbol{\varepsilon}$ is decomposed into the elastic strain $\boldsymbol{\varepsilon}^{\mathrm{el}} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}} - \boldsymbol{\varepsilon}^{\mathrm{ve}} - \boldsymbol{\varepsilon}^{\mathrm{f}} - \boldsymbol{\varepsilon}^{\mathrm{shr}}$, the plastic strain $\boldsymbol{\varepsilon}^{\mathrm{p}}$, the (due to aging only partially recoverable) viscoelastic strain $\boldsymbol{\varepsilon}^{\mathrm{ve}}$, the flow strain $\boldsymbol{\varepsilon}^{\mathrm{f}}$ and the shrinkage strain $\boldsymbol{\varepsilon}^{\mathrm{shr}}$. The nominal stress tensor $\boldsymbol{\sigma}$ is related to the effective stress tensor $\bar{\boldsymbol{\sigma}}$ by the isotropic scalar variable ω as

$$\boldsymbol{\sigma} = (1 - \omega)\bar{\boldsymbol{\sigma}}. \tag{2}$$

To account for the evolution of material properties of shotcrete, especially during the hydration, several extensions and improvements are presented. They comprise the modification of the volume function $v(t)$ of the solidification theory and of the ductility formulation of the CDP model and the proposal of a law for the evolution of the material strength.

2.1. Damage Plasticity Framework

The plastic response, governed by the yield function for delimiting the elastic domain, the plastic potential function for controlling the evolution of the plastic strain and the hardening law are adopted from the CDP model [18]. In contrast to the latter, the uniaxial compressive strength $f_{\mathrm{cu}}(t)$, the uniaxial yield stress $f_{\mathrm{cy}}(t)$, the biaxial compressive strength $f_{\mathrm{cb}}(t)$, the uniaxial tensile strength $f_{\mathrm{tu}}(t)$ and the plastic strain $\varepsilon^{\mathrm{p}}_{\mathrm{cpu}}(t)$ at uniaxial compressive peak stress are assumed to be dependent on time t. Hence, the time-independent yield function f_{p} and the plastic potential g_{p} of the CDP model are replaced by

$$\begin{aligned} f_{\mathrm{p}}(\bar{\sigma}_{\mathrm{m}}, \bar{\rho}, \theta, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) = & \left((1 - q_{\mathrm{h}}(\alpha_{\mathrm{p}})) \left(\frac{\bar{\rho}}{\sqrt{6} f_{\mathrm{cu}}(t)} + \frac{\bar{\sigma}_{\mathrm{m}}}{f_{\mathrm{cu}}(t)} \right)^2 + \sqrt{\frac{3}{2}} \frac{\bar{\rho}}{f_{\mathrm{cu}}(t)} \right)^2 \\ & + m_0\, q_{\mathrm{h}}^2(\alpha_{\mathrm{p}}) \left(\frac{\bar{\rho}}{\sqrt{6} f_{\mathrm{cu}}(t)} r(\theta) + \frac{\bar{\sigma}_{\mathrm{m}}}{f_{\mathrm{cu}}(t)} \right) - q_{\mathrm{h}}^2(\alpha_{\mathrm{p}}), \end{aligned} \tag{3}$$

$$\begin{aligned} g_{\mathrm{p}}(\bar{\sigma}_{\mathrm{m}}, \bar{\rho}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) = & \left((1 - q_{\mathrm{h}}(\alpha_{\mathrm{p}})) \left(\frac{\bar{\rho}}{\sqrt{6} f_{\mathrm{cu}}(t)} + \frac{\bar{\sigma}_{\mathrm{m}}}{f_{\mathrm{cu}}(t)} \right)^2 + \sqrt{\frac{3}{2}} \frac{\bar{\rho}}{f_{\mathrm{cu}}(t)} \right)^2 \\ & + q_{\mathrm{h}}^2(\alpha_{\mathrm{p}}) \left(\frac{m_0\, \bar{\rho}}{\sqrt{6} f_{\mathrm{cu}}(t)} + \frac{m_{\mathrm{g}}(\bar{\sigma}_{\mathrm{m}})}{f_{\mathrm{cu}}(t)} \right). \end{aligned} \tag{4}$$

They are formulated in terms of three invariants of the effective stress tensor, i.e., the effective mean stress $\bar{\sigma}_{\mathrm{m}}$, the effective deviatoric radius $\bar{\rho}$ and the Lode angle θ, and the stress-like internal variable q_{h}, the latter in terms of the strain-like internal variable α_{p}. $r(\theta)$ describes the shape of the yield surface in deviatoric planes, and m_0 and m_{g} are model parameters, dependent on the material parameters $f_{\mathrm{cu}}(t)$, $f_{\mathrm{cy}}(t)$, $f_{\mathrm{cb}}(t)$ and $f_{\mathrm{tu}}(t)$. The rate of the plastic strain is given as

$$\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}} = \dot{\lambda}\, \frac{\partial g_{\mathrm{p}}(\bar{\sigma}_{\mathrm{m}}, \bar{\rho}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t)}{\partial \bar{\boldsymbol{\sigma}}} \tag{5}$$

and the modified evolution law for α_{p} is adopted from [22]:

$$\dot{\alpha}_{\mathrm{p}} = \|\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}}(t)\|\, \frac{1}{x_{\mathrm{h}}(\bar{\sigma}_{\mathrm{m}}, t)} \left(1 + 3 \frac{\bar{\rho}^2}{\bar{\rho}^2 + \vartheta_{\mathrm{h}}} \cos^2(1.5\,\theta) \right), \tag{6}$$

with ϑ_{h} as a small disturbance parameter to avoid division by zero in the case of hydrostatic stress. The ductility function $x_{\mathrm{h}}(\bar{\sigma}_{\mathrm{m}})$ controls the magnitude of $\dot{\alpha}_{\mathrm{p}}$ dependent on the acting hydrostatic stress:

$$x_h(\bar{\sigma}_m) = \begin{cases} A_h - (A_h - B_h)\exp\left(\dfrac{-R_h(\bar{\sigma}_m)}{C_h}\right) & \text{if } R_h(\bar{\sigma}_m) \geq 0, \\ (B_h - D_h)\exp\left(\dfrac{R_h(\bar{\sigma}_m)(A_h - B_h)}{(B_h - D_h)\,C_h}\right) + D_h & \text{otherwise,} \end{cases} \tag{7}$$

$$R_h(\bar{\sigma}_m) = -\frac{\bar{\sigma}_m}{f_{cu}} - \frac{1}{3},$$

in which A_h, B_h, C_h and D_h are model parameters, which are calibrated from uniaxial tensile and uniaxial and triaxial compressive tests. Assuming fixed values of A_h, C_h and D_h, the material ductility in the pre-peak regime of the stress–strain relation is governed by the model parameter B_h. Although the actual relation between ε^p_{cpu}—the plastic strain at peak stress in uniaxial compression—and B_h is nonlinear, an approximative linear relation is proposed by Grassl and Jirásek. However, this linear relation is valid only for a limited range of ε^p_{cpu}, from 0 to approximately -0.002. To extend the valid range of the approximative law, an extension is proposed as

$$B_h(\varepsilon^p_{cpu}) = \begin{cases} -2.29\,\varepsilon^p_{cpu} + 0.000\,46 & \text{if } \varepsilon^p_{cpu} > -0.002, \\ -1.55\,(\varepsilon^p_{cpu} + 0.002) + 0.005\,04 & \text{if } \varepsilon^p_{cpu} \leq -0.002. \end{cases} \tag{8}$$

The evolution law for $\varepsilon^p_{cpu}(t)$ is adopted from the model by Schädlich and Schweiger [4], as experimental results for a more refined law are not available: It is based on linear interpolation of ε^p_{cpu} between the values at the age of 1 h, 8 h and 24 h, denoted as $\varepsilon^{p(1)}_{cpu}$, $\varepsilon^{p(8)}_{cpu}$ and $\varepsilon^{p(24)}_{cpu}$, respectively. For $t < 1$ h and $t > 24$ h, ε^p_{cpu} is assumed to be constant.

The softening behavior is adopted from the CDP model [18], relating the isotropic damage variable ω to the internal strain-like softening variable α_d by the exponential softening law

$$\omega = 1 - \exp\left(-\frac{\alpha_d}{\varepsilon_f}\right). \tag{9}$$

ε_f is the softening modulus, controlling the slope of the softening curve. The evolution of the strain-like internal variable α_d is related to the rate of the volumetric plastic strain $\dot{\varepsilon}^p_V$ by

$$\dot{\alpha}_d = \begin{cases} 0 & \text{if } \alpha_p < 1, \\ \dot{\varepsilon}^p_V / x_s(\dot{\boldsymbol{\varepsilon}}^p) & \text{otherwise.} \end{cases} \tag{10}$$

Function $x_s(\dot{\boldsymbol{\varepsilon}}^p)$ represents a ductility measure:

$$x_s(\dot{\boldsymbol{\varepsilon}}^p) = \begin{cases} 1 + A_s\,R_s^2(\dot{\boldsymbol{\varepsilon}}^p) & \text{if } R_s(\dot{\boldsymbol{\varepsilon}}^p) < 1, \\ 1 - 3\,A_s + 4\,A_s\sqrt{R_s(\dot{\boldsymbol{\varepsilon}}^p)} & \text{otherwise.} \end{cases} \tag{11}$$

A_s is a model parameter to be calibrated from uniaxial tensile tests and $R_s(\dot{\boldsymbol{\varepsilon}}^p)$ denotes the ratio of the *negative* volumetric plastic strain rate to the total volumetric plastic strain rate:

$$R_s(\dot{\boldsymbol{\varepsilon}}^p) = \frac{\dot{\varepsilon}^{p\ominus}_V}{\dot{\varepsilon}^p_V} \quad \text{with} \quad \dot{\varepsilon}^{p\ominus}_V = \sum_{i=\mathrm{I}}^{\mathrm{III}} \left\langle -\dot{\varepsilon}^p_{(i)} \right\rangle, \tag{12}$$

the latter computed from the principal values of the plastic strain rate tensor.

2.2. Evolution of Material Strength

The evolution of $f_{\mathrm{cu}}(t)$ is described by a modified version of the law originally proposed in [23] for the evolution of the Young's modulus:

$$f_{\mathrm{cu}}(t) = f_{\mathrm{cu}}^{(28)}\,\beta_{\mathrm{f}}(t), \qquad \beta_{\mathrm{f}}(t) = \begin{cases} \beta_{\mathrm{f}}^{\mathrm{I}} = r_{\mathrm{f}} + c_{\mathrm{f}}t + d_{\mathrm{f}}t^2 & \text{if } t \leq t_{\mathrm{f}}, \\ \beta_{\mathrm{f}}^{\mathrm{II}} = \left(a_{\mathrm{f}} + \dfrac{b_{\mathrm{f}}}{t - \Delta t_{\mathrm{f}}}\right)^{-0.5} & \text{if } t_{\mathrm{f}} < t \leq 28\,\mathrm{d}, \\ \beta_{\mathrm{f}}^{\mathrm{III}} = 1 & \text{otherwise.} \end{cases} \tag{13}$$

Supplementing the original formulation in [23], a residual parameter r_{f} ensures a non-zero compressive strength already at zero age. A default value is proposed as $r_{\mathrm{f}} = 10^{-2}$, resulting in a uniaxial compressive strength of 1 % of $f_{\mathrm{cu}}^{(28)}$ at $t = 0$. The model parameters are given as

$$\begin{aligned} a_{\mathrm{f}} &= \frac{1 - \frac{28-\Delta t_{\mathrm{f}}}{1-\Delta t_{\mathrm{f}}}\left(f_{\mathrm{cu}}^{(1)}/f_{\mathrm{cu}}^{(28)}\right)^2}{\left(1 - \frac{28-\Delta t_{\mathrm{f}}}{1-\Delta t_{\mathrm{f}}}\right)\left(f_{\mathrm{cu}}^{(1)}/f_{\mathrm{cu}}^{(28)}\right)^2}, & b_{\mathrm{f}} &= (28 - \Delta t_{\mathrm{f}})(1 - a_{\mathrm{f}}), \\ c_{\mathrm{f}} &= \left.\frac{\mathrm{d}\beta_{\mathrm{f}}^{\mathrm{II}}}{\mathrm{d}t}\right|_{t=t_{\mathrm{f}}} - 2d_{\mathrm{f}}t_{\mathrm{f}}, & d_{\mathrm{f}} &= \left(\left.\frac{\mathrm{d}\beta_{\mathrm{f}}^{\mathrm{II}}}{\mathrm{d}t}\right|_{t=t_{\mathrm{f}}} t_{\mathrm{f}} - (\beta_{\mathrm{f}}^{\mathrm{II}} - r_{\mathrm{f}})\right)/t_{\mathrm{f}}^2. \end{aligned} \tag{14}$$

$f_{\mathrm{cu}}^{(1)}$ and $f_{\mathrm{cu}}^{(28)}$ denote the experimentally determined values for $f_{\mathrm{cu}}(t)$ at the age of 1 day and 28 days, respectively. Parameters t_{f} and Δt_{f} (given in days) control the delayed start of the evolution of compressive strength. Default values are proposed as $t_{\mathrm{f}} = 0.25\,\mathrm{d}$ and $\Delta t_{\mathrm{f}} = 0.18\,\mathrm{d}$. Employing these values, monotonic growth of $f_{\mathrm{cu}}(t)$ is ensured only for ratios $f_{\mathrm{cu}}^{(1)}/f_{\mathrm{cu}}^{(28)} \geq 0.16$, which are usually met in practical applications.

Regarding $f_{\mathrm{cy}}(t)$, $f_{\mathrm{cb}}(t)$ and $f_{\mathrm{tu}}(t)$, it is assumed that ratios $f_{\mathrm{cy}}(t)/f_{\mathrm{cu}}(t)$, $f_{\mathrm{cb}}(t)/f_{\mathrm{cu}}(t)$ and $f_{\mathrm{tu}}(t)/f_{\mathrm{cu}}(t)$ remain constant throughout the hydration process, and thus, they are equal to the respective ratios at the age of 28 days.

2.3. Aging Material Behavior and Creep Strain

The evolution laws for the elastic strain, the viscoelastic strain and the flow strain are adopted from the solidification theory for concrete aging and creep [20]. Since they are incorporated into the damage plasticity model, they are formulated in the effective stress space:

$$\dot{\boldsymbol{\varepsilon}}^{\mathrm{el}}(t) = q_1\,\mathbb{C}_{\nu}^{-1} : \dot{\bar{\boldsymbol{\sigma}}}(t), \tag{15a}$$

$$\dot{\boldsymbol{\varepsilon}}^{\mathrm{ve}}(t) = \frac{F(\bar{\boldsymbol{\sigma}}(t))}{v(t)} \int_0^t \dot{\Phi}(t - t')\,\mathbb{C}_{\nu}^{-1} : \mathrm{d}\bar{\boldsymbol{\sigma}}(t'), \tag{15b}$$

$$\dot{\boldsymbol{\varepsilon}}^{\mathrm{f}}(t) = \frac{q_4\,F(\bar{\boldsymbol{\sigma}}(t))}{t}\,\mathbb{C}_{\nu}^{-1} : \bar{\boldsymbol{\sigma}}(t). \tag{15c}$$

$\mathbb{C}_{\nu}$ denotes the elastic unit stiffness tensor [24]. The volume function $v(t)$, describing the evolution of the load bearing volume fraction of the hydrated material, and function $\Phi(t - t')$ are given as

$$v(t) = \left(\left(\frac{1}{t}\right)^{0.5} + \frac{q_3}{q_2}\right)^{-1}, \tag{16}$$

$$\Phi(t - t') = q_2\,\ln(1 + (t - t')^{0.1}). \tag{17}$$

q_1, q_2, q_3 and q_4 in (15)–(17) are the compliance parameters.

Function $F(\bar{\sigma}(t))$ in (15) controls the nonlinear dependency of the creep strain rate on the acting effective stress. Adapting the proposed expression for $F(\sigma(t))$ for uniaxial compressive states in [20] to effective stresses yields

$$F(\bar{\sigma}(t)) = 1 + s(t)^2, \tag{18}$$

with $s(t) = \bar{\sigma}(t)/f_{\mathrm{cu}}(t)$ denoting the ratio of the acting effective uniaxial compressive stress $\bar{\sigma}(t)$ over $f_{\mathrm{cu}}(t)$. Equation (18) is sufficient for the numerical simulation of the uniaxial creep tests, presented in Section 3. However, the extension to multi-axial stress states is pending. The compliance parameters q_1 to q_4 may be computed according to the estimation procedures presented in [25,26], or alternatively they are calibrated on the basis of experimental data. However, even if q_2 and q_3 are identified from experimental results, the evolution of the Young's modulus of shotcrete at very early ages is not well represented by $v(t)$. Especially up to the age of approximately 3 h, the Young's modulus is overestimated, whereas beyond the age of 3 h the rapid evolution of Young's modulus is heavily underestimated. Indeed, this fact is not surprising as neither the solidification theory nor the respective parameter estimation procedure are intended to represent the material behavior at such early material ages. Furthermore, usually shotcrete exhibits an increased hydration speed compared to normal concrete due to an added accelerator. Hence, $v(t)$ is modified based on a time transformation, by analogy to the methods used in [27] to account for temperature effects on the evolution of the creep strain. To this end, $v(t)$ is replaced by $v(\tau(t))$ with $\tau(t)$ as a time transformation function.

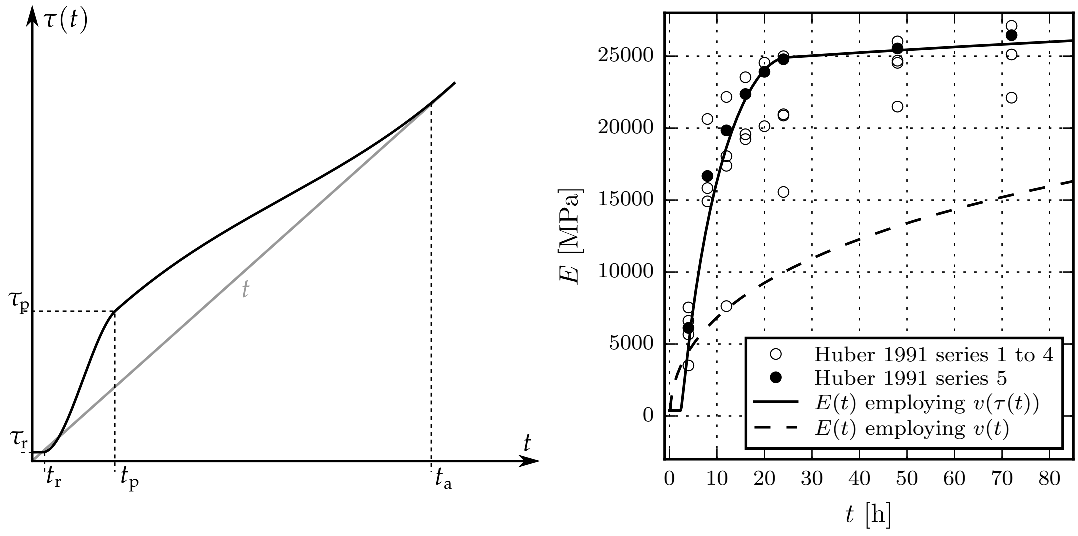
The proposed modification is characterized by (i) an improved representation of the initially low stiffness up to approximately 3 h, which is overestimated by the original volume function; (ii) an increased hydration speed after the initially delayed hydration speed; and (iii) the introduction of a new material parameter τ_{p} to calibrate the volume growth on the basis of early age experimental data.

With the transformation function only affecting the early age behavior of the material, with ongoing hydration the modified volume function $v(\tau(t))$ approaches the original volume function $v(t)$. Hence, the calibration scheme proposed in [25] is valid for ages beyond the period of accelerated hydration.

The time transformation function $\tau(t)$ is defined as

$$\tau(t) = \begin{cases} \tau_{\mathrm{r}} & \text{if } t \leq t_{\mathrm{r}}, \\ \dfrac{-2\tau_{\mathrm{p}} + 2\tau_{\mathrm{r}} + \Delta t_{\mathrm{p}}}{\Delta t_{\mathrm{p}}^3}(t - t_{\mathrm{r}})^3 + \dfrac{3\tau_{\mathrm{p}} - 3\tau_{\mathrm{r}} - \Delta t_{\mathrm{p}}}{\Delta t_{\mathrm{p}}^2}(t - t_{\mathrm{r}})^2 + \tau_{\mathrm{r}} & \text{if } t_{\mathrm{r}} < t \leq t_{\mathrm{p}}, \\ \dfrac{2(\tau_{\mathrm{p}} - 1)}{\Delta t_{\mathrm{a}}^3}(t - t_{\mathrm{p}})^3 - \dfrac{3(\tau_{\mathrm{p}} - 1)}{\Delta t_{\mathrm{a}}^2}(t - t_{\mathrm{p}})^2 + t - t_{\mathrm{p}} + \tau_{\mathrm{p}} & \text{if } t_{\mathrm{p}} < t \leq t_{\mathrm{a}}, \\ t & \text{otherwise}, \end{cases} \tag{19}$$

with $\Delta t_{\mathrm{p}} = t_{\mathrm{p}} - t_{\mathrm{r}}$ and $\Delta t_{\mathrm{a}} = t_{\mathrm{a}} - t_{\mathrm{p}}$. To ensure monotonic growth of $\tau(t)$, condition $t_{\mathrm{a}} > 1.5\,(\tau_{\mathrm{p}} - 1) + t_{\mathrm{p}}$ must be satisfied. $\tau(t)$ and the parameters $(t_{\mathrm{r}}, \tau_{\mathrm{r}})$, $(t_{\mathrm{p}}, \tau_{\mathrm{p}})$ and t_{a}, controlling its behavior, are illustrated in Figure 1a.



(**a**) (**b**)

Figure 1. (**a**) Illustration of the time transformation function $\tau(t)$ (black curve) versus time t (straight gray line); (**b**) Evolution of the Young's modulus $E(t)$, based on $v(t)$ and $v(\tau(t))$, and comparison with experimental data by Huber [15].

The parameters of the transformation function are determined as follows: By assuming $t_\mathrm{p} = 1\,\mathrm{d}$, $\tau_\mathrm{p} = \tau(1\,\mathrm{d})$ can be related to the Young's modulus $E^{(1)}$ at the age of one day, which can be determined in laboratory tests. $E^{(1)}$ can be approximated by the inverse compliance, computed for a short time interval Δt according to [25] as

$$E^{(1)} \approx J(1\,\mathrm{d} + \Delta t, 1\,\mathrm{d})^{-1} \tag{20}$$

with the compliance function $J(t + \Delta t, t)$ given according to [25] as

$$J(t,t') = q_1 + q_2\, Q(t,t') + q_3\, \ln(1 + (t-t')^{0.1}) + q_4\, \ln\left(\frac{t}{t'}\right). \tag{21}$$

$J(t,t')$ relates a uniaxial constant stress $\bar{\sigma}$, applied at time t' to the uniaxial total strain at time t as

$$\varepsilon(t) = \varepsilon^{\mathrm{ve}}(t) + \varepsilon^{\mathrm{f}}(t) + \varepsilon^{\mathrm{el}}(t) = J(t,t')\,\bar{\sigma}, \tag{22}$$

assuming linear viscoelastic material behavior and neglecting the shrinkage strain. Equation (22) is based on the integration of (15) for the special case of uniaxial stress, with the initial condition $\varepsilon(t') = q_1\,\bar{\sigma}$.

$Q(t,t')$ may be approximated for $t - t' \ll t'$ according to [20] as

$$Q(t,t') \approx t'^{-0.5}\, \ln\left(1 + (t-t')^{0.1}\right). \tag{23}$$

For short durations $\Delta t = t - t'$, the term referring to the evolution of the flow strain, i.e., the last term in (21), may be neglected. Substituting (23) into (21) and replacing $v(t)$ by $v(\tau(t))$ in (15), $J(1\,\mathrm{d} + \Delta t, 1\,\mathrm{d})$ can be rewritten as

$$J(1\,\mathrm{d} + \Delta t, 1\,\mathrm{d}) = q_1 + q_2\, \tau(1\,\mathrm{d})^{-0.5}\, \ln\left(1 + \Delta t^{0.1}\right) + q_3\, \ln\left(1 + \Delta t^{0.1}\right), \tag{24}$$

in which t' is replaced by $\tau(t') = \tau(1\,\text{day})$. Making use of (20) and (24) and solving for $\tau(1\,\text{d})$ leads to

$$\tau(1\,\text{d}) = \tau_{\text{p}} = \left(\frac{1/E^{(1)} - q_1}{q_2 \ln(1 + \Delta t^{0.1})} - \frac{q_3}{q_2} \right)^{-2}. \quad (25)$$

Regarding the remaining parameters, experimental data on shotcrete, provided in [15], indicates good agreement by choosing $t_{\text{r}} = 0.1\,\text{d}$ and $\tau_{\text{r}} = 10^{-2}\,\text{d}$. For t_{a}, the relation $t_{\text{a}} = \max(28, 3\,(\tau(1)\text{-}\,1)\,{+}1)$ ensures a smooth transition to the original function $v(t)$ at a shotcrete age of 28 days or beyond.

Figure 1b shows a comparison of the evolution of the Young's modulus employing the original volume function and the modified volume function. The compliance parameters, identified from test series 5 in [15], are $q_1 = 16.43$, $q_2 = 206.34$ and $q_3 = 2.61$ (all in $10^{-6}\,\text{MPa}^{-1}$) and the effective Young's modulus at the age of 1 day is computed for a time period of $\Delta t = 10^{-3}\,\text{d}$ as $E^{(1)} = 24{,}780\,\text{MPa}$. In addition, in Figure 1b, the experimental results for test series 1 to 4 in [15] are shown, which serve for validation of the proposed approach.

2.4. Shrinkage

The shrinkage strain is computed by means of the law for concrete, proposed by Bažant and Panula [21], as

$$\boldsymbol{\varepsilon}^{\text{shr}}(t) = \mathbf{I}\,\varepsilon^{\text{shr}}_{\infty}\,k_{\text{h}}\,S(t - t_0, \tau_{\text{shr}}), \quad (26)$$

in which $\varepsilon^{\text{shr}}_{\infty}$ denotes the ultimate shrinkage strain, k_{h} a humidity-dependent scaling factor, $S(t - t_0, \tau_{\text{shr}})$ the square-root hyperbolic law, dependent on time t, start time of drying t_0 and shrinkage half time τ_{shr}, and $\mathbf{I}$, the second order unit tensor. An estimation procedure for determining the material parameters, based on environmental conditions and the concrete mixture, is presented in [21].

3. Comparison of the New Shotcrete Model with Other Shotcrete Models

3.1. Brief Review of the Shotcrete Models Considered for the Comparison

3.1.1. Viscoplastic Shotcrete Model by Meschke

The total strain $\boldsymbol{\varepsilon}$ is decomposed into the elastic strain $\boldsymbol{\varepsilon}^{\text{el}}$, the aging induced irrecoverable strain $\boldsymbol{\varepsilon}^{\text{t}}$, the shrinkage strain $\boldsymbol{\varepsilon}^{\text{shr}}$, the (visco)plastic strain $\boldsymbol{\varepsilon}^{\text{vp}}$ and the thermally induced strain $\boldsymbol{\varepsilon}^{\theta}$. The evolution of stiffness is modeled by hyperelastic constitutive relations. In this context, the strain is decomposed into a recoverable elastic part $\boldsymbol{\varepsilon}^{\text{el}}$ and an aging induced irrecoverable part $\boldsymbol{\varepsilon}^{\text{t}}$. The hyperelastic constitutive relations are specified in total form, relating the stress to the elastic strain by the stiffness tensor determined at the age of 28 days. The evolution of the Young's modulus is approximated by a modified version of the recommendation of the CEB-FIP model code 1990 [28]. It is based on the material parameters $E^{(1)}$ and $E^{(28)}$, denoting the Young's modulus determined at the age of 1 day and 28 days, respectively, and two parameters t_{E} and Δt_{E}, controlling the shape of the evolution function. The evolution of the uniaxial compressive strength is described by the ÖVBB recommendation [29] up to the shotcrete age of 24 h and by the relation proposed in [30] afterwards, based on the material parameters $f^{(1)}_{\text{cu}}$ and $f^{(28)}_{\text{cu}}$, denoting the uniaxial compressive strength at the age of 1 day and 28 days, respectively. The evolution of the uniaxial tensile strength is also based on the proposal in [30].

Nonlinear mechanical behavior of both hardening and hardened shotcrete is described on the basis of multisurface viscoplasticity theory. A hardening Drucker–Prager model is used for predominant compressive stress states and mixed stress states and a softening Rankine criterion for predominant tensile stress states to model cracking. The plastic strain rate is determined by means of an associated flow rule.

Shrinkage of shotcrete is taken into account on the basis of the semi-empirical model proposed by Bažant and Panula [21], identical to the SCDP model. Creep of shotcrete is modeled by a Duvaut–Lions type viscoplastic formulation. Basically, the viscoplastic strain rate depends on the difference between the stress, computed by assuming elastic material behavior, and the stress, determined by rate-independent plasticity, and a viscosity parameter η. Hence, it is assumed that no viscous deformation occurs in the case of elastic material behavior. The model is denoted here as the Meschke model.

3.1.2. Shotcrete Model by Schädlich and Schweiger

The total strain $\boldsymbol{\varepsilon}$ is decomposed into the elastic strain $\boldsymbol{\varepsilon}^{\mathrm{el}}$, the shrinkage strain $\boldsymbol{\varepsilon}^{\mathrm{shr}}$, the plastic strain $\boldsymbol{\varepsilon}^{\mathrm{p}}$ and the creep strain $\boldsymbol{\varepsilon}^{\mathrm{cr}}$. Aging of shotcrete is considered by evolution equations for stiffness and strength. To this end, the respective equations in the CEB-FIP model code 1990 [28], EN 14487-1 [30,31] are recommended. The evolution laws are based on the values of the Young's modulus and the uniaxial compressive strength determined at the age of 1 day and 28 days, $E^{(1)}$ and $f_{\mathrm{cu}}^{(1)}$, and $E^{(28)}$ and $f_{\mathrm{cu}}^{(28)}$, respectively.

Nonlinear mechanical behavior of both hardening and hardened shotcrete is described on the basis of multisurface plasticity theory. A hardening and softening Mohr–Coulomb model is used for predominant compressive stress states and mixed stress states and a softening Rankine criterion for predominant tensile stress states to model cracking. The plastic strain rate is determined by means of a non-associated flow rule.

Shrinkage of shotcrete is taken into account on the basis of the model proposed by the ACI committee 209 [32]. It describes the evolution of the shrinkage strain in terms of the ultimate shrinkage strain $\varepsilon_{\infty}^{\mathrm{shr}}$ and a time-dependent function dependent on a shrinkage half-time parameter t_{50}^{shr}. Creep of shotcrete is modeled on the basis of the theory of viscoelasticity. The evolution of the creep strain is formulated in terms of a creep coefficient φ^{cr}, the acting permanent stress and a time-dependent function dependent on a creep half-time parameter t_{50}^{cr}. Nonlinear creep, encountered for higher stress levels, is taken into account. The model is denoted here as the Schädlich model.

3.1.3. Multi-field Shotcrete Model Based on the Hygro–Thermal–Chemo-Mechanical Concrete Model by Gawin et al.

The total strain $\boldsymbol{\varepsilon}$ is decomposed into the elastic strain $\boldsymbol{\varepsilon}^{\mathrm{el}}$, the creep strain $\boldsymbol{\varepsilon}^{\mathrm{cr}}$, the strain induced by chemical reactions $\boldsymbol{\varepsilon}^{\mathrm{ch}}$, and the thermally induced strain $\boldsymbol{\varepsilon}^{\theta}$. In contrast to the models discussed in the previous subsections, the shrinkage strain is not explicitly contained in this split. It is rather taken into account by the assumption of a multi-phase constitutive effective stress. In order to avoid confusion with the effective stress introduced in the damage-plasticity framework earlier, this constitutive effective stress will be referred to as generalized Bishop stress $\boldsymbol{\sigma}^{\mathrm{Bishop}}$ in the following. Assuming a passive gas phase, it is defined by

$$\boldsymbol{\sigma}^{\mathrm{Bishop}} = \boldsymbol{\sigma} - \mathbf{I} \left(a^{\mathrm{Bishop}} \, S_{\mathrm{w}} \left(p^{\mathrm{c}} \right) + b^{\mathrm{Bishop}} \right) p^{\mathrm{c}}. \tag{27}$$

Parameters a^{Bishop} and b^{Bishop} are obtained from drying shrinkage tests. Capillary pressure is denoted by p^{c}, and S_{w} is the degree of water saturation. The elastic law as well as the creep response are formulated in terms of this generalized Bishop stress. Thus, drying shrinkage manifests in both, contributions to the elastic strain and the creep strain.

Hygral behavior is governed by the van Genuchten law [33], which establishes a relation between the degree of water saturation and capillary pressure. The latter is related to the relative humidity φ in the pores based on the Kelvin–Laplace relationship. The two parameters used to describe the sorption characteristics in the van Genuchten law are the air entry value a^{VanGe} and a dimensionless fitting parameter b^{VanGe}.

The main parameter governing Darcy-type flow through a porous medium is the intrinsic permeability K. It is converted to the permeability with respect to the fluid phases using the viscosities of air and water, respectively. For partially saturated conditions, the permeability with respect to a fluid is reduced based on the current degree of saturation. Therefore, relative permeabilities are defined based on the van Genuchten sorption characteristics using Mualem's approach [34], as usual.

While assuming a constant Poisson's ratio ν, the evolution of other important material properties is described in terms of the degree of hydration Γ [9], such as the uniaxial compressive strength is

$$f_{\mathrm{cu}}(\Gamma) = f_{\mathrm{cu}}^{(\infty)} \left(\frac{\Gamma - \Gamma_0}{1 - \Gamma_0} \right)^{a_{\mathrm{fc}}}, \tag{28}$$

Formulated in terms of the ultimate uniaxial compressive strength $f_{\mathrm{cu}}^{(\infty)}$, the initial degree of hydration Γ_0, and a power-law exponent a_{fc}. The asymptotic elastic modulus $E_0(\Gamma)$ is defined in the same way, using the ultimate asymptotic elastic modulus $E_0^{(\infty)}$ and a power-law exponent b_{E}. Based on a law proposed by Cervera et al. [35] and extended in [9] to take into account the influence of relative humidity on the chemical reaction rate, the rate of the hydration degree is described by an Arrhenius-type law:

$$\dot{\Gamma} = \frac{\left(\hat{A}_1 + \hat{A}_2 \Gamma\right)(1 - \Gamma)}{1 + 625\,(1 - \varphi)^4} \exp\left(-\overline{\eta}\,\Gamma - \frac{5000\,\mathrm{K}}{T} \right). \tag{29}$$

It is stated here using a fixed ratio of activation energy and ideal gas constant of 5000 K. The quantities $\hat{A}_1$, $\hat{A}_2$, and $\overline{\eta}$ are model parameters and T denotes the temperature. Due to the released heat of hydration, the chemical reaction also affects the temperature distribution.

Creep is modeled by the microprestress-solidification theory [36] in a generalized Bishop stress formulation. The short-term, age-dependent, viscoelastic creep response is modeled on the basis of solidification theory. It is based on the same viscoelastic power creep curves with an exponent of 0.1 and parameter q_2 which are on display in the description of the SCDP model. Since the model proposed in [10] does not have an age-independent contribution to the creep compliance, as it is present in the solidification theory for single-phase materials, the parameter q_2 in the multi-field model and the SCDP model are not identical in general. To highlight this fact, q_2 is replaced by q_2^*. Long-term creep is described by means of the microprestress theory. It is characterized by the relaxation of self-equilibriated stresses, i.e., the microprestress, in the cement gel during hydration. If the macroscopic viscosity of the flow creep is assumed to be proportional to the microprestress, the long-term creep is described by three parameters c^{mps}, c_0^{mps}, and c_1^{mps}. The parameter c^{mps} governs the viscous flow, while c_0^{mps} controls the evolution of the microprestress in time. They will be fitted from the available shotcrete creep data. A useful interpretation of the ratio $2\,c^{\mathrm{mps}}/c_0^{\mathrm{mps}}$, given for instance in [37], is that under standard conditions, this ratio corresponds to the parameter q_4 present in the solidification theory, which controls the influence of the viscous dashpot part with age-dependent viscosity. Again, q_4 is replaced by q_4^* indicating that creep in the multi-field context is based on a different formulation than in the single field models. The influence of changing pore humidity on the evolution of the microprestress is taken into account via c_1^{mps}. Since the shotcrete creep data investigated here does not allow to calibrate this value, the value 1.98 MPa determined for concrete in [10,38] is used. A nonlinear dependency of creep on the effective stress at high stress levels is included in the model using the amplification function $F(s) = (1 + s^2)/(1 - s^{10})$ present for instance in [10,20]. In the present context, due to drying and temperature effects, the stress in the shotcrete specimens is not uniaxial. Therefore, the stress ratio s is computed as the ratio between the norm of the (effective) von Mises shear stress and the compressive strength evolving in time.

3.2. Evaluation of the Shotcrete Models on the Basis of Experimental Data

In the following, the performance of the SCDP model is evaluated by experimental data from the literature and compared to the performance of the reviewed shotcrete models. Investigated phenomena comprise the evolution of material stiffness and strength, shrinkage and creep. Time-independent nonlinear material behavior for uniaxial and multi-axial stress paths is not addressed here, as it was considered in the respective publications of the models, e.g., based on the biaxial tests by Kupfer et al. [39].

The presented numerical results for the Meschke model, the Schädlich model and the SCDP model are obtained at material point level, prescribing the stress in incremental-iterative simulations based on the respective stress update algorithms. The evolution laws within the framework of plasticity theory are integrated by means of the fully implicit Euler backward scheme within the return mapping algorithm. The evolution laws for creep behavior of the SCDP model and the Gawin model within the framework of viscoelasticity are approximated by Kelvin chains, and are integrated based on the exponential algorithm as proposed in [40]. The respective coefficients of the Kelvin chains are determined from the retardation spectra of the creep compliance functions as presented in [41].

In the multi-field model, mechanical, thermal, and hygral equilibrium is obtained simultaneously in a monolithic way. Since in this approach the variables are not uniformly distributed throughout a specimen, coupled three-dimensional finite element analyses of the considered specimens for determining the displacements, fluid pressures, and temperature have to be performed. For the latter, one-eighth of a cubic specimen is meshed, exploiting symmetry, using 512 elements with quadratic shape functions for displacements and temperature and linear shape functions for capillary pressure. A higher resolution is chosen near the drying surfaces. Thermal and hygral boundary conditions are governed by the ambient temperature T_∞ and the ambient relative humidity φ_∞. The convective heat transfer coefficient is denoted as α_c and the convective mass transfer coefficient is named β_c. In the following figures, the results at the center of the specimens are shown.

3.2.1. Comparison of Model Response with Experimental Results by Huber

The experimental data by Huber [15] is chosen for comparing the predicted and measured evolution of the Young's modulus and the uniaxial compressive strength. The respective experimental data is characterized by a good documentation and a comparatively small scatter of experimental data. In total, five test series, conducted on specimens with dimensions of 0.10 m × 0.10 m × 0.40 m, are reported. Test series 5 is chosen for the calibration of the shotcrete models, and the numerical results, computed on the basis of those parameters, are compared to the experimental results of the test series 1 to 4. The shotcrete composition, listed in Table 1, is identical for each series.

Table 1. Composition of shotcrete for the tests reported by Huber [15].

Property	Quantity	Unit
aggregate content (0/12 mm)	1800	kg/m^3
cement content	350	kg/m^3
water content	160	L/m^3
accelerator	5 to 7	%

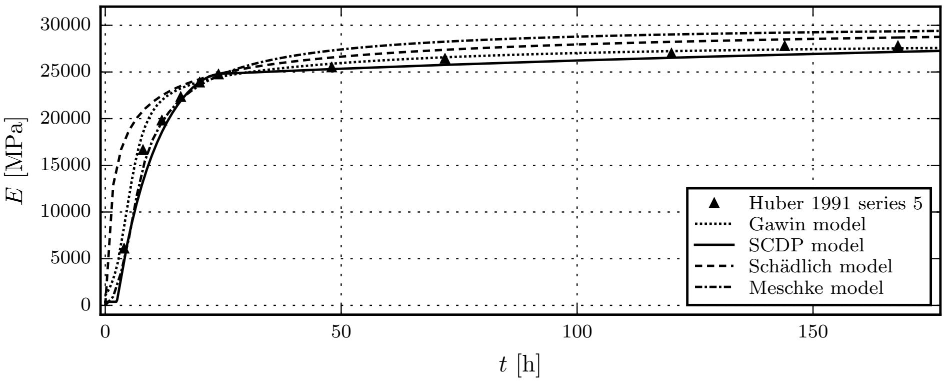
The measured Young's modulus at the age of one day in test series 5 is specified in [15] as $E^{(1)}$ = 24,780 MPa. Unfortunately, no values for the Young's modulus at the age of 28 days $E^{(28)}$ are reported for any of the test series. For this reason, $E^{(28)}$ is estimated as $E^{(28)}$ = 30,000 MPa. Parameters t_E and Δt_E of the Meschke model, which govern the early age evolution of stiffness, are identified as t_E = 5.65 h and Δt_E = 4.08 h.

The viscoelastic compliance parameters q_2 and q_3 of the SCDP model are computed according to the estimation procedure [25] as $q_2 = 206.34 \times 10^{-6}\,\mathrm{MPa}^{-1}$ and $q_3 = 2.61 \times 10^{-6}\,\mathrm{MPa}^{-1}$. To recover $E^{(28)} = 30{,}000$ MPa at the age of 28 days, q_1 is determined as $q_1 = 16.43 \times 10^{-6}\,\mathrm{MPa}^{-1}$ by computing the effective Young's modulus based on a duration of $\Delta t = 10^{-3}$ d. Flow compliance parameter q_4 is not required in the present context.

The simulations using the Gawin model are based on the reported ambient temperature of 23 °C and ambient relative humidity of 50%. The heat and mass transfer coefficients are $\alpha_c = 5\,\mathrm{W/m^2K}$

and $\beta_c = 0.0002\,\mathrm{m/s}$. Thermal conductivity, the densities of solid, air, and water as well as their heat capacities are set to match typical values for ordinary concrete. An intrinsic permeability of $3 \times 10^{-19}\,\mathrm{m}^2$ is used, which is considered to be a reasonable guess since it is in the range of values presented in [9] for concrete. The porosity in the fully matured state is assumed to be 12%. The shotcrete specific values calibrated from test series 5 are the parameters governing hydration, the evolution of uniaxial compressive strength and effective Young's modulus. The values are $\Gamma_0 = 0.02$, $\hat{A}_1 = 0.03836\ \mathrm{s}^{-1}$, $\hat{A}_2 = 4480.5\,\mathrm{s}^{-1}$, $\overline{\eta} = 4.0$, $E_0^{(\infty)} = 55{,}000\ \mathrm{MPa}$, $b_{\mathrm{E}} = 0.2$, and $q_2^* = 44 \times 10^{-6}\,\mathrm{MPa}^{-1}$. The effective Young's modulus at the center of the specimen is measured for the same time period Δt as for the SCDP model.

Figure 2a shows the results of the calibration of the evolution laws for the Young's modulus based on test series 5 of the experimental data in [15] and Figure 2b contains the validation of the respective evolution laws by means of test series 1 to 4 in [15].



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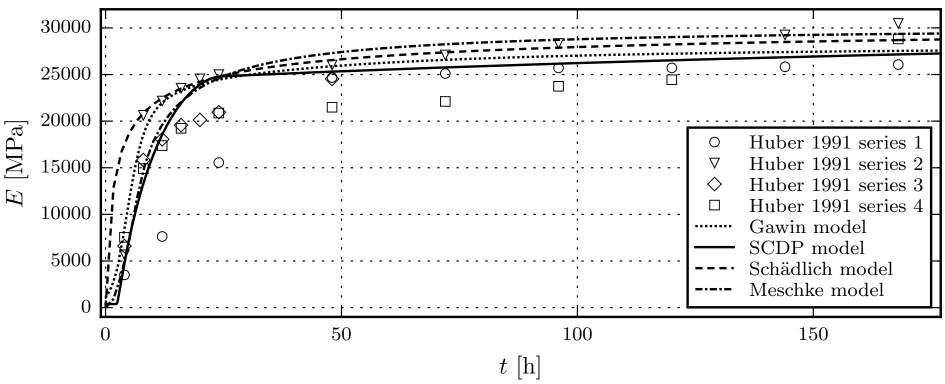


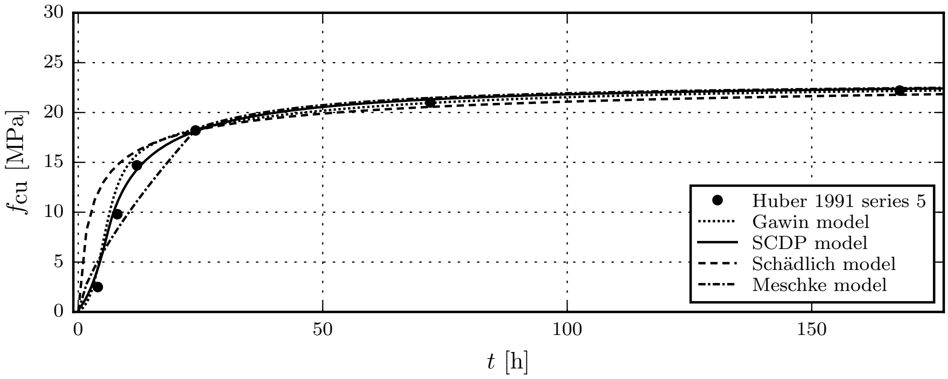
(**b**)

Figure 2. Evolution of the Young's modulus: (**a**) Results of the calibration of the shotcrete models based on test data of test series 5 by Huber [15]; (**b**) Validation of the shotcrete models by means of the data of test series 1 to 4 in [15].

It can be seen that prior to the shotcrete age of 24 h, the evolution of the Young's modulus, predicted by the Meschke model and the SCDP model, is closer to the experimental data than the one of the Schädlich model. This is mainly due to the considered delayed start of the hydration by both models. Although being based on several assumptions on the hygral and thermal parameters, the predictions using the Gawin model are of a similar quality as the SCDP model and the Meschke model.

The uniaxial compressive strength at the age of one day, $f_{\mathrm{cu}}^{(1)}$, measured in test series 5, is reported in [15] as 18.2 MPa. Again, as for the Young's modulus, data for the uniaxial compressive strength at the age of 28 days $f_{\mathrm{cu}}^{(28)}$ is not available. For this reason, a value of $f_{\mathrm{cu}}^{(28)} = 23\,\mathrm{MPa}$ is estimated. For the Gawin model, $f_{\mathrm{cu}}^{(\infty)} = f_{\mathrm{cu}}^{(28)}$ and $a_{\mathrm{fc}} = 1.0$ is assumed.

Figure 3a shows the results of the calibration of the evolution laws for the uniaxial compressive strength based on test series 5 of the experimental data in [15] and Figure 3b contains the validation of the respective evolution laws by means of test series 1 to 4 in [15].



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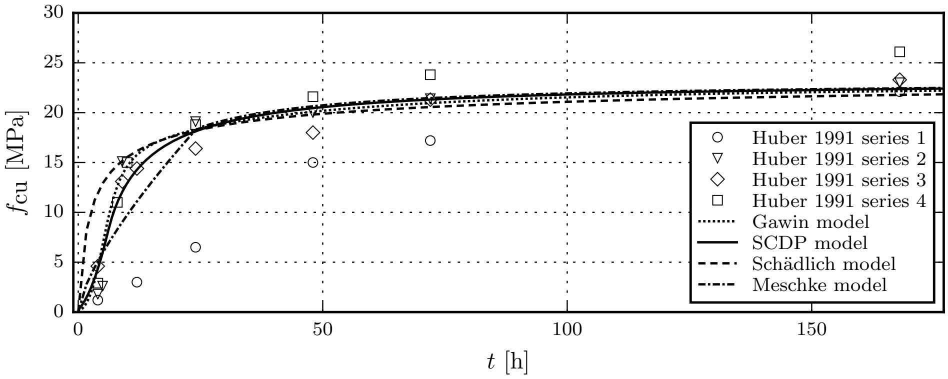

(**b**)

Figure 3. Evolution of the uniaxial compressive strength: (**a**) Results of the calibration of the shotcrete models based on test data of test series 5 by Huber [15]; (**b**) Validation of the shotcrete models by means of the data of test series 1 to 4 in [15].

It can be seen that the compressive strength of shotcrete, younger than 24 h, is overestimated by the Schädlich model, while the Meschke model underestimates the strength between 8 and 24 h. The SCDP model is able to represent the delayed start of the evolution of material strength with respect to the casting time, which agrees well with experimental results. Again, the Gawin model and the SCDP model perform equally well. After the age of 24 h, all models predict similar compressive strengths.

3.2.2. Comparison of Model Response with Experimental Results by Müller

The experimental data by Müller [17] is chosen for comparison of the shrinkage and creep behavior of the shotcrete models. Müller presented five series of experimental tests on shotcrete, consisting of four laboratory test series which are compared to one in situ test series. The shotcrete composition is listed in Table 2.

Table 2. Composition of shotcrete for the tests presented by Müller [17].

Property	Quantity	Unit
aggregate content (0/8 mm)	1768	kg/m^3
cement content *SBM W&P*	340	kg/m^3
water content	150	L/m^3

Creep and shrinkage tests were carried out on unsealed specimens from steel molds with dimensions of 0.15 m × 0.15 m × 0.30 m. Ambient conditions, such as relative humidity or temperature, are not reported.

For the subsequently presented comparison, test series 3 and 4 are chosen because they are characterized by the smallest scatter of data. While test series 3 includes one creep test series, in test series 4, two different loading sequences were applied on two specimens each. The two test schemes are denoted as creep test series 4/1 and creep test series 4/2, which is consistent with the notation used in [17]. The recorded strain from creep tests includes the combined autogenous and drying shrinkage strain as well as the basic and drying creep strain. Hence, shrinkage has to be considered for simulating the creep tests.

The material parameters referring to shrinkage and creep are calibrated on the basis of test series 4 including only creep test series 4/2. Subsequently, the performance of the shotcrete models is evaluated by simulating creep test series 4/1 and 3 employing the identified parameters.

Parameter identification from the shrinkage test on the unsealed specimen in test series 4 by the method of least squares yields the ultimate shrinkage strain $\varepsilon_\infty^{\text{shr}} = -0.0019$ (with $k_\text{h} = 1.0$) and the shrinkage half time $\tau_\text{shr} = 32\,\text{d}$ for the Meschke model and the SCDP model. The respective values for the Schädlich model are obtained as $\varepsilon_\infty^{\text{shr}} = -0.0015$ and $t_{50}^{\text{shr}} = 8.3\,\text{d}$. Although $\varepsilon_\infty^{\text{shr}}$ as well as τ_shr and t_{50}^{shr} have the identical physical meaning for all models, different values are estimated. This is the consequence of the different temporal evolution laws of the shrinkage models, which together with the short time span of the experimental shrinkage data, result in the identification of different parameters, depending on the employed shrinkage law.

The environmental conditions, required for the Gawin model, are assumed as $T_\infty = 23\,^\circ\text{C}$ and $\varphi_\infty = 60\%$. The same heat and mass transfer coefficients, porosity, hygral and thermal parameters as described in the previous subsection for the experiments by Huber are applied. Since drying induces creep in the Gawin model, the drying behavior cannot be calibrated independently from the creep parameters, which are presented for the subsequent numerical simulation of the creep tests.

Figure 4a shows the results of the calibration of the shotcrete models based on test series 4 in [17]. Figure 4b contains the validation of the shotcrete models based on test series 3 in [17].

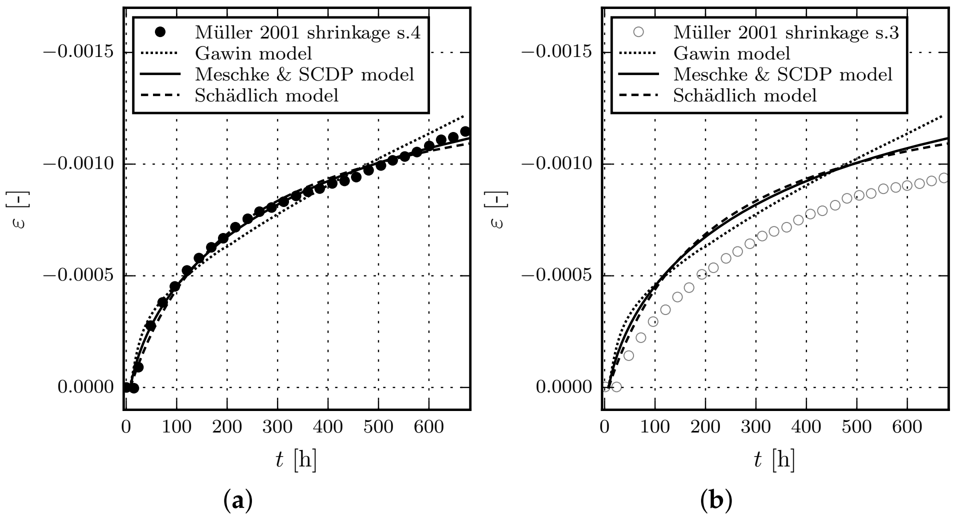

Figure 4. Evolution of the total strain in shrinkage tests: (**a**) Results of the calibration of the shotcrete models based on test data of test series 4 by Müller [17]; (**b**) Validation of the shotcrete models by means of the data of test series 3 in [17].

Creep tests on unsealed specimens were conducted simultaneously with the shrinkage tests. The applied load was increased multiple times during the tests lasting approximately 28 days in total.

Periods of constant stress levels lasted only for a few hours or days before the load was increased further. The start time of initial loading and the loading sequence was different for each test series. The individual loading sequences are listed in Table 3. Test series 3 includes identical creep tests on three specimens, here denoted as specimens a, b and c. In test series 4, the two different loading sequences 4/1 and 4/2 were applied to two specimens each. The four specimens of test series 4/1 and 4/2 are denoted as a and b, and c and d, respectively. A third specimen tested in test series 4/1 is not considered here, as the excessive measured total strain indicates a faulty specimen.

Table 3. Loading sequences for the creep test series 3, 4/1 and 4/2 [17], beginning at casting of the specimens.

Step	Test Series 3		Test Series 4/1		Test Series 4/2	
	Duration	Stress	Duration	Stress	Duration	Stress
-	8 h	0 MPa	7 h	0 MPa	48 h	0 MPa
1	16 h	−1 MPa	17 h	−1 MPa	120 h	−4 MPa
2	144 h	−2.5 MPa	144 h	−4 MPa	168 h	−10 MPa
3	168 h	−7.5 MPa	168 h	−10 MPa	168 h	−15 MPa
4	168 h	−10 MPa	168 h	−15 MPa	168 h	−0.6 MPa
5	168 h	−0.6 MPa	168 h	−0.6 MPa		

The creep parameters of the three models are exclusively identified from test series 4/2, and are subsequently used to simulate test series 3 and 4/1. The Young's modulus at the age of one day, $E^{(1)}$, is specified in [17] as $E^{(1)}$ = 7690 MPa for test series 4. The Young's modulus at the age of 28 days, $E^{(28)}$, is computed as the mean value of the two values presented in [17] as $E^{(28)}$ = 11,580 MPa. The uniaxial compressive strength at the age of one day is reported as $f_{\rm cu}^{(1)}$ = 8.72 MPa and the compressive strength at the age of 28 days was measured in tests series 4 on three specimens, from which the mean value is computed as $f_{\rm cu}^{(28)}$ = 16.8 MPa.

The ratio of the yield stress over the compressive strength $f_{\rm cy}/f_{\rm cu}$, determined from uniaxial compression tests, strongly influences the creep behavior of the Meschke model due to the viscoplastic formulation. Since identification from the creep tests leads to a ratio $f_{\rm cy}/f_{\rm cu}$ tending to zero, $f_{\rm cy}/f_{\rm cu}$ is estimated as $f_{\rm cy}/f_{\rm cu} = 0.1$ for all models, which is the lower bound of the range proposed in [23]. The plastic strains at uniaxial compressive peak stress for different shotcrete ages, $\varepsilon_{\rm cpu}^{\rm p(1)}$, $\varepsilon_{\rm cpu}^{\rm p(8)}$ and $\varepsilon_{\rm cpu}^{\rm p(24)}$, are chosen as the default values proposed in [4]: $\varepsilon_{\rm cpu}^{\rm p(1)} = -30.0 \times 10^{-3}$, $\varepsilon_{\rm cpu}^{\rm p(8)} = -1.5 \times 10^{-3}$ and $\varepsilon_{\rm cpu}^{\rm p(24)} = -0.7 \times 10^{-3}$.

Regarding the creep half-time parameter $t_{50}^{\rm cr}$ of the Schädlich model, identification simultaneously with the creep coefficient $\varphi^{\rm cr}$ leads to an ill-posed problem due to the short time span of the available experimental data. Hence, $t_{50}^{\rm cr}$ is chosen as $t_{50}^{\rm cr} = 24$ h, supported by the evaluation of different values for $t_{50}^{\rm cr}$ employed for the present parameter identification scheme. For comparison, $t_{50}^{\rm cr}$ = 36 h is proposed in [42]. The viscoelastic compliance parameters q_2 and q_3 of the SCDP model are computed according to the guidelines [25], which results in $q_2 = 269.82 \times 10^{-6}\,\mathrm{MPa}^{-1}$ and $q_3 = 3.84 \times 10^{-6}\,\mathrm{MPa}^{-1}$, whereas q_1 follows from the measured value of $E^{(28)}$ as $q_1 = 59.5 \times 10^{-6}\,\mathrm{MPa}^{-1}$, with the effective Young's modulus computed for a time period of $\Delta t = 10^{-2}$ d. Further creep parameters are identified from both specimens of test series 4/2 considering only the time-dependent strain during the first level of the sustained stress, yielding the viscosity parameter η = 16.1 h for the Meschke model, the creep coefficient $\varphi^{\rm cr}$ = 1.21 for the Schädlich model, and the flow compliance parameter $q_4 = 54.2 \times 10^{-6}\,\mathrm{MPa}^{-1}$ for the SCDP model. For the Gawin model, the estimated parameters governing hydration, the evolution of uniaxial compressive strength and effective Young's modulus are $\Gamma_0 = 0.1$, $\hat{A}_1 = 0.01387\ s^{-1}$, $\hat{A}_2 = 1577.7\ s^{-1}$, $\overline{\eta} = 4.0$, $f_{\rm cu}^{(\infty)} = 17.67$ MPa, $a_{\rm fc} = 1.05$, $E_0^{(\infty)} = 28{,}450$ MPa, $b_{\rm E} = 0.16$, and $q_2^* = 103 \times 10^{-6}\,\mathrm{MPa}^{-1}$. Using this parameter set, the effective Young's modulus evaluated for a time period of $\Delta t = 10^{-2}$ d is in the range of 7600 MPa to 7780 MPa at the age of 1 day and 11,440 MPa to 11,700 MPa at the age of 28 days. The corresponding values for the uniaxial compressive strength are in the range of 8.57 MPa to 8.9 MPa at the age of 1 day and 16.41 MPa to 17.06 MPa at the age of 28 days. Maximum values are always obtained at the center of the sample, while minimum values are located at the corners where drying and cooling slows down the hydration process. Parameters governing the viscous creep are $c^{\mathrm{mps}} = 3.1\,\mathrm{MPa}^{-2}\mathrm{d}^{-1}$ and $c_0^{\mathrm{mps}} = 0.104\,\mathrm{MPa}^{-1}\mathrm{d}^{-1}$. The parameters in the effective stress relationship (27) are estimated as $a^{\mathrm{Bishop}} = 0.01$ and $b^{\mathrm{Bishop}} = 0.16$. They are obtained based on the drying shrinkage test. In the range above 50% water saturation, which is present in the computation, these values model a response which is close to the response of ordinary concrete shown in [10].

Figure 5 shows the results of the calibration of the shotcrete models, based on the experimental data of the first sustained stress level of test series 4/2. Due to the viscoplastic formulation of the Meschke model, the viscosity parameter η only controls the rate of the creep strain in the hardening regime, but not its magnitude. This explains the discrepancies between measured and predicted total strain for the Meschke model. The validation of the shotcrete models based on the further load levels of test series 4/2 is shown in Figure 6. Figures 7 and 8 contain the validation of the shotcrete models based on the experimental data of test series 4/1 and 3, respectively.

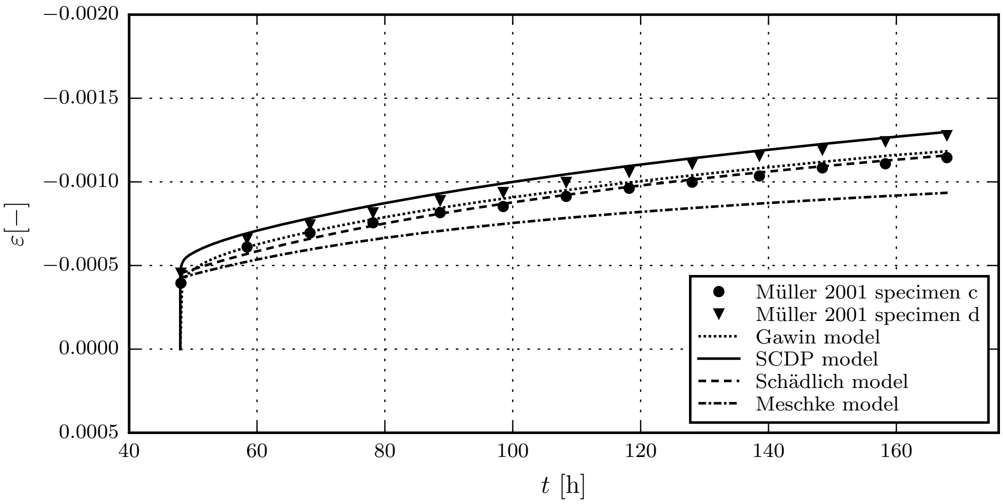

Figure 5. Evolution of the total strain in creep tests on unsealed specimens: Results of the calibration of the shotcrete models based on the test data from creep test series 4/2, load step 1, by Müller [17].

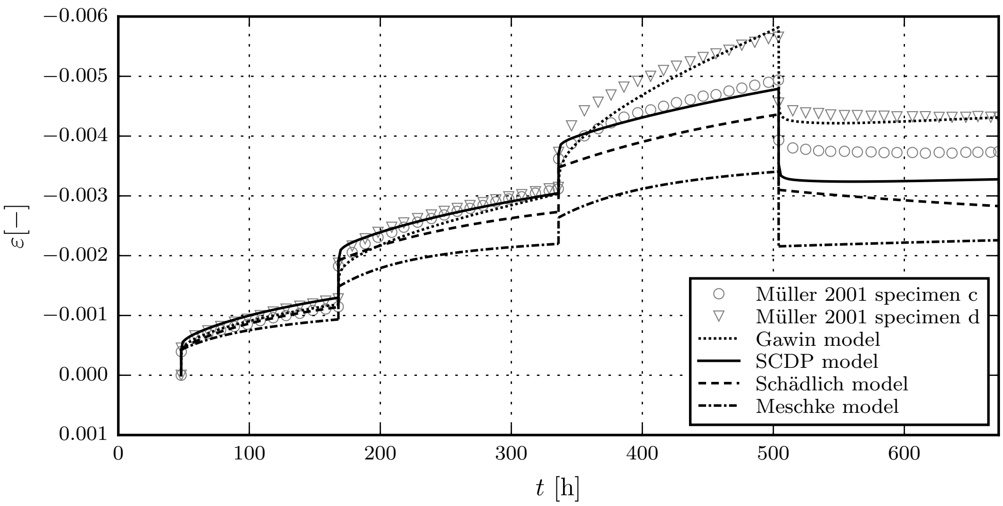

Figure 6. Evolution of the total strain in creep tests on unsealed specimens: Validation of the shotcrete models by means of the test data from creep test series 4/2 by Müller [17].

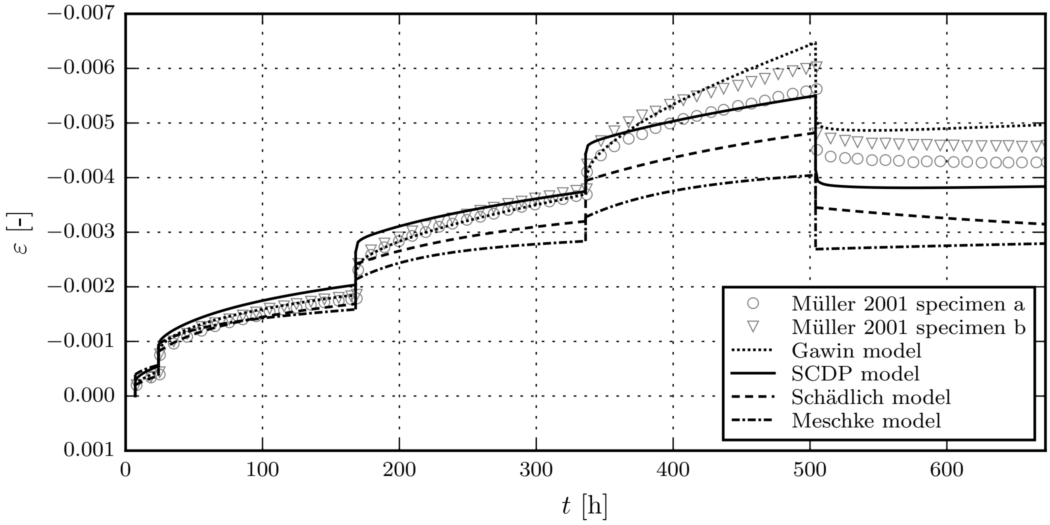

Figure 7. Evolution of the total strain in creep tests on unsealed specimens: Validation of the shotcrete models by means of the test data from creep test series 4/1 by Müller [17].

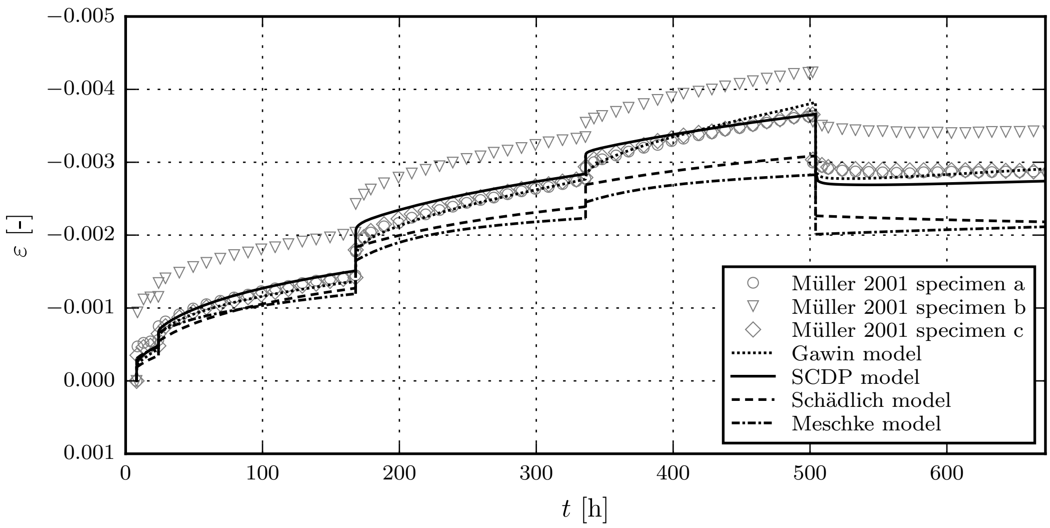

Figure 8. Evolution of the total strain in creep tests on unsealed specimens: Validation of the shotcrete models by means of the test data from creep test series 3 by Müller [17].

It can be seen that the numerical results by the SCDP model and the Gawin model agree very well with the experimental data, while the Schädlich model and the Meschke model underestimate the measured total strain. In the Gawin model, the stress-dependent amplification factor for the creep strain at high stress levels, which causes the fast strain increase in the highest load steps of series 4, is very sensitive to the value of uniaxial compressive strength.

4. Summary and Outlook

An extended damage plasticity model for shotcrete, able to represent nonlinear time-dependent material behavior, as well as material aging, creep and shrinkage, was proposed. Three well-established constituents serve as the starting point, which are the CDP model by Grassl and Jirásek [18], the solidification theory by Bažant and Prasannan [20] and the shrinkage model by Bažant and Panula [21]. An extension of the CDP model to time-dependent material behavior, as well as a modification of the solidification theory for shotcrete were presented. The material parameters of the new model can be identified from standard experimental tests. Moreover, the identification can be supported by the two parameter estimation procedures presented in [21,25]. The proposed modification of the solidification theory allows for an improved representation of the early age behavior of shotcrete, but maintains the applicability of the calibration scheme proposed in [25] for matured shotcrete.

Based on experimental tests published by Huber [15] and Müller [17], the performance of the model was compared to other models for shotcrete, i.e., the models by Meschke [2], by Schädlich and Schweiger [4] and the multi-field model by Gawin et al. [9,10]. It was shown that the extended damage plasticity model is able to represent the time-dependent material behavior of shotcrete very well.

For future research on the constitutive behavior of shotcrete, it seems to be worthwhile to focus on the nonlinear creep behavior. To identify the nonlinear creep behavior from experimental tests, data of multiple creep tests with different stress levels conducted on specimens of the same age are necessary. However, the experimental data for shotcrete available in the literature is very limited, and many of the available test sets suffer from several shortcomings: (i) Frequently, experimental data on the evolution of the Young's modulus and the uniaxial compressive strength data are characterized by large scatter. Certainly, this is the consequence of faulty shotcrete specimens caused by the complicated and error-prone in situ casting process; (ii) compressive creep tests frequently consist of several load steps, characterized by only short periods of constant stress levels. This approach seems to be advantageous for the validation of a constitutive model, but it is rather disadvantageous for the identification of creep parameters; (iii) regarding nonlinear creep, to the authors' best knowledge, no comprehensive experimental data is currently available in the literature; (iv) in general, the measurement periods for the creep tests are too short; it would be beneficial for the calibration process to extend the measurement periods; (v) for many experimental sets, the ambient conditions are not reported, which makes the calibration of a multi-field model virtually impossible. A more accurate calibration of multi-field models would furthermore require shrinkage and creep tests on sealed and unsealed specimens, the determination of the sorption and permeability characteristics as well as convective transfer coefficients, and calorimetric experiments.

Hence, further experimental programs on shotcrete are required for resolving the described deficiencies.

Author Contributions: M.N. and P.G. implemented the constitutive models and conducted the numerical simulations; M.N., P.G. and G.H. wrote the paper.

Conflicts of Interest: The authors declare no conflict of interest.

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3 Study of the Influence of the Shotcrete Model in Numerical Simulations of Deep Tunneling

In Publication II, *Influence of the Constitutive Model for Shotcrete on the Predicted Structural Behavior of the Shotcrete Shell of a Deep Tunnel*, the assessment of the SCDP model, the Meschke model, and the Schädlich model is pursued. The numerical simulations presented in Publication I were performed at integration point level of a finite element, and the presented assessment concentrated only on the capabilities of the models to represent the behavior of shotcrete observed in experimental tests on shotcrete specimens.

For this reason, a finite element benchmark of deep tunnel advance, derived from a stretch of the Brenner Base Tunnel, is developed for studying the influence of the shotcrete model in finite simulations of deep tunnel advance, and for comparing the investigated shotcrete models at structural level. For representing the surrounding rock mass, a linear-elastic perfectly-plastic model based on the Hoek-Brown criterion (Hoek and Brown, 1980) is employed. The material parameters of the three shotcrete models are adopted from Publication I, based on the experimental results by M. Müller (2001). In addition, a linear-elastic shotcrete model, following the practical approach proposed by Pöttler (1990b) based on the artificially low, time-independent HME (Hypothetical Modulus of Elasticity) is included in the comparison. The latter model is considered as a representative for commonly employed models in engineering practice.

It is demonstrated that the material model for shotcrete has a considerable impact on the predicted mechanical response, and it is shown that based on the linear-elastic HME a considerably stiffer mechanical response of the structure is predicted compared to the advanced material models.

The scientific contribution by the author comprises

- the development of the numerical model, jointly with Magdalena Schreter,
- the performance of the simulations, jointly with Magdalena Schreter,
- the evaluation of the simulation results, jointly with Magdalena Schreter,
- the preparation of the manuscript, jointly with Magdalena Schreter and Günter Hofstetter.

materials

Article

Influence of the Constitutive Model for Shotcrete on the Predicted Structural Behavior of the Shotcrete Shell of a Deep Tunnel

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Abstract: The aim of the present paper is to investigate the influence of the constitutive model for shotcrete on the predicted displacements and stresses in shotcrete shells of deep tunnels. Previously proposed shotcrete models as well as a new extended damage plasticity model for shotcrete are evaluated in the context of 2D finite element simulations of the excavation of a stretch of a deep tunnel by means of the New Austrian Tunneling Method. Thereby, the behavior of the surrounding rock mass is described by the commonly used Hoek–Brown model. Differences in predicted evolutions of displacements and stresses in the shotcrete shell, resulting from the different shotcrete models, are discussed and simulation results are compared to available in situ measurement data.

Keywords: shotcrete; tunneling; finite element model; numerical simulation; constitutive model; damage model; plasticity model

1. Introduction

The use of shotcrete plays an essential role within the New Austrian Tunneling Method (NATM). Subsequent to each excavation step, shotcrete is directly sprayed onto the surrounding rock mass, forming a supporting structure which is usually loaded already several hours after casting due to further advance of the tunnel. The prediction of the structural behavior of the rock-support system is a complex time-dependent problem and requires realistic constitutive modeling of both shotcrete and the surrounding rock mass. However, in numerical simulations, the constitutive behavior of shotcrete is often approximated using simple material models, like linear-elastic models. In contrast to normal concrete, only a few constitutive models for shotcrete representing the highly nonlinear, time-dependent material behavior have been proposed in the literature so far. Frequently used representatives of the latter are:

- The viscoplastic shotcrete model by Meschke [1]: Nonlinear mechanical behavior of shotcrete is described on the basis of the multisurface viscoplasticity theory using an associated flow rule. A hardening Drucker–Prager model is used for predominant compressive stress states and mixed stress states and a softening Rankine criterion for predominant tensile stress states to model cracking. The evolution of stiffness is described by hyperelastic constitutive relations. Aging of shotcrete is considered by evolution laws for the Young's modulus, the uniaxial compressive strength and the uniaxial tensile strength. Shrinkage of shotcrete is taken into account on the basis of the semi-empirical model proposed by Bažant and Panula [2]. Creep of shotcrete is modeled by a Duvaut–Lions type viscoplastic formulation.

- The viscoelastic-plastic shotcrete model by Schädlich and Schweiger [3]: Nonlinear mechanical behavior of shotcrete is described on the basis of multisurface plasticity theory using a non-associated flow rule. A hardening and softening Mohr–Coulomb model is used for predominant compressive stress states and mixed stress states and a softening Rankine criterion for predominant tensile stress states to model cracking. Aging of shotcrete is considered by evolution equations for stiffness and strength. Shrinkage of shotcrete is taken into account on the basis of the model proposed by the American Concrete Institute (ACI) committee 209 [4]. Nonlinear creep of shotcrete is modeled on the basis of the theory of viscoelasticity.
- The thermo-chemo-mechanical shotcrete model proposed by Hellmich et al. [5]: An alternative to single-field shotcrete models are multi-field models. Considering the dependency of shotcrete properties on the degree of hydration, i.e., on the time- and temperature-dependent chemical reaction between cement and water, it is based on thermo-chemo-plasticity theory along the lines of the thermo-chemo-mechanical framework, developed by Ulm and Coussy [6]. This model was developed further in [7] to consider early age cracking of shotcrete.
- The shotcrete damage plasticity (SCDP) model [8]: It is based on (i) the damage plasticity model by Grassl and Jirásek [9] to describe hardening and softening material behavior; (ii) the solidification theory by Bažant and Prasannan [10] to represent aging and creep; and (iii) the shrinkage model by Bažant and Panula [2]. To account for the early age behavior of shotcrete, a modification of the solidification theory is employed. Time-dependent material properties are represented by evolution laws for material strength and for ductility.

Applications of shotcrete models in numerical simulations of tunneling can be found, e.g., in the following papers:

- Pöttler [11] used a linear-elastic shotcrete model in the simulation of the advance of a shallow railway tunnel. In order to take into account the influence of time-dependent shotcrete behavior due to hydration and creep and the time-dependent process of tunneling in a simplified manner, a reduced Young's modulus of shotcrete, denoted as Hypothetical Modulus of Elasticity (HME), was employed. The HME was derived from a parameter study on a 2D finite element simulation of tunnel advance using a viscoelastic material model for shotcrete and it was concluded that a good approximation of the complex nonlinear time-dependent numerical model is achieved by a linear-elastic material model for shotcrete with an HME of 7000 MPa. However, the derivation of the HME was restricted to axisymmetric solutions and linear-elastic behavior of the rock mass.
- Meschke et al. [12] applied their viscoplastic shotcrete model in combination with an extended version of the soil model by DiMaggio and Sandler [13] in 3D finite element simulations of a shallow tunnel and compared the results to in situ measurement data. The tunnel was driven according to a partial excavation scheme, consisting of sequential excavation and securing of crown, bench and invert.
- Schädlich and Schweiger [14] applied their viscoelastic-plastic shotcrete model in 2D finite element simulations of the advance of a shallow tunnel with temporary side drift walls using a Mohr–Coulomb yield criterion for the surrounding soil. Focus was on the realistic prediction of the relaxation of bending moments in the shotcrete shell due to creep.
- Lackner and Mang [7] analyzed the stress state in the shotcrete shell during tunnel advance using an extended version of the thermo-chemo-mechanical shotcrete model by Hellmich et al. [5]. A hybrid method based on prescribed displacements at the rock-shotcrete interface, available from in situ measurement data, in combination with a structural model of the shotcrete shell, was used. Special attention was paid to cracking of shotcrete due to bending moments resulting from heterogeneous soil and rock conditions and due to tensile loading induced by shrinkage and thermal gradients. In contrast to isotropic softening models, two Rankine criteria in longitudinal

and circumferential direction were used. Hence, the tensile strength was affected only by the respective crack direction.

Most of the case studies in the literature refer to tunnels with low overburden [7,11,12,14–17]. To the authors' best knowledge, no comparison of constitutive models for shotcrete based on numerical simulations of tunneling has been presented so far. This is the motivation for complementing the comparative investigation of shotcrete models at material point level in [8] by a comparative investigation of the influence of the respective shotcrete models on the structural behavior of the shotcrete shell by means of a benchmark example of deep tunnel advance. In particular, the performance of selected shotcrete models, i.e., (i) the SCDP model [8]; (ii) the model by Schädlich and Schweiger [3]; (iii) the model by Meschke [1] and (iv) the approach proposed by Pöttler [11] are evaluated by means of 2D finite element simulations of deep tunnel advance, employing the commonly used Hoek–Brown model for describing the behavior of the surrounding rock mass.

The paper is organized as follows: in Section 2, the 2D initial boundary value problem, serving as the benchmark example for the simulation of deep tunnel advance, is presented and the material parameters of the constitutive models for shotcrete and rock mass are specified. In Section 3, simulation results on the basis of the respective shotcrete models, in combination with the Hoek–Brown model for the rock mass, are presented. The shotcrete models are compared by means of the predicted temporal evolution of displacements and stresses in the shotcrete shell. Finally, in Section 4, a summary and conclusions are presented.

2. Finite Element Model of Deep Tunnel Advance

A deep tunnel with circular cross-section, driven in Innsbruck quartz phyllite by the New Austrian Tunneling Method (NATM), is considered. The numerical model is derived from a stretch of the Brenner Basetunnel for which in situ measurement data is available. For a detailed description the reader is referred to [18]. To account for 3D effects due to the excavation process in 2D simulations, an equivalent plane strain model using the convergence confinement method as discussed in, e.g., [19,20], is developed.

2.1. Geometry and Boundary Conditions

The analyzed tunnel section is characterized by a circular excavation profile with a diameter of 8.5 m and an overburden of 950 m measured from the tunnel axis. The shotcrete shell has a thickness of 0.2 m. The prevailing initial geostatic stress state is assumed as a constant hydrostatic pressure of $p_i^{(0)} = 25.7$ MPa within the discretized domain of rock mass. It follows from the geometric properties, the assumed initial stress state, the boundary conditions and the full face excavation that axial symmetry with respect to the tunnel axis can be exploited leading to a reduced finite element model consisting of a single row of finite elements, highlighted in black in Figure 1, in contrast to the full 2D model, illustrated in Figure 1 in gray. The discretized domain covers 100 m of the surrounding rock mass around the tunnel axis. The rock mass and the shotcrete shell are discretized by 8-node quadrilateral continuum elements, the latter by four elements through the thickness. Along the tunnel perimeter, a pressure boundary condition is applied, which will be described in the next subsection. Homogeneous Dirichlet boundary conditions are prescribed perpendicular to the remaining part of the boundary of the single row of finite elements.

This simple numerical model, representing an ideal situation, has been chosen because it allows a straightforward comparison of the different shotcrete models regarding the time-dependent response of the shotcrete shell. Since the assumed axisymmetric conditions idealize the actual situation at the tunnel construction site, in addition to axial forces, bending moments in the shotcrete shells are present, which are reduced by cracking and creep (or relaxation) of the shotcrete. However, the influence of bending moments on the response of a shotcrete shell has been the focus of several previous studies, e.g., [7,14].

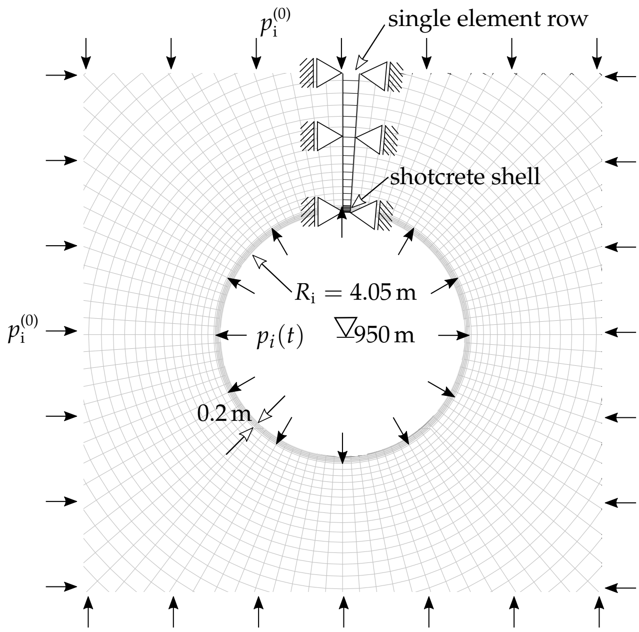


Figure 1. Schematic view of the benchmark problem together with the 2D axisymmetric finite element model.

2.2. Simulation of Tunnel Advance

The influence of the advance of the tunnel face on the considered cross section in the 2D analysis is expressed by reducing the fictitious internal pressure $p_i(t)$ acting on the tunnel perimeter, illustrated in Figure 1, by means of

$$p_i(t) = (1 - \lambda(t))\ p_i^{(0)}. \tag{1}$$

Herein, t represents the time with $t = 0$ referring to the installation of the shotcrete shell, $p_i^{(0)}$ is the initial hydrostatic pressure, which was specified in the previous subsection, and $\lambda(t)$ denotes the stress release ratio ranging from 0% to 100%. The stress release ratio at installation of the shotcrete shell is defined by the initial stress release ratio λ_0. For the idealization of the 3D problem of tunnel advance by means of a 2D plane strain model, λ_0 has to be determined either from corresponding 3D simulations or from in situ measurements, if available. Based on the observations by Pöttler [11], a parabolic decline of the remaining internal pressure $p_i(t)$ is assumed during the further excavation process, i.e., for $t > 0$. Approximating the drill and blast sequence of tunnel advance, $p_i(t)$ is assumed as a stepwise stress release function according to Figure 2.

The numerical simulations are structured as follows: (i) application of the initial hydrostatic stress field; (ii) consideration of the initial stress release, corresponding to measured pre-displacements, i.e., the displacements due to excavation of preceding segments and of the current segment; (iii) installation of the shotcrete shell and (iv) stepwise release of the remaining internal pressure due to the subsequent excavation steps according to the stress release function. The full release of the internal pressure, i.e., $p_i = 0$, is assumed at a progressed distance of the tunnel face of one tunnel diameter from the investigated cross section. Considering an advance length of 1 m, the internal pressure is completely released after nine excavation steps, including 8 h of rest period between two successive advance steps. Hence, the considered excavation process after installation of the shotcrete shell extends over 72 h.

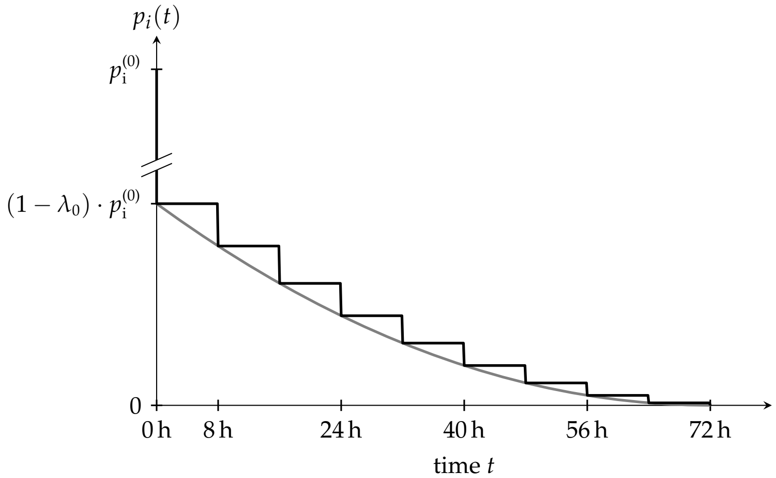


Figure 2. Stepwise stress release function $p_{\mathrm{i}}(t)$ assumed in the finite element simulations (black curve), which is derived from the idealized parabolic stress release function, proposed by Pöttler [11] (gray curve).

2.3. Material Parameters for the Shotcrete Models

Material parameters for the shotcrete models are calibrated based on experimental tests conducted by Müller [21], who presented a set of experimental data on the evolution of material stiffness, uniaxial compressive strength, creep and shrinkage of shotcrete. The material parameters and the respective calibration procedure for the shotcrete models are described in detail in [8]. Some of the material parameters are common for more than one model, i.e., the Poisson's ratio ν, the uniaxial compressive strength at the age of 1 d and 28 d, $f_{\mathrm{cu}}^{(1)}$ and $f_{\mathrm{cu}}^{(28)}$, the Young's modulus at the age of 1 d and 28 d, $E^{(1)}$ and $E^{(28)}$, the ratios of uniaxial yield stress to uniaxial compressive strength, $f_{\mathrm{cy}}/f_{\mathrm{cu}}$, of biaxial compressive strength to uniaxial compressive strength, $f_{\mathrm{cb}}/f_{\mathrm{cu}}$, and of uniaxial tensile strength to uniaxial compressive strength, $f_{\mathrm{tu}}/f_{\mathrm{cu}}$. In lack of experimental data, the latter is assumed as $f_{\mathrm{tu}}/f_{\mathrm{cu}} = 0.1$. Common parameters related to shotcrete ductility are the plastic strain at peak stress in uniaxial compression at the age of 1 h, 8 h and 24 h, $\varepsilon_{\mathrm{cpu}}^{\mathrm{p}(1)}$, $\varepsilon_{\mathrm{cpu}}^{\mathrm{p}(8)}$ and $\varepsilon_{\mathrm{cpu}}^{\mathrm{p}(24)}$, and the specific mode I fracture energy at the age of 28 d, $G_{\mathrm{fI}}^{(28)}$. Common parameters related to shrinkage are the ultimate shrinkage strain, $\varepsilon_{\infty}^{\mathrm{shr}}$, the shrinkage half time, τ_{shr} (SCDP model and Meschke model) and t_{shr}^{50} (Schädlich model), and the humidity dependent parameter k_{h}.

Time-dependent stiffness and creep behavior of the SCDP model, described by a modified version of the solidification theory by Bažant and Prasannan [10], are governed by the four compliance parameters q_1, q_2, q_3 and q_4, provided in Table 1. The softening behavior of the SCDP model is regularized by means of the crack-band theory [22], based on the characteristic element length, the specific mode I fracture energy G_{fI} and the uniaxial tensile strength f_{tu}. The regularization scheme is described in detail in [23]. Experimental data by Brameshuber and Hilsdorf [24] indicate that the temporal evolution of the specific mode I fracture energy can be considered approximately proportional to the increase of strength. Since the regularization scheme depends only on the ratio of fracture energy to tensile strength, the softening modulus can be computed from the values determined at the age of 28 d, i.e., $G_{\mathrm{fI}}^{(28)}$ and $f_{\mathrm{tu}}^{(28)}$. In lack of experimental data by Müller on the specific mode I fracture energy, $G_{\mathrm{fI}}^{(28)} = 0.1\,\mathrm{N/mm}$ is assumed for all models, which is in good agreement with commonly assumed values.

The creep behavior of the Meschke model is governed by the viscosity parameter η. Parameters Δt_E and t_E of the Meschke model control the early age evolution of the Young's modulus. The respective parameters are summarized in Table 2.

In the Schädlich model, ψ denotes the dilatancy angle within the framework of a hardening and softening Mohr–Coulomb model. Softening material behavior is represented by a linearly decreasing compressive and tensile strength, specified by the ratios f_{cfn}, f_{cun} and f_{tun}, and the compressive fracture energy $G_c^{(28)}$ at the age of 28 d. The creep behavior of the Schädlich model is formulated on the basis of the Eurocode 2 creep model [25] with the parameters representing the creep half-time t_{50}^{cr} and the creep coefficient φ^{cr}. They are specified in Table 3.

Table 1. Material parameters for the SCDP model.

q_1 (MPa^{-1})	q_2 (MPa^{-1})	q_3 (MPa^{-1})	q_4 (MPa^{-1})	ν (−)	$E^{(1)}$ (MPa)	$f_{cu}^{(1)}$ (MPa)	$f_{cu}^{(28)}$ (MPa)	f_{cy}/f_{cu} (−)
59.5×10^{-6}	269.82×10^{-6}	3.84×10^{-6}	54.2×10^{-6}	0.21	7690	8.72	16.8	0.1
f_{cb}/f_{cu}(−)	f_{tu}/f_{cu}(−)	ε_∞^{shr} (−)	k_h (−)	τ_{shr} (−)	$\varepsilon_{cpu}^{p(1)}$ (d)	$\varepsilon_{cpu}^{p(8)}$ (−)	$\varepsilon_{cpu}^{p(24)}$ (−)	$G_{fl}^{(28)}$(N/mm)
1.16	0.1	−0.0019	1.0	32	−0.03	−0.0015	−0.0007	0.1

Table 2. Material parameters for the Meschke model.

$E^{(1)}$ (MPa)	$E^{(28)}$ (MPa)	ν (−)	$f_{cu}^{(1)}$ (MPa)	$f_{cu}^{(28)}$ (MPa)	f_{cy}/f_{cu} (−)	f_{cb}/f_{cu} (−)	f_{tu}/f_{cu} (−)
7690	11 580	0.21	8.72	16.8	0.1	1.16	0.1
η (h)	ε_∞^{shr} (−)	k_h (−)	τ_{shr} (d)	Δt_E (h)	t_E (h)	$G_{fl}^{(28)}$ (N/mm)	
16.1	−0.0019	1.0	32	6	8	0.1	

Table 3. Material parameters for the Schädlich model.

$E^{(1)}$ (MPa)	$E^{(28)}$ (MPa)	ν (−)	ψ (°)	t_{50}^{cr} (h)	φ^{cr} (−)	$f_{cu}^{(1)}$ (MPa)	$f_{cu}^{(28)}$ (MPa)	$f_{tu}^{(28)}$ (MPa)	f_{cy}/f_{cu} (−)
7690	11 580	0.21	0	24	1.21	8.72	16.8	1.68	0.1
f_{cfn} (−)	f_{cun} (−)	f_{tun} (−)	ε_∞^{shr} (−)	t_{50}^{shr} (d)	$\varepsilon_{cpu}^{p(1)}$ (−)	$\varepsilon_{cpu}^{p(8)}$ (−)	$\varepsilon_{cpu}^{p(24)}$ (−)	$G_{fl}^{(28)}$ (N/mm)	$G_c^{(28)}$ (N/mm)
0.1	0.1	0.1	−0.0015	8.3	−0.03	−0.0015	−0.0007	0.1	30

Material parameters of the SCDP model and the models by Schädlich and Meschke, which are not considered in the calibration procedure, are chosen as the default values, proposed in the papers on the respective shotcrete models. In particular, they comprise $\varepsilon_{cpu}^{p(1)}$, $\varepsilon_{cpu}^{p(8)}$ and $\varepsilon_{cpu}^{p(24)}$ (SCDP model and Schädlich model), ψ, f_{cfn}, f_{cun}, f_{tun}, $G_c^{(28)}$ (Schädlich model) and Δt_E and t_E (Meschke model).

Figure 3 [8] depicts the evolution of the total strain of the two specimens of creep test series No. 4/2 by Müller (consistent with the notation used in [21]) and the predicted evolutions by the shotcrete models considered in the present investigation. Creep test series No. 4/2 was selected for parameter identification because of the high sustained stress levels.

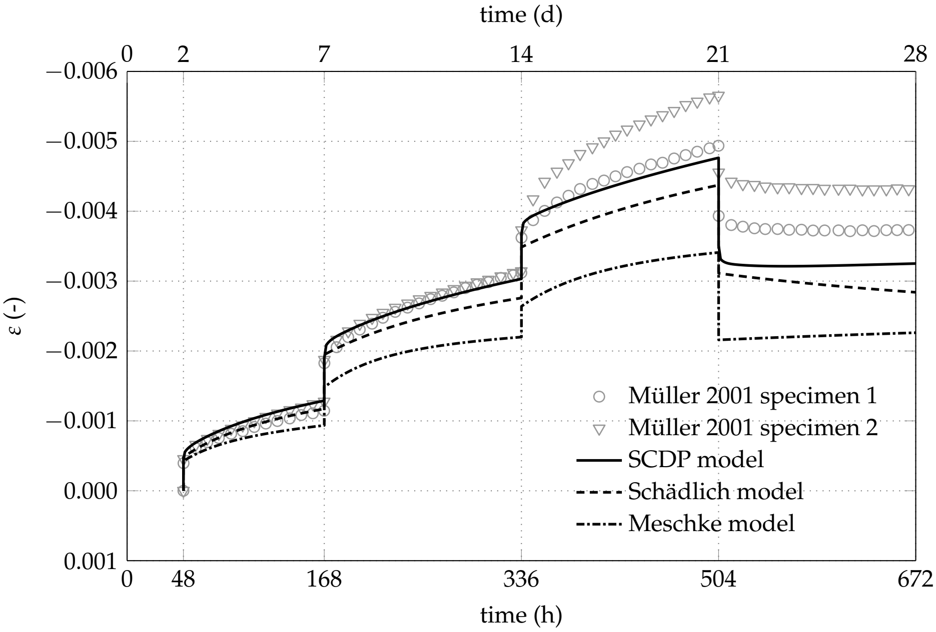


Figure 3. Measured and predicted evolution of the total strain of the two specimens of creep test series No. 4/2 in creep tests on unsealed specimens by Müller [21].

For the linear-elastic shotcrete model, the hypothetical modulus of elasticity of 7000 MPa is adopted from Pöttler [11], which takes into account the time-dependent material behavior of shotcrete and time-dependent effects due to tunnel advance in a simplified manner.

2.4. Material Parameters for the Hoek–Brown Model for Rock Mass

The mechanical behavior of rock mass is described by the linear-elastic perfectly-plastic constitutive law with the failure criterion for rock mass by Hoek and Brown [26,27] and the Mohr–Coulomb type plastic potential function, both in the smooth version proposed by Menétrey and Willam [28]. Main simplifying assumptions inherent in the Hoek–Brown model are the neglect of hardening in the pre-peak region and softening behavior in the post peak region of the stress–strain relations.

The parameters of the Hoek–Brown model were determined from triaxial compression tests on small-scale intact rock specimens of Innsbruck quartz phyllite from drill cores sampled from the tunnel site [18]. Since the Hoek–Brown model is restricted to isotropic material behavior, average values of the identified parameters for the different loading angles in the triaxial compression tests are applied in the numerical simulations. The parameters for the Hoek–Brown model are summarized in Table 4. E, ν, f_{cu}, m_0, ψ and e denote the Young's Modulus, the Poisson's ratio, the uniaxial compressive strength, the friction parameter, the dilatancy angle and the eccentricity parameter for the Menétrey–Willam formulation of the yield criterion and the plastic potential function, respectively. The down-scaling factors for the transition from intact rock to rock mass, i.e., the geological strength index GSI and the disturbance factor D [27], are chosen according to the geological survey at the tunnel site [18].

Table 4. Material parameters of Innsbruck quartz phyllite for the Hoek–Brown model.

E(MPa)	ν (−)	f_{cu} (MPa)	m_0 (−)	ψ (°)	e (−)	GSI (−)	$\boldsymbol{D}$ (−)
56,670	0.21	42	12	11.6	0.51	40	0

3. Finite Element Simulations of Tunnel Advance

In the 2D finite element simulations, the structural behavior of the shotcrete shell is simulated for initial stress release ratios λ_0 (Figure 2) of 85% and 95% at installation of the shotcrete shell. The corresponding pre-displacements at the rock-shotcrete interface are obtained from the ground response curve, computed on the basis of the Hoek–Brown model for the rock mass with the parameters according to Table 4, resulting in 19 mm and 34 mm, respectively, illustrated in Figure 4. For comparison, at the addressed tunnel site, preplaced measurement devices located approximately 1 m above the tunnel roof recorded in situ pre-displacements between 11 mm and 43 mm [18].

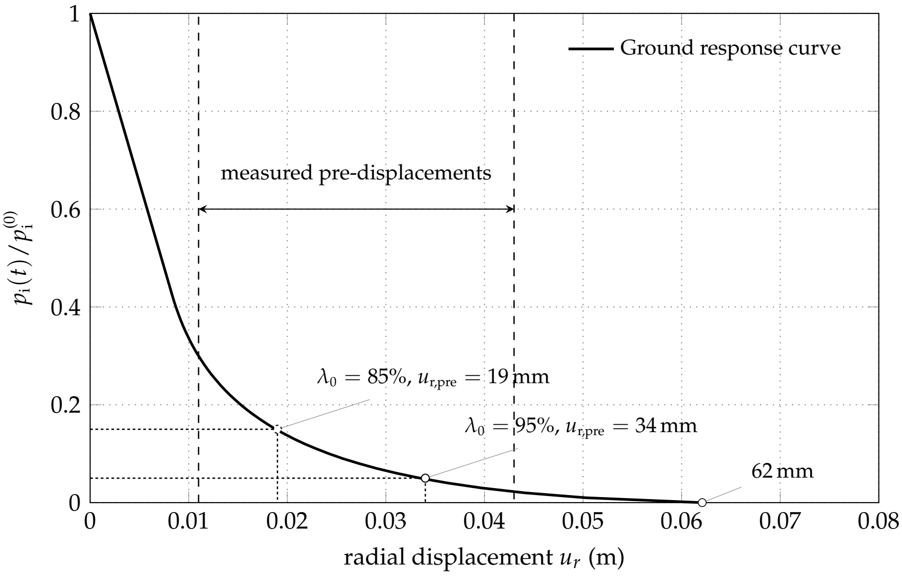


Figure 4. Ground response curve computed on the basis of the Hoek–Brown model.

In addition, from the ground response curve, the maximum displacement of the tunnel surface of 62 mm at vanishing internal pressure, corresponding to the solution without the supporting shotcrete shell, is obtained. It can serve in the subsequent investigation as a reference for assessing the supporting effect achieved by the shotcrete shell.

The simulations are conducted by the finite element software Abaqus/Standard v6.14-1 (Dassault Systèmes: Providence, RI, USA) [29]. The constitutive models for rock and shotcrete are implemented as user material (UMAT) subroutines, providing the updated stress and consistent tangent stiffness at each integration point within the framework of the return mapping algorithm.

3.1. Predicted Displacements of the Shotcrete Shell

The computed uniform radial displacement along the tunnel perimeter at the shotcrete-rock interface is used for comparing the influence of the shotcrete models on the structural behavior of the shotcrete shell. To this end, a period of 408 h (17 d) after installation of the shotcrete shell is considered. It includes the duration of the nine excavation and securing steps, extending over 72 h (3 d) according to Figure 2, plus additional 336 h (14 d) for the completed tunnel. The predicted evolution of the radial displacement includes (i) the pre-displacement, which is related to the initial stress release before installation of the shotcrete shell; (ii) the instantaneous and time-dependent displacement during the subsequent nine excavation and securing steps and (iii) the time-dependent displacement of the completed tunnel, resulting from creep and shrinkage of the shotcrete.

Figure 5a shows a comparison of the predicted evolution of the radial displacement at the shotcrete-rock interface for the initial stress release ratio of 85%. It can be seen that the linear-elastic shotcrete model predicts the smallest displacement, i.e., the stiffest response of the shotcrete shell. In contrast, the Schädlich model predicts the largest instantaneous displacement during the nine excavation and securing steps, which is due to the larger plastic strains predicted by this shotcrete model. It will be shown in Section 3.2 that this is a consequence of the employed Mohr–Coulomb yield criterion. Compared to the Schädlich model, the SCDP model predicts smaller instantaneous displacements during the excavation and securing steps, but a stronger increase of the displacement due to creep, approaching the radial displacement of the Schädlich model by the end of the investigated time period at $t = 17$ d. Displacements predicted by the Meschke model are in between the results using the SCDP model and the Schädlich model on the one hand, and the linear-elastic model on the other hand.

The respective comparison of the predicted evolution of the radial displacement for the initial stress release ratio of 95% is shown in Figure 5b. Due to the higher initial stress release, smaller instantaneous displacements during the excavation and securing steps are predicted by all models, compared to the results shown in Figure 5a. Again, the SCDP model predicts the strongest increase of the time-dependent displacement due to creep, which is in good agreement with the simulations of the creep tests [21] by the investigated shotcrete models, shown in Figure 3.

The differences between the SCDP model and the Schädlich model concerning the predicted time-dependent displacements need some further explanations. The creep law of the Schädlich model is formulated by means of a temporally decreasing ultimate creep strain, which is related to the evolution of the Young's modulus. As a consequence, creep slows down already several days after casting of the shotcrete shell. Since the SCDP model [8] is based on a modified version of the solidification theory by Bažant and Prasannan [10], which is partially based on the log-double power law by Bažant and Chern [30], the evolution of the long-term creep strain is unbounded. The different creep behavior, predicted by the Schädlich model and the SCDP model, is illustrated on the basis of uniaxial creep tests on shotcrete specimens with two different levels of sustained compressive stresses of −4 MPa and −8 MPa in Figure 6. It can be seen that the creep strains predicted by the SCDP model increase faster than the ones predicted by the Schädlich model and the differences between the predicted creep strains increase with increasing stress levels.

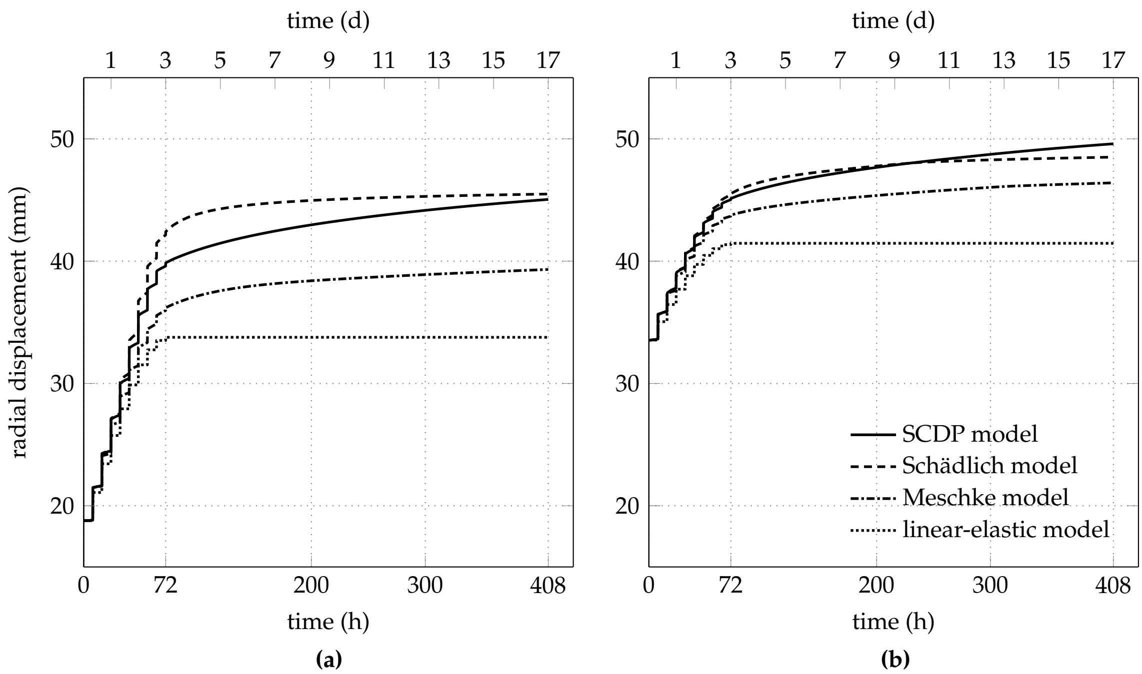


Figure 5. Comparison of the predicted radial displacement at the shotcrete-rock interface for initial stress release ratios of (**a**) 85% and (**b**) 95%.

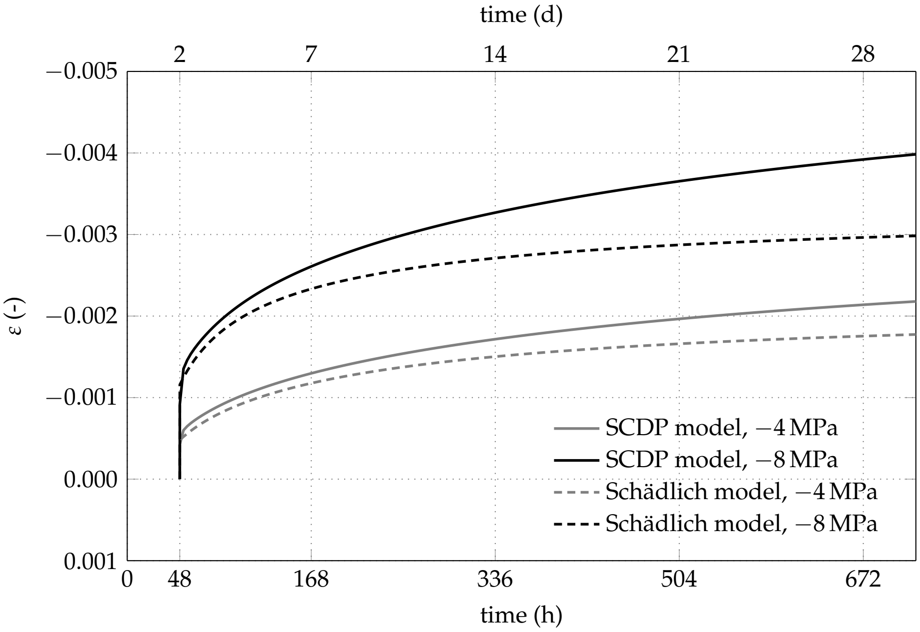


Figure 6. Comparison of the creep behavior, based on the evolution of the total strain, predicted by the SCDP model and the Schädlich model, for sustained compressive stresses of −4 MPa and −8 MPa.

At the tunnel site, radial displacements of the shotcrete shell were recorded by geodetic measurements in the range of 14 mm to 38 mm two weeks after excavation of the corresponding segment. According to Figure 5a, in simulations based on the initial stress release ratio of 85%, displacements induced by the subsequent excavation process after installation of the shotcrete shell are predicted in the range of 15 mm (linear-elastic model) to 27 mm (Schädlich model and SCDP model). For the initial stress release ratio of 95%, the predicted displacements induced by the subsequent excavation process after installation of the shotcrete shell of 8 mm (linear-elastic model) to 16 mm (SCDP model), according to Figure 5b, are smaller than the measured displacements. Thus, the initial stress release ratio of 85% results in reasonable agreement of the predicted displacements after installation of the shotcrete shell with the geodetic measurements.

3.2. Predicted Stress Response

The degree of loading of the shotcrete shell is evaluated by means of the stress state at the integration point located closest to the free surface of the shotcrete shell. For each shotcrete model, the evolution of the principal stress components of the biaxial stress state, i.e., the circumferential stress σ_C and the longitudinal stress σ_L perpendicular to the considered cross-section, are investigated.

Figure 7a shows the predicted evolution of σ_C for the initial stress release ratio of 85%. Both the linear-elastic shotcrete model and the Meschke model predict circumferential stresses that are larger than the experimentally determined strength of young shotcrete. In case of the Meschke model, this shortcoming is attributed to the viscoplastic formulation, which is characterized by limiting rate-dependent material behavior, in particular stress relaxation, to stress states outside the yield surface. In contrast, the SCDP model and the Schädlich model predict considerably smaller circumferential stresses, resulting from fast stress relaxation of the young shotcrete.

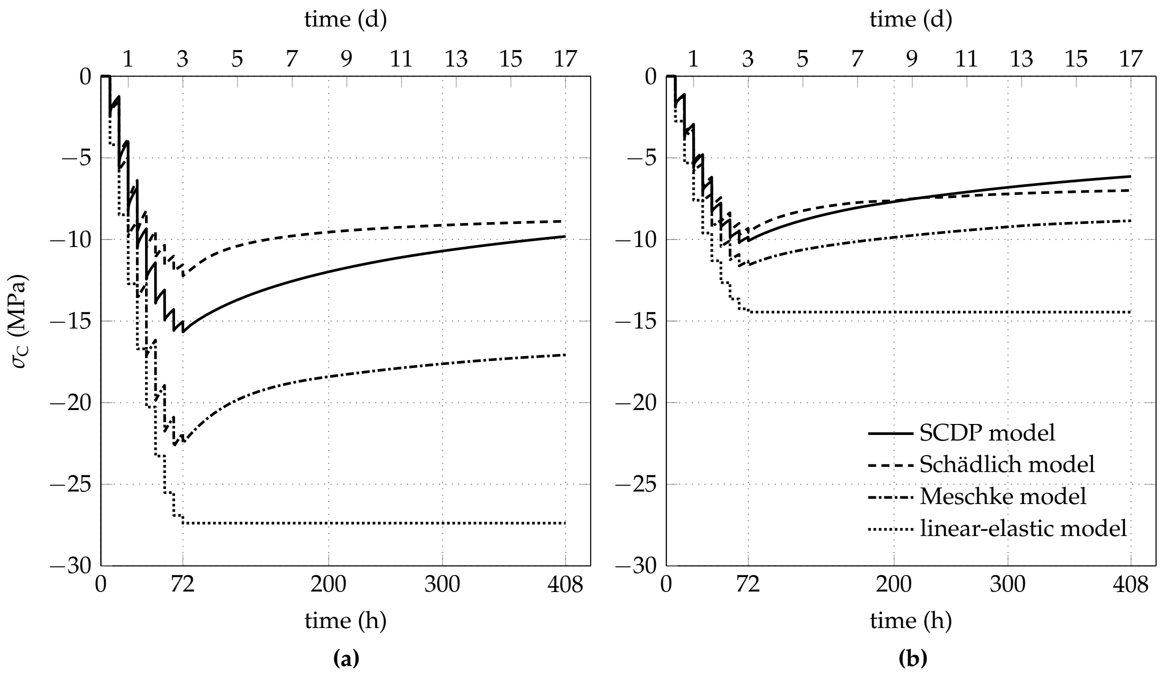
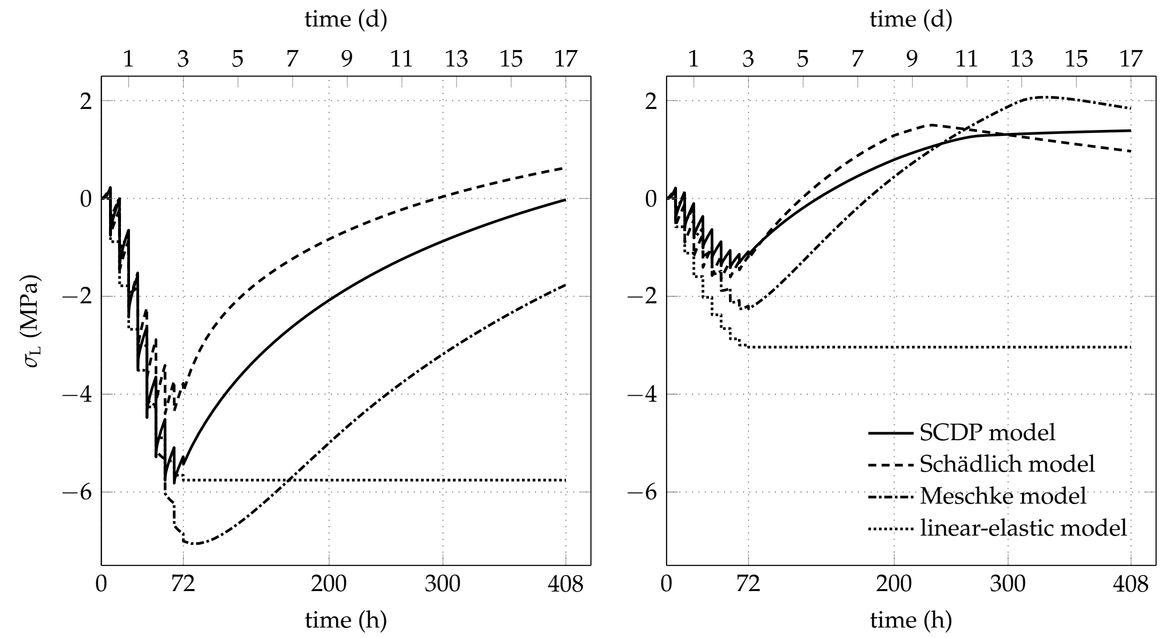


Figure 7. Predicted evolution of the circumferential stress in the shotcrete shell for initial stress release ratios of (**a**) 85% and (**b**) 95%.

The predicted evolution of σ_L for the initial stress release ratio of 85% is shown in Figure 8a. Because of the plane strain conditions, the strain in longitudinal direction of the tunnel, induced by shrinkage of shotcrete, is prevented by an evolving tensile stress. Hence, the nonlinear shotcrete models predict a decreasing compressive longitudinal stress due to shrinkage, gradually changing to a tensile stress in the case of the Schädlich model. The effect of shrinkage on the evolution of the longitudinal stress is also reported in [16]. For comparison, Figure 9a shows the evolution of σ_L, if shrinkage is neglected in the three models. Expectedly, in this case the longitudinal stresses remain in the compressive regime.

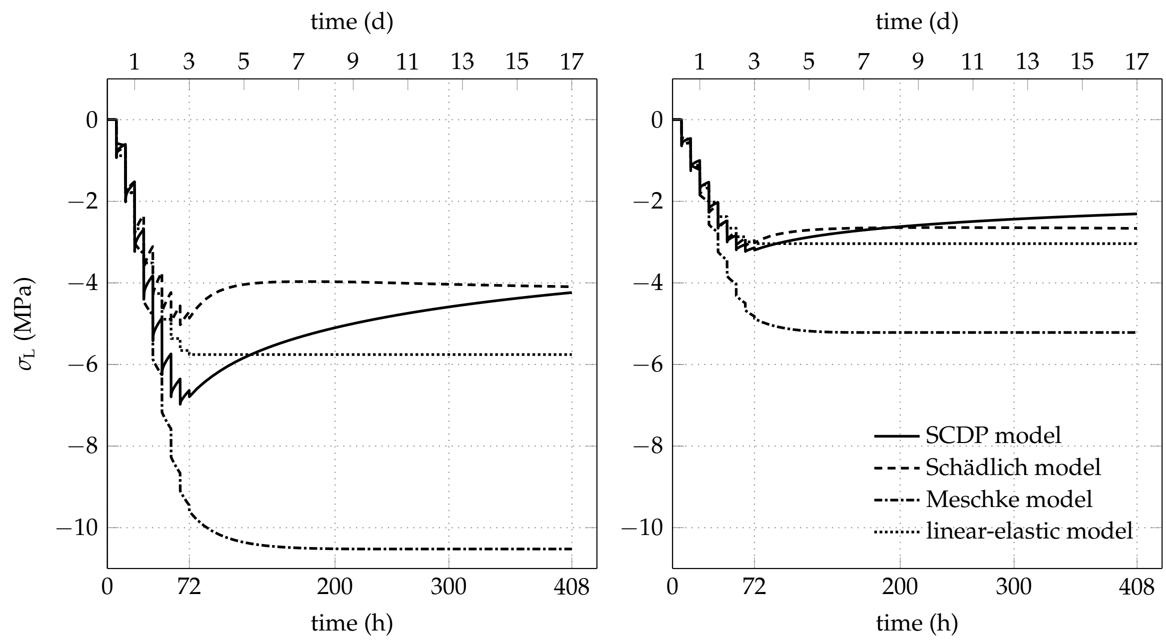
For the initial stress release ratio of 95%, the evolution of σ_C and σ_L is shown in Figures 7b and 8b, respectively. The higher initial stress release results in smaller circumferential stresses, accompanied by tensile stresses in longitudinal direction due to shrinkage. For all three nonlinear shotcrete models, the tensile stresses in longitudinal direction attain the tensile strength, leading to tension softening. The SCDP model and the Meschke model predict a minor effect of softening, indicated for the SCDP model by the maximum value of the isotropic scalar damage variable of $\omega = 2\%$ only and for the Meschke model by a reduction of the uniaxial tensile yield stress of 13% with respect to the tensile strength. In contrast, in the case of the Schädlich model, the uniaxial tensile strength is reduced due to softening by 27%. Figure 9b depicts the evolution of the longitudinal stress, if shrinkage is neglected in the shotcrete models.

The utilization of the shotcrete shell is visualized by plotting the evolution of the biaxial strength envelope together with the stress path. Figure 10a,b show the evolution of the biaxial strength envelope and the stress paths, predicted by the SCDP model for the initial stress release ratios of 85% and 95%, respectively. In the case of 85% initial stress release, during the last steps of the excavation process ($t \leq 72$ h), the stress state approaches the biaxial strength envelope, whereas, for $t > 72$ h, the compressive stresses are reduced due to stress relaxation and shrinkage. Thus, the outermost biaxial strength envelope approached by the stress state is the one at $t = 72$ h. In contrast, considering the initial stress release ratio of 95%, the stress path approaches the strength envelope due to shrinkage in the tensile-compressive region towards the end of the time period considered. Thus, the outermost biaxial strength envelope, relevant for the numerical simulation, is the one at $t = 408$ h.



(a) **(b)**

Figure 8. Predicted evolution of the longitudinal stress in the shotcrete shell for initial stress release ratios of (**a**) 85% and (**b**) 95%.



(a) **(b)**

Figure 9. Predicted evolution of the longitudinal stress, if shrinkage of shotcrete is neglected, for initial stress release ratios of (**a**) 85% and (**b**) 95%.

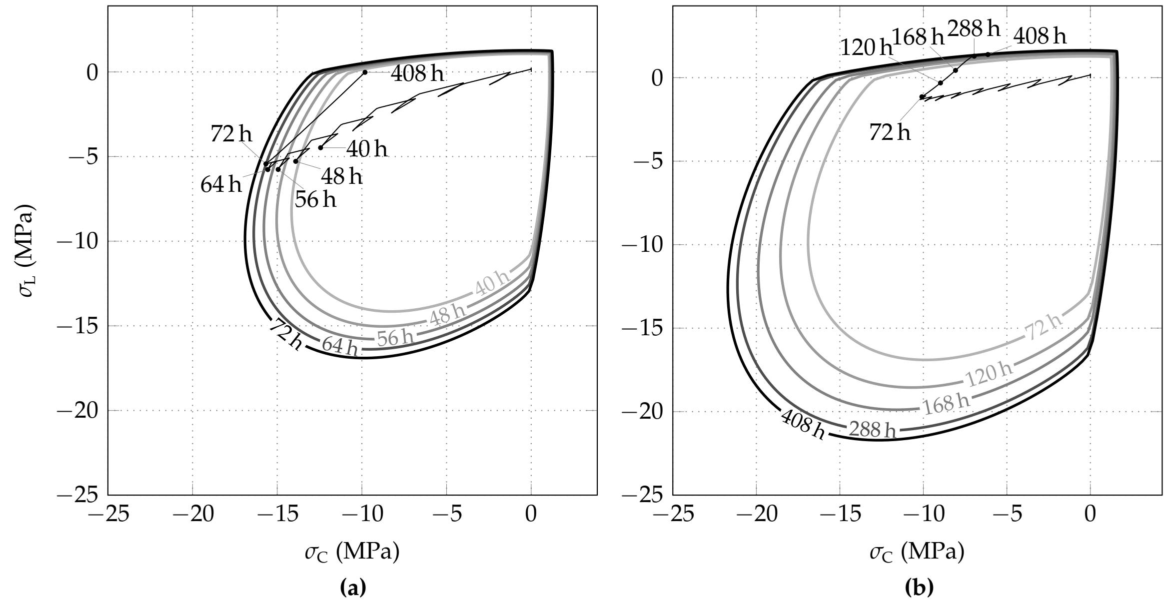


Figure 10. Evolution of biaxial strength envelopes of shotcrete and stress paths, predicted by the SCDP model for initial stress release ratios of (**a**) 85% and (**b**) 95%.

The evolution of the biaxial strength envelope of shotcrete and the stress path, predicted on the basis of the Schädlich model, are shown in Figure 11a,b for the initial stress release ratios of 85% and 95%, respectively. Contrary to the SCDP model, for the initial stress release ratio of 85%, the stress state attains the biaxial strength envelope. Hence, compressive softening behavior occurs as a consequence of the employed Mohr–Coulomb yield criterion, which underestimates the biaxial compressive strength by assuming it equal to the uniaxial compressive strength. However, it follows from Figure 11a, that the predicted compressive softening is negligible.

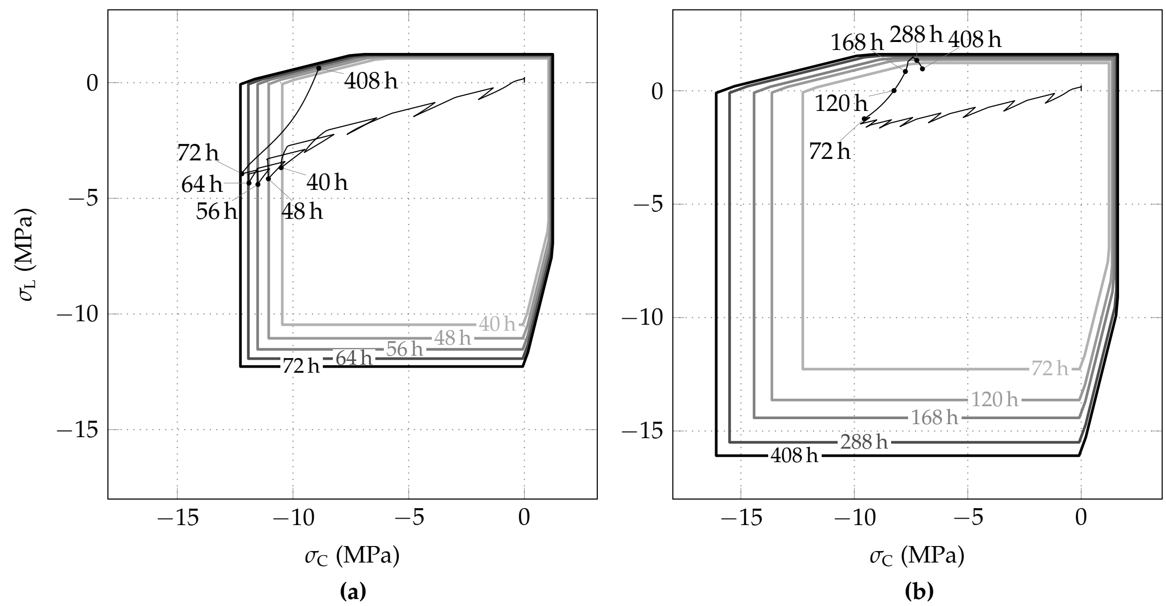


Figure 11. Evolution of biaxial strength envelopes of shotcrete and stress paths, predicted by the Schädlich model for initial stress release ratios of (**a**) 85% and (**b**) 95%.

Figure 12a,b show the evolution of the biaxial strength envelope and stress path, predicted by the Meschke model for the initial stress release ratios of 85% and 95%, respectively. Note that temporally limited stress states located outside of the biaxial strength envelope are typical for the theory of viscoplasticity, applied in the Meschke model.

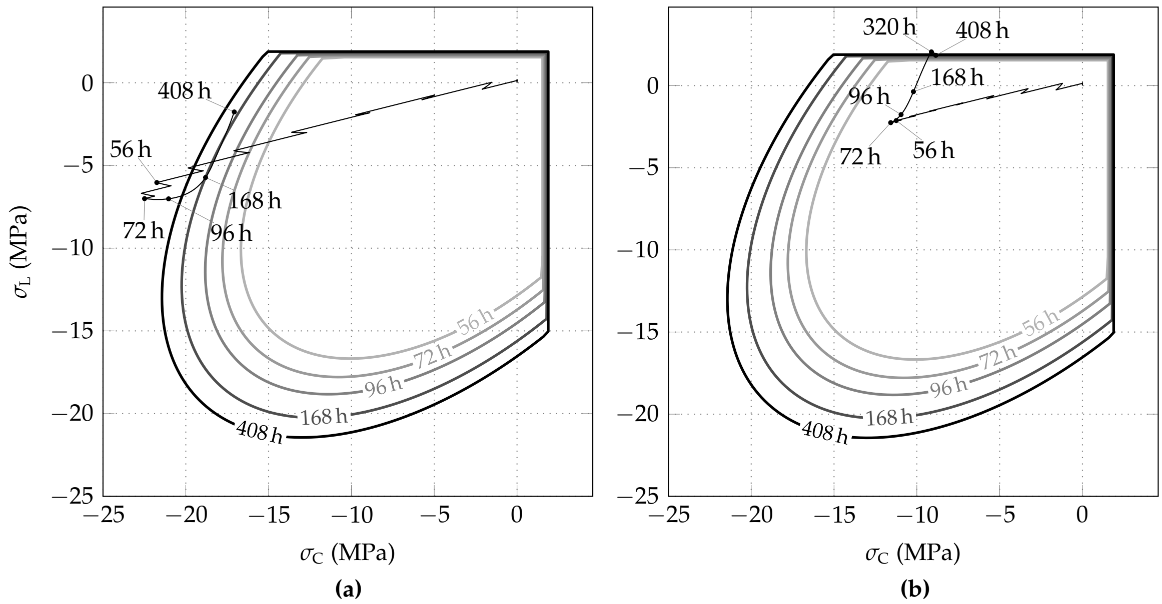


Figure 12. Evolution of biaxial strength envelopes of shotcrete and stress paths, predicted by the Meschke model for initial stress release ratios of (**a**) 85% and (**b**) 95%.

3.3. Long-Term Behavior Employing the SCDP Model

Since in contrast to the shotcrete models by Meschke and by Schädlich, the SCDP model predicts an unbounded evolution of the long-term creep strain, special attention is paid to the long-term behavior predicted by the SCDP model. For this purpose, the investigated time period is extended to two years, i.e., 17,520 h.

In Figure 13, the long-term evolutions of the radial displacement and the circumferential stress are shown in a semi-logarithmic scale for both initial stress release ratios. After two years, radial displacements of 52 mm and 55 mm are obtained for the initial stress release ratios of 85% and 95%, respectively. Hence, despite the different initial stress release ratios, resulting in different pre-displacements, the predicted long-term displacements are getting closer. In contrast to the radial displacements, after two years, the circumferential stresses of −4.2 MPa ($\lambda_0 = 85\%$) and −2.35 MPa ($\lambda_0 = 95\%$) are still quite different, but are approaching each other slowly.

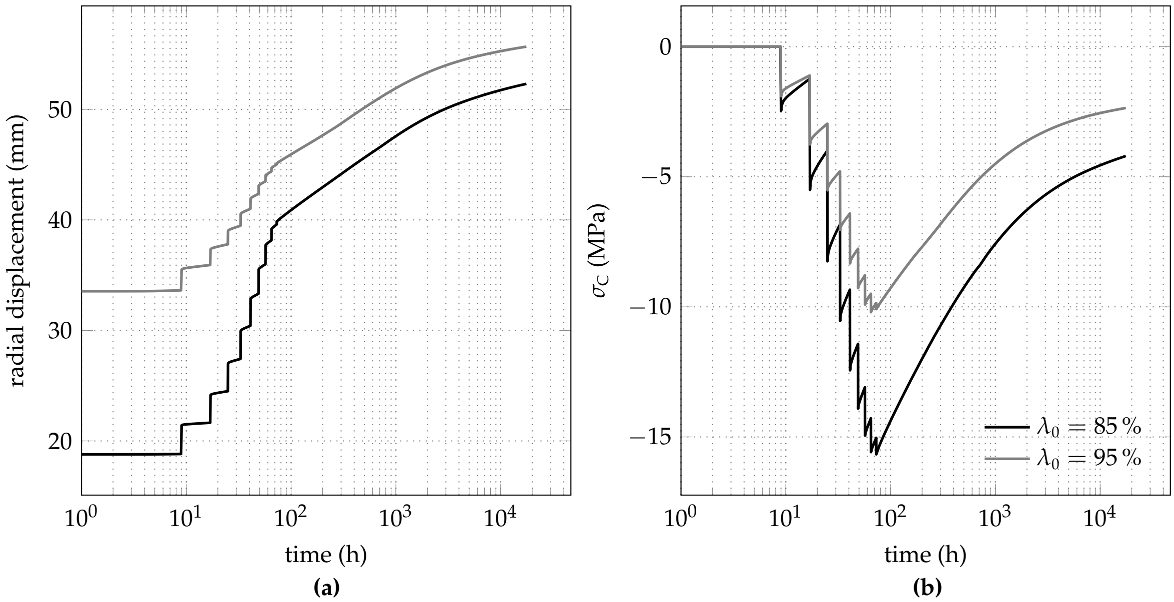


Figure 13. Long-term behavior predicted by the SCDP model for both initial stress release ratios: evolution of (**a**) the radial displacement and (**b**) the circumferential stress.

4. Conclusions

Based on finite element simulations of deep tunnel advance, the influence of the constitutive model for shotcrete on the predicted structural behavior of the shotcrete shell was evaluated. For this purpose, different shotcrete models were employed in a benchmark study derived from a stretch of the Brenner Basetunnel, driven in Innsbruck quartz phyllite by the New Austrian Tunneling Method. In particular, the linear-elastic shotcrete model with a reduced Young's modulus proposed by Pöttler, the viscoplastic shotcrete model by Meschke, the viscoelastic-plastic shotcrete model with the creep model according to the Eurocode 2 by Schädlich and Schweiger and the shotcrete damage plasticity (SCDP) model, characterized by representing aging and creep by the solidification theory by Bažant and Prasannan and by the shrinkage model by Bažant and Panula, were considered.

Since the focus of this contribution was the evaluation of different shotcrete models, the following assumptions for the numerical model were made: The 3D problem of tunnel advance was approximated by a 2D plane strain model, employing the commonly used Hoek–Brown model for describing the material behavior of the surrounding rock mass. In addition, the assumed hydrostatic initial stress state and the considered full-face excavation allowed the reduction to an axisymmetric problem with respect to the tunnel axis. In the 2D finite element simulations, two different initial stress release ratios of 85% and 95% at installation of the shotcrete shell were considered according to in situ measurements for the addressed tunnel stretch. The influence of the shotcrete models on the predicted structural behavior of the shotcrete shell was compared with respect to the evolution of the predicted radial displacement and the circumferential and longitudinal stresses in the shell. From the results, the following conclusions are drawn:

- The assumption of linear-elastic material behavior of shotcrete with a reduced Young's modulus to account for time-dependent effects in a simplified manner results in an overestimation of the stiffness of the shotcrete shell compared to shotcrete models, which consider time-dependent material behavior. Certainly, the results could be improved by assuming a smaller value of the reduced Young's modulus; however, estimation of the latter relies on engineering judgement.
- Compared to the SCDP model and the Schädlich model, a very stiff response of the shotcrete shell is also predicted by the Meschke model, which is caused by describing time-dependent shotcrete

behavior by a viscoplastic formulation. A shortcoming of the latter is the restriction of creep to stress states located outside of the elastic domain.

- The SCDP model and the Schädlich model predict similar results concerning the short-term response of the shotcrete shell. Concerning the predicted long-term response of the shotcrete shell, substantially larger creep strains are predicted by the SCDP model because of the unbounded long-term creep strain of the employed modified version of the solidification theory by Bažant and Prasannan, which is partially based on the log-double power law by Bažant and Chern.
- Shrinkage has a considerable influence on the overall behavior of the shotcrete shell, as it counteracts the compressive longitudinal stresses, induced by the excavation process, due to evolving tensile stresses resulting from constrained deformations in longitudinal direction of the tunnel. Because of shrinkage, a higher initial stress release ratio does not necessarily lead to a lower degree of loading of the shotcrete shell. As demonstrated for the high initial stress release ratio of 95 %, the assumed zero-strain condition in the longitudinal direction results in tensile stresses up to the tensile strength and, consequently, in tension softening.

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4 Experimental Study of Concrete Creep, Shrinkage, and Water Content Evolution

In Publication III, *Comprehensive Study of Concrete Creep, Shrinkage, and Water Content Evolution under Sealed and Drying Conditions*, an experimental program on concrete is presented.

The experimental program focuses on the creep and shrinkage behavior of concrete, which is determined in tests on cylindrical specimens. Furthermore, the influence of sealed and unsealed conditions, as well as the water content evolution within those specimens are investigated. Three series of creep tests were performed, in which sealed and unsealed specimens were loaded at different ages of the material, i.e., 2 days, 7 days and 28 days after casting. For the creep tests, three hydraulic creep test benches were designed, each able to test one sealed and one unsealed specimen simultaneously. The shrinkage tests on unsealed specimens were started simultaneously to the creep tests, whereas the shrinkage test on a sealed specimen was started already at a material age of 1 day.

The experiences gained during the experimental program, and the methods presented in this publication form the basis for a new experimental program on shotcrete, which is presented subsequently in Publication IV.

For the present publication, the scientific contribution by the author comprises the development and the documentation of the creep test benches.

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FULL PAPER

Comprehensive study of concrete creep, shrinkage, and water content evolution under sealed and drying conditions

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Abstract

A comprehensive study of the time-dependent behaviour of concrete, enhanced by measuring the evolution of the respective mass water content, is presented. Compressive creep as well as shrinkage are investigated on sealed and unsealed specimens in terms of the concrete age at loading of 2, 7, and 28 days, respectively, at a loading level of 30% of the compressive strength at loading. In addition, on the basis of the collected measurement data, the impact of load application on the moisture content is studied and the Pickett effect is re-examined. The obtained set of material data consists of basic and drying creep strains for different concrete ages at loading, autogenous shrinkage strains as well as combined autogenous and drying shrinkage strains, the evolution of the mass water content, the water desorption isotherm, and the time-dependent evolution of Young's modulus and the compressive strength. It provides a consistent set of material data for a particular concrete grade and will be available for calibrating and validating concrete models.

KEYWORDS

basic and drying creep, concrete, mass water content, Pickett effect, shrinkage

1 | INTRODUCTION

Creep of concrete can be subdivided into basic creep and drying creep.[1] Basic creep is defined as the load-induced time-dependent increase in strain in a sealed specimen under sustained constant load, whereas drying creep is defined as the creep component developing additionally to basic creep in a loaded specimen exposed to a drying environment.[2] In the studies,[3–7] the drying creep effect, also called the Pickett effect,[8,9] is explained by two different mechanisms: (a) the apparent mechanism due to surface microcracking[10] and (b) a true mechanism denoted as stress-induced shrinkage.[11] Hence, according to Altoubat and Lange,[6] total creep can be divided into three major components: basic creep, microcracking, and stress-induced shrinkage.

Although creep is mainly a long-term effect caused by sustained loading, it is also important for early age concrete, as reported by De Schutter and Taerwe.[12] A large body of literature exists on the time-dependent behaviour of early age and mature concrete in compression, considering both sealed and drying conditions. For instance, in Hubler et al.,[13] a comprehensive database[14] assembled during 2010 and 2013 at Northwestern University is presented. It contains 636 tests of compressive basic creep and 734 tests of compressive total creep. Moreover, the influence of moisture content on creep of concrete was studied by several researchers in the past[8,11,15–29] and is still subject of recent investigations.[4,5,30–36] Already in the early 1940s of the last century, Pickett[8] mentioned that "changes in moisture content do more than cause volume changes". Lorman[15] and other researchers[11,22,28] suggested that "shrinkage or swelling due to loss or gain of moisture and creep due to seepage are interrelated phenomena". Various physical mechanisms to explain the creep phenomena in terms of the role played by water were proposed but later disproved.

A popular theory is the seepage theory, in which volume changes due to application of an external load are assumed to be caused by squeezing pore water out of the specimen.[11,19] In his pioneering work, Powers[11] suggested the gradual movement of water between pores of different sizes (gel pores and capillary pores) to be responsible for basic creep. Another widely accepted theory is the viscous shear theory proposed by Ali et al.[21] and Ruetz,[22] assuming creep to be purely a shear process through slip between calcium-silicate hydrates (C-S-H), lubricated by water. According to the viscous shear theory, moisture movement is not essential but may facilitate creep, as summarized by Hannant.[25] Feldman and Sereda[23] found the gradual crystallization or aging process of the layered C-S-H phases to be responsible for creep. Thereby, the role of adsorbed water is scarcely significant, as mentioned in Feldman et al.[23] and Feldman.[28] According to Wittmann,[26] water influences the creep mechanism only by lowering the surface energy and, hence, does not play an active role in the deformation process. More recently, Bažant et al.[31] proposed the so-called microprestress-solidification theory. The latter is a further development of the solidification theory,[37] in which concrete aging due to hydration is explained by the growth in volume of solidified hydration products, modelled by nonaging viscoelasticity. The microprestress concept is the basis for describing long-term creep and drying creep. The microprestress is generated by the disjoining pressure of the hindered adsorbed water in the micropores and by the very large and highly localized volume changes caused by hydration or drying.[31] It is generated at early stages of the formation of the microstructure and is then gradually reduced by relaxation processes. Bažant pointed out that "due to the idea of microdiffusion, some sort of the seepage theory, and due to the equivalence of the stress-induced shrinkage with the dependence of the viscosity on the relative humidity (RH), some sort of the viscous shear theory appears to be involved in the microprestress-solidification theory".[9,31]

Among practical scientists, the seepage theory gained widest acceptance. The water redistribution mechanism is still supposed to contribute to the drying creep effect, i.e., Pickett effect.[36] Wyrzykowski et al.[36] agree with Acker and Ulm[5] that moisture content influences the creep phenomena. However, the water redistribution mechanism was critically questioned in the research works carried out by Young[38] and Tamtsia et al.[30]

Hence, subsequently, seepage theory will be briefly reviewed, complemented by reporting the wide variety of opinions expressed by researchers during the past 70 years regarding the role played by water in connection with the creep phenomena, in particular regarding the effect of external loading on the hygral state.

According to Powers,[11] in the narrow spaces between the hydration products that are in the micropores or gel pores, the full thickness of the adsorption layer cannot develop, i.e., the adsorption is hindered. Hence, a compressive stress, the so-called disjoining pressure, is exerted on the layers with hindered adsorption, which pushes components of the solid framework apart.[11,30,31,36] For a given relative humidity, the water adsorbed in areas with hindered adsorption is in equilibrium with the one adsorbed in free adsorption areas as well as with the condensed capillary water.[11,36] Due to an external compressive load, which initiates a time-dependent diffusion process, some of the load-bearing water were squeezed out from the hindered adsorption areas into areas of unhindered adsorption. In order to restore the balance in free energy between the adsorbed water layers, water molecules in the gel pores are transferred to adjacent larger pores, i.e., the capillary pores at the microscale.[11,31,36] Since for a sealed specimen, the displaced water must remain in the specimen, either on adsorbed unhindered layers or perhaps as capillary water, and the effect is an increase in the internal relative humidity.[11,36]

Bažant et al.[31] excluded a change of the pore space and, thus, of the internal relative humidity by more than a fraction of 1% due to an applied compressive load. According to Bažant et al.,[31] the solid framework of concrete as well as the cement paste is so stiff that the walls of the micropores cannot come into contact.

However, in the recent experimental investigation on basic creep,[36] Wyrzykowski et al. observed an increase in internal relative humidity due to uniaxial compression. The instantaneous RH increase at loading, with a stress/strength ratio of 30%, measured on sealed mortar samples, was about 2%. During sustained loading, the initial RH was gradually regained whereas upon unloading, an instantaneous RH decrease was observed. According to Wyrzykowski et al., the measurement data presented in a recent study[36] is the first experimental validation of the water redistribution mechanism between pores of different sizes proposed by Powers.[11]

For unsealed specimens, the seepage theory proposed by Powers[11] predicts loss of the excess water to the atmosphere in order to restore equilibrium with ambient relative humidity, which had been upset due to the application of a compressive load. The latter, however, contradicts Bažant and Chern,[9] pointing out that compressive stress causes no increase in mass loss, i.e., water loss of concrete. This opinion is confirmed by the experimental investigation recently published by Cagnon et al.,[39] in which the mass loss measured on loaded samples at the end of compressive drying creep was compared with the one measured on unloaded samples at the end of drying, considering both constant and cyclic drying conditions. Similar to Cagnon et al.,[39] it was reported in the studies[40–42] that the mass transport properties of concrete are generally not affected by uniaxial compressive loading.

Regarding moisture diffusion one needs to distinguish between macrodiffusion and microdiffusion. As mentioned by other researchers,[9,31,36,43] only microdiffusion, i.e., the water migration between gel pores and capillary pores, affects the deformation rate of the solid framework, i.e., increases creep, whereas the effect of water transport between capillary pores has no measurable effect on deformation. As described in the investigations,[31,43,44] the latter is confirmed by the famous experiments of Hansen.[17,18]

A re-examination of the role of water in the creep process of hardened cement paste, carried out recently by Tamtsia et al.[30] using a thermogravimetric technique, showed the microsliding between adjacent particles of C-S-H to be responsible for creep. Thus, the water redistribution mechanism was critically questioned by Tamtsia et al.[30]

Among the experimental investigations focusing on the influence of moisture content on the creep of concrete, briefly summarized above, only the one recently carried out by Wyrzykowski et al.[36] provides time-dependent moisture data, namely the time-dependent evolution of internal relative humidity during basic creep, however, determined in an artificial pore.

The time-dependent evolution of moisture content and its influence on deformations is fundamental for a sound interpretation of the role played by water in the creep process. Hence, in the present contribution, the time-dependent evolution of the mass water content in both sealed and unsealed concrete specimens, subjected to compressive stress, and in the respective load-free companion specimens, is presented. Basic creep and drying creep and the respective mass water content evolution are investigated in terms of the concrete age at loading at 2, 7, and 28 days.

The presented measurement data will contribute to a better understanding of the role played by water in the creep process and, hence, of the impact of load application on the moisture content as well as of the Pickett effect. A comprehensive dataset for a particular concrete grade is provided, including basic and drying creep strains for different concrete ages at loading, autogenous shrinkage strains, combined autogenous and drying shrinkage strains, time-dependent evolution of mass water contents for creep specimens and companion load-free specimens, the water desorption isotherm, and time-dependent evolution of selected material parameters. It is intended for calibrating and validating concrete models for creep and shrinkage within the framework of single-phase and multi-phase approaches.[45,46] For this purpose, the results of the present investigation will be made available to the scientific community in a database assembled by Hubler et al. at Northwestern University.[13,14]

2 | EXPERIMENTAL PROCEDURES

Creep in compression was investigated in terms of the concrete age at loading of 2, 7, and 28 days on sealed and unsealed specimens, the latter at ambient relative humidity of (65 ± 2.5)%, and at a temperature of (20 ± 1)°C. During the test duration of 1 year, the applied stress was 30% of the average compressive strength, measured at the respective concrete age at loading. Whereas the autogenous shrinkage strains were measured on one load-free companion specimen, starting 1 day after casting, the combined autogenous and drying shrinkage strains were measured on the respective load-free companion specimens, i.e., starting after unsealing at concrete ages of 2, 7, and 28 days. For determining the time-dependent mass water content, a multi-ring-sensor (MRS)[47] was placed at the centre of each cylindrical test specimen. Moreover, shrinkage tests on thin concrete slices served for determining the water desorption isotherm in terms of ambient relative humidity.

2.1 | Material characterization

The investigated concrete can be classified as normal strength concrete, as indicated by the mean value of the cylinder compressive strength f_{cm} = 35.9 MPa at a concrete age of 28 days.[48]

The mix proportions are given in Table 1. To achieve good workability and compaction of the concrete mixture, we used a lignosulphonate plasticizer (Proplast 200 by TAL Betonchemie Handel). Furthermore, an air-entraining admixture (Proair NVX by TAL Betonchemie Handel) was added to the concrete mixture.

The mean values of the compressive strength f_{cm} and the Young's modulus E_{cm} at different concrete ages, determined on specimens sealed immediately after unmolding, are provided in Table 2. In the latter, the material parameters represent the respective mean values determined on three specimens. The term "concrete age" is referred to the beginning of the mixing procedure, i.e., according to Le Chatelier[49] and referenced in Sant et al.,[50] when water comes in contact with the cement.

As shown in Figure 1, and also confirmed by other researchers,[51,52] at the very beginning, the time-dependent increase of the Young's modulus ratio, i.e., the ratio of the actual Young's modulus over the one measured at a concrete age of 28 days, is much faster than the time-dependent increase of the compressive strength ratio, i.e., the ratio of the actual compressive strength over the one measured at a concrete age of 28 days. For instance, the Young's modulus ratio measured at a concrete age of 2 days amounts to about 76%, whereas the respective compressive strength ratio amounts to about 51%. For a concrete age of 7 days, the

TABLE 1 Composition of the investigated concrete mix (mass proportions)

Cement type: CEM II AM (S-L) 42,5 N	**375 kg/m^3**
Water-cement ratio	0.44
Additives:	
Type: Problast 200	0.600% referred to the cement weight
Type: Proair NVX	0.045% referred to the cement weight
Aggregate size distribution	0/4 4/8 8/16 16/32 mm 45 10 25 20%

TABLE 2 Mean values of material parameters at different concrete ages in N/mm^2

	2 days	7 days	28 days	56 days	112 days	365 days
f_{cm}	18.3	29.0	35.9	37.8	42.3	43.1
E_{cm}	25,780	29,120	33,853	34,857	35,793	37,823

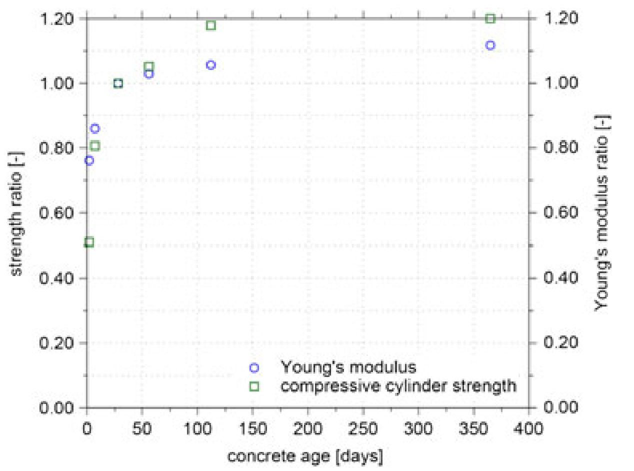

FIGURE 1 Evolution of compressive strength ratio and of Young's modulus ratio, referred to the respective values measured at a concrete age of 28 days

FIGURE 2 Compressive creep tests with two cylindrical specimens mounted on top of each other

Young's modulus ratio yields about 86%; the respective compressive strength ratio amounts to about 81%.

2.2 | Testing program

2.2.1 | Creep tests and companion shrinkage tests

The creep tests in compression were performed on cylinders of 150 mm diameter and 450 mm height. The selected specimen size complies with the respective provisions stated in the American Society for Testing and Materials Designation C512/C512M-10[53] ensuring tests on representative volumes of the material.

The companion shrinkage tests were performed on cylindrical specimens with the same dimensions. The material components summarized in Table 1 were mixed in a rotary pan mixer for 3 min. Afterwards, the concrete mixture was cast in plastic cylinder molds and compacted by external vibration.

All specimens were unmolded after 24 hr. Thereafter, the specimens for the creep and shrinkage tests were immediately protected by one layer of cling film, covered by one layer of bitumen-coated aluminium foil, similar to the sealing procedure described in recent studies[54,55] and recommended by Hubler et al.[13] Hence, at the beginning of the tests for drying creep and combined autogenous and drying shrinkage, the protection could easily be removed in order to obtain unsealed specimens. Both, drying creep and drying shrinkage tests were carried out under the same environmental conditions, i.e., at constant ambient relative humidity of (65 ± 2.5)% and at a temperature of (20 ± 1)°C.

In each of the compressive creep tests, started at concrete ages at loading of 2, 7, and 28 days, one sealed and one unsealed specimen were mounted on top of each other in the testing machine (Figure 2).

According to Figure 2, the force is applied by a hydraulic jack acting on the lower thick metallic plate. In order to limit the oil pressure variations, the hydraulic jack is connected to an oil/nitrogen accumulator. Both the pressure sensor, mounted on the hydraulic jack, and the strain gauges, applied on the spindles, allow permanent monitoring of the applied load.

The loading level was chosen as 30% of the average cylinder compressive strength f_{cm} measured at the respective concrete age at loading. By sudden opening of a valve, the respective load was applied within about 3 s.

The longitudinal strains of both basic and drying creep specimens were determined from elongations measured by means of three inductive displacement transducers WA-2 of Hottinger Baldwin Messtechnik with a measurement length of 200 mm, fixed at 0°, 120°, and 240° (Figure 3 [left]), respectively, along the perimeter of a specimen. The linearity error of the inductive displacement transducers is 0.1%.

The sealed load-free companion specimen for investigating autogenous shrinkage as well as the unsealed load-free

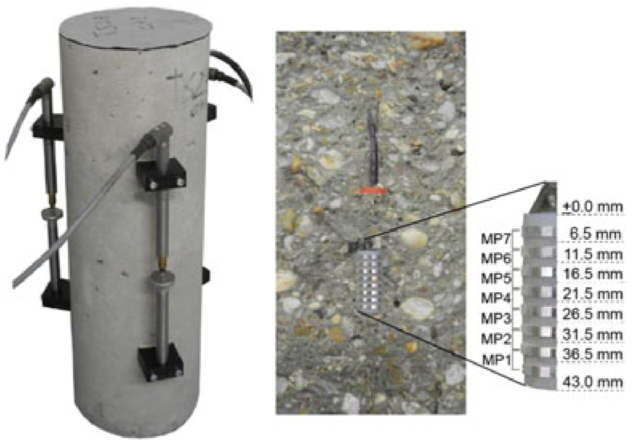

FIGURE 3 Cylindrical concrete specimen with three inductive displacement transducers along the perimeter (left) and multi-ring-sensor placed at the centre of the specimen (right) with the respective measuring points (MPs) and elevations

specimens for investigating combined autogenous and drying shrinkage was equipped with identical measurement devices as the creep test specimens.

The complete testing program, consisting of compressive creep tests with different concrete ages at loading and companion shrinkage tests, is summarized in Table 3.

2.2.2 | Measurement of mass water contents

The water desorption isotherm was determined on thin concrete slices, which were obtained from drill cores by wet sawing. The equilibrium mass water content, representing the ratio of the mass of water in the specimen to the mass of the dry specimen (desiccated at 105°C), was determined by weighing. Thereby, five values of ambient relative humidity, i.e., RH of 97%, 85%, 75%, 59%, and 43%, were considered.

Each cylindrical specimen was equipped with a MRS,[47] placed in axial direction at the centre of the specimen. From the electrolytic resistances, measured at seven measuring points (MPs) extending over a length of 30 mm of the MRS (Figure 3 [right]), the mass water contents were determined by means of the calibration curve for the respective concrete. Tests and numerical simulations confirmed that the MRSs do not influence the measured longitudinal strains. The authors of the present contribution are aware of the fact that determining the mass water content in a very young concrete by means of the calibration curve for MRSs is questionable; however, it is an approved method for hardened concrete. The measuring accuracy of MRSs was investigated in previous studies by comparing measurement data of mass water contents determined by MRSs with the respective data obtained by the gravimetric method. The measuring accuracy of the MRS is about 0.30%. In addition, measurements of mass water contents by MRSs were conducted by the authors and described in previous papers,[56,57] dealing with the evaluation of the effects of drying shrinkage on the behaviour of concrete structures strengthened by overlays, where more details concerning nondestructive measurements with MRSs are provided.

3 | EXPERIMENTAL RESULTS

In this section, the evolution of the autogenous shrinkage strain and combined autogenous and drying shrinkage strain as well as the evolution of the respective mass water content will be presented. For interpretation of the latter, the water desorption isotherm of the investigated concrete is provided. Moreover, the evolution of total longitudinal strains determined on sealed and unsealed specimens, loaded at different concrete ages, is shown. The effect of load application on the moisture content is studied by means of the evolution of the mass water content of the respective specimens. The Pickett effect is re-examined by comparison of time-dependent strains in sealed and unsealed loaded and load-free specimens. In addition, compliance functions for basic creep and for drying creep are provided. Creep and shrinkage measurement data as well as compliance functions, available in the open literature for similar experimental procedures, are compared with the present ones. For reporting creep and shrinkage data, the recommendations provided by Hubler et al.[13] are considered.

It is a common practice to determine the creep strain of a loaded specimen by subtracting the instantaneous elastic strain and the shrinkage strain from the total strain. Thereby, creep and shrinkage are assumed to be independent of each other,[2] and the principle of superposition of creep and shrinkage strains is assumed to be valid.[58–60] Because the loading level for the creep tests in the present investigation amounts to 30% of the average compressive strength, measured at the respective concrete ages at loading, linear viscoelastic behaviour is expected.[54,61,62]

3.1 | Tests on load-free specimens

In Figure 4, the evolution of the autogenous shrinkage strain ε^{as} for the sealed load-free specimen is plotted. Moreover, for comparison, the autogenous shrinkage strains, provided by Lee et al.[60] and by Baroghel-Bouny et al.[63] for similar concrete mixtures, are shown. Whereas the water to cement ratio as well as the aggregate to cement ratio of the concrete mixtures investigated in the studies[60],[63] and in the present study are quite similar, the maximum aggregate size of the respective concrete mixtures is quite different, ranging from 19 and 20 mm for the mixtures described in the studies[60,63] to 32 mm (Table 1) for the one investigated in the present study. The size and shape of the specimens used in Lee et al.[60] and in Baroghel-Bouny et al.[63] and in the present study are different. However, according to Lee et al.,[60] the effect of

TABLE 3 Overview of the testing program

	Basic Creep			Drying Creep			Autogenous Shrinkage	Combined Autogenous and Drying Shrinkage		
Concrete age at unmolding and sealing of the specimen (h)		24			24		24		24	
Concrete age at unsealing t_s in days	-	-	-	2	7	28	-	2	7	28
Concrete age at start of measurements (days)	2	7	28	2	7	28	1	2	7	28
Concrete age at loading t_0 in days	2	7	28	2	7	28	-		-	
Applied compressive stress (N/mm^2)	5.5	8.7	10.8	5.5	8.7	10.8	-		-	
Test duration (months)		12			12		12		12	

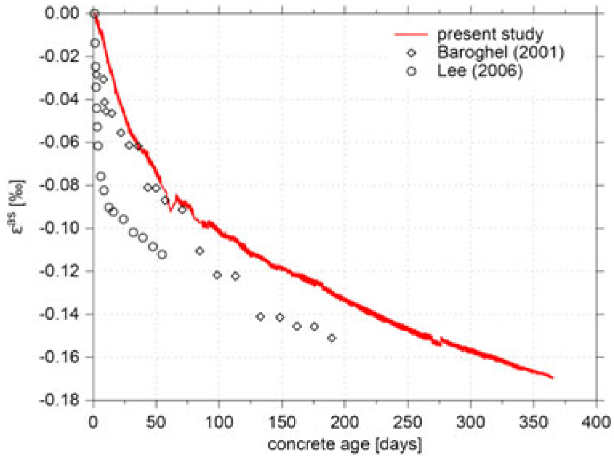

FIGURE 4 Comparison of autogenous shrinkage strain of the present study with test data according to Lee et al.[60] and Baroghel-Bouny et al.[63]

the specimen shape on autogenous shrinkage is insignificant. Determination of the autogenous shrinkage strain, reported by Lee et al.,[60] was started just after casting. Comparison of the present test data, representing the mean value of three measurements along the perimeter of the specimen (cf. Figure 3 [left]), starting 1 day after casting, with the ones described in Lee et al.[60] and in Baroghel-Bouny et al.[63] is only possible in a qualitative manner.

The evolution of combined autogenous and drying shrinkage strains ε^{cads}, determined on the load–free specimens after removal of the bitumen-coated aluminium foil and the cling film at concrete ages of t_s = 2, 7, and 28 days, respectively, is shown in Figure 5. Moreover, the combined autogenous and drying shrinkage strains, determined on specimens of similar concrete mixtures and exposed to equal drying conditions at different concrete ages, i.e., at 1 day,[63] 3, 7, and 28 days,[64] respectively, are displayed for comparison. Because the experimental investigations described in Baroghel-Bouny et al. [63] and in Wesche et al. [64] were carried out on specimens of different shape and size, the present test data can be compared with the latter only in a qualitative manner. The comparison between the combined autogenous and drying shrinkage strains, determined by Wesche et al.[64] on specimens, unsealed at 3, 7, and 28 days, and the present ones, shows quite good agreement. The combined autogenous and drying shrinkage strain for the specimen exposed to drying 2 days after casting is larger than the ones measured for the specimens exposed to drying 7 and 28 days after casting. This can be explained by the capillary pressure, which, with increasing concrete age at exposure to drying, is acting on a stiffer solid skeleton resulting in smaller negative volumetric strain.[65] The shrinkage strain provided by Baroghel-Bouny et al.[63] increases faster than in the present study, because exposure to drying conditions started already 1 day after casting.

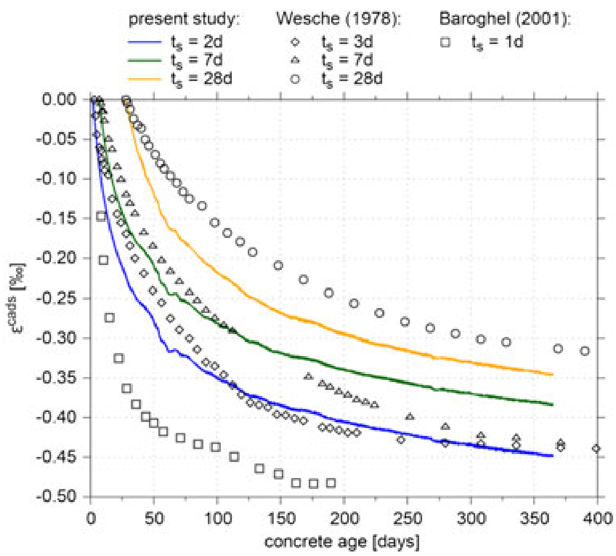

FIGURE 5 Combined autogenous and drying shrinkage strains, determined on concrete specimens exposed to drying at t_s = 2, 7, and 28 days, and comparison with the test data according to Wesche et al.[64] and Baroghel-Bouny et al.[63]

Figure 6 shows the evolution of the mass water content for the load-free specimens. On the basis of a calibration curve, relating the measured specific electrolytic resistance

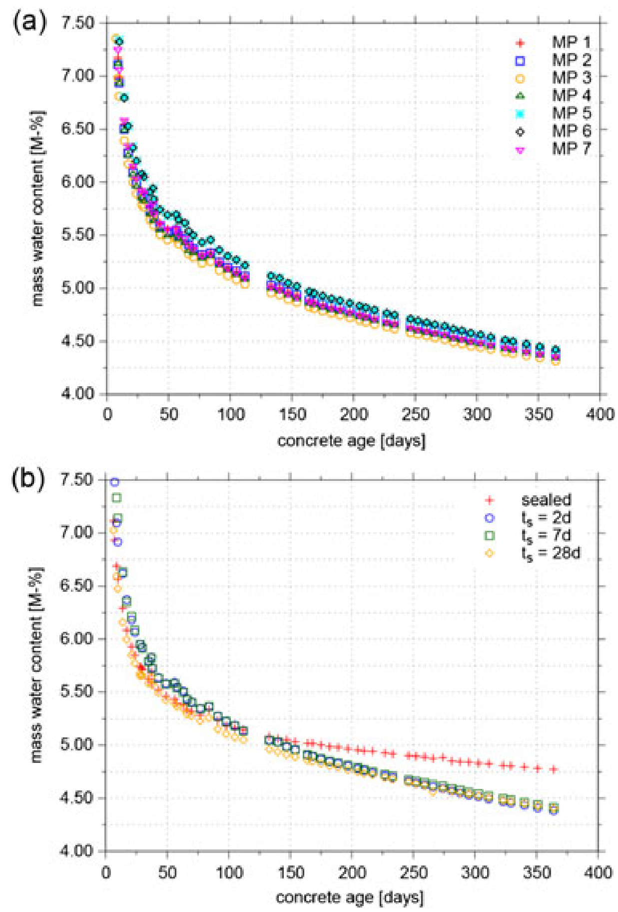

FIGURE 6 Evolution of (a) the mass water contents measured at the seven measuring points (MPs) of the multi-ring-sensor embedded in the load-free specimen exposed to drying at t_s = 2 days and (b) the mean values of seven MPs of single multi-ring-sensors embedded in the sealed load-free specimen and in the load-free specimens exposed to drying at t_s = 2, 7, and 28 days after casting

to the mass water content,[47] the seven measured values of the specific electrolytic resistance for each MRS were converted to mass water contents as a function of depth. Figure 6a shows the mass water content, i.e., the ratio of the mass of water contained in the load-free specimen exposed to drying 2 days after casting, to the mass of the dry specimen, determined at the seven MPs (Figure 3 [right]) of the respective MRS, as a function of drying time. The standard deviation of the measurement data from the seven MPs of the MRS is very small, namely 0.09%, 0.11%, and 0.04% at concrete ages of 7, 28, and 365 days. The depicted mass water contents in Figure 6b represent the mean values of seven MPs of a single MRS, embedded in the sealed concrete cylinder for investigating autogenous shrinkage and in the load-free specimens exposed to drying 2, 7, and 28 days after casting, respectively. According to Figure 6b, the mass water contents measured at the centre of the load-free specimens, exposed to drying at different concrete ages, are almost identical after 1 year of drying. The standard deviation of the mass water contents, determined on the sealed specimen and on the specimens exposed to drying 2, 7, and 28 days after casting, is provided in Table 4 for concrete ages of 7, 28, and 365 days.

About 100 days after casting, a difference of the gradient of mass water content between the sealed specimen and the unsealed specimens emerges. The steeper gradients observed for the unsealed specimens with progressing drying time can be explained by the moisture migration in the specimens and moisture transfer to the environment. To establish the relationship between the mass water content and relative humidity, the mass loss until achieving sorption equilibrium, at different values of ambient relative humidity, as well as the water desorption isotherm of the employed concrete, is shown in Figure 7a and Figure 7b (mean values and standard deviation).

At the end of drying time, at constant ambient relative humidity of (65 ± 2.5)% and at a temperature of (20 ± 1)°C, a mass water content of about 3.0% is expected for the unsealed specimens, based on the water desorption isotherm provided in Figure 7b. For the sealed specimen, a mass water content of about 4.7% is predicted at the end of the self-desiccation process, assuming thereby a relative humidity of about 95%. For comparison, the internal relative humidity measured on samples immediately enclosed into sealed cells upon casting, described by Baroghel-Bouny et al.,[66] amounts to 94% after 6 months. The concrete mixture investigated in this investigation study[66] is similar to the present one.

TABLE 4 Standard deviation of mass water contents determined on the sealed specimen and the specimens exposed to drying at different concrete ages

Concrete age	Sealed Specimen	Specimens Exposed to Drying at		
		2 days	7 days	28 days
7 days	0.31%	0.10%	0.10%	0.31%
28 days	0.15%	0.12%	0.13%	0.21%
365 days	0.08%	0.04%	0.07%	0.07%

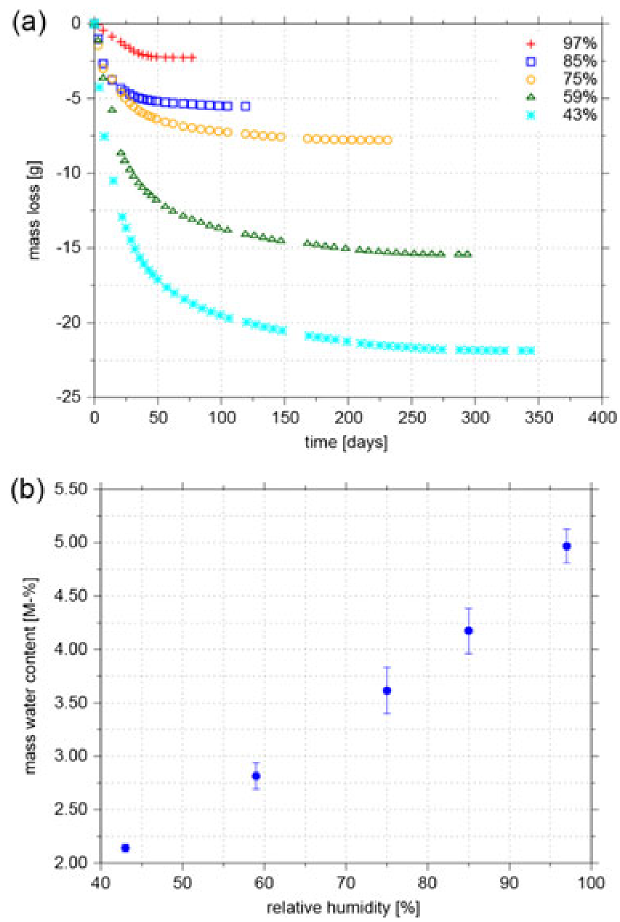

FIGURE 7 Results of the tests on thin concrete slices: (a) mass loss until achieving sorption equilibrium at RH of 97%, 85%, 75%, 59%, and 43% and (b) water desorption isotherm (mean values and standard deviation)

3.2 | Tests on sealed loaded specimens

For the three sealed specimens, loaded at concrete ages of $t_0 = 2$, 7, and 28 days to 30% of the respective compressive strength, the evolution of the total longitudinal strains $\varepsilon_{\text{sealed}}^{\text{total}}$ is depicted in Figure 8. The latter consists of the instantaneous elastic strains ε^{ci}, the basic creep strains ε^{bc}, and the autogenous shrinkage strains ε^{as}:

$$\varepsilon_{\text{sealed}}^{\text{total}} = \varepsilon^{ci} + \varepsilon^{bc} + \varepsilon^{as}. \tag{1}$$

Similar to Tamtsia et al.,[30] ε^{ci} was determined from stress–strain plots. The authors of the present investigation are aware of the fact that during load application, concrete already exhibits creep deformations, as also described in the recommendations by RILEM committee TC 107.[67,68]

From comparison of the total strains of the three sealed specimens, increasing absolute values of ε^{ci} (indicated by red dots) for increasing concrete ages at loading can be observed due to the different time-dependent development of Young's modulus and compressive strength, as shown in Figure 1.

Moreover, Figure 8 shows the evolution of total longitudinal strains provided by Rossi et al.[54] for a concrete age

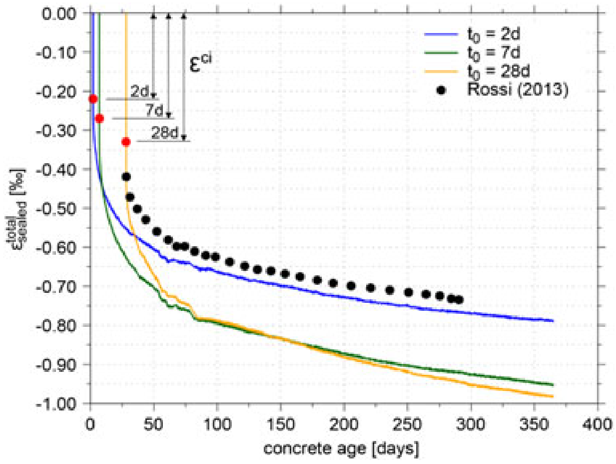

FIGURE 8 Evolution of the total longitudinal strains for the sealed specimens, loaded at t_0 = 2, 7, and 28 days, and comparison with test data of Rossi et al.[54]

at loading of 28 days. The loading level in the latter investigation is the same as in the present one; however, both Young's modulus and compressive strength of the concrete at the age of 28 days are higher than the present ones. Hence, the evolution of total longitudinal strains of the present study with the one provided in Rossi et al.[54] can be compared only in a qualitative manner.

The compliance functions determined on sealed specimens, loaded at concrete ages t_0 = 2, 7, and 28 days after casting to 30% of the respective compressive strength, are shown in Figure 9. According to the American Concrete Institute (ACI) code,[2] the compliance function is defined as the total load-induced strain, i.e., $\varepsilon_{sealed}^{total} - \varepsilon^{as} = \varepsilon^{ci} + \varepsilon^{bc}$ (see Equation [1]), per unit of the applied stress $\sigma(t_0)$. Expectedly, the compliance of sealed specimens decreases with increasing concrete ages at loading. Comparison of the present measurement data with the one provided in Lee et al.[60] for a similar concrete mixture and an equal loading level, however with concrete ages at loading of 1, 3, 7, and 28 days, respectively, shows very good agreement.

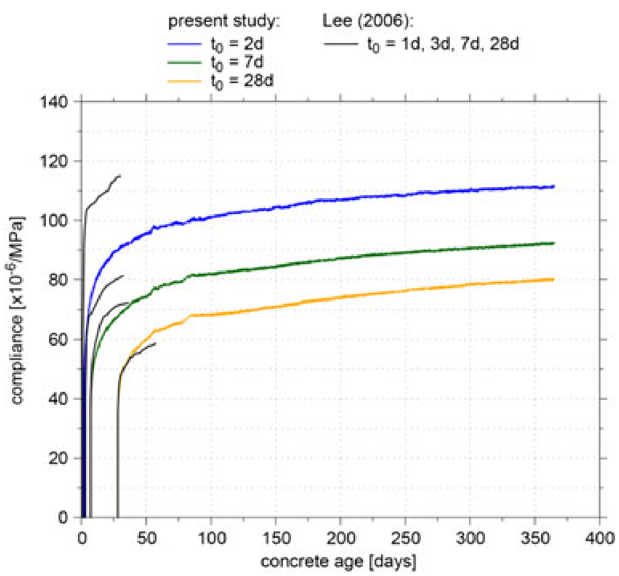

FIGURE 9 Compliance functions determined on sealed specimens, loaded at t_0 = 2, 7, and 28 days, and comparison with test data of Lee et al.[60]

One year after casting, the ratio between the compliance of specimens loaded at concrete ages of 2 and 7 days after casting is about 1.2, whereas the respective ratio for specimens loaded at concrete ages of 2 and 28 days after casting is about 1.4.

Comparison of the evolution of the mass water contents measured on the sealed specimens, loaded at concrete ages t_0 = 2, 7, and 28 days to 30% of the respective compressive strength (Figure 10), does not show any effect of load application on the hygral state. Moreover, they match quite well with the mass water content of the sealed load-free companion specimen. Whereas the standard deviation of the mass water contents, determined on sealed specimens, loaded 2, 7, and 28 days after casting, is provided in Table 5 for concrete ages of 7, 28, and 365 days, the respective standard deviation of the sealed load-free companion specimen is shown in Table 4.

The small differences of less than 0.30% between the depicted mass water contents are within the measuring accuracy of the MRSs.

Whereas the MRSs, used in the present investigation, were placed in the interior of the cylindrical specimens before concreting, the miniature sensor HC2-C05, used in Wyrzykowski et al.[36] for measuring the relative humidity in the mortar samples, was placed in a precast hole of 10 mm diameter and 70 mm length. Thus, only the MRS, with dimensions similar to the maximum grain size, is completely integrated in the concrete. Based on the water desorption isotherm for the present concrete (Figure 7b), which is typical for an ordinary concrete,[66] an increase of internal relative humidity from about 89% to about 91%, as observed by Wyrzykowski et al.[36] due to load-application, yields a negligible effect on the mass water content as well.

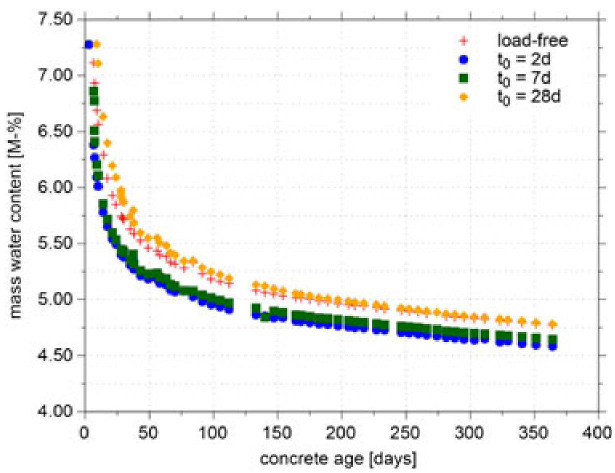

FIGURE 10 Evolution of mass water contents for sealed specimens loaded at t_0 = 2, 7, and 28 days, and for the load-free companion specimen

TABLE 5 Standard deviation of mass water contents determined on sealed specimens loaded at different concrete ages

Concrete age	Sealed Specimens Loaded at		
	2 days	7 days	28 days
7 days	0.18%	0.11%	0.07%
28 days	0.07%	0.03%	0.03%
365 days	0.01%	0.04%	0.04%

3.3 | Tests on unsealed loaded specimens

The evolution of total longitudinal strains determined on unsealed specimens $\varepsilon_{\text{unsealed}}^{\text{total}}$, loaded at concrete ages $t_0 = 2$, 7, and 28 days to 30% of the respective cylindrical compressive strength, is presented in Figure 11. Moreover, the evolution of total longitudinal strains, reported in one study[64] for concrete ages at loading of $t_0 = 3$, 7, and 28 days, is displayed. The specimens, used by Wesche et al.,[64] were exposed to equal drying conditions; however, they were of different size compared to the present ones. Furthermore, the loading level in Wesche et al.[64] was specified as one-third of the respective cubic compressive strength at loading, i.e., it was considerably higher than the present one. Hence, only a qualitative comparison between the experimental results, described by Wesche et al.,[64] with the present ones is possible.

$\varepsilon_{\text{unsealed}}^{\text{total}}$ consists of instantaneous elastic strain ε^{ci}, basic creep strain ε^{bc}, drying creep strain ε^{dc}, and combined autogenous and drying shrinkage strain ε^{cads}:

$$\varepsilon_{\text{unsealed}}^{\text{total}} = \varepsilon^{ci} + \varepsilon^{bc} + \varepsilon^{dc} + \varepsilon^{cads}. \tag{2}$$

Equation 2 is based on the assumptions that the basic creep strains are identical for sealed and unsealed conditions and the combined autogenous and drying shrinkage strains are equal for loaded and load-free specimens.

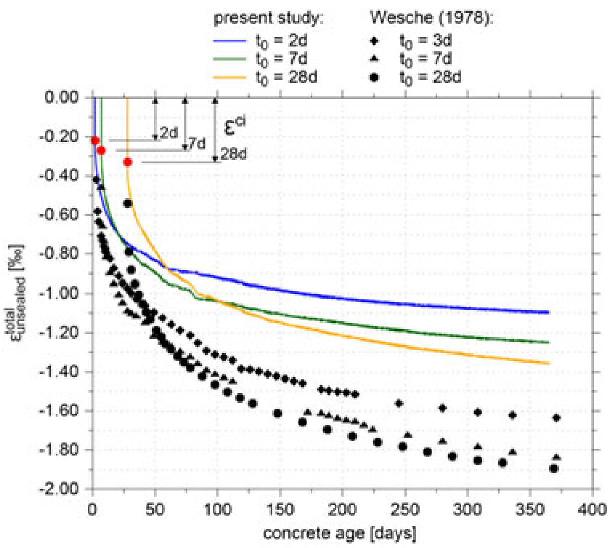

FIGURE 11 Evolution of the total longitudinal strains for unsealed specimens loaded at $t_0 = 2$, 7, and 28 days, and comparison with test data of Wesche et al.[64]

As expected, ε^{ci} (indicated by red dots) is equal for sealed (Figure 8) and unsealed (Figure 11) specimens, since the respective load was applied within a few seconds and, consequently, drying effects during load application are negligible.

According to Figure 11, the total longitudinal strains increase faster for increasing concrete ages at loading due to the larger sustained compressive stress. Compared to the present investigation, Wesche et al.[64] reported larger total longitudinal strains. The differences are related to lower values of Young's modulus at the respective concrete ages at loading and to the higher loading level. Rearranging Equation 2 yields

$$\varepsilon_{\text{unsealed}}^{\text{total}} - \varepsilon^{ci} - \varepsilon^{bc} = \varepsilon^{dc} + \varepsilon^{cads}. \tag{3}$$

$\varepsilon_{\text{unsealed}}^{\text{total}}$ and ε^{ci} are depicted in Figure 11, ε^{bc} is known from subsection 3.2, and ε^{cads} is known from Figure 5. Hence, the only unknown in Equation 3 is ε^{dc}. In Figure 12, $\varepsilon_{\text{unsealed}}^{\text{total}} - \varepsilon^{ci} - \varepsilon^{bc}$, representing the strain associated with drying of the loaded specimens, is plotted against ε^{cads}, denoting the respective strain of the load-free companion specimen, plotted in Figure 5.

As shown in Figure 12, in good approximation, a linear relationship between $\varepsilon_{\text{unsealed}}^{\text{total}} - \varepsilon^{ci} - \varepsilon^{bc}$ and ε^{cads} can be assumed. For a concrete age at loading of 2 days, the respective regression line is characterized by a slope close to one and, hence, ε^{dc} is very small. For increasing concrete ages at loading, i.e., at 7 and 28 days, the respective regression lines are characterized by increasingly steeper slopes and, hence, the drying creep strain ε^{dc} or, equivalently, the Pickett effect is increasing. As already mentioned, the latter is explained by two different mechanisms: (a) the apparent mechanism due to surface microcracking[10] and (b) a true mechanism denoted as stress-induced shrinkage.[11] The experimental results are in line with Gamble and Parrott,[69]

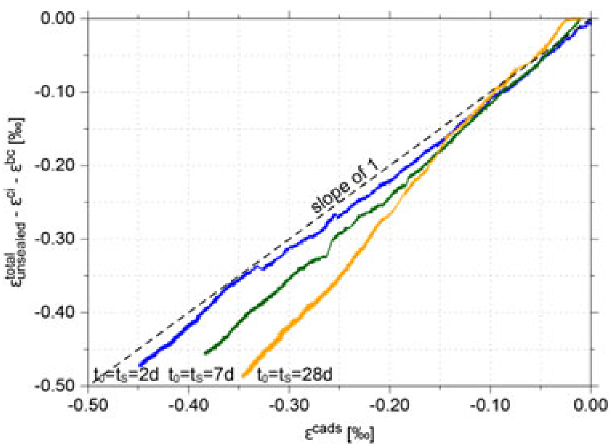

FIGURE 12 Relationship between strains associated with drying of specimens, loaded at $t_0 = 2$, 7, and 28 days, and the respective combined autogenous and drying shrinkage strains for load-free specimens, exposed to drying at $t_s = 2$, 7, and 28 days

who found that the development of strain in excess of basic creep is linearly related to that of combined autogenous and drying shrinkage independent of the age at loading.

The increasingly larger strain associated with drying of the loaded specimen compared to the respective strain of the load-free specimen for increasing concrete ages at loading, depicted in Figure 12, is also confirmed by the findings of Wittmann and Roelfstra.[10] They attributed this effect mainly to microcracking due to nonuniform drying in unloaded specimens, which results in smaller strains. Since microcracking is suppressed by compressive stresses, combined autogenous and drying shrinkage strains increase. As in the present study, the applied compressive stress is larger at higher concrete ages at loading, microcracking is suppressed to a larger extent in specimens with higher concrete ages at loading. Thus, the strain, associated with drying of the loaded specimens, increases with increasing age at loading. Furthermore, according to Bažant et al.,[3] the microcracking effect decreases with the moisture distribution approaching a uniform state, whereas stress-induced shrinkage increases with time.

In addition to Figure 12, the Pickett effect is also shown by comparing the total time-dependent strain $\varepsilon_{\text{unsealed}}^{\text{total}} - \varepsilon^{ci}$ with the sum of the basic creep strain and the combined autogenous and drying shrinkage strain $\varepsilon^{bc} + \varepsilon^{cads}$. In Figure 13, $\varepsilon_{\text{unsealed}}^{\text{total}} - \varepsilon^{ci}$ is depicted by solid lines for specimens loaded at concrete ages $t_0 = 2$, 7, and 28 days and the sum $\varepsilon^{bc} + \varepsilon^{cads}$ is shown by dashed lines with ε^{bc} determined on the sealed specimens, loaded at $t_0 = 2$, 7, and 28 days, and ε^{cads} obtained from the load-free specimens, exposed to drying at $t_s = 2$, 7, and 28 days. From Equation 2 follows

$$\varepsilon_{\text{unsealed}}^{\text{total}} - \varepsilon^{ci} = \varepsilon^{bc} + \varepsilon^{dc} + \varepsilon^{cads}. \tag{4}$$

Hence, for a nonvanishing drying creep strain, from Equation 4, one obtains the inequality

$$\left|\varepsilon_{\text{unsealed}}^{\text{total}} - \varepsilon^{ci}\right| > \left|\varepsilon^{bc} + \varepsilon^{cads}\right|. \tag{5}$$

Figure 13 confirms this inequality. Whereas for a concrete age at loading of 2 days the Pickett effect is very small, it increases with increasing concrete ages at loading as explained above.

In Figure 14, the compliance functions determined on unsealed specimens, loaded at concrete ages $t_0 = 2$, 7, and 28 days after casting to 30% of the respective compressive strength, are compared with the ones determined on sealed specimens (Figure 9). Whereas the former are displayed in color, the latter are shown in grey.

According to the ACI code,[2] the compliance function of unsealed specimens is defined as the total load-induced strain, i.e., $\varepsilon_{\text{unsealed}}^{\text{total}} - \varepsilon^{cads} = \varepsilon^{ci} + \varepsilon^{bc} + \varepsilon^{dc}$ (see Equation [2]), per unit of the applied stress $\sigma(t_0)$.

As shown in Figure 14, expectedly, the compliance of specimens exposed to drying decreases with increasing concrete ages at loading, similar to the compliance of sealed specimens. The difference between the compliance, determined on unsealed specimens, and the one, obtained on sealed specimens, increases for increasing concrete ages at loading. For instance, 1 year after casting, the respective differences for concrete ages at loading of 2, 7, and 28 days amount to about 4%, 9%, and 16%, respectively. This finding can be explained by stress-induced shrinkage and surface microcracking.

Figure 15 displays the evolution of the mass water contents, measured by means of MRSs at the centre of unsealed specimens, loaded at concrete ages $t_0 = 2$, 7, and 28 days to 30% of the respective compressive strength, and of the load-free companion specimens exposed to drying at the respective concrete ages t_s (cf. Figure 6). Whereas the standard deviation of the mass water contents, determined on unsealed specimens, loaded at the respective concrete ages, is provided in Table 6 for concrete ages of 7, 28, and 365 days,

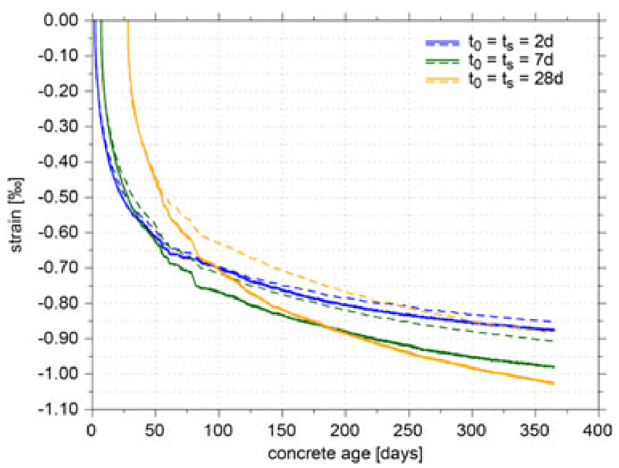

FIGURE 13 Comparison of the total time-dependent strains (solid lines) with the sum of basic creep strains of specimens, loaded at $t_0 = 2$, 7, and 28 days, and combined autogenous and drying shrinkage strains (dashed lines) of specimens, exposed to drying at $t_s = 2$, 7, and 28 days

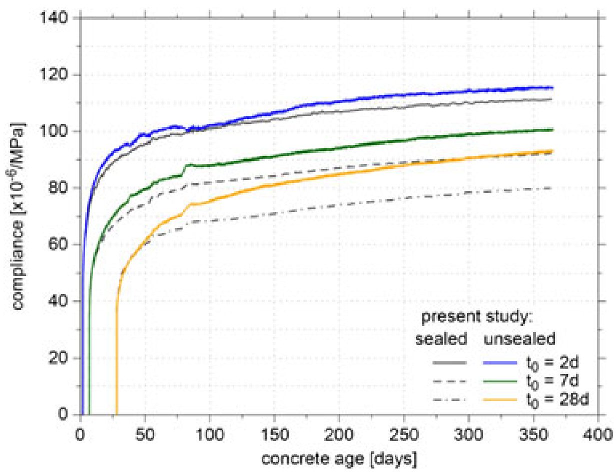

FIGURE 14 Comparison of the compliance functions determined on unsealed specimens (displayed in color), loaded at $t_0 = 2$, 7, and 28 days, with the respective ones, determined on sealed specimens (displayed in grey)

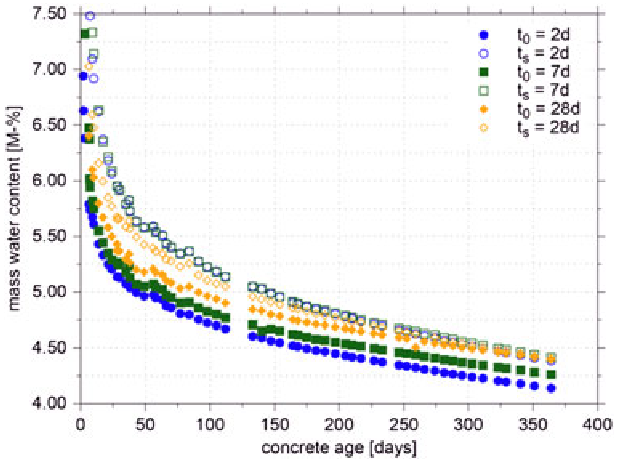

FIGURE 15 Evolution of the mass water contents for unsealed specimens, loaded at $t_0 = 2$, 7, and 28 days to 30% of the respective compressive strength (filled dots), and for the load-free companion specimens, exposed to drying at the respective concrete ages t_s (open dots)

TABLE 6 Standard deviation of mass water contents determined on unsealed specimens loaded at different concrete ages

Concrete age	Unsealed Specimens Loaded at		
	2 days	7 days	28 days
7 days	0.11%	0.22%	0.15%
28 days	0.09%	0.10%	0.11%
365 days	0.04%	0.05%	0.07%

the standard deviation of the load-free companion specimens is shown in Table 4.

Within the first few months after casting slightly smaller mass water contents are observed in the loaded specimens. However, the differences are decreasing with progressing time, and after 1 year, they are within the measuring accuracy of the MRSs. Similar to the results obtained on sealed specimens, shown in Figure 10, no measurable change in mass water content during load application and immediately thereafter occurs.

The present measurement data are in agreement with the study by Bažant and Chern.[9] In the latter, water loss of concrete due to compressive stress is excluded, as also observed in Cagnon et al.[39] In the present investigation, from the water desorption isotherm (Figure 7b) at the end of drying to 65% RH, a mass water content of about 3.0% is expected, regardless of whether the specimen is loaded or load-free.

As is well known and has been confirmed by the experimental investigations,[56,57,66] due to the low permeability of concrete, moisture migration in the specimen and moisture transfer to the environment are extremely slow processes. For instance, according to Bažant[70] moisture equilibrium in a specimen of 25 mm of thickness will be reached after about 2 years of drying from 100% ambient relative humidity to 50%.

Expectedly, the mass water contents measured at the centre of sealed (Figure 10) and unsealed (Figure 15) specimens, loaded at different concrete ages, are quite similar during the first year after casting with only slightly smaller mass water contents in the unsealed specimens.

4 | CONCLUSIONS

The present contribution aimed at the experimental investigation of the time-dependent behaviour of normal strength concrete, enhanced by measuring the evolution of the respective mass water content. Creep in compression was investigated in terms of the concrete age at loading of 2, 7, and 28 days on sealed and unsealed cylindrical specimens, the latter at ambient relative humidity of (65 ± 2.5)% and at a temperature of (20 ± 1)°C. During the test duration of 1 year, the applied stress was 30% of the compressive strength at the respective concrete age at loading.

Whereas the autogenous shrinkage strains were measured on one load-free sealed specimen, starting 1 day after casting, the combined autogenous and drying shrinkage strains were measured on load-free specimens, starting after unsealing at concrete ages of 2, 7, and 28 days. The evolution of the mass water contents for both sealed loaded and unsealed loaded specimens as well as for the load-free companion specimens was determined by means of MRS. The latter were placed at the centre of each cylindrical test specimen. To establish the relationship between mass water content and relative humidity, we determined the water desorption isotherm on thin concrete slices. Moreover, the time-dependent increase of the compressive strength and of the Young's modulus was determined during the first year after casting. The major findings of the present investigation can be summarized as follows:

- Compared to the combined autogenous and drying shrinkage strain, the autogenous shrinkage strain is not negligible.
- Due to the different time-dependent development of Young's modulus and compressive strength, increasing absolute values of instantaneous elastic strains for increasing concrete ages at loading, determined on both sealed and unsealed specimens, were observed.
- Comparison of the evolution of mass water contents for sealed loaded and sealed load-free specimens did not show any effect of load-application on the hygral state.
- For unsealed specimens, independent of the concrete age at loading at 2, 7, and 28 days, a nearly linear relationship between the strains associated with drying of the loaded specimens and the combined autogenous and drying shrinkage strain of the load-free specimens was observed.
- For a concrete age at loading of 2 days, the Pickett effect was found to be very small because of the low sustained compressive stress amounting to 30% of the respective compressive strength at loading. However, the Pickett effect increases with increasing concrete age at loading

due to the larger sustained compressive stress, resulting in stress-induced shrinkage and the suppression of microcracking due to nonuniform drying. Similarly, the difference between the compliance of unsealed and sealed specimens increases for increasing concrete age at loading.

- Comparison of the evolution of mass water contents for unsealed loaded and unsealed load-free specimens did not show any significant impact of load-application on the hygral state.

All measured quantities of the present study, like the autogenous shrinkage strain, the combined autogenous and drying shrinkage strain, the mass water contents for sealed and unsealed, loaded and load free specimens, and the total strains of loaded sealed and unsealed specimens, were compared to experimental results of similar previous studies. However, this comparison was only feasible in a qualitative manner by reverting to several previous studies on specimens with similar but somewhat different material properties and different loading conditions; as to the best knowledge of the authors, no previous single study, focusing on all the above mentioned quantities, is available.

The results of the present investigation provide a consistent set of material data for a particular concrete grade for calibrating and validating improved creep and shrinkage prediction models, in particular for multiphase concrete models. They will be made available to the scientific community via the database assembled by Hubler et al. at Northwestern University.[13,14]

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5 Study of Time-Dependent Material Properties of Shotcrete

In Publication IV, *Time-Dependent Material Properties of Shotcrete: Experimental and Numerical Study*, a novel experimental program for shotcrete is presented.

For the numerical simulations presented in Publication I and Publication II, experimental results from the literature, i.e., those presented by Huber (1991) and M. Müller (2001), were employed for calibrating and validating the material models. However, the shotcrete compositions investigated in those publications do not represent the state of the art of modern shotcrete anymore. Motivated by this fact, a new experimental program on the evolution of stiffness, uniaxial compressive strength, creep and shrinkage was designed.

The creep and shrinkage tests were planned based on the experiences gained during the experimental program presented in Publication III. For the experimental tests, shotcrete specimens were sampled directly at a construction site of the Brenner Base Tunnel, and they were subsequently transported and tested in the laboratory at the University of Innsbruck. Two different techniques were employed for sampling the specimens, aiming at an early start of the experimental tests. Based on the new experimental results, the SCDP model, the Meschke model and the Schädlich model are calibrated, and their performance in representing the experimentally observed material behavior of a modern shotcrete composition is studied and discussed.

The scientific contribution by the author comprises

- the design of the new experimental program,
- the preparation of the specimens, jointly with Tobias Cordes,
- the conduction of the experimental tests, jointly with Martin Drexel,
- the evaluation of the experimental results,
- the calibration of the SCDP model, the Schädlich model and the Meschke model on the basis of the new experimental results,
- the preparation of the manuscript, jointly with Günter Hofstetter.

materials

Article

Time-Dependent Material Properties of Shotcrete: Experimental and Numerical Study

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Abstract: A new experimental program, focusing on the evolution of the Young's modulus, uniaxial compressive strength, shrinkage and creep of shotcrete is presented. The laboratory tests are, starting at very young ages of the material, conducted on two different types of specimens sampled at the site of the Brenner Basetunnel. The experimental results are evaluated and compared to other experiments from the literature. In addition, three advanced constitutive models for shotcrete, i.e., the model by Meschke, the model by Schädlich and Schweiger, and the model by Neuner et al., are validated on the basis of the test data, and the capabilities of the models to represent the observed shotcrete behavior are assessed. Hence, the gap between the the outdated experimental data on shotcrete available in the literature on the one hand and the nowadays available advanced shotcrete models, on the other hand, is closed.

Keywords: shotcrete; sprayed concrete; material tests; constitutive model; creep; shrinkage; damage model; plasticity model

1. Introduction

The proper representation of the constitutive behavior of shotcrete is an important prerequisite for realistic numerical simulations of deep tunnels driven by the New Austrian Tunneling Method. In the past 30 years, several approaches for describing the complex, time-dependent material behavior of shotcrete have been presented in the literature. Besides the commonly employed approach in practice, based on linear elasticity with a hypothetical Young's modulus for shotcrete, to consider time-dependent material behavior and loading in an approximate manner [1], two of the most widely used shotcrete models are the viscoplastic model by Meschke [2] and the viscoelastic-plastic model by Schädlich and Schweiger [3]. Recently, a new damage plasticity model for shotcrete and a comparison of the latter with the models by Meschke and by Schädlich and Schweiger were published by Neuner et al. [4] and the performance of the aforementioned models was compared on the basis of a 2D benchmark example of deep tunnel advance in [5].

However, most of the papers on constitutive models for shotcrete rely on rather old experimental data for validation and evaluation of the proposed models. These quite old experimental programs comprise (i) the evolution of the Young's modulus and the uniaxial compressive strength up to the age of 28 days and short-term uniaxial and triaxial compression tests on shotcrete specimens of different age by Sezaki et al. [6], (ii) the evolution of the uniaxial compressive strength up to the age of 28 days and stress–strain relations from uniaxial compression tests and the evolution of the total strain in creep and shrinkage tests by Aldrian [7] and Golser et al. [8], (iii) the evolution of temperature due to hydration, of the Young's modulus and of the uniaxial compressive strength up to the age of 7 days as well as the evolution of the total strain in shrinkage and creep tests, reported by Huber [9], (iv) the evolution of the Young's modulus and uniaxial compressive strength and the results of relaxation and shrinkage tests up to the age of 7 days by Fischnaller [10] and (v) the evolution of stiffness and strength, short-term uniaxial compression tests and experiments for determining the total strain in shrinkage and creep tests by Müller [11].

More recently, experimental studies on the bond strength between shotcrete and sandblasted concrete [12] and on the bond strength between shotcrete and hard rock [13] were published. In the former, additionally the evolution of the uniaxial compressive strength up to the age of 28 days and the drying shrinkage behavior up to the age of 112 days is described, whereas in the latter only the evolution of the uniaxial compressive strength up to the age of 72 h is reported. In addition, a detailed summary of experimental results on shotcrete available in the literature and a discussion can be found in [14].

However, the available experimental data on the time-dependent material behavior of shotcrete is limited and many of the reported experimental data sets are outdated by today, since nowadays different shotcrete compositions accompanied by improved technology are used. These shortcomings motivated the design of a new a experimental program.

The paper is structured as follows: in the next two sections, the experimental program for determining material data of shotcrete and the test results are described. In the subsequent sections, the shotcrete models by Meschke [2], Schädlich and Schweiger [3] and Neuner et al. [4] are reviewed briefly, the calibration of the shotcrete models on the basis of the presented experimental data is described and their ability to reproduce the experimental data is evaluated.

2. Experimental Program

The experimental program on shotcrete comprises the evolution of (i) the Young's modulus and (ii) uniaxial compressive strength and the time-dependent development of the total strain in (iii) shrinkage tests and (iv) creep tests, both on sealed specimens.

The employed shotcrete composition, designated as *Sp C25/30/(56)/ÜK3/J2/XC4/XF3/GK8* according to the Austrian guideline for shotcrete in [15], is listed in Table 1. *Sp C25/30/(56)* characterizes the shotcrete strength class, *ÜK3* the monitoring class, *J2* the early age strength class according to [16], *XC4* and *XF3* the exposure classes, and *GK8* a maximum aggregate size of 8 mm. The composition investigated in the present study is representative for commonly employed modern shotcrete compositions used within the New Austrian Tunneling Method.

Table 1. Composition of the shotcrete.

Property	Quantity	Unit
Water content	203	kg/m^3
Cement content *CEM I* 52.5*N(sb)*	380	kg/m^3
Limestone sand (0/4)	1031	kg/m^3
Crushed limestone aggregates (4/8)	694	kg/m^3
Additive *Fluasit*	40	kg/m^3
Accelerator *Mapequick 043 FFG*	7.5–8.5	%

All specimens were sampled at the site of the Brenner Basetunnel during the securing measures of tunnel advance with a shotcrete shell. The specimens were taken from three different locations of the tunnel face in order to capture the variability of shotcrete properties due to the adjustment of the amount of the added accelerator at the tunnel site according to current requirements and ambient conditions.

The tests were conducted on two different types of specimens: (i) drill cores sampled from spray boxes, which is the commonly employed type of specimen for determining material properties of shotcrete, and (ii) cylindrical specimens directly sampled from tubular molds, denoted as *sprayed* specimens. For the latter longitudinally slotted PVC tubes, mounted on wooden rigs, were used as molds (Figure 1). They were closed in circumferential direction by metal pipe clamps and the operator sprayed the shotcrete directly into the PVC tubes. The clamps were removed at unmolding of the specimens. The sprayed specimens were used for tests on shotcrete younger than 24 h, since drill cores cannot be sampled for this early shotcrete age due to the low material strength. The quality of the sprayed specimens was evaluated by means of visible cavities at the surface. Faulty specimens due to shotcrete rebound and air pockets were sorted out. Suitable specimens were obtained if the spraying direction was nearly coaxial with the axis of the tubular molds. For the present shotcrete composition, sprayed specimens could be unmolded safely already 6 h after casting.

Figure 1. Molds for sprayed specimens: slotted tubes, fixed with a clamp on the upper side and mounted on a wooden rig.

A basic requirement for experiments are smooth top and bottom surfaces of the specimens, which are parallel to each other. Since cutting was impossible at young shotcrete ages, compensation layers, consisting of a fast-curing epoxy resin, were applied onto the top surfaces of the specimens. No treatment of the bottom surfaces was necessary.

Both drill cores and sprayed specimens had a diameter of 100 mm. The length of specimens for determining the Young's modulus and the uniaxial compressive strength was 200 mm, while specimens for shrinkage and creep tests had a length of 300 mm. The number of intact specimens for each type of test is provided in Table 2.

All specimens were sealed and stored in an air-conditioned chamber with an ambient temperature of (20 ± 1) °C until testing. The same constant ambient conditions were ensured during the long-term shrinkage and creep tests.

Experiments on the evolution of the uniaxial compressive strength were started at the shotcrete age of 6 h, while tests on the evolution of the Young's modulus were started at the shotcrete age of 24 h. Both tests were performed up to the shotcrete age of 28 days. For shotcrete younger than 24 h, exclusively sprayed specimens were used. For tests on shotcrete older than 24 h, both types of specimens were used. Between the two specimen types, no significant differences of test results were detected and, thus, no distinction between them is made in the following. In contrast, Müller [11] observed differences in the determined material properties for different types of specimens.

Tests for determining shrinkage and creep of shotcrete were only performed on sealed, sprayed specimens. After unmolding, the specimens were sealed using a bitumen–aluminum foil in order to prevent drying. The creep tests were started at different shotcrete ages, i.e., 8 h, 24 h and 27 h, in order to investigate the influence of the material age at load application on the creep behavior. A shotcrete age of 27 h (instead of 24 h) had to be chosen for scheduling reasons. Creep tests were carried out with a hydraulic creep test bench (Figure 2). The load was applied within 2 s, and was held constant for 56 days.

This is in contrast to many creep tests on shotcrete reported in the literature, which are characterized by a stepwise increase of the sustained stress in order to simulate the further loading of shotcrete due to tunnel advance. However, it is in accordance with the recommendation by Sercombe et al. [17], who criticized the commonly employed stepwise increase of the sustained load because it complicates the calibration of shotcrete models. The magnitude of the applied compressive stress was chosen such that linear viscoelastic shotcrete behavior was ensured. For each creep test, the shotcrete age at application of the sustained compressive stress and its magnitude are provided in Table 2. Each creep test was accompanied by one shrinkage test, which was started simultaneously. In the shrinkage and creep tests, the time-dependent length changes were measured by three inductive displacement transducers with a measurement length of 200 mm, fixed to the specimens at 0°, 120°, and 240° along the perimeter of a specimen. The reported strain of a specimen is computed from the mean value of the three measured length changes and, thus, it represents the total strain at the center of the specimen.

Table 2. Experimental program.

Tests for Determining the Evolution of	Number of Specimens	Mean Value (MPa)	Standard Deviation (SD) (MPa)
• shotcrete stiffness			
tests at the age of 1 day	8	13,943	1692
tests at the age of 2 days	5	18,054	954
tests at the age of 3 days	2	19,410	390
tests at the age of 7 days	2	21,485	295
tests at the age of 28 days	5	21,537	376
• shotcrete strength			
tests at the age of 6 h	3	5.46	0.51
tests at the age of 8 h	3	6.01	0.16
tests at the age of 16 h	3	14.85	0.48
tests at the age of 1 day	4	18.57	2.84
tests at the age of 2 days	3	20.64	1.69
tests at the age of 3 days	1	28.63	–
tests at the age of 7 days	3	32.90	0.80
tests at the age of 28 days	7	40.84	1.68
• shrinkage strains of sealed specimens			
started at the age of 8 h	1		
started at the age of 24 h	1		
started at the age of 27 h	1		
• creep strains of sealed specimens			
1.9 MPa applied at the age of 8 h	1		
3.9 MPa applied at the age of 24 h	1		
3.7 MPa applied at the age of 27 h	1		

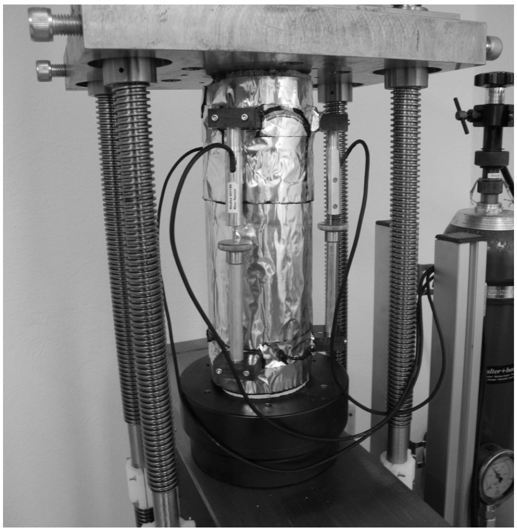


Figure 2. Sealed specimen in the creep test bench with displacement transducers fixed to the specimen.

3. Experimental Results

Figure 3 shows the measured Young's modulus at shotcrete ages according to Table 2, and, in addition, for comparison, the respective experimental results on the evolution of the Young's modulus presented by Müller [11], determined in two test series, designated as test series 3 and 4. The shotcrete composition investigated by Müller is characterized by a water content of 150 kg/m^3, an aggregate content of 1768 kg/m^3 (with a maximum aggregate size of 8 mm) and a cement content (manufacturer designation *SBM W&P)* of 340 kg/m^3. Comparing the results presented by Müller and those of the present study reveals the improved performance of modern shotcrete compositions: compared to the shotcrete tested by Müller, the specimens sampled during the current experimental program exhibit a higher Young's modulus throughout the considered time period. Furthermore, in the present experimental program, the stiffness at the age of seven days is already equal to the one at the age of 28 days. From the experimental data mean values of the Young's modulus at the ages of one day and 28 days are identified as $E^{(1)} = 13,943$ MPa and $E^{(28)} = 21,537$ MPa, respectively.

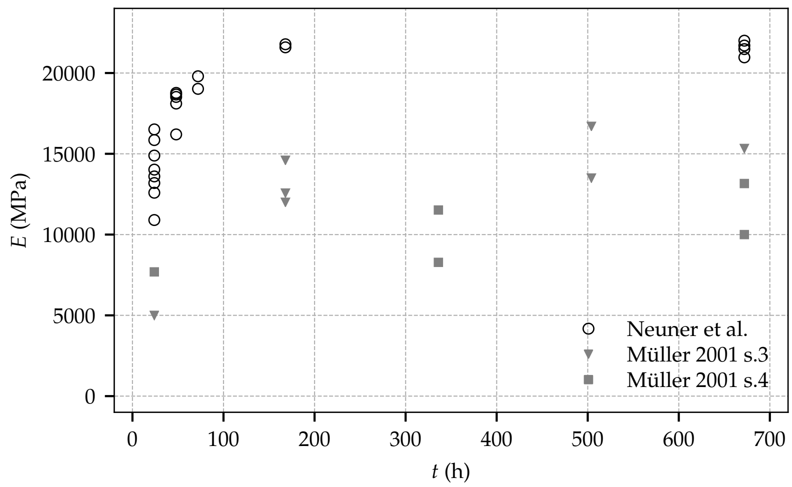


Figure 3. Evolution of the Young's modulus: experimental results from the present experimental program and test data by Müller.

Figure 4 contains the measured uniaxial compressive strength at shotcrete ages according to Table 2, and, for comparison, the respective experimental data from test series 3 and 4 by Müller [11]. Similar to the tests on the evolution of the Young's modulus, the observed uniaxial compressive strength in the present experimental program is substantially higher compared to the one determined by Müller. Contrary to the evolution of material stiffness and in contrast to the experimental data on material strength by Müller, for the present experimental program, a considerable increase of strength can be observed between the age of seven days and 28 days. From the experimental data, mean values of the uniaxial compressive strength at the ages of one day and 28 days are identified as $f_{cu}^{(1)} = 18.57$ MPa and $f_{cu}^{(28)} = 40.84$ MPa, respectively.

Comparing the material properties determined in the present experimental program and in the one presented by Müller reveals a noticeable higher stiffness and strength of the shotcrete employed in the present experimental program. These differences can be attributed to further developed shotcrete technology and accordingly improved shotcrete compositions.

The mean values and the standard deviations (SD) of the Young's modulus and the uniaxial compressive strength at different shotcrete ages are provided in Table 2, and are illustrated in Figure 5. For both material properties, the largest scatter is observed at the shotcrete age of 24 h. Remarkably, the determined values of the uniaxial compressive strength at even younger ages are in very good agreement with each other. Regarding the evolution of material stiffness, the standard deviation of the Young's modulus is decreasing gradually up to the age of seven days.

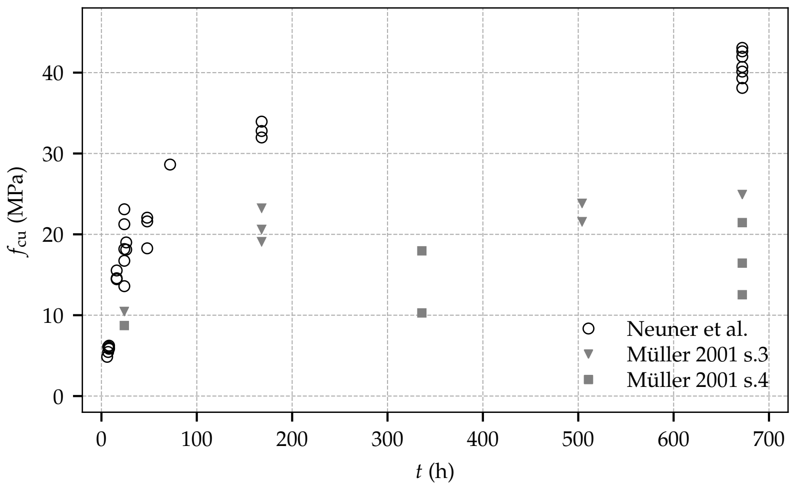


Figure 4. Evolution of the uniaxial compressive strength: experimental results from the present experimental program and test data by Müller.

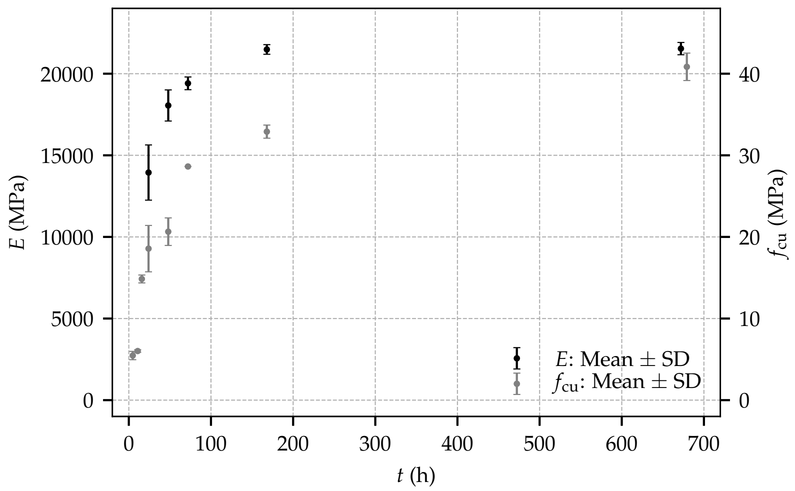


Figure 5. Mean values and standard deviations (SD) of the Young's modulus and the uniaxial compressive strength for the present experimental program.

Figure 6 shows the evolution of the total strain determined in the shrinkage tests on sealed specimens. The total strain consists of the autogenous shrinkage strain and the thermally induced strain due to hydration. It follows from this figure that, for one specimen, an initial delay of the evolution of the identified total strain is observed, whereas, for the other two specimens, the expected evolution of the total strain right after the start of the experiments occurred. The origin of the initial delay remains unclear; however, the measured rate of the shrinkage strains is similar for the three specimens. Furthermore, the determined total strain is similar to the respective strain observed by Huber [9] in shrinkage tests on sealed prismatic specimens with dimensions of 100 mm × 100 mm × 400 mm.

Figure 7 depicts the evolution of the total strain determined in the creep tests on sealed specimens loaded by sustained compressive stresses of 1.9 MPa, 3.9 MPa and 3.7 MPa, at shotcrete ages of 8 h, 24 h and 27 h, respectively. The total strain consists of the instantaneous strain due to load application, the autogenous shrinkage strain, the thermally induced strain due to hydration and the basic creep strain, as explained in [18].

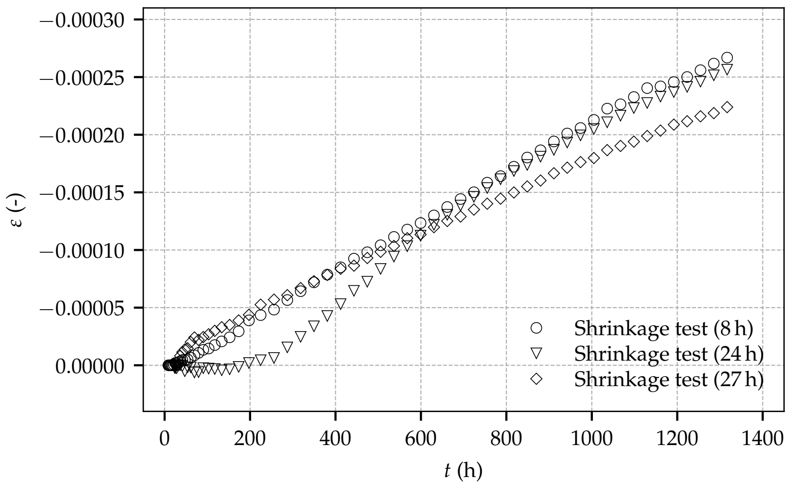


Figure 6. Evolution of the total strain determined on sealed, load-free specimens: experimental results for the tests on three specimens of the present experimental program, started at shotcrete ages of 8 h, 24 h and 27 h.

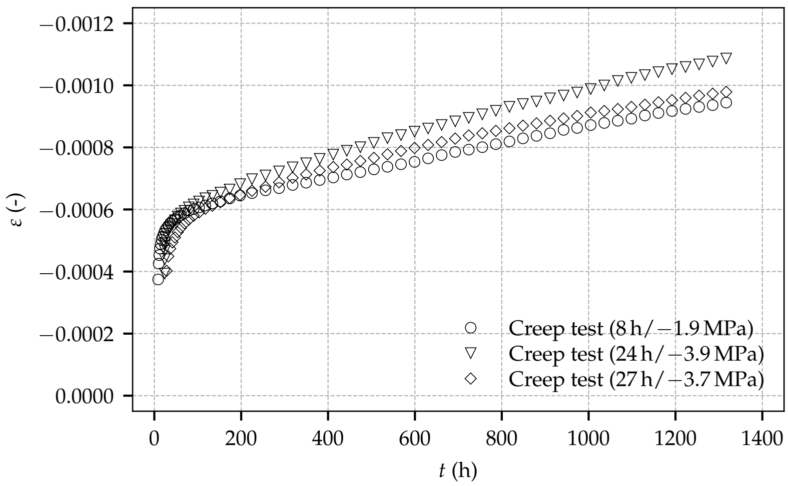


Figure 7. Evolution of the total strain determined on sealed, loaded specimens: experimental results for the tests on three specimens of the present experimental program, started at shotcrete ages of 8 h, 24 h and 27 h.

The curves representing the total strain determined on the loaded specimens are diverging slightly due to different magnitudes of the applied sustained stresses. From the total strain determined in a creep test, the compliance function, defined as load-induced time-dependent strain per unit stress [19], can be computed by subtracting the total strain determined in the respective shrinkage test and by dividing the so-obtained result by the stress applied in the creep test. Figure 8 shows the computed compliance functions for the three creep tests and, for comparison, the compliance functions obtained from the experimental data of the first load step of creep test 4/2 on two identical specimens by Müller [11], which were loaded at the shotcrete age of 48 h, keeping the load constant for 120 h. Only this first load step can be used for computing the compliance functions, since nonlinear creep effects due to high applied stresses occurred during further load steps. It can be seen that the computed compliance functions from the creep tests of the present experimental program exhibit the same slope after loading. Furthermore, the curves obtained from the creep tests, started at shotcrete ages of 24 h and 27 h, are nearly identical. By contrast, the compliance functions obtained from the creep tests by Müller are diverging and the respective curves in Figure 8 reveal the duration of the first load step to be too short for the assessment of the material compliance functions.

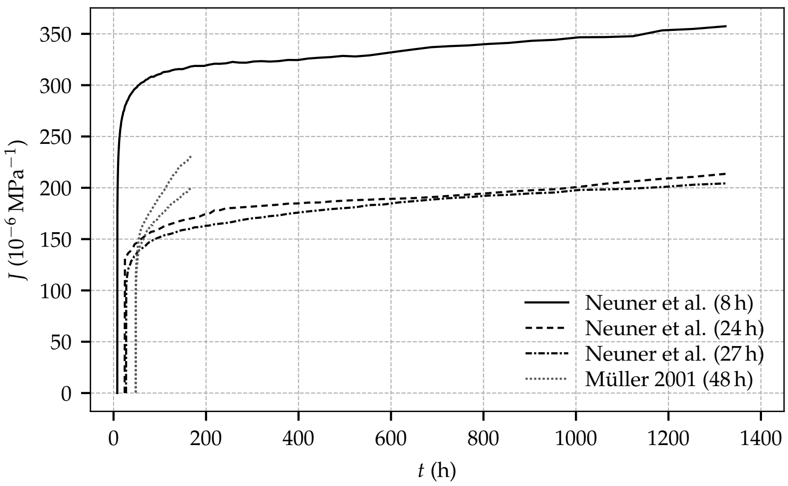


Figure 8. Compliance functions determined from the creep tests of the present experimental program, started at shotcrete ages of 8 h, 24 h and 27 h, and compliance functions from creep test series 4/2, conducted on two specimens by Müller, started at the shotcrete age of 48 h.

4. Constitutive Models for Shotcrete

Three advanced shotcrete models are considered in the present paper: (i) the model by Meschke [2], (ii) the model by Schädlich and Schweiger [3] and (iii) the SCDP (Shotcrete Damage Plasticity) model [4].

4.1. The Meschke Model

A three-dimensional constitutive model for shotcrete based on multisurface viscoplasticity theory was proposed by Meschke [2]. Validation based on experimental data from the literature and application of the model are reported in [20]. The model is capable of representing the evolution of stiffness and strength, shrinkage, creep, hardening material behavior in compressive and mixed loading and softening behavior in tensile loading. Associated flow rules are employed to describe viscoplastic flow. The evolution of material stiffness is based on an empirical law given as

$$
\begin{aligned}
&E(t) = E^{(28)}\,\beta_{\mathrm{E}}(t),\\
&\beta_{\mathrm{E}}(t) = \begin{cases} \beta_{\mathrm{E}}^{\mathrm{I}} = c_{\mathrm{E}} t + d_{\mathrm{E}} t^2, & \text{if } t \leq t_{\mathrm{E}},\\ \beta_{\mathrm{E}}^{\mathrm{II}} = \left(a_{\mathrm{E}} + \dfrac{b_{\mathrm{E}}}{t - \Delta t_{\mathrm{E}}}\right)^{-0.5}, & \text{if } t_{\mathrm{E}} < t \leq 672\,\mathrm{h},\\ \beta_{\mathrm{E}}^{\mathrm{III}} = 1, & \text{otherwise,} \end{cases}
\end{aligned} \tag{1}
$$

in which t is the shotcrete age in hours, and t_{E} and Δt_{E} are model parameters for the early age behavior to describe an initially delayed start of the evolution of the Young's modulus. Parameters a_{E}, b_{E}, c_{E} and d_{E} are model parameters to ensure a smooth transition at $t = t_{\mathrm{E}}$:

$$
\begin{aligned}
a_{\mathrm{E}} &= \frac{1 - \frac{28-\Delta t_{\mathrm{E}}/24}{1-\Delta t_{\mathrm{E}}/24}\left(\frac{E^{(1)}}{E^{(28)}}\right)^2}{\left(1 - \frac{28-\Delta t_{\mathrm{E}}/24}{1-\Delta t_{\mathrm{E}}/24}\right)\left(\frac{E^{(1)}}{E^{(28)}}\right)^2}, \quad & c_{\mathrm{E}} &= \frac{2\beta_{\mathrm{E}}^{\mathrm{II}}}{t_{\mathrm{E}}} - \left.\frac{\mathrm{d}\beta_{\mathrm{E}}^{\mathrm{II}}}{\mathrm{d}t}\right|_{t=t_{\mathrm{E}}},\\
b_{\mathrm{E}} &= (672 - \Delta t_{\mathrm{E}})(1 - a_{\mathrm{E}}), \quad & d_{\mathrm{E}} &= \frac{\left.\frac{\mathrm{d}\beta_{\mathrm{E}}^{\mathrm{II}}}{\mathrm{d}t}\right|_{t=t_{\mathrm{E}}} t_{\mathrm{E}} - \beta_{\mathrm{E}}^{\mathrm{II}}}{t_{\mathrm{E}}^2}.
\end{aligned} \tag{2}
$$

Therein, $E^{(1)}$ and $E^{(28)}$ denote the Young's modulus at the age of one day and 28 days, respectively.

The evolution of the uniaxial compressive strength $f_{\text{cu}}(t)$ is formulated for two time regimes: prior to the shotcrete age of 24 h, the recommendation reported in [15] is followed, whereas, after 24 h, the empirical approach by Oluokon et al. [21] is employed:

$$f_{\text{cu}}(t) = \begin{cases} f_{\text{cu}}^{(1)} \left(\dfrac{t + 0.12}{24} \right)^{0.72453}, & \text{if } t \leq 24\,\text{h}, \\ a_{\text{c}} \exp(-b_{\text{c}}/t), & \text{otherwise.} \end{cases} \tag{3}$$

Parameters a_{c}, b_{c} are computed from the uniaxial compressive strength at the age of one day and 28 days, i.e., $f_{\text{cu}}^{(1)}$ and $f_{\text{cu}}^{(28)}$:

$$a_{\text{c}} = \frac{f_{\text{cu}}^{(28)}}{\exp\left(\ln\left(\frac{f_{\text{cu}}^{(1)}}{f_{\text{cu}}^{(28)}}\right)/27\right)}, \quad b_{\text{c}} = -(672/27)\ln\left(\frac{f_{\text{cu}}^{(1)}}{f_{\text{cu}}^{(28)}}\right). \tag{4}$$

Creep of shotcrete is modeled by a Duvaut–Lions type viscoplastic formulation, governed by the viscosity parameter η. Since in this model creep strains are viewed as temporally delayed plastic strains, it follows that creep due to sustained stresses within the elastic domain is neglected.

Shrinkage of shotcrete is described by means of the semi-empirical model proposed by Bažant and Panula [22], formulated on the basis of the three material parameters shrinkage half time τ_{shr}, ultimate shrinkage strain $\varepsilon_{\infty}^{\text{shr}}$ and humidity dependent parameter k_{h}.

4.2. The Schädlich Model

Schädlich and Schweiger [3] proposed a viscoelastic-plastic material model for shotcrete, able to represent the evolution of material stiffness and strength, creep, shrinkage and plastic material behavior including hardening and softening, the latter formulated using non-associated plastic flow rules.

The evolution of material stiffness is modeled by a time-dependent Young's modulus according to the CEB-FIP model code 1990 [23]:

$$E(t) = E^{(28)} \exp\left(s_{\text{stiff}}\left(1 - \sqrt{28/t}\right)\right), \tag{5}$$

in which $E^{(28)}$ is the Young's modulus at the age of 28 days, t represents the time in days and s_{stiff} is a model parameter computed as

$$s_{\text{stiff}} = \frac{-\ln(E^{(1)}/E^{(28)})}{\sqrt{28} - 1}, \tag{6}$$

with $E^{(1)}$ as the Young's modulus at the age of one day.

The evolution of the uniaxial compressive strength $f_{\text{cu}}(t)$ is described similarly to the evolution of the Young's modulus by

$$f_{\text{cu}}(t) = f_{\text{cu}}^{(28)} \exp\left(s_{\text{strength}}\left(1 - \sqrt{28/t}\right)\right), \tag{7}$$

in which $f_{\text{cu}}^{(28)}$ is the uniaxial compressive strength at the age of 28 days and s_{strength} is a model parameter computed as

$$s_{\text{strength}} = \frac{-\ln(f_{\text{cu}}^{(1)}/f_{\text{cu}}^{(28)})}{\sqrt{28} - 1}, \tag{8}$$

with $f_{\text{cu}}^{(1)}$ as the uniaxial compressive strength at the age of one day.

The evolution of material ductility is represented by a time-dependent reduction of the plastic strain at peak stress in uniaxial compression, $\varepsilon_{\text{cpu}}^{\text{p}}$. It is formulated in terms of linear interpolation between the values of $\varepsilon_{\text{cpu}}^{\text{p}}$ at the age of 1 h, 8 h and 24 h, i.e., $\varepsilon_{\text{cpu}}^{\text{p}(1)}$, $\varepsilon_{\text{cpu}}^{\text{p}(8)}$ and $\varepsilon_{\text{cpu}}^{\text{p}(24)}$.

Shrinkage of shotcrete is modeled by the American Concrete Institute (ACI) committee 209 recommendation [19], based on the two material parameters ultimate shrinkage strain $\varepsilon_{\infty}^{\mathrm{shr}}$ and shrinkage half time t_{50}^{shr}.

Nonlinear creep is described using a viscoelastic approach derived from the Eurocode 2 guideline [24]. It is formulated by means of the creep coefficient φ^{cr} and creep half time t_{50}^{cr}.

4.3. The SCDP Model

The SCDP model is based on (i) the damage plasticity model for concrete by Grassl and Jirásek [25], (ii) the shrinkage model by Bažant and Panula [22] and (iii) a modified version of the solidification theory by Bažant and Prasannan [26,27]. Employing the combination of plasticity theory and isotropic continuum damage theory, non-associated plastic flow and isotropic hardening are modeled by the plasticity part of the model, and stiffness and strength degradation are described by means of the continuum damage part of the model.

The evolution of material stiffness and creep of shotcrete are described by a modified version of the solidification theory. The respective compliance function $J(t,t')$, which relates the load-induced strain at time t (given in days) to a constant stress applied at time t', is expressed as

$$J(t,t') = q_1 + q_2\,Q(t,t') + q_3\,\ln\left(1+(t-t')^{0.1}\right) + q_4\,\ln\left(\frac{t}{t'}\right). \tag{9}$$

The compliance function contains the four compliance parameters q_1, q_2, q_3 and q_4. Parameter q_1 is equal to the instantaneous, asymptotic material compliance, q_2 and q_3 are associated with short-term creep, and thus govern the evolution of material stiffness, and q_4 is related to long-term creep. For normal concrete, an estimation scheme for the compliance parameters based on the concrete composition is reported in [28] and extended in [29]. Although this estimation procedure was not developed for shotcrete, the obtained parameters may serve as a rough first estimate. Function $Q(t,t')$ represents a binomial integral, which cannot be expressed in closed form, as discussed in [26]. However, it can be approximated for short time periods $t-t'$ as $Q(t,t') \approx \tau(t)^{-0.5}\,\ln\left(1+(t-t')^{0.1}\right)$ (see [4]).

The solidification theory is modified by means of the time transformation function $\tau(t)$ in the SCDP model in order to take into account both an initial delay in the evolution of material stiffness and the subsequent faster hydration of shotcrete compared to normal concrete. Function $\tau(t)$ is defined as

$$\tau(t) = \begin{cases} \tau_{\mathrm{r}}, & \text{if } t \le t_{\mathrm{r}}, \\ \dfrac{-2\tau_{\mathrm{p}}+2\tau_{\mathrm{r}}+\Delta t_{\mathrm{p}}}{\Delta t_{\mathrm{p}}^3}(t-t_{\mathrm{r}})^3 + \dfrac{3\tau_{\mathrm{p}}-3\tau_{\mathrm{r}}-\Delta t_{\mathrm{p}}}{\Delta t_{\mathrm{p}}^2}(t-t_{\mathrm{r}})^2 + \tau_{\mathrm{r}}, & \text{if } t_{\mathrm{r}} < t \le t_{\mathrm{p}}, \\ \dfrac{2(\tau_{\mathrm{p}}-1)}{\Delta t_{\mathrm{a}}^3}(t-t_{\mathrm{p}})^3 - \dfrac{3(\tau_{\mathrm{p}}-1)}{\Delta t_{\mathrm{a}}^2}(t-t_{\mathrm{p}})^2 + (t-t_{\mathrm{p}}) + \tau_{\mathrm{p}}, & \text{if } t_{\mathrm{p}} < t \le t_{\mathrm{a}}, \\ t, & \text{otherwise}, \end{cases} \tag{10}$$

with $t_{\mathrm{p}} = 1\,\mathrm{d}$, $t_{\mathrm{r}} = 0.1\,\mathrm{d}$, $t_{\mathrm{a}} = \max(28, 3\,\tau_{\mathrm{p}} - 2)$, $\Delta t_{\mathrm{p}} = t_{\mathrm{p}} - t_{\mathrm{r}}$, $\Delta t_{\mathrm{a}} = t_{\mathrm{a}} - t_{\mathrm{p}}$, $\tau_{\mathrm{r}} = 10^{-2}\,\mathrm{d}$ and τ_{p} is given as

$$\tau_{\mathrm{p}} = \left(\frac{1/E^{(1)} - q_1}{q_2\ln(1+\Delta t^{0.1})} - \frac{q_3}{q_2}\right)^{-2}. \tag{11}$$

Therein, $E^{(1)}$ denotes the Young's modulus at the age of 1 day, which can be considered as the fifth material parameter of the modified solidification theory, and Δt denotes a short time period for computing the effective stiffness: unlike for the Schädlich model and the Meschke model, the concept of a distinct time-dependent Young's modulus is not applicable for the SCDP model. Rather, the Young's modulus is approximated by the effective stiffness, computed for a short time period Δt from the compliance function $J(t+\Delta t, t)$ of the modified solidification theory in Equation (9) as

$E(t) \approx J(t+\Delta t, t)^{-1}$. Neglecting the effect of long-term creep on the effective stiffness over short time periods, departing from Equation (9), the Young's modulus at time t can be approximated as

$$E(t) \approx \left(q_1 + \ln\left(1 + \Delta t^{0.1}\right) \left(q_2\, \tau(t)^{-0.5} + q_3\right)\right)^{-1}. \tag{12}$$

Recommended values for Δt are given in the literature within the range of 10^{-3} d [28] to 10^{-2} d [29].

For describing the evolution of material strength, an empirical formulation is employed, which is derived from the relation used by Meschke [2] for describing the evolution of the Young's modulus. It is formulated by means of the shotcrete age t (given in days) and the uniaxial compressive strength at the age of one day and 28 days, $f_{\mathrm{cu}}^{(1)}$ and $f_{\mathrm{cu}}^{(28)}$, respectively:

$$\begin{aligned} f_{\mathrm{cu}}(t) &= f_{\mathrm{cu}}^{(28)}\, \beta_{\mathrm{f}}(t), \\ \beta_{\mathrm{f}}(t) &= \begin{cases} \beta_{\mathrm{f}}^{\mathrm{I}} = r_{\mathrm{f}} + c_{\mathrm{f}} t + d_{\mathrm{f}} t^2, & \text{if } t \le t_{\mathrm{f}}, \\ \beta_{\mathrm{f}}^{\mathrm{II}} = \left(a_{\mathrm{f}} + \dfrac{b_{\mathrm{f}}}{t - \Delta t_{\mathrm{f}}}\right)^{-0.5}, & \text{if } t_{\mathrm{f}} < t \le 28\,\mathrm{d}, \\ \beta_{\mathrm{f}}^{\mathrm{III}} = 1, & \text{otherwise.} \end{cases} \end{aligned} \tag{13}$$

The model parameters are given as $r_{\mathrm{f}} = 10^{-2}$ and

$$\begin{aligned} a_{\mathrm{f}} &= \frac{1 - \frac{28-\Delta t_{\mathrm{f}}}{1-\Delta t_{\mathrm{f}}} \left(\frac{f_{\mathrm{cu}}^{(1)}}{f_{\mathrm{cu}}^{(28)}}\right)^2}{\left(1 - \frac{28-\Delta t_{\mathrm{f}}}{1-\Delta t_{\mathrm{f}}}\right) \left(\frac{f_{\mathrm{cu}}^{(1)}}{f_{\mathrm{cu}}^{(28)}}\right)^2}, \qquad c_{\mathrm{f}} = \left.\frac{\mathrm{d}\beta_{\mathrm{f}}^{\mathrm{II}}}{\mathrm{d}t}\right|_{t=t_{\mathrm{f}}} - \frac{2\beta_{\mathrm{f}}^{\mathrm{II}}(t_{\mathrm{f}})}{t_{\mathrm{f}}}, \\ b_{\mathrm{f}} &= (28 - \Delta t_{\mathrm{f}})(1 - a_{\mathrm{f}}), \qquad d_{\mathrm{f}} = \frac{\left.\frac{\mathrm{d}\beta_{\mathrm{f}}^{\mathrm{II}}}{\mathrm{d}t}\right|_{t=t_{\mathrm{f}}} t_{\mathrm{f}} - \beta_{\mathrm{f}}^{\mathrm{II}}(t_{\mathrm{f}}) + r_{\mathrm{f}}}{t_{\mathrm{f}}^2}. \end{aligned} \tag{14}$$

Therein, $f_{\mathrm{cu}}^{(1)}$ and $f_{\mathrm{cu}}^{(28)}$ denote the experimentally determined values for $f_{\mathrm{cu}}(t)$ at the age of one day and 28 days, respectively, and parameters t_{f} and Δt_{f} are proposed as $t_{\mathrm{f}} = 0.25\,\mathrm{d}$ and $\Delta t_{\mathrm{f}} = 0.18\,\mathrm{d}$. Identical to the model by Schädlich and Schweiger [3], the evolution of material ductility is modeled by a temporal decrease of the plastic strain at peak stress in uniaxial compression, $\varepsilon_{\mathrm{cpu}}^{\mathrm{p}}$.

Shrinkage is described by the semi-empirical model by Bažant and Panula [22], identical to the shotcrete model by Meschke.

An overview of the formulations governing the time-dependent material behavior in the three shotcrete models is provided in Table 3.

Table 3. Summary and comparison of the shotcrete models.

Model	Evolution of Stiffness	Evolution of Strength	Shrinkage	Creep
Meschke	Empirical law	Guideline [15] and Oluokon et al. [21]	Bažant–Panula [22]	Viscoplastic formulation
Schädlich	CEB-FIP [23]	CEB-FIP [23]	ACI [19]	Eurocode 2 [24] based formulation
SCDP	Modified solidification theory [4,26]	Empirical law	Bažant–Panula [22]	Modified solidification theory [4,26]

5. Comparison of Experimental and Computed Shotcrete Behavior

The identification of the material parameters for the shotcrete models closely follows the scheme presented in [4]. It is characterized by identifying parameters related to a material property exclusively from the respective experimental test on that property. In the following, all presented numerical results were obtained at material point level in incremental iterative simulations of the experimental tests. The constitutive models are integrated numerically by means of the fully implicit Euler scheme within the return mapping algorithm. The solidification theory employed by the SCDP model is implemented by means of a discrete Kelvin chain, as described in [30].

For validation of the shotcrete models with respect to the measured evolution of stiffness and strength, the mean values of the Young's modulus and the uniaxial compressive strength at the age of one day and 28 days, specified in Table 2, are employed. In addition, for the SCDP model, following the parameter estimation procedure by Bažant and Baweja, compliance parameter q_3 is computed as $q_3 = 3.03 \times 10^{-6}$ MPa^{-1}. By employing a least square optimization, parameters q_1 and q_2 are identified as $q_1 = 42.20 \times 10^{-6}$ MPa^{-1} and $q_2 = 41.10 \times 10^{-6}$ MPa^{-1}. They assure a good agreement between the predicted and the observed evolution of the Young's modulus between one day and 28 days. For comparison of the predicted evolution of the stiffness by the SCDP model, a time period of 10^{-3} d is chosen to determine the effective stiffness, in accordance with [28] and the approximate duration of load application during the experimental tests. Parameters Δt_{E} and t_{E} of the Meschke model are chosen as their default values proposed in [20] as $t_{\mathrm{E}} = 8$ h and $\Delta t_{\mathrm{E}} = 6$ h, since identification of them requires sufficient experimental data at a material age of a couple of hours. Figures 9 and 10 show the predicted results for the evolution of stiffness and strength and for comparison the respective experimental data. While the SCDP model and the Meschke model predict very similar values of the Young's modulus between shotcrete ages of one day and 28 days, the Schädlich model slightly underestimates the material stiffness in that time period. Furthermore, it can be seen that in contrast to the Meschke model and the Schädlich model, the SCDP model assumes a certain stiffness already at zero age, governed by model parameter τ_{r}.

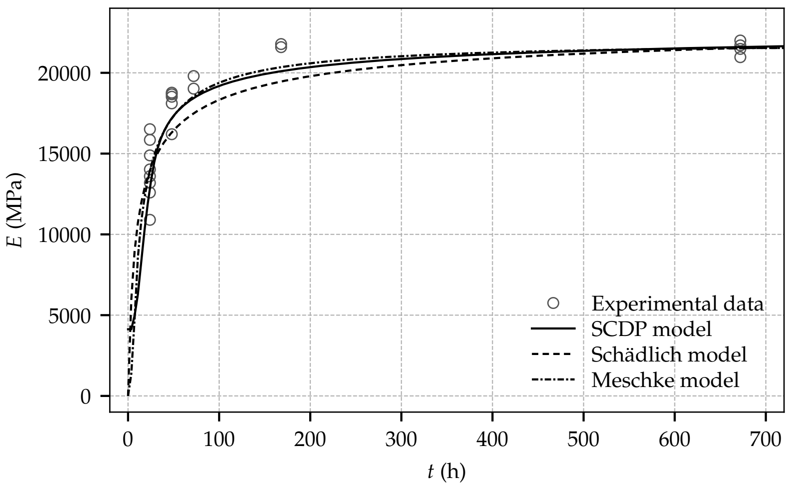


Figure 9. Evolution of the Young's modulus: experimental results and computed numerical results based on the calibrated shotcrete models.

Very similar evolutions of the uniaxial compressive strength are obtained by the SCDP model and the Schädlich model, which are in good agreement with the experimental data. In contrast, the evolution predicted by the Meschke model is somewhat overestimating the actual material strength observed in the experiments.

A least square optimization is employed to identify the shrinkage parameters from the test data of the shrinkage tests started at shotcrete ages of 8 h and 27 h. The experimental data of the shrinkage test started at 24 h is neglected due to the initial delay of the evolution of the strain. Accordingly, parameters $\varepsilon_{\infty}^{\mathrm{shr}}$ and t_{50}^{shr} of the Schädlich model are obtained as $\varepsilon_{\infty}^{\mathrm{shr}} = -0.0019$ and $t_{50}^{\mathrm{shr}} = 8645$ h. In contrast to the ACI model, which is employed by the Schädlich model, the Bažant–Panula model, employed by the SCDP model and the Meschke model, is not able to represent the low curvature of the experimental data: For this reason, for the SCDP model and the Meschke model, an ultimate shrinkage strain $\varepsilon_\infty^{\rm shr}$ is assumed as $\varepsilon_\infty^{\rm shr} = -0.002$ and the humidity dependent parameter $k_{\rm h}$ is set to $k_{\rm h} = 1.0$. Thereby, a shrinkage half time of $\tau_{\rm shr} = 4082$ h is identified. Experimental results of the shrinkage tests and the computed numerical results by the calibrated shotcrete models are shown in Figure 11. It can be seen that the Schädlich model is able to represent the evolution of the shrinkage strain better than the SCDP model and the Meschke model.

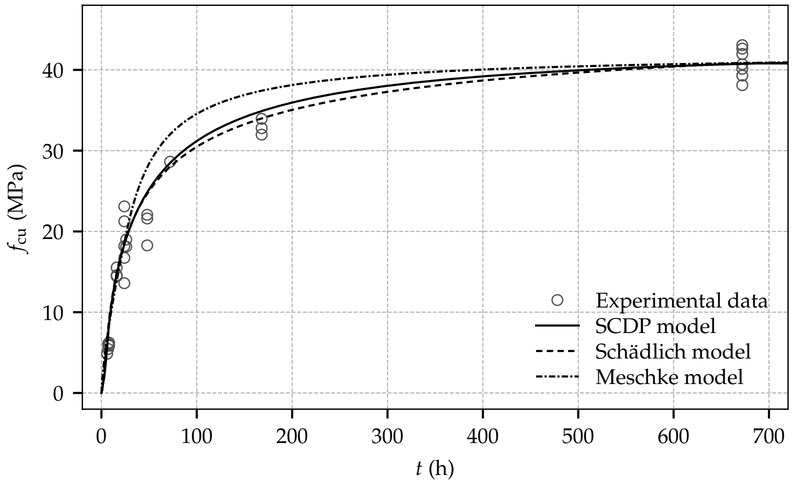


Figure 10. Evolution of the uniaxial compressive strength: experimental results and computed numerical results based on the calibrated shotcrete models.

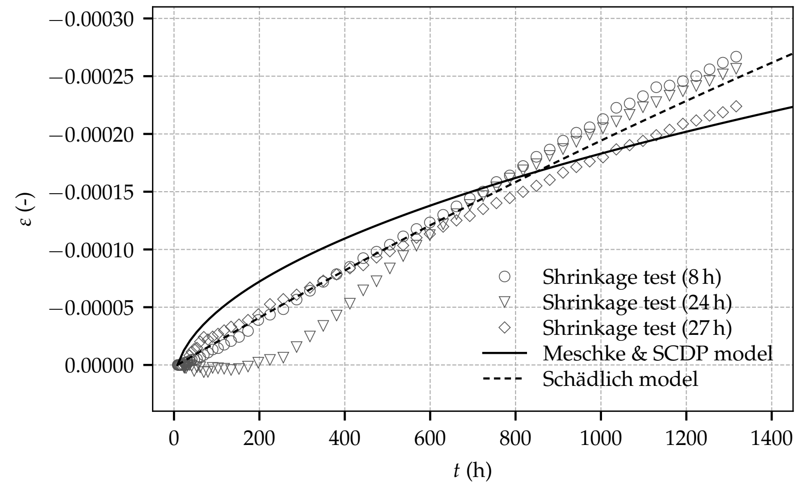
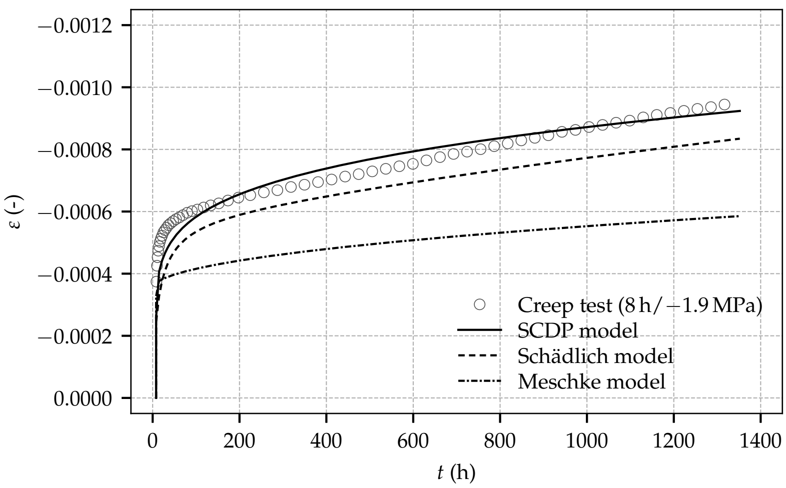


Figure 11. Evolution of the total strain determined on sealed, load-free specimens: experimental results and computed numerical results based on the calibrated shotcrete models.

For calibration of the creep behavior, the creep parameters are identified from the test data of the three creep tests employing a least square optimization. However, the respective parameters are exclusively identified from the time-dependent strain after application of the load, i.e., the instantaneous strain due to load application is not considered in the identification procedure. The motivation for this approach becomes clear especially for the Meschke model and the Schädlich model: if the instantaneous strain was considered in the calibration procedure, an erroneous compensation of the discrepancy in the instantaneous strain between experimental results and predicted numerical results through overly high creep rates would be the consequence of the employed least square optimization. Due to the viscoplastic formulation of the Meschke model, the ratio of uniaxial compressive yield stress to uniaxial compressive strength, $f_{\rm cy}/f_{\rm cu}$, has a considerable influence on the predicted creep behavior. Similar to the identification procedure presented in [4] and in agreement with the proposed parameter range in [20], $f_{cy}/f_{cu} = 0.1$ is assumed for all three shotcrete models. For the SCDP model and the Schädlich model, the material parameters governing the evolution of material ductility, $\varepsilon_{cpu}^{p(1)}$, $\varepsilon_{cpu}^{p(8)}$ and $\varepsilon_{cpu}^{p(24)}$ are chosen as the default values proposed in [3]: $\varepsilon_{cpu}^{p(1)} = -30.0 \times 10^{-3}$, $\varepsilon_{cpu}^{p(8)} = -0.7 \times 10^{-3}$ and $\varepsilon_{cpu}^{p(24)} = -0.7 \times 10^{-3}$. For the SCDP model, the long-term creep compliance parameter q_4 is identified as $q_4 = 33.95 \times 10^{-6}\,\mathrm{MPa}^{-1}$. For the Schädlich model, the creep half time t_{50}^{cr} is assumed as the value proposed in [3] as $t_{50}^{cr} = 36\,\mathrm{h}$. Thereby, φ^{cr} is identified from the experimental data as $\varphi^{cr} = 2.62$. Remarkably, it is nearly identical to the one proposed as $\varphi^{cr} = 2.6$ in [3], calibrated on the basis of the creep tests by Aldrian [7]. For the Meschke model, the viscosity parameter η of the Meschke model is assumed as $\eta = 0.25$ h, in accordance with [20]. Figure 12 shows the experimental results together with the predicted numerical results by the shotcrete models.

The Schädlich model underestimates the total strain for the creep test started at 8 h, and overestimates the total strain for the creep tests started at 24 h and 27 h. For all creep tests, the SCDP underestimates the total strain immediately after loading and overestimates it thereafter. Long-term creep is represented by the Schädlich model and the SCDP model equally well, indicated by a similar slope of the predicted and measured long-term creep curves. Both the SCDP model and the Schädlich model are in good agreement with the experimental results, with the SCDP model slightly performing better.



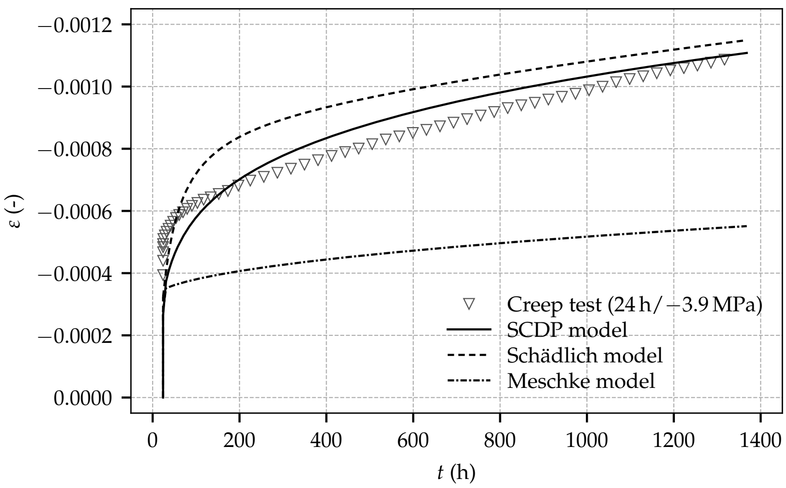


Figure 12. *Cont.*

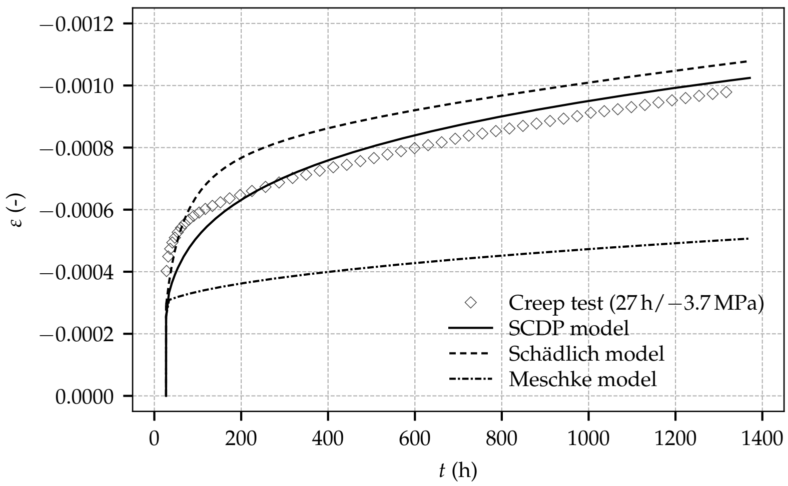


Figure 12. Evolution of the total strain determined on sealed, loaded specimens: Experimental results from the creep tests started at shotcrete ages of 8 h (**top**), 24 h (**middle**) and 27 h (**bottom**) and the respective computed numerical results based on the calibrated shotcrete models.

In contrast, similarly as reported in [4] for the experimental tests by Müller [11], the viscoplastic formulation of the Meschke model is not able to represent the magnitude of the creep strain observed in the experiments. Creep, represented by viscoplastic flow, occurs only at the beginning of a creep test during a short time period after loading, and vanishes as soon as the uniaxial yield stress exceeds the applied stress due to the evolving material strength. Hence, identification of creep parameter η of the Meschke model is not possible on the basis of the present experimental data because of the low sustained stress levels in the present creep tests. As demonstrated in [20] on the basis of the creep tests by Huber [9] and in [5] by means of a finite element simulation of deep tunnel advance, the Meschke model represents creep of shotcrete very well for comparatively high prevailing stress levels.

6. Conclusions

A new experimental program on shotcrete and the evaluation of three advanced constitutive models for shotcrete, i.e., the Meschke model, the Schädlich model and the SCDP model, on the basis of the test data, were presented. The present paper closes the gap between the outdated experimental data on shotcrete available in the literature on the one hand and the nowadays available advanced constitutive models for representing the complex material behavior of shotcrete on the other hand. The new experimental program concentrated on the evolution of material properties of young shotcrete up to the age of 28 days as well as on the shrinkage and creep behavior, tested on sealed specimens up to the age of 56 days. Experiments were performed on two different types of specimens, i.e., drill cores sampled from spray boxes and directly sprayed specimens, the latter especially for experiments at very early shotcrete ages. The following conclusions can be drawn from the experimental program:

- The application of directly sprayed specimens from tubular molds proved to be a suitable alternative to drill cores from spray boxes. Furthermore, using sprayed specimens, experiments for determining the uniaxial compressive strength could be performed already six hours after casting.
- The tests on both the evolution of stiffness and strength are characterized by a remarkably small scatter of the determined material properties, especially for matured shotcrete.
- The observed evolution of the Young's modulus indicates the ultimate stiffness to be fully developed at a material age of seven days.
- In contrast to material stiffness, the observed evolution of the uniaxial compressive strength shows a substantial increase of material strength between shotcrete ages of seven days and 28 days.

- Comparison of the experimental results of the present study to those presented by Müller, considering the latter as representative for shotcrete compositions used in the last decades, reveals the considerably improved performance of modern shotcrete compositions.

The presented calibration of three shotcrete models and the comparison of experimental and predicted results revealed the following insights:

- The determination of the material parameters for the three shotcrete models is characterized by identifying parameters related to a particular material property exclusively from the respective experimental test on that property. Hence, a simple least square approach can be employed for parameter identification from the standard lab tests presented in this paper. The only exception is the viscosity parameter η of the Meschke model, which must be calibrated on the basis of creep tests at higher stress levels, since creep due to stresses in the elastic domain is neglected by this model.
- All shotcrete models represent the evolution of material stiffness very well. The Young's modulus is underestimated slightly by all models between shotcrete ages of one day and 28 days.
- Regarding the evolution of the uniaxial compressive strength, for all models, the computed response is in very good agreement with the experimental data. Both the SCDP model and the Schädlich model predict very similar results. A slight overestimation is obtained by the Meschke model between one day and 28 days.
- While the Schädlich model represents the shrinkage behavior very well, discrepancies between experimental results and the results predicted by the SCDP model and the Meschke model are striking.
- Both the Schädlich model and the SCDP model are able to represent the experimentally determined creep behavior very well. In contrast, the observed creep behavior cannot be reproduced using the viscoplastic Meschke model.

Although the presented experimental program covers a quite large variety of material properties of shotcrete, some material phenomena have received less attention so far, especially for nonlinear creep, which is considered as a very important subject in the context of tunneling. However, no profound experimental data is available in the literature. For this reason, a follow up experimental program, concentrating on that complex subject, is currently in preparation.

Author Contributions: The focus of the paper was jointly derived by M.N. and G.H.; M.N., T.C. and M.D. conducted the experiments; M.N. conducted the numerical simulations and evaluated the results; M.N. and G.H. jointly wrote the paper.

Conflicts of Interest: The authors declare no conflict of interest.

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6 Further Development and Application of the SCDP Model

In Publication V, *Comparative Investigation of Constitutive Models for Shotcrete Based on Numerical Simulations of Deep Tunnel Advance*, a summary of Publication I, Publication II and Publication IV is presented, and the respective scientific results are placed in a common context.

In addition, an improvement of the viscoelastic part of the SCDP model is proposed, which aims at a simplified calibration procedure. To this end, the formulation of the function for describing the evolution of the solidified volume within the solidification theory is modified, and three of the compliance material parameters are substituted by a single new material parameter, which is the Young's modulus at the age of 28 days.

Consequently, the viscoelastic part of the model can be calibrated by means of the Young's modulus at the age of 1 day and 28 days, which can be determined in standard laboratory tests. A further improvement of the long-term creep behavior is introduced based on the experimental results from the creep tests presented in Publication IV in terms of a modification of the flow creep formulation.

Finally, the benchmark presented in Publication II is employed to compare the Meschke model, the Schädlich model and the extended SCDP model, using the material parameters presented in Publication IV.

The scientific contribution by the author comprises

- the proposal of the extended SCDP model,
- the performance of the numerical simulations,
- the evaluation of the simulation results,
- the preparation of the manuscript, jointly with Günter Hofstetter.

Comparative Investigation of Constitutive Models for Shotcrete Based on Numerical Simulations of Deep Tunnel Advance

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ABSTRACT: According to the New Austrian Tunneling Method (NATM), subsequent to each excavation step shotcrete is directly applied onto the surrounding rock mass, forming a supporting structure which is usually loaded already several hours after application due to further advance of the tunnel. Only a few advanced material models for shotcrete for representing the highly nonlinear, time-dependent material behavior have been proposed in the literature so far. Recently, in (Neuner, Gamnitzer, & Hofstetter 2017) a new constitutive model for shotcrete, denoted as the SCDP model, based on a combination of the theory of plasticity, the theory of continuum damage mechanics, and a modified version of the solidification theory (Bažant & Prasannan 1989) was presented. The model represents time-dependent material behavior, as well as hardening and softening material behavior of shotcrete. In the present contribution, an extension of the SCDP model, and a comparison with two selected, frequently used shotcrete models, i.e., the models proposed by Meschke (1996), and by Schädlich and Schweiger (2014), are presented. The comparison is based on a benchmark example of a deep tunnel driven by the NATM, which is derived from a stretch of the Brenner Basetunnel.

1 INTRODUCTION

Shotcrete linings, installed immediately after an excavation step, serve as a securing measure during tunneling according to the New Austrian Tunneling Method. Since these shotcrete shells are loaded already at a very early age of the material due to further tunnel advance, i.e., usually a few hours after application of the shell, the representation of the early age material behavior is an important aspect in numerical simulations of tunnel advance. In addition, shotcrete exhibits nonlinear material behavior like hardening and softening, creep and shrinkage, which have to be taken into account properly.

Only a few advanced material models for representing these nonlinear, time-dependent material phenomena were proposed in the literature so far. Among them are the viscoplastic model by Meschke (1996), the viscoelastic-plastic model by Schädlich and Schweiger (2014), and a recently proposed damage-plasticity model presented in (Neuner, Gamnitzer, & Hofstetter 2017), denoted as the SCDP model. In (Neuner, Schreter, Unteregger, & Hofstetter 2017), the three models were evaluated and compared by means of a benchmark example of deep tunnel advance. For calibrating the models, experimental data by Huber (1991) and Müller (2001) was used. However, since these experimental results were obtained from dry-mix shotcrete compositions which do not represent the state-of-the-art of modern shotcrete technology, a new experimental program on a modern wet-mix shotcrete composition was designed and presented in (Neuner, Cordes, Drexel, & Hofstetter 2017).

In the present contribution, the experimental program and the investigated material models are summarized briefly, and an extension of the SCDP model is presented. Finally, the application of the three models to the benchmark example of deep tunnel advance, supported by the novel experimental test results is presented.

2 EXPERIMENTAL PROGRAM

The new experimental program on the early age material behavior of a modern wet-mix shotcrete composition focused on the evolution of the Young's modulus and the evolution of the uniaxial compressive strength, creep and shrinkage, with the latter two tested on sealed specimens to exclude drying effects. The specimens of the investigated composition (manufacturer designation *Sp C25/30/(56)/ÜK3/J2/XC4/XF3/GK8* according to the Austrian guideline for shotcrete (ÖVBB 2009)) were sampled directly at a construction site of the Brenner Base Tunnel. To this end, two different tech-

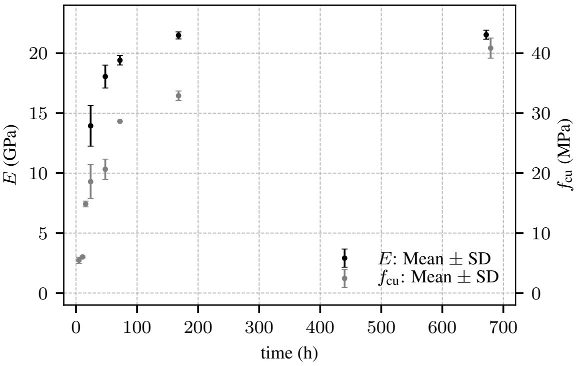

Figure 1: Evolution of the Young's modulus (E) and the uniaxial compressive strength (f_{cu}): Mean values and standard deviations (SD) form shotcrete ages of 6 hours to 672 hours (28 days).

niques were employed: For tests on shotcrete younger than 24 hours, *sprayed* specimens directly sampled from tubular molds were used, while for tests on shotcrete older than 24 hours both sprayed specimens as well as drill cores sampled from spray boxes were used. Sprayed specimens could already be unmolded safely 6 hours after casting, and experimental tests on the evolution of the uniaxial compressive strength were started immediately afterwards. Although sprayed specimens are a suitable alternative to drill cores, special care has to be taken of the spraying direction, which must be aligned coaxial with the axis of the tubular molds in order to minimize the number of faulty specimens due to shotcrete rebound and air pockets.

The evolution of the Young's modulus was investigated between a material age of 24 hours and 28 days, and the uniaxial compressive strength was tested from 6 hours to 28 days. The mean values of both the Young's modulus and the uniaxial compressive strength, as well as the respective standard deviations (SD) are shown in Figure 1. Compared to the uniaxial compressive strength, the ultimate Young's modulus is already attained at a material age of 7 days, while a substantial increase of material strength still can be observed between 7 days and 28 days.

Shrinkage and creep tests were started simultaneously at three different ages of the material, i.e., 8 hours, 24 hours and 27 hours, and lasted for 56 days each. During the shrinkage and creep tests, the time-dependent strain at the center of the specimens was determined by measuring the time-dependent displacements over a distance of 200 mm, using three displacement transducers which were arranged along the perimeter of each specimen. Creep tests were conducted on a hydraulic creep test bench (Figure 2), and the magnitude of the applied compressive load was chosen to ensure linear viscoelastic material behavior. Accordingly, the specimens tested at 8 hours, 24 hours and 27 hours were loaded with compressive stresses of 1.9 MPa, 2.9 MPa and 2.7 MPa. After application of the load, it was held constant throughout the complete testing period of 56 days.

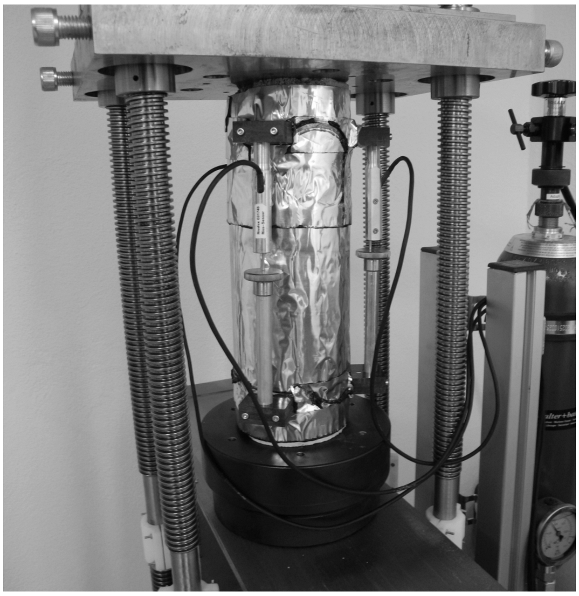

Figure 2: Sealed specimen during a creep test in the hydraulic creep test bench.

3 MATERIAL MODELS

In the following, the assessed shotcrete models are described briefly, and an extension of the SCDP model is presented.

3.1 *Viscoplastic Model by Meschke*

In the viscoplastic model by Meschke (1996), non-linear mechanical behavior of shotcrete is described on the basis of multisurface viscoplasticity theory. A hardening Drucker-Prager criterion is used for predominantly compressive stress states and mixed stress states, and a softening Rankine criterion for tensile stress states. Creep of shotcrete is described by the evolution of viscoplastic strains, employing associated flow rules. Aging of shotcrete is considered by empirical time functions for the Young's modulus, the uniaxial compressive strength and the uniaxial tensile strength. Shrinkage of shotcrete is taken into account on the basis of the model proposed by Bažant and Panula (1978).

3.2 *Viscoelastic-Plastic Model by Schädlich and Schweiger*

In the shotcrete model by Schädlich and Schweiger (2014) hardening and softening material behavior of shotcrete is described on the basis of non-associated multisurface plasticity. To delimit the elastic domain, a hardening and softening Mohr-Coulomb criterion for predominantly compressive stress states and mixed stress states, and a softening Rankine criterion for tensile stress states is employed. Aging of shotcrete is considered by empirical time functions for stiffness, strength and ductility. Shrinkage of shotcrete is taken into account on the basis of the model proposed by the ACI committee 209 (1992), and nonlinear creep of shotcrete is modeled on the basis of the theory of viscoelasticity, derived from the model proposed in the Eurocode 2 guidelines (2004).

3.3 *SCDP Model by Neuner et al.*

The SCDP model for shotcrete presented in (Neuner, Gamnitzer, & Hofstetter 2017) is based on three material models for concrete, which are the damage plasticity model for concrete by Grassl and Jirásek (2006), the shrinkage model by Bažant and Panula (1978) and the solidification theory by Bažant and Prasannan (1989). The latter model is used to describe the evolution of material stiffness and nonlinear creep. To model the evolution of material strength and ductility, empirical time functions are employed.

The stress–strain relation in total form is expressed as

$$\boldsymbol{\sigma} = (1-\omega)\,\mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}} - \boldsymbol{\varepsilon}^{\mathrm{ve}} - \boldsymbol{\varepsilon}^{\mathrm{f}} - \boldsymbol{\varepsilon}^{\mathrm{shr}}), \tag{1}$$

in which $\boldsymbol{\sigma}$ denotes the nominal stress (force per total area), ω is the isotropic scalar damage parameter and $\mathbb{C}$ is the fourth order stiffness tensor. The total strain $\boldsymbol{\varepsilon}$ is decomposed into the instantaneous elastic strain $\boldsymbol{\varepsilon}^{\mathrm{el}} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}} - \boldsymbol{\varepsilon}^{\mathrm{ve}} - \boldsymbol{\varepsilon}^{\mathrm{f}} - \boldsymbol{\varepsilon}^{\mathrm{shr}}$, the plastic strain $\boldsymbol{\varepsilon}^{\mathrm{p}}$, the viscoelastic strain $\boldsymbol{\varepsilon}^{\mathrm{ve}}$, the flow (viscous) strain $\boldsymbol{\varepsilon}^{\mathrm{f}}$ and the shrinkage strain $\boldsymbol{\varepsilon}^{\mathrm{shr}}$.

The evolution of $\boldsymbol{\varepsilon}^{\mathrm{el}}$, $\boldsymbol{\varepsilon}^{\mathrm{ve}}$ and $\boldsymbol{\varepsilon}^{\mathrm{f}}$ is described according to the solidification theory and formulated within the framework of coupled continuum damage mechanics and plasticity theory as

$$\begin{aligned}
\dot{\boldsymbol{\varepsilon}}^{\mathrm{el}}(t) &= q_1\,\mathbb{C}_\nu^{-1} : \dot{\bar{\boldsymbol{\sigma}}}(t),\\
\dot{\boldsymbol{\varepsilon}}^{\mathrm{ve}}(t) &= \frac{F(\bar{\boldsymbol{\sigma}}(t))}{v(t,q_2,q_3)} \int_0^t \dot{\Phi}(t-t')\,\mathbb{C}_\nu^{-1} : \mathrm{d}\bar{\boldsymbol{\sigma}}(t'),\\
\dot{\boldsymbol{\varepsilon}}^{\mathrm{f}}(t) &= \frac{q_4\,F(\bar{\boldsymbol{\sigma}}(t))}{t}\,\mathbb{C}_\nu^{-1} : \bar{\boldsymbol{\sigma}}(t).
\end{aligned} \tag{2}$$

Therein, $\bar{\boldsymbol{\sigma}}$ is the effective stress (force per undamaged area), $\mathbb{C}_\nu$ the fourth order unit stiffness tensor, $v(t,q_2,q_3)$ denotes the solidified volume at time t (in days), $\dot{\Phi}(t-t')$ is the unit viscoelastic compliance rate with its antiderivative given as

$$\Phi(t-t') = q_2\,\ln(1+(t-t')^{0.1}), \tag{3}$$

q_1 to q_4 are compliance parameters, and $F(\bar{\boldsymbol{\sigma}}(t)) = 1 + s(\bar{\boldsymbol{\sigma}})^2$ is an amplifying function dependent on the degree of loading $s(\bar{\boldsymbol{\sigma}})$ to account for nonlinear creep. To determine the degree of loading for multiaxial stress states in $F(\bar{\boldsymbol{\sigma}}(t))$, the invariant J_2 of the effective stress tensor and the uniaxial compressive strength are used for the SCDP model. The volume function $v(t,q_2,q_3)$ of the solidification theory is given as

$$v(t,q_2,q_3) = \frac{1}{t^{-0.5} + \frac{q_3}{q_2}}. \tag{4}$$

Based on these relations, the compliance function of the solidification theory for a constant unit stress $\bar{\boldsymbol{\sigma}}$ applied at time t' is expressed as

$$J(t,t') = q_1 + F(\bar{\boldsymbol{\sigma}})\,\int_{t'}^{t} \frac{\dot{\Phi}(\tau-t')}{v(\tau,q_2,q_3)}\mathrm{d}\tau + F(\bar{\boldsymbol{\sigma}})\,q_4 \ln\left(\frac{t}{t'}\right). \tag{5}$$

In (Neuner, Gamnitzer, & Hofstetter 2017), a time transformation function $\tau(t)$ is introduced to account for the hydration behavior of shotcrete, and $v(t,q_2,q_3)$ in (2) is replaced accordingly by $v(\tau(t),q_2,q_3)$. Time transformation function $\tau(t)$ is calibrated by the experimentally determined value of the Young's modulus at the age of 1 day, $E^{(1)}$, to represent the early age material behavior. For matured material stages, it is characterized by $\tau(t) \to t$ and thus, the original formulation is recovered.

Although the SCDP model is in good agreement with the experimentally observed material behavior, two shortcomings can be identified for the present approach: While $\tau(t)$ is formulated by means of $E^{(1)}$, the respective 28 days value $E^{(28)}$ does not enter the model, and hence compliance parameters q_1, q_2 and q_3 must be calibrated to represent the evolution of material stiffness properly. The second shortcoming is related to q_3, which governs that part of the viscoelastic creep strain rate which is independent of the material age. Since calibration of q_3 is rather difficult and requires comprehensive experimental results which are usually not available, it is proposed to eliminate this parameter from the formulation, and replace the formulation by a simplified one which can be parameterized more easily by $E^{(1)}$ and $E^{(28)}$. Accordingly, the modified volume function $v(\tau(t),q_2,q_3)$ is substituted by a function $v_{\mathrm{sc}}(t)$, specifically designed to represent the hydration behavior of shotcrete.

The relation between $v_{\mathrm{sc}}(t)$ and $E^{(1)}$ and $E^{(28)}$ is introduced as follows: As described in (Bažant & Baweja 1995), the experimentally determined Young's modulus can be approximated by computing the effective stiffness for a short time period Δt as $E(t) = J(t+\Delta t, t)^{-1}$. Approximating the integral in (5) for a short Δt as

$$\int_t^{t+\Delta t} \frac{\dot{\Phi}(\tau - t)}{v_{\mathrm{sc}}(\tau)}\,\mathrm{d}\tau \approx \frac{\Phi(\Delta t)}{v_{\mathrm{sc}}(t)}, \tag{6}$$

assuming linear viscoelastic behavior only, i.e., $F(\boldsymbol{\sigma}(t)) = 1$, and neglecting the effect of flow creep for short Δt, $E(t)$ can be approximated as

$$E(t) = \left(q_1 + q_2\,v_{\mathrm{sc}}(t)^{-1}\,\ln(1+\Delta t^{0.1})\right)^{-1}. \tag{7}$$

Based on this approximation and assuming that $v_{sc}(28\,\mathrm{d}) = 1$, $E^{(1)}$ and $E^{(28)}$ are given as

$$
\begin{aligned}
E^{(1)} &= \left(q_1 + q_2\, v_{sc}(1\,\mathrm{d})^{-1}\,\chi)\right)^{-1}, \\
E^{(28)} &= (q_1 + q_2\,\chi)^{-1},
\end{aligned}
\tag{8}
$$

with $\chi = \ln(1 + \Delta t^{0.1})$.

Accordingly, q_2 and $v_{sc}(1\,\mathrm{d})$ are computed as

$$
\begin{aligned}
q_2 &= \frac{1/E^{(28)} - q_1}{\chi}, \\
v_{sc}(1\,\mathrm{d}) &= \frac{q_2\,\chi}{1/E^{(1)} - q_1}.
\end{aligned}
\tag{9}
$$

The remaining unknown compliance parameter q_1 is estimated as $q_1 = 0.6/E^{(28)}$ and Δt is assumed as 10^{-3} d, as proposed by Bažant and Baweja (1995).

Derived from the evolution law for the uniaxial compressive strength in (Neuner, Gamnitzer, & Hofstetter 2017), the improved solidified volume function for shotcrete $v_{sc}(t)$ is proposed as

$$
v_{sc}(t) = \begin{cases} v^{\mathrm{I}}_{sc}(t) = r_v + a_v\, t + b_v\, t^2 & \text{if } t < t_{\mathrm{T}}, \\ v^{\mathrm{II}}_{sc}(t) = \dfrac{c_v}{(t - t_{\mathrm{D}})^{e_v} + d_v} & \text{otherwise,} \end{cases}
\tag{10}
$$

in which $t - t_{\mathrm{D}}$ is a time period to model the delayed start of hydration, t_{T} is the time of transition between the two expressions v^{I}_{sc} and v^{II}_{sc}, r_v is a residual value at $t = 0$, and a_v, b_v, c_v, d_v, and e_v are model parameters.

To ensure a smooth transition at $t = t_{\mathrm{T}}$, parameters a_v and b_v are computed as

$$
a_v = \left.\frac{\mathrm{d}v^{\mathrm{II}}_{sc}}{\mathrm{d}\,t}\right|_{t=t_{\mathrm{T}}} - 2\,b_v\,t_{\mathrm{T}}
\tag{11}
$$

and

$$
b_v = \left(r_v + t_{\mathrm{T}} \left.\frac{\mathrm{d}v^{\mathrm{II}}_{sc}}{\mathrm{d}\,t}\right|_{t=t_{\mathrm{T}}} - \left. v^{\mathrm{II}}_{sc}\right|_{t=t_{\mathrm{T}}}\right) / t^2_{\mathrm{T}}.
\tag{12}
$$

To satisfy $v_{sc}(28\,\mathrm{d}) = 1$ in conjunction with the prescribed value of $v_{sc}(1\,\mathrm{d})$, parameters c_v and d_v follow as

$$
c_v = (28\,\mathrm{d} - t_{\mathrm{D}})^{e_v} + d_v
\tag{13}
$$

and

$$
d_v = \frac{v_{sc}(1\,\mathrm{d})\,(1\,\mathrm{d} - t_{\mathrm{D}})^{e_v} - (28\,\mathrm{d} - t_{\mathrm{D}})^{e_v}}{1 - v_{sc}(1\,\mathrm{d})}\,.
\tag{14}
$$

The remaining model parameters t_{D}, t_{T}, r_v and e_v are proposed as $t_{\mathrm{D}} = 0.3\,\mathrm{d}$, $t_{\mathrm{T}} = 0.5\,\mathrm{d}$, $r_v = 0.01$, and $e_v = -0.65$, calibrated based on the experimentally observed evolution of the Young's modulus. While the proposed improved formulation predicts a very similar evolution of the material stiffness compared to the original formulation for shotcrete in (Neuner, Gamnitzer, & Hofstetter 2017), it can be calibrated more easily by means of the commonly experimentally determined values of $E^{(1)}$ and $E^{(28)}$.

A further extension of the SCDP model is proposed for the evolution law of the flow strain, based on the experimental results from the long-term creep tests: By introducing an additional exponential material parameter e_{flow}, the prediction of the long-term creep behavior can be improved slightly. Departing from (2), the extended expression for the flow creep strain rate is given as

$$
\dot{\boldsymbol{\varepsilon}}^{\mathrm{f}}(t) = \frac{q_4\,F(\bar{\boldsymbol{\sigma}}(t))}{t^{\,e_{\mathrm{flow}}}}\,\mathbb{C}^{-1}_{\nu} : \bar{\boldsymbol{\sigma}}(t).
\tag{15}
$$

A least square optimization based on the experimental results from the three long term creep tests yields $e_{\mathrm{flow}} = 1.22$, together with $q_4 = 34 \times 10^{-6}\,\mathrm{MPa}^{-1}$.

Finally, concerning the evolution of the shrinkage strain, in (Neuner, Cordes, Drexel, & Hofstetter 2017) based on the experimental results from the shrinkage tests it was concluded that the time function proposed by the ACI committee 209 (1992) is in better agreement with the observed material behavior than the one of the Bažant and Panula model. For this reason, for the subsequently presented numerical results, the time function proposed by the ACI committee 209 is employed for the SCDP model to represent the evolution of the shrinkage strain, identical to the Schädlich model.

3.4 *Calibration of the Material Models*

The parameter identification procedure for the three shotcrete models is described in (Neuner, Gamnitzer, & Hofstetter 2017), and the identification from the data of the present experimental program is reported in (Neuner, Cordes, Drexel, & Hofstetter 2017). The resulting material parameters are listed in Tables 1, 2 and 3. For the sake of brevity, a detailed description of the parameters is omitted here.

The evolution of the Young's modulus (Figure 3) is predicted very well by all models, although it is slightly underestimated between a material age of 1 day and 28 days by all models.

The evolution of the uniaxial compressive strength and the computed values are shown in Figure 4. Both the Schädlich model and the SCDP model are in good agreement with the experimental results, while the compressive strength is overestimated by the Meschke model between 1 day and 28 days.

The experimental results from the creep tests on sealed specimens loaded at material ages of 8 hours, 24 hours and 27 hours with compressive stresses of 1.9 MPa, 2.9 MPa and 2.7 MPa are shown together

Table 1: Material parameters for the SCDP model.

$E^{(1)}$ (MPa)	$E^{(28)}$ (MPa)	q_4 (MPa^{-1})	ν $(-)$	e^{flow} $(-)$	$f_{\text{cu}}^{(1)}$ (MPa)	$f_{\text{cu}}^{(28)}$(MPa)	$f_{\text{cy}}/f_{\text{cu}}$ $(-)$
13 943	21 537	34×10^{-6}	0.21	1.22	18.56	40.85	0.1
$f_{\text{cb}}/f_{\text{cu}}(-)$	$f_{\text{tu}}/f_{\text{cu}}(-)$	$\varepsilon_{\infty}^{\text{shr}}$ $(-)$	t_{50}^{shr} (h)	$\varepsilon_{\text{cpu}}^{\text{p(1)}}$ $(-)$	$\varepsilon_{\text{cpu}}^{\text{p(8)}}$ $(-)$	$\varepsilon_{\text{cpu}}^{\text{p(24)}}$ $(-)$	$G_{\text{fI}}^{(28)}$(N/mm)
1.16	0.1	−0.0019	8645	−0.03	−0.0007	−0.0007	0.1

Table 2: Material parameters for the Meschke model.

$E^{(1)}$ (MPa)	$E^{(28)}$ (MPa)	ν $(-)$	$f_{\text{cu}}^{(1)}$ (MPa)	$f_{\text{cu}}^{(28)}$ (MPa)	$f_{\text{cy}}/f_{\text{cu}}$ $(-)$	$f_{\text{cb}}/f_{\text{cu}}$ $(-)$	$f_{\text{tu}}/f_{\text{cu}}$ $(-)$
13 943	21 537	0.21	18.56	40.85	0.1	1.16	0.1
η (h)	$\varepsilon_{\infty}^{\text{shr}}$ $(-)$	k_{h} $(-)$	τ_{shr} (d)	Δt_{E} (h)	t_{E} (h)	$G_{\text{fI}}^{(28)}$ (N/mm)	
5	−0.002	1.0	4082	6	8	0.1	

Table 3: Material parameters for the Schädlich model.

$E^{(1)}$ (MPa)	$E^{(28)}$ (MPa)	ν $(-)$	ψ (°)	t_{50}^{cr} (h)	φ^{cr} $(-)$	$f_{\text{cu}}^{(1)}$ (MPa)	$f_{\text{cu}}^{(28)}$ (MPa)	$f_{\text{tu}}^{(28)}$ (MPa)	$f_{\text{cy}}/f_{\text{cu}}$ $(-)$
13 943	21 537	0.21	0	36	2.62	18.56	40.85	4.08	0.1
f_{cfn} $(-)$	f_{cun} $(-)$	f_{tun} $(-)$	$\varepsilon_{\infty}^{\text{shr}}$ $(-)$	t_{50}^{shr} (h)	$\varepsilon_{\text{cpu}}^{\text{p(1)}}$ $(-)$	$\varepsilon_{\text{cpu}}^{\text{p(8)}}$ $(-)$	$\varepsilon_{\text{cpu}}^{\text{p(24)}}$ $(-)$	$G_{\text{fI}}^{(28)}$ (N/mm)	$G_{\text{c}}^{(28)}$ (N/mm)
0.1	0.1	0.1	−0.0019	8645	−0.03	−0.0007	−0.0007	0.1	30

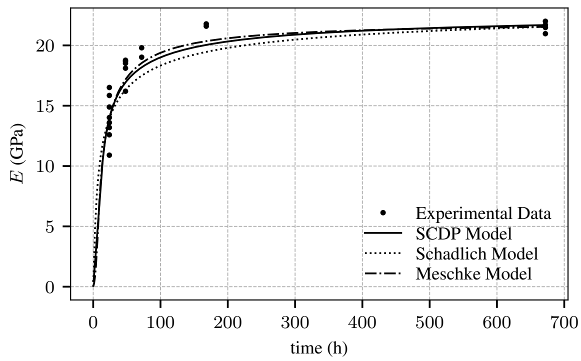

Figure 3: Measured and computed evolution of the Young's modulus E.

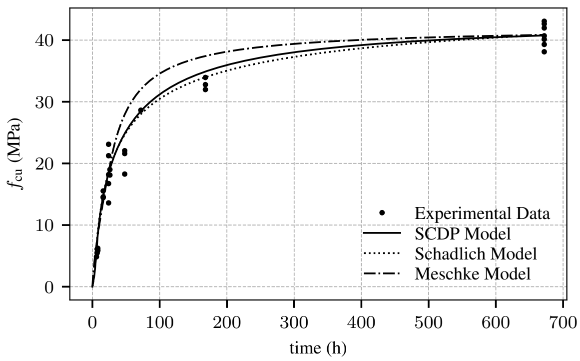

Figure 4: Measured and computed evolution of the uniaxial compressive strength f_{cu}.

with the predicted results in Figure 5. It can be concluded that compared to the Meschke model the creep behavior is represented better by the Schädlich model and the SCDP model. As pointed out in (Neuner, Cordes, Drexel, & Hofstetter 2017), this is a consequence of the applied low stress levels in combination with the viscoplastic formulation of the Meschke model. Since calibrating viscosity parameter η on the basis of the present creep test results is not reasonable, the default viscosity parameter $\eta = 5$ h as proposed in (Meschke, Kropik, & Mang 1996) is assumed for the subsequent numerical simulations.

4 APPLICATION

The structural benchmark example for comparing the shotcrete models was presented in (Neuner, Schreter, Unteregger, & Hofstetter 2017). It is derived from a stretch of the Brenner Basetunnel, for which in-situ measurement results of deformations in the surrounding rock mass during tunnel advance are available. The investigated stretch is characterized by a circular full face excavation profile with a diameter of 8.5 m and an overburden of 950 m measured from the tunnel axis. The installed shotcrete shell has a thickness of 0.2 m.

The problem is analyzed by means of a simplified 2D finite element model (Figure 6), assuming plane strain conditions. For representing the surrounding rock mass, the linear-elastic perfectly-plastic Hoek-Brown criterion (Hoek & Brown 1980), calibrated for representing the material behavior of Innsbruck Quartz-phyllite (Schreter, Neuner, Unteregger, Hofstetter, Reinhold, Cordes, & Bergmeister 2017), characteristic for this specific stretch, is employed. The employed material parameters for the Hoek-Brown

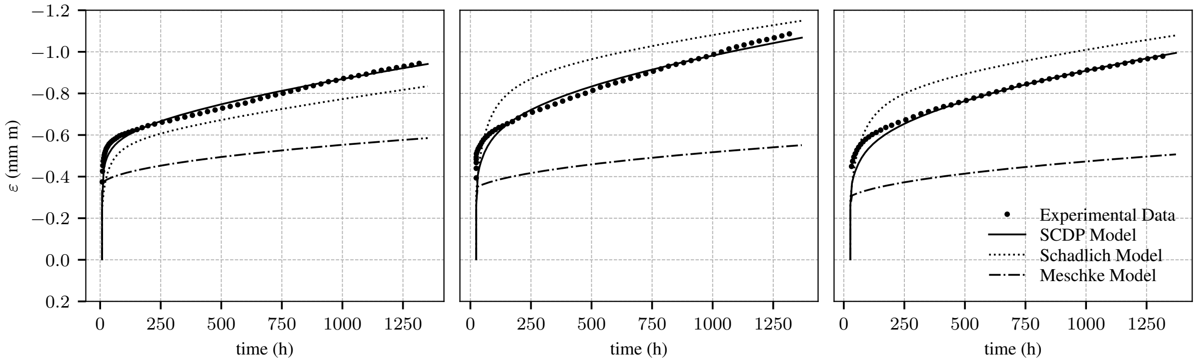

Figure 5: Measured and computed evolution of the total strain during the creep tests on sealed specimens loaded at material ages of 8 hours (left), 24 hours (center) and 27 hours (right).

Table 4: Material parameters for the Hoek-Brown model representing the rock mass.

E(MPa)	ν $(-)$	f_{cu} (MPa)	m_0 $(-)$
56 670	0.21	42	12
ψ (°)	e $(-)$	GSI $(-)$	$\boldsymbol{D}$ $(-)$
11.6	0.51	40	0

model are summarized in Table 4.

Due to the geometry, boundary conditions, an assumed initial hydrostatic geostatic stress state, and full face excavation, axial symmetry can be exploited, leading to a reduced finite element model consisting of a single element row. Accordingly, homogeneous Dirichlet boundary conditions are assumed perpendicular to the remaining part of the boundary of the single row of finite elements. For discretizing the rock mass and the shotcrete shell, 8-node quadrilateral continuum elements are used, with the shotcrete shell modeled by four elements through its thickness. Figure 6 shows the full and the reduced (highlighted in black) 2D finite element model. As a consequence of the employed geometry and boundary conditions, in the numerical simulations only radial displacements of the shotcrete shell are arising, and bending moments in the shotcrete shell are excluded. Obviously, this is a simplification of the actual deformations of the shotcrete shell during tunneling, however it allows for an easy comparison of the shotcrete models.

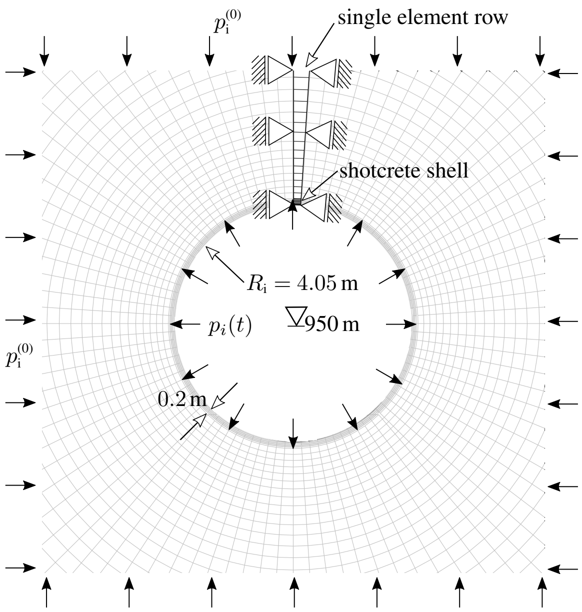

Figure 6: Schematic view of the benchmark example together with the 2D axisymmetric finite element model.

In the numerical simulations, an initial hydrostatic stress state in the rock mass of $p_{\text{i}}^{(0)} = 25.7\,\text{MPa}$ is assumed. Within the convergence-confinement method, the excavation of the investigated cross section is modeled by reducing an equivalent fictitious internal pressure $p_{\text{i}}(t)$ acting on the tunnel lining by means of a stress release ratio $\lambda(t)$ ranging from $0\,\%$ (no reduction) to $100\,\%$ (full removal), expressed by the relation

$$p_{\text{i}}(t) = (1 - \lambda(t))\, p_{\text{i}}^{(0)}\,. \tag{16}$$

Before installing the shotcrete shell in the numerical simulations, the stress is released according to an initial stress release ratio. Subsequently, the shotcrete shell is installed and the remaining fictitious internal pressure is removed employing a stepwise, time-dependent stress release function representing the sequence of *drill, blast and idle periods* during further tunnel advance. This stepwise stress release is assumed according to a parabolic decline, as suggested by Pöttler (1990). In total, 9 excavation steps are considered, separated by idle periods of 8 hours. After the last excavation step, an additional time period of two weeks, i.e. 336 hours, is investigated for observing the relaxation behavior of the shotcrete shell.

Two different initial stress release ratios are considered: 85 % and 95 %. They were derived according to the in-situ measured range of predeformations in the rock mass, as reported in (Schreter, Neuner, Unteregger, Hofstetter, Reinhold, Cordes, & Bergmeister 2017) and (Neuner, Schreter, Unteregger, & Hofstetter 2017).

The time-dependent mechanical response of the shotcrete shell is assessed by comparing the time-

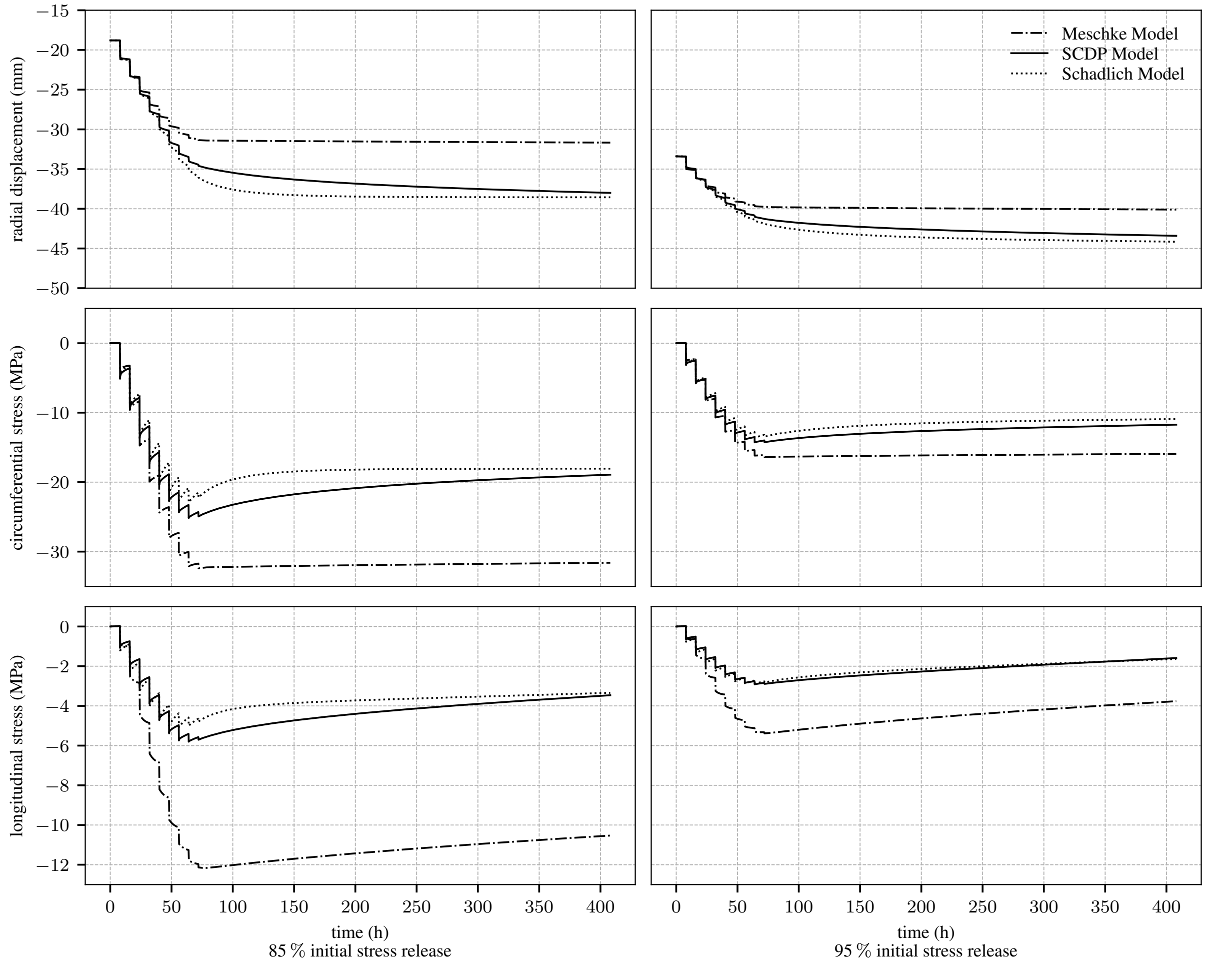

Figure 7: Evolution of the radial displacement (top), circumferential stress in the shotcrete shell (center) and longitudinal stress in the shotcrete shell (bottom), for two different initial stress releases 85 % (left column) and 95 % (right column).

dependent evolution of the radial displacement of an arbitrary node at the inner surface of the shotcrete shell, as well as the time-dependent evolution of the stresses in circumferential and longitudinal direction. The latter are taken at an integration point close to the inner surface of the shotcrete shell. Since the stress state in the shotcrete shell can be classified as nearly biaxial, the radial stress in thickness direction of the shotcrete shell is neglected in the present assessment. This observation was already reported in (Meschke 1996), and confirmed in (Neuner, Schreter, Unteregger, & Hofstetter 2017).

Figure 7 shows the predicted time-dependent evolution of the radial displacement, the circumferential stress and the longitudinal stress for both initial stress release ratios. Therein, the time-dependent radial displacement consists of the instantaneous displacement of the unsupported rock mass due to initial stress release, the instantaneous displacements due to the 9 excavation steps after installation of the shotcrete shell, and the displacements due to the time-dependent deformations due to shrinkage and creep of shotcrete. Comparing the obtained radial displacements for the initial stress release ratio of 85 %, it can be seen that during the first excavation steps the predicted responses by the Schädlich model and the SCDP model are very similar, with larger displacements predicted by the Schädlich model during the last excavations steps. However, due to the stronger relaxation predicted by the SCDP model, the radial displacement and stresses attain those predicted by the Schädlich model at the end of the investigated time period. A stiffer material behavior, and thus smaller radial displacements are predicted by the Meschke model throughout the complete investigated time period. As explained in (Neuner, Schreter, Unteregger, & Hofstetter 2017), this is attributed to the viscoplastic formulation of the Meschke model. Both the circumferential and longitudinal stress predicted by the Meschke model are considerably higher, which again is a consequence of the employed viscoplastic formulation, neglecting creep in the elastic domain.

For the initial stress release of 95 %, nearly identical displacements and stresses are predicted by the Schädlich model and the SCDP model, and again, a stiffer response is predicted by the Meschke model due to the viscoplastic formulation.

An interesting detail can be observed for the Meschke model: In contrast to the SCDP model and the Schädlich model, no relaxation of the longitudinal stress during the idle periods occurs, but conversely, it is increasing. This is a consequence of the employed associated viscoplastic formulation, assuming dilatant material behavior during creep.

5 CONCLUSIONS

Three different constitutive models for shotcrete, i.e., the Meschke model, the Schädlich model and the SCDP model were described briefly, and an extension of the SCDP model was presented. The proposed improvements of the SCDP model comprise the reformulation of the evolution law of the viscoelastic strain aiming at an easier calibration procedure on the basis of experimental data, as well as an additional material parameter for the evolution law of the flow creep strain was introduced for an improved agreement with the experimentally observed material behavior.

Finally, a comparison of the influence of the different shotcrete models on the evolution of displacements and stresses by means of a benchmark example of deep tunnel advance was presented. Thereby, it was shown that despite of the different formulations for the evolution of stiffness and strength, hardening material behavior, shrinkage and creep, similar displacements and stresses are predicted by the Schädlich model and the SCDP model.

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7 Implicit Gradient-Enhancement of the SCDP Model

In Publication VI, *Evaluation of a Gradient Enhanced Damage Plasticity Model for Shotcrete*, an extension of the SCDP model is proposed, aiming at an improved and more robust treatment of softening material behavior in finite element simulations compared to the hitherto employed crack band approach by Bažant and Oh (1983). To this end, the over-nonlocal implicit gradient-enhanced formulation presented by Poh and Swaddiwudhipong (2009b) for the concrete model by Grassl and Jirásek is adopted for the SCDP model. The gradient-enhanced SCDP model is applied in finite element simulations of an experimental test on an L-shaped concrete panel presented by Winkler et al. (2004). Different refinements and orientations of the finite element mesh are investigated, and it is demonstrated that the gradient-enhanced SCDP model is able to represent softening material behavior in an objective manner with respect to the employed finite element mesh for mode I failure.

In Publication VII, *Study of the Regularization Scheme of an Advanced Rock Model*, the same gradient-enhanced damage formulation is employed for extending a damage plasticity mode for rock mass, proposed by Unteregger (2015). The extended model is applied to finite element simulations of biaxial compression tests on rock specimens, and the capabilities of the gradient-enhanced formulation for regularizing softening material behavior in mode II failure are assessed.

The scientific contribution by the author comprises

- the numerical implementation of the implicit gradient-enhanced damage framework (Publication VI and Publication VII), jointly with Magdalena Schreter,
- the extension of the SCDP model to the gradient-enhanced damage formulation (Publication VI),
- the performance of the numerical simulations (Publication VI),
- the preparation of the manuscript (Publication VI), jointly with Günter Hofstetter.

Since the implicit-gradient enhancement is only described very briefly in the respective publications, in the following a more detailed introduction to this subject is presented.

7.1 Motivation

Standard continuum models describing softening material behavior, e.g., models based on the theory of plasticity or the theory of continuum damage mechanics, exhibit several theoretical deficiencies, as reported and summarized by Bažant (1976):

- The softening process zone is infinitesimally small, i.e., it has the size of a material point,
- at structural level, snapback behavior due to the infinitesimally small softening zone is observed,
- and the amount of dissipated energy during the failure process is zero due to the infinitesimal zone in which energy is dissipated.

In a mathematical sense, those issues are related to the loss of ellipticity of the initial boundary value problem (IBVP) in static and quasi-static analyses.

In finite element analyses, those deficiencies lead to mesh-dependent results, which is commonly referred to as a *pathological* mesh sensitivity. Failure patterns like cracks tend to localize into the smallest possible bandwidth, i.e., usually a single layer of finite elements, and upon mesh refinement the localization zone is decreasing into an arbitrarily small domain. Accordingly, the obtained results are not objective with respect to the finite element mesh, i.e., the results are sensitive to the numerical discretization scheme.

Objective, i.e., mesh insensitive, representation of softening material behavior is a vast subject of research, and in the past decades different approaches for regularizing softening material behavior in finite element simulations have been proposed. A popular and widely used method for continuum models is the crack band approach, based on a mesh adjusted softening modulus. This approach was adopted and investigated by Valentini (2011) for the concrete damage plasticity model by Grassl and Jirásek (2006a), which forms a constituent of the SCDP model, and it is also employed for the SCDP model, as reported in Publication II.

7.2 Deficiencies of the Crack Band Approach

Following the crack band approach, the softening modulus is adjusted by means of a characteristic size of a finite element and the specific mode I fracture energy, which is considered as a material parameter.

However, as pointed out by Jirásek and Bauer (2012), the crack band approach involves several issues:

(i) In finite element simulations employing continuum models based on the crack band approach, crack paths tend to be sensitive to the orientation of the finite element mesh , i.e., crack paths tend to be aligned with the grid lines of the finite element mesh.
(ii) For higher order finite elements, localization into subdomains may occur, and hence, the adjustment of the softening modulus based on the size of the complete element yields inaccurate results.
(iii) The determination of the characteristic length of an element is complicated, especially for distorted or elongated elements, and the proper characteristic length depends on the orientation of the crack.

Issues (i) and (ii) are related to the fact that deformations in consequence of material failure localize usually into the smallest possible domain if no proper countermeasures are employed. Concerning issue (iii), different methods of different complexity for estimating the characteristic length of a finite element were proposed, for instance by Oliver (1989). However, to the author's best knowledge an approach which is universally applicable still does not exist.

Furthermore, due to the assumption of a softening process zone attributed to the size of a finite element, the crack band approach is not suitable for diffuse material failure patterns, i.e., failure patterns which do not localize into a single layer of finite elements. A discussion on that issue was presented by Grassl and Jirásek (2006b), based on an example of compressive failure of an eccentrically loaded concrete column.

7.3 Localization Limiters

To prevent mesh dependent localization into the smallest possible domain, several advanced techniques were proposed in the literature. Commonly, they are referred to as *localization limiters* (Jirásek and Bažant, 2002). Different methods proposed so far are, for instance

- nonlocal models of the integral type,
- gradient-enhanced models,
- models based on Cosserat continua,
- and viscoplastic models.

Nonlocal models of the integral type and gradient-enhanced models enrich the constitutive relations for an infitesimal material point by nonlocal terms, i.e., quantities which are influenced by a finite or even infinite neighborhood of the considered material point. Both approaches are widely used and are therefore presented briefly in the subsequent subsections.

Models based on the Cosserat continuum (E. Cosserat and F. Cosserat, 1909) incorporate an additional field of micropolar rotations, and were applied successfully to mode II failure (de Borst, 1991). However, these models are not suitable for pure mode I failure.

Viscoplastic models enrich the constitutive relations by a rate-dependent viscous term which has a regularizing effect, but which also introduces a dependency on the rate of loading.

All of those approaches have in common that a length scale is incorporated into the constitutive relations. This holds also for the viscoplastic approach, as explained by Needleman (1988).

In this context, phase-field models are mentioned, which have been developed extensively in the past years (Miehe et al., 2010): Phase-field models regularize sharp crack discontinuities by means of pure continuum formulations. Although phase-field models originate from a completely different motivation, they might be considered as localization limiters in the aforementioned sense. In fact, phase-field models are closely related to implicit gradient-enhanced damage models in a mathematical sense, as demonstrated by means of a comparison of both approaches by de Borst and Verhoosel (2016).

7.3.1 Nonlocal Models of the Integral Type

According to nonlocal models of the integral type, the stress-strain relation for a material point is governed not exclusively by its local state, but also by the state of its neighborhood (Jirásek and Bažant, 2002).

The nonlocal character is introduced in the constitutive relations by replacing one or more quantities by their nonlocal counterparts. The nonlocal counterpart of a quantity is defined as the spatial average of that local quantity, which is computed in terms of an integral expression employing a nonlocal weight function.

For instance, assuming a local quantity $f(\boldsymbol{x})$ for each material point $\boldsymbol{x}$ of a domain Ω, its nonlocal counterpart $\bar{f}(\boldsymbol{x})$ is computed as

$$\bar{f}(\boldsymbol{x}) = \int_{\Omega} w(\boldsymbol{x}, \boldsymbol{\xi})\, f(\boldsymbol{\xi})\, \mathrm{d}\boldsymbol{\xi}\,. \tag{7.1}$$

Therein, the nonlocal weight function $w(\boldsymbol{x}, \boldsymbol{\xi})$ defines the interaction between two material points $\boldsymbol{x}$ and $\boldsymbol{\xi}$, dependent on their spatial distance. Although the zone of interaction can embrace the complete domain, for numerical implementations usually only a finite neighborhood is considered in the nonlocal weight function. Commonly employed weight functions are Gaussian bell functions or parabolic functions.

The integral nonlocal formulation is basically applicable to any type of constitutive model, including models based on the theory of plasticity or the theory of continuum damage mechanics. A drawback of integral nonlocal models is that in finite element simulations due to the increased interaction radius of each degree of freedom the global finite element stiffness matrix has an increased bandwidth, and in general is nonsymmetric.

7.3.2 Gradient-Enhanced Models

Gradient-enhanced models are considered as the differential complement to non-local models of the integral type: The nonlocal character is introduced in the constitutive relations by incorporating higher order gradients of one or more quantities. Basically, two families exist within this class of models: *Explicit* and *implicit* gradient-enhanced models.

Explicit gradient-enhanced models incorporate higher order gradients directly into the constitutive relations, whereas for implicit gradient-enhanced models the constitutive relations are enriched by nonlocal quantities which are defined implicitly as the solution of a higher order partial differential equation, with the local quantity acting as the driving force.

For instance, by analogy to (7.1), the nonlocal counterpart $\bar{f}(\boldsymbol{x})$ of a local quantity $f(\boldsymbol{x})$ may be described according to the explicit approach in terms of a second order expression as

$$\bar{f}(\boldsymbol{x}) = f(\boldsymbol{x}) + l^2 \nabla^2 f(\boldsymbol{x}), \tag{7.2}$$

and according to the implicit approach employing a second order partial differential equation as

$$\bar{f}(\boldsymbol{x}) - l^2 \nabla^2 \bar{f}(\boldsymbol{x}) = f(\boldsymbol{x}). \tag{7.3}$$

In both expressions, l denotes the interaction radius, which is considered as an internal length scale material parameter.

Despite their superficial similarities, the regularization and localization properties of both approaches are substantially different. A comparison of explicit and implicit gradient-enhanced models based on continuum damage mechanics was presented by Peerlings et al. (2001). It is demonstrated that despite their theoretical simplicity, the numerical implementation of explicit gradient-enhanced models is complicated, and in general, higher order finite elements are required. For the explicit approach, boundary conditions must be formulated explicitly in terms of higher order derivatives of the displacement field. In contrast, implicit gradient-enhanced models are amenable to a more simplified and robust implementation, at the price of a fully coupled partial differential equation system. Furthermore, Peerlings et al. demonstrated that the implicit approach is equivalent to the nonlocal approach of the integral type if the Green's function of the higher order partial differential equation is employed as the nonlocal weight function of the integral approach. A comparison of different implicit and explicit gradient-enhanced models employing strain gradients and gradients of internal variables, and an assessment of them was presented by Jirásek and Rolshoven (2009a,b).

Most gradient-enhanced models proposed in the literature, including gradient-enhanced plasticity models as well as gradient-enhanced damage models, were postulated more or less in an ad-hoc fashion, neglecting any thermodynamic considerations. An attempt to incorporate models based on the implicit gradient approach into a thermodynamic sound framework, i.e., the *micromorphic* framework, was presented by Forest (2009). In addition, a discussion and the reformulation of selected gradient-enhanced models within the micromorphic framework were presented.

7.3.3 Over-Nonlocal Formulations

For particular continuum models representing softening material behavior, it is reported that a mere nonlocal formulation, employing either the integral approach or the gradient-enhanced formulation, fails to achieve a proper regularization (Geers et al., 1998; Simone et al., 2004). Di Luzio and Bažant (2005) investigated the localization properties of different nonlocal plasticity and damage models by means of spectral analyses of wave propagation in a body exhibiting softening material behavior. They demonstrated a necessary requirement for a proper regularization of softening material behavior. It is given by the fact that the propagation speed of stress acceleration waves does not become imaginary. In case of plasticity models formulated in terms of strain-like internal damage driving variables, this requirement is fulfilled only if the so called *over-nonlocal* formulation is employed.

The over-nonlocal approach was proposed by Vermeer and Brinkgreve (1994), and introduces a weighted nonlocal quantity $\hat{f}(\boldsymbol{x})$ for formulating the constitutive relations. Weighted variable $\hat{f}(\boldsymbol{x})$ is computed by means of a scalar weighting parameter m from the local field $f(\boldsymbol{x})$ and the nonlocal field $\bar{f}(\boldsymbol{x})$:

$$\hat{f}(\boldsymbol{x}) = m\,\bar{f}(\boldsymbol{x}) + (1-m)\,f(\boldsymbol{x}). \tag{7.4}$$

Choosing $m = 0$ leads to a pure local formulation, i.e., $\hat{f}(\boldsymbol{x}) = f(\boldsymbol{x})$, and choosing $m = 1$ leads to a pure nonlocal formulation, i.e., $\hat{f}(\boldsymbol{x}) = \bar{f}(\boldsymbol{x})$. To obtain the over-nonlocal formulation, weighting parameter m is chosen larger than unity, i.e., $m > 1$. Examples of application of the over-nonlocal formulation to damage plasticity models were presented by Poh and Swaddiwudhipong (2009a,b) and Hosseini et al. (2015).

7.4 Over-Nonlocal Gradient-Enhancement of the SCDP Model

Grassl and Jirásek (2006b) presented an application of the nonlocal formulation of the integral type to their damage plasticity model for concrete (Grassl and Jirásek, 2006a). It is shown that the integral nonlocal approach yields mesh-insensitive results for both cracking in tension and diffuse failure in compression. Poh and Swaddiwudhipong (2009b) proposed an implicit gradient-enhanced version of that concrete model, using the over-nonlocal weighting for computing the damage driving variable. For simplicity and for a clear distinction, the nonlocal approach of the integral type is henceforth simply denoted as the *integral* approach, and the implicit gradient-enhanced formulation is denoted simply as the *gradient* approach.

Comparing both approaches, following differences are observed:

- According to the gradient approach, the nonlocal field is computed implicitly by solving a fully coupled partial differential equation system, whereas for the integral approach the computation of the spatially averaged quantity in terms of the spatial distance between individual material points requires a specific algorithmic treatment.
- In finite element simulations, no additional degrees of freedom are introduced for models based on the integral approach. From a computational viewpoint, this is advantageous for constitutive models with higher order tensorial nonlocal quantities and for models based on multiple nonlocal quantities. For instance, in (Xenos and Grassl, 2016) an extension of the concrete model by Grassl and Jirásek to cyclic material behavior adopting the integral approach is proposed, based on altogether six scalar nonlocal variables.
- Models based on the integral approach require a special treatment of the nonlocal weight function near boundaries, i.e., if the support of the nonlocal weight function is located outside the domain. In contrast, boundary conditions are treated in a natural and mathematical rigorous sense within the gradient approach (Peerlings et al., 2001).
- Models based on the integral approach predict spurious interactions between material points near non-convex boundaries, for instance across notches. These spurious interactions are not predicted by models based on the gradient approach.
- For models based on the gradient approach, the computation of the algorithmic tangent stiffness by means of consistent linearization of the constitutive relations is straightforward, and the global system matrix in finite element simulations preserves its sparse structure.
- Regarding the numerical implementation into a finite element framework, models based on the gradient approach can be implemented at element level. In contrast, models based on the integral approach require operations at *higher level*, since for each material point nonlocal quantities are influenced by a finite neighborhood, i.e., usually across several elements. As a consequence, implementation of such models is excluded for finite element frameworks which do not provide a programming interface for that purpose.

Because of the advantages of the gradient approach over the integral approach, and considering that the SCDP model is formulated in terms of a single scalar damage driving internal variable, an extension of the SCDP model adopting the over-nonlocal implicit gradient-enhanced formulation proposed by Poh and Swaddiwudhipong (2009b) is presented.

Following that approach, the local damage driving variable α_d in the damage evolution law of the SCDP model, defined in Publication I (9) as

$$\omega = 1 - \exp\left(-\frac{\alpha_\mathrm{d}}{\varepsilon_\mathrm{f}}\right), \tag{7.5}$$

is replaced by

$$\hat{\alpha}_{\mathrm{d}} = m\,\bar{\alpha}_{\mathrm{d}} + (1-m)\,\alpha_{\mathrm{d}}, \tag{7.6}$$

which is an weighted average of the local variable α_{d} and its nonlocal counterpart $\bar{\alpha}_{\mathrm{d}}$, with m denoting the weighting parameter.

Accordingly, the over-nonlocal evolution law for the damage parameter is given as

$$\omega = 1 - \exp\left(-\frac{\hat{\alpha}_{\mathrm{d}}}{\varepsilon_{\mathrm{f}}}\right). \tag{7.7}$$

As explained in (Grassl and Jirásek, 2006b), for over-nonlocal formulations negative values of ω can arise due to the negative weighting of α_{d} in (7.6), which is thermodynamically inadmissible. Hence, for the SCDP model an additional constraint ensuring a strictly non-negative damage parameter is employed.

The field of the nonlocal variable $\bar{\alpha}_{\mathrm{d}}$ is defined by the second order partial differential equation of the Helmholtz type

$$\bar{\alpha}_{\mathrm{d}} - l^2\,\nabla^2\bar{\alpha}_{\mathrm{d}} = \alpha_{\mathrm{d}}, \tag{7.8}$$

in which the local strain-like internal variable acts as the driving force. Together with the equilibrium equation

$$\nabla\boldsymbol{\sigma} + \boldsymbol{f} = \boldsymbol{0}, \tag{7.9}$$

in which $\boldsymbol{f}$ denotes the body forces, (7.8) forms the fully coupled system of partial differential equations.

The standard boundary conditions for the equilibrium equation are given as

$$\begin{aligned} \boldsymbol{u} &= \bar{\boldsymbol{u}} && \text{on } \Gamma_u \\ \boldsymbol{\sigma n} &= \bar{\boldsymbol{t}} && \text{on } \Gamma_t\ , \end{aligned} \tag{7.10}$$

in which $\boldsymbol{n}$ denotes the unit vector perpendicular onto the boundary $\Gamma = \Gamma_u \cap \Gamma_t$ of the domain of interest, $\boldsymbol{u}$ the displacement vector, $\bar{\boldsymbol{u}}$ the prescribed displacement vector on Γ_u and $\bar{\boldsymbol{t}}$ the prescribed traction vector on Γ_t.

For the Helmholtz type equation (7.8), homogeneous Neumann boundary conditions are assumed for the complete domain:

$$\nabla\bar{\alpha}_{\mathrm{d}}^T\,\boldsymbol{n} = 0 \quad \text{on } \Gamma. \tag{7.11}$$

An interpretation of this boundary condition in the context of phase-field models was presented by de Borst and Verhoosel (2016) in terms of the condition that cracks must be oriented perpendicular to the surface of a body.

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Evaluation of a Gradient Enhanced Damage Plasticity Model for Shotcrete

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Micro Abstract
A damage plasticity model representing the time-dependent and nonlinear material behavior of shotcrete is discussed. In order to obtain mesh-insensitive numerical results upon strain softening, an over-nonlocal implicit gradient enhancement is employed. The capabilities of the model are demonstrated by means of finite element simulations, employing meshes of different size and orientations.

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Introduction

The use of shotcrete (sprayed concrete) is an essential securing measure within the New Austrian Tunneling Method for stabilizing the surrounding rock mass during tunnel advance. In the context of numerical simulations of tunnel advance, the complex material behavior shotcrete, characterized by time-dependent evolution of material properties like stiffness and strength, hardening and softening behavior, creep and shrinkage is described using advanced constitutive models. One recently proposed constitutive model for shotcrete is the SCDP model, presented in [5]. In order to obtain objective results in case of softening material behavior in finite element simulations with respect to the employed mesh, an appropriate regularization technique has to be employed. In the present contribution, the over-nonlocal gradient enhancement for regularizing the softening material behavior, presented in [6], is adopted and validation by means of numerical simulations of the experimental tests by Winkler [7] is presented.

1 The SCDP model

The SCDP model (Shotcrete Damage Plasticity model) is based on three well established constitutive models for concrete, which are the damage plasticity model for concrete by Grassl and Jirásek [3], the solidification theory for concrete creep by Bažant and Prasannan [2] and the shrinkage model by Bažant and Panula [1], with the latter two models being incorporated into the framework of the former model. Accordingly, the stress–strain relation in total form is expressed as

$$\boldsymbol{\sigma} = (1-\omega)\,\mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}} - \boldsymbol{\varepsilon}^{\mathrm{ve}} - \boldsymbol{\varepsilon}^{\mathrm{f}} - \boldsymbol{\varepsilon}^{\mathrm{shr}}), \tag{1}$$

in which $\boldsymbol{\sigma}$ denotes the nominal stress (force per total area), ω the isotropic scalar damage parameter and $\mathbb{C}$ the fourth order stiffness tensor. The total strain $\boldsymbol{\varepsilon}$ is decomposed into the instantaneous elastic strain $\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}} - \boldsymbol{\varepsilon}^{\mathrm{ve}} - \boldsymbol{\varepsilon}^{\mathrm{f}} - \boldsymbol{\varepsilon}^{\mathrm{shr}}$, the viscoelastic strain $\boldsymbol{\varepsilon}^{\mathrm{ve}}$, the flow (viscous) strain $\boldsymbol{\varepsilon}^{\mathrm{f}}$, the shrinkage strain $\boldsymbol{\varepsilon}^{\mathrm{shr}}$ and the plastic strain $\boldsymbol{\varepsilon}^{\mathrm{p}}$.

The evolution of the viscoelastic strain and the flow strain is described by means of the solidification theory, which is modified in order to account for the accelerated hydration of shotcrete compared to concrete, and the evolution of the shrinkage strain is treated by the model by Bažant and Panula. To delimit the elastic domain, a single yield surface is employed, and a nonassociated flow rule is used for describing the evolution of the plastic strain. Isotropic hardening material behavior is modeled by a single, scalar strain-like internal variable α_p, with a fully hardened state attained at $\alpha_\mathrm{p} = 1$.

The evolution of ω is driven by a strain like internal variable α_d by means of an exponential softening law given as

$$\omega = 1 - \exp\left(-\frac{\alpha_\mathrm{d}}{\varepsilon_\mathrm{f}}\right), \tag{2}$$

with ε_f representing the softening modulus. The evolution of α_d is related to the plastic strain rate $\dot{\boldsymbol{\varepsilon}}^\mathrm{p}$:

$$\dot{\alpha}_\mathrm{d} = \begin{cases} 0 & \text{if } \alpha_\mathrm{p} < 1, \\ \dot{\varepsilon}^\mathrm{p}_\mathrm{V}/x_\mathrm{s}(\dot{\boldsymbol{\varepsilon}}^\mathrm{p}) & \text{otherwise.} \end{cases} \tag{3}$$

Therein, $\dot{\varepsilon}^\mathrm{p}_\mathrm{V}$ denotes the first invariant of the plastic strain rate tensor and $x_\mathrm{s}(\dot{\boldsymbol{\varepsilon}}^\mathrm{p})$ is a measure of ductility.

2 Over-Nonlocal Gradient Enhancement of the SCDP model

It is a well known issue that without proper regularization softening material behavior leads to a pathological mesh sensitivity in finite element simulations, associated with the loss of ellipticity of the boundary value problem. A commonly employed remedy is a mesh adjusted softening modulus, based on the characteristic size of a finite element. However, sensitivity with respect to the orientation of the mesh and determination of the characteristic size for higher order elements remain unresolved issues [4]. One possible remedy to overcome these problems are nonlocal approaches, which incorporate gradients of internal variables. In this context, a gradient enhancement for the damage plasticity model by Grassl and Jirásek was proposed by Poh and Swaddiwudhipong [6]. Adopting this approach for the SCDP model, the local damage driving variable α_d in (2) is replaced by an averaged strain-like variable $\hat{\alpha}_\mathrm{d}$ as

$$\omega = 1 - \exp\left(-\frac{\hat{\alpha}_\mathrm{d}}{\varepsilon_\mathrm{f}}\right). \tag{4}$$

Therein, $\hat{\alpha}_\mathrm{d}$ is defined as

$$\hat{\alpha}_\mathrm{d} = m\,\bar{\alpha}_\mathrm{d} + (1-m)\,\alpha_\mathrm{d}, \tag{5}$$

in which $\bar{\alpha}_\mathrm{d}$ denotes the nonlocal counterpart of α_d, implicitly defined by the solution of the partial differential equation

$$\bar{\alpha}_\mathrm{d} - l^2\,\nabla^2\bar{\alpha}_\mathrm{d} = \alpha_\mathrm{d}. \tag{6}$$

The latter is in the context of nonlocal gradient enhanced constitutive models commonly denoted as the Helmholtz-like partial differential equation and leads together with the equilibrium equation to a fully coupled problem. Parameter l represents a length scale for describing the width of the softening process zone and is considered as a material parameter. In (5), m denotes the weighting parameter. A value of m larger than unity yields the so-called over-nonlocal formulation, which is employed in order to prevent a spurious broadening of the process zone with increasing damage, as reported in [6].

3 Application

The regularization capabilities of the over-nonlocal gradient approach are evaluated by means of finite element simulations of the experimental test on a L-shaped concrete specimen, presented by Winkler [7]. The corresponding experimental test setup is illustrated in Figure 1.

The L-shaped specimen with a uniform thickness of 100 mm is fixed at the bottom edge and a displacement of 1 mm in vertical direction is applied at the right end of the specimen. A crack is expected to emanate from the central reentrant corner, and to evolve horizontally across the specimen to the opposite side. To assess the regularization capabilities of the gradient

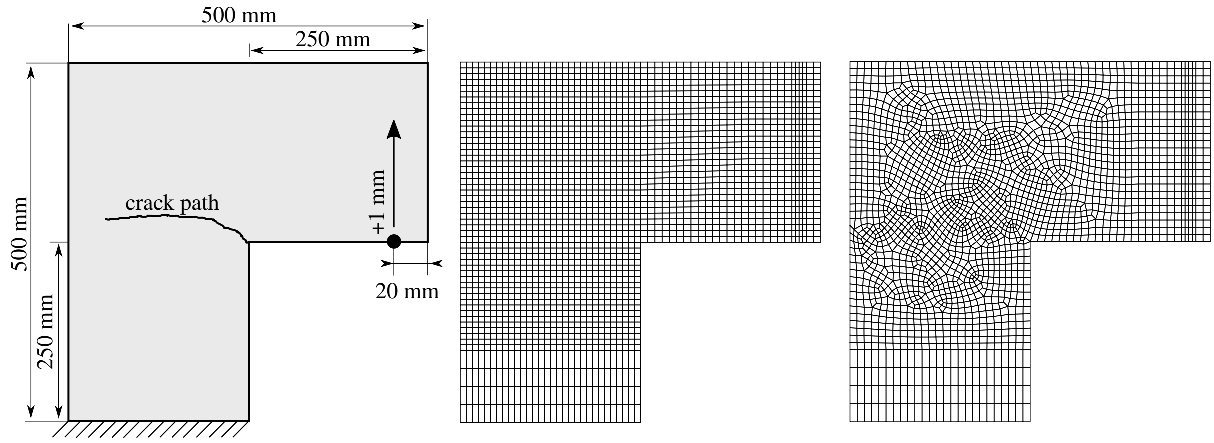

Figure 1. L-shaped specimen investigated by Winkler [7], fixed at the bottom and subjected to a vertical displacement of 1 mm at the right end: Test setup (left), the 8 mm structured mesh (center), and the 8 mm unstructured mesh (right).

q_1 (MPa^{-1})	q_2 (MPa^{-1})	q_3 (MPa^{-1})	q_4 (MPa^{-1})	ν (−)
23.2×10^{-6}	169.3×10^{-6}	6.6×10^{-6}	6.8×10^{-6}	0.18
$f_{\mathrm{cu}}^{(28)}$ (MPa)	$f_{\mathrm{cy}}/f_{\mathrm{cu}}$ (−)	$f_{\mathrm{cb}}/f_{\mathrm{cu}}$(−)	$f_{\mathrm{tu}}/f_{\mathrm{cu}}$(−)	$\varepsilon_{\mathrm{cpu}}^{\mathrm{p}(24)}$ (−)
31.0	0.3	1.16	0.1	−0.0011

Table 1. Material parameters for the SCDP model [5] for representing the concrete composition employed by Winkler [7].

enhancement for mode I fracture, three different refinements of the finite element mesh in the vicinity of the crack path are investigated, i.e., elements with sizes of 8 mm, 4 mm and 2 mm. Additionally, structured meshes, aligned with the boundaries of the specimen, and unstructured meshes are investigated, resulting in a total of 6 finite element meshes to be studied. Fully integrated quadrilateral elements with linear shape functions are used, and plane stress conditions are assumed. The 8 mm structured mesh and the 8 mm unstructured mesh are shown in Figure 1.

The parameters of the nonlocal approach are chosen as $l = 10$ mm and $m = 1.05$, and a softening modulus of $\varepsilon_{\mathrm{f}} = 0.0017$ is employed. The remaining parameters of the SCDP model are chosen to resemble to material behavior of the concrete composition employed in the experimental tests, and are listed in Table 1. For the sake of brevity, the reader is referred to [5] for a description of the model and its material parameters. Due to the time-dependency of the constitutive model, a material age of 28 d is assumed in the numerical simulations, and the displacement is applied within 0.1 h. Homogeneous Neumann boundary conditions are employed for the Helmholtz-like equation (6).

The resulting load–displacement curves are shown in Figure 2. Comparing the obtained load–displacement curves for the structured and unstructured meshes of the same size, it can be seen that the results can be considered as objective, i.e., they are independent of the mesh orientation. Furthermore, for sufficiently fine meshes, i.e., for element sizes of 4 mm and 2 mm, the obtained curves are nearly identical, while the peak load is somewhat overestimated for the coarse 8 mm mesh. In general, it can be seen that the predicted peak load decreases slightly with decreasing element size.

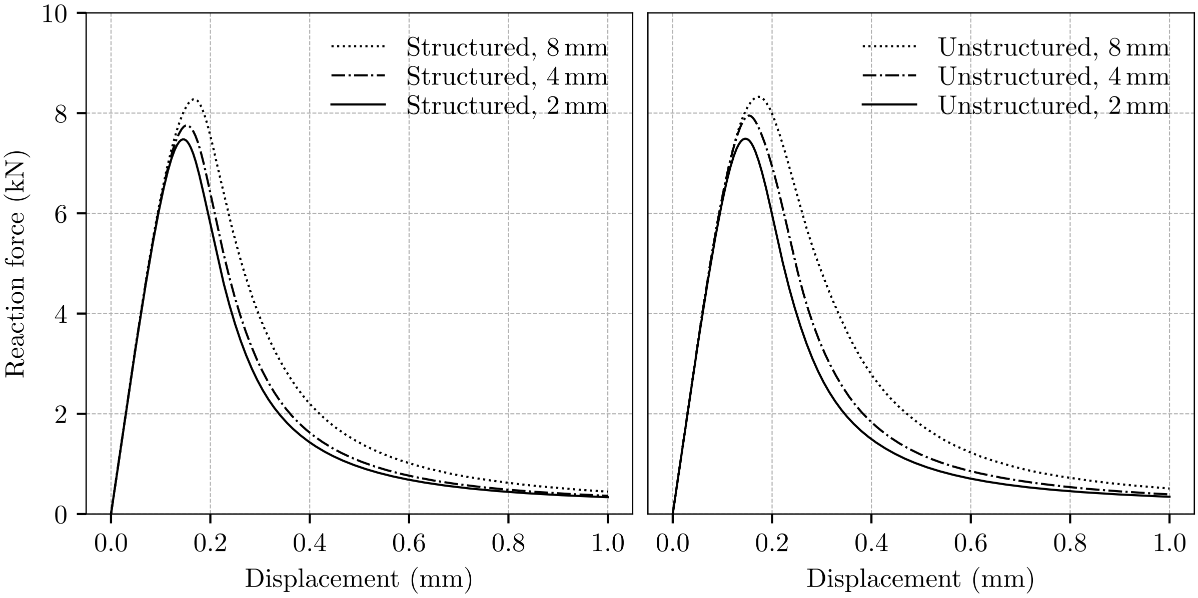

Figure 2. Load–displacement curves employing structured meshes (left) and unstructured meshes (right) with three different refinements each.

Conclusions

The over-nonlocal gradient enhancement presented by Poh and Swaddiwudhipong was employed for regularizing the softening behavior of the SCDP model, and the capabilities of the approach were demonstrated by means of numerical simulations of the experimental tests by Winkler employing finite element meshes of different refinements and orientations. It was demonstrated that the gradient enhancement is able to regularize the softening material behavior very well, leading to mesh insensitive results. While a time-independent softening modulus was assumed in the presented simulations, an extension of the constitutive model to time-dependent softening behavior is currently pending.

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**Proceedings of the 7th GACM Colloquium on Computational Mechanics
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Study of the Regularization Scheme of an Advanced Rock Model

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Micro Abstract
Quasi-brittle materials such as rock exhibit strain softening in the post-peak region leading to failure. The application of an advanced constitutive model for rock predicting irreversible deformation, strain hardening and strain softening is discussed. The aim is to the study the regularization scheme of the rock model based on the over-nonlocal implicit gradient enhancement in numerical simulations of a biaxial compression test.

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Introduction

During the construction of a deep tunnel, reliable estimates of displacements of the tunnel surface and of loads acting on the support structure are of major interest. The complex mechanical behavior of the rock-support system during tunnel advance can be analyzed by means of numerical simulation methods, e.g., the finite element method, where modeling the highly nonlinear mechanical behavior of rock plays a significant role. Rock is classified as a frictional cohesive material characterized by nonlinear stress-strain behavior including softening in the post-peak region leading to failure. In [10], a damage plasticity model for intact rock formulated in the framework of continuum mechanics was proposed. This rock model captures irreversible deformation, strain hardening, stiffness degradation and strain softening. It is common knowledge that strain softening is a structural phenomenon and may yield mesh-dependent results in finite element simulations without regularization of the softening behavior. In order to ensure objective results upon mesh refinement, in the aforementioned rock model the concept of the mesh-adjusted softening modulus from the crack-band theory of Bažant and Oh [1] was adopted. Nevertheless, the consistent determination of the characteristic element length especially for higher order or distorted elements remains questionable and localization zones are biased by the orientation of the finite element mesh, as discussed in [5]. Hence, in the present contribution, the regularization scheme of the aforementioned rock model is modified by means of the adoption of the over-nonlocal implicit gradient-enhancement, as presented in [8]. The capability of predicting shear failure in a mesh-insensitive manner is assessed in finite element simulations of a biaxial compression test.

1 Constitutive Model for Intact Rock and Rock Mass

An isotropic damage plasticity model describing the nonlinear mechanical behavior of intact rock was proposed in [10]. The rock model adopts the framework of plasticity theory in the effective stress space in conjunction with the theory of damage mechanics and is able to represent linear elastic and non-associated plastic material behavior, nonlinear isotropic strain hardening and nonlinear isotropic strain softening, the latter modeled by means of a scalar isotropic damage variable. The nominal stress $\boldsymbol{\sigma}$ is calculated from the effective stress $\bar{\boldsymbol{\sigma}}$ by means of $\boldsymbol{\sigma} = (1-\omega)\,\bar{\boldsymbol{\sigma}}$, where ω denotes the scalar isotropic damage variable ranging from $\omega = 0$

(undamaged material) to $\omega = 1$ (fully damaged material). The effective stress $\bar{\boldsymbol{\sigma}}$ is related to the elastic strain $\boldsymbol{\varepsilon}^{\mathrm{e}} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}}$, determined from the total strain $\boldsymbol{\varepsilon}$ and the plastic strain $\boldsymbol{\varepsilon}^{\mathrm{p}}$, by the elastic stiffness tensor $\mathbb{C}$ leading to the stress-strain law in the total form

$$\boldsymbol{\sigma} = (1-\omega)\,\mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}})\,. \tag{1}$$

The evolution law of ω, defined as

$$\omega(\alpha_{\mathrm{d}}) = 1 - \exp\left(-\alpha_{\mathrm{d}}/\varepsilon_{\mathrm{f}}\right)\,, \tag{2}$$

depends on a strain-like internal softening variable α_{d}, whose evolution is linked to the plastic strain rate, and the softening modulus ε_{f}. To account for the nonlinear mechanical behavior in predominantly hydrostatic compression, the initial yield surface is characterized by a cap. Nonlinear isotropic hardening is described by an evolving yield surface in the effective stress space driven by a strain-like internal hardening variable. The attained yield surface in the ultimate hardening state corresponds to the failure criterion of Hoek and Brown [3] in the smooth version proposed by Menétrey and Willam [6]. Furthermore, the intact rock model can be extended to rock mass using empirical down-scaling factors of the rock mass classification system of Hoek and Brown [4].

2 Regularization by the Over-Nonlocal Gradient Enhancement of the Rock Model

To provide objective results in terms of mesh refinement, mesh orientation and interpolation order of finite elements, the original damage plasticity model for intact rock in [10] is enriched by a nonlocal field of the damage-driving variable $\bar{\alpha_{\mathrm{d}}}$, which is implicitly defined as the solution of a Helmholtz-like partial differential equation with the local field of the strain-like softening variable α_{d} as the source term. Hence, the governing equations in the strong form yield a fully coupled system of second-order partial differential equations

$$\boldsymbol{\nabla} \cdot \boldsymbol{\sigma} + \boldsymbol{f} = \boldsymbol{0}\,, \tag{3}$$

$$\bar{\alpha_{\mathrm{d}}} - l^2\,\boldsymbol{\nabla}^2\bar{\alpha_{\mathrm{d}}} = \alpha_{\mathrm{d}}\,, \tag{4}$$

composed of the equilibrium equation and the Helmholtz-like partial differential equation, respectively. The material parameter l describes a length scale which controls the size of the fracture process zone and acts as a localization limiter. In addition, an averaged strain-like softening variable $\hat{\alpha_{\mathrm{d}}}$ is calculated from the weighted sum of the local and nonlocal softening variable by the weighting factor m and is expressed as

$$\hat{\alpha_{\mathrm{d}}} = m\,\bar{\alpha_{\mathrm{d}}} + (1-m)\alpha_{\mathrm{d}}\,. \tag{5}$$

Values for $m > 1$ correspond to the over-nonlocal formulation, which was originally introduced to provide full regularization, as has been shown in, e.g., [2]. Finally, the averaged strain-like softening variable replaces the local strain-like softening variable in (2) leading to the modified damage law written as

$$\omega(\hat{\alpha_{\mathrm{d}}}) = 1 - \exp(-\hat{\alpha_{\mathrm{d}}}/\varepsilon_{\mathrm{f}})\,. \tag{6}$$

The general mathematical derivation and the detailed implementation framework for the adopted over-nonlocal implicit gradient enhancement can be found in, e.g., [7, 8].

3 Finite Element Simulations of a Biaxial Compression Test

The over-nonlocal implicit gradient enhancement of the damage plasticity model for intact rock is verified in finite element simulations of a biaxial compression test under plane-strain conditions, which was analyzed in [10] on the basis of the mesh-adjusted softening modulus.

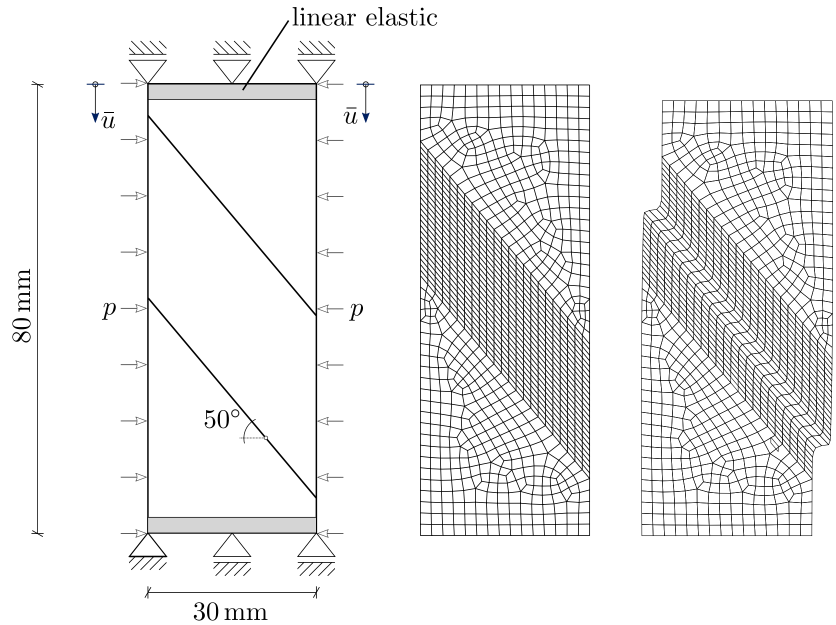

Figure 1. Model of the biaxial compression test (left), finite element discretization with element size of 1 mm in the vicinity of the localization zone in the undeformed (center) and the deformed configuration (right).

In this benchmark test for predicting the formation of shear bands, a rectangular specimen of non-burst-prone Donbass sandstone is considered. The employed material parameters of the rock model are taken from [10]. An associated plastic flow rule is considered to avoid localization due to non-associated plastic flow before entering the softening domain. The additional parameters required for the over-nonlocal implicit gradient-enhanced model, l, m and ε_f, are chosen as 2.5 mm, 1.05 and 0.0067 respectively. The model geometry and boundary conditions are depicted in Figure 1. The top and bottom row of finite elements are modeled by linear elastic material behavior. For the nonlocal damage-driving variable homogeneous Neumann boundary condition ($\boldsymbol{n} \cdot \boldsymbol{\nabla} \bar{\alpha}_\mathrm{d} = 0$) is assumed at the boundary surface, as proposed in [8].

As this test is simulated under plane strain conditions, the element edges in the vicinity of the localization zone are aligned with the expected shear band to avoid spurious locking, as reported in [9]. The specimen is discretized by fully integrated quadrilateral elements using linear shape functions. Three different sizes of finite elements in the localization zone are investigated: 1 mm, 2 mm and 4 mm. To trigger localization, the material parameters of a small zone of elements on the right boundary of the inclined mesh are slightly reduced.

In the first simulation step, hydrostatic initial stresses of $\sigma_{11}^{(0)} = \sigma_{22}^{(0)} = \sigma_{33}^{(0)} = -50\,\mathrm{MPa}$ are applied, which are in equilibrium with the lateral pressure $p = 50\,\mathrm{MPa}$. In the second simulation step, an incrementally increasing uniform vertical displacement at the top boundary of the specimen is prescribed up to $\bar{u} = 2\,\mathrm{mm}$. The deformed configuration of the specimen with finite element size of 1 mm is illustrated in Figure 1, right.

In Figure 2, the vertical reaction force at the top surface divided by the cross section of the specimen is shown for the different meshes in terms of the vertical displacement. Upon mesh refinement the obtained load-displacement curves converge towards a distinct result. For the coarse mesh, the element size of 4 mm is larger than the interaction radius l, which prohibits that the gradient of the nonlocal damage-driving variable can be represented sufficiently.

Conclusions

The damage plasticity model for intact rock and rock mass proposed in [10] was extended by the over-nonlocal implicit gradient formulation presented in [8] to ensure mesh-independent results.

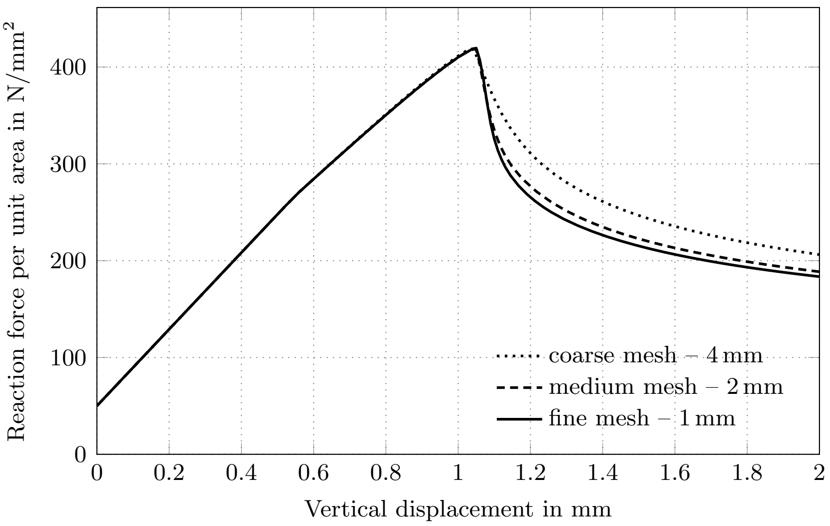

Figure 2. Comparison of the load-displacement curves of the biaxial compression test computed based on different sizes of finite elements.

The capability of the model to predict shear failure in an objective manner was shown based on finite element simulations of a biaxial compression test. As a next step the rock model enhanced by the over-nonlocal implicit gradient formulation will be employed in simulations of deep tunnel advance.

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8 Numerical Implementation

In the following chapter, the implementation of the SCDP model and of the implicit gradient-enhanced formulation into a finite element framework is explained. Details on the implementation of the Meschke model and the Schädlich model are omitted here, since the numerical implementation of the Meschke model is reported in (Meschke, 1996), and due to the similarities of the implementation of those models compared to the SCDP model.

The material models are implemented in the C++ programming language, following an object-oriented design approach. Since C++ provides no linear algebra functionality like vector-matrix operations, the Eigen template library (Guennebaud, Jacob, et al., 2010) for linear algebra is used.

The numerical simulations presented in the publications of this thesis were performed with different software tools:

- For the numerical simulations presented in Publication II and Publication V, the finite element software package Abaqus 6.14-1 (Abaqus, 2015) was employed. Abaqus provides programming interfaces for user defined subroutines, including the UMAT interface for constitutive models expressed in terms of stress-strain relations, and the UEL interface for user defined elements, directly providing the contribution by that element to the system matrix and to the right hand side of the global linear equation system. For the numerical simulations presented in those publications, the UMAT interface was used to implement the SCDP model, the Schädlich model and the Meschke model.
- The implicit gradient-enhanced formulation is implemented in terms of user defined finite elements according to the UEL interface. However, since Abaqus provides only very limited post processing functionality for user defined elements, for the numerical simulations presented in Publication VI and Publication VII a custom finite element framework written in the Python programming language was used. For combining Python and C++ code, the Cython (Behnel et al., 2011) language is used, which allows to connect these two programming languages in an efficient and straightforward manner. For solving the global equation system, the SuperLU solver for sparse, nonsymmetric systems of linear equations (Li, 2005) is employed, which is provided by the SciPy package (Jones et al., 2001).
- For the numerical simulations of the experimental tests at integration point level presented in Publication I, Publication IV and Publication V, a material

driver framework, also denoted as an *integration point solver*, written in Python was used. Similar to the aforementioned Python finite element framework, the Cython language is used for interfacing with the material models written in C++. This material driver allows to prescribe arbitrary stress-strain paths, and it was also employed for calibrating the constitutive models. For parameter identification, a least square optimization method based on the trust region reflective algorithm provided by the optimization toolbox of the SciPy Package was used.

8.1 Implementation of the SCDP model

The SCDP model is implemented in terms of a strain-driven stress update algorithm, computing for a given time interval $[t_\mathrm{n}, t_\mathrm{n+1} = t_\mathrm{n} + \Delta t]$, a given strain increment $\Delta\boldsymbol{\varepsilon}$, and based on the known state of the material at time t_n the updated stress $\boldsymbol{\sigma}(t_\mathrm{n+1})$ and the updated set of internal variables $\boldsymbol{\alpha}(t_\mathrm{n+1})$. The set of internal variables $\boldsymbol{\alpha}$ of the SCDP model comprises the hardening variable α_p, the damage driving variable α_d, the damage parameter ω, and the internal variables $\gamma_1, \dots, \gamma_M$ of the viscoelastic part of the model:

$$\boldsymbol{\alpha} = \{\alpha_\mathrm{p}, \alpha_\mathrm{d}, \omega, \gamma_1, \dots, \gamma_M\}\,. \tag{8.1}$$

Variables $\gamma_1, \dots, \gamma_M$ are related to the numerical approximation scheme for the viscoelastic formulation of the SCDP model, and their purpose is explained in the subsequent sections.

In addition, the algorithmic tangent stiffness $\mathbb{A}_\mathrm{n+1}$ is computed within the stress update algorithm. The algorithmic tangent stiffness is required for computing the finite element stiffness matrix and consequently for the Jacobian of the global system of nonlinear algebraic equations, which is solved iteratively by means of the Newton-Raphson scheme, ensuring a quadratic rate of convergence. In the following, subscript $(\bullet)_\mathrm{n+1}$ denotes the value of a variable at the end of the interval, whereas subscript $(\bullet)_\mathrm{n}$ denotes the respective value at the beginning of the interval.

Summarizing, the input parameters of the stress update algorithm comprise

- the interval $[t_\mathrm{n}, t_\mathrm{n+1}]$ with time increment $\Delta t = t_\mathrm{n+1} - t_\mathrm{n}$,
- the total strain increment $\Delta\boldsymbol{\varepsilon}$,
- the nominal stress $\boldsymbol{\sigma}_\mathrm{n}$ at the beginning of the interval,
- the set of internal state variables $\boldsymbol{\alpha}_\mathrm{n} = \{\alpha_{\mathrm{p}_\mathrm{n}}, \alpha_{\mathrm{d}\mathrm{n}}, \omega_\mathrm{n}, \gamma_{1_\mathrm{n}}, \dots, \gamma_{M_\mathrm{n}}\}$ at the beginning of the interval,

whereas the output parameters of the algorithm comprise

- the updated nominal stress $\boldsymbol{\sigma}_\mathrm{n+1}$,
- the updated set of internal state variables $\boldsymbol{\alpha}_\mathrm{n+1} = \{\alpha_{\mathrm{p}_\mathrm{n+1}}, \alpha_{\mathrm{d}\mathrm{n+1}}, \omega_\mathrm{n+1}, \gamma_{1_\mathrm{n+1}}, \dots, \gamma_{M_\mathrm{n+1}}\}$,
- the algorithmic tangent stiffness $\mathbb{A}_\mathrm{n+1}$.

The SCDP model describes the relation between the nominal stress and the total strain according to Publication I (1) as

$$
\begin{aligned}
\boldsymbol{\sigma} &= (1-\omega)\,\mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{p}} - \boldsymbol{\varepsilon}^{\mathrm{ve}} - \boldsymbol{\varepsilon}^{\mathrm{f}} - \boldsymbol{\varepsilon}^{\mathrm{shr}}) \\
&= (1-\omega)\,\bar{\boldsymbol{\sigma}}\,,
\end{aligned}
\tag{8.2}
$$

and hence, the nominal stress $\boldsymbol{\sigma}_{\mathrm{n}}$ at the beginning of the interval is given as

$$
\boldsymbol{\sigma}_{\mathrm{n}} = (1-\omega_{\mathrm{n}})\,\bar{\boldsymbol{\sigma}}_{\mathrm{n}} \tag{8.3}
$$

and the nominal stress $\boldsymbol{\sigma}_{\mathrm{n+1}}$ at the end of the interval is computed as

$$
\begin{aligned}
\boldsymbol{\sigma}_{\mathrm{n+1}} &= (1-\omega_{\mathrm{n+1}})\,\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \\
&= (1-\omega_{\mathrm{n+1}})\,(\bar{\boldsymbol{\sigma}}_{\mathrm{n}} + \Delta\bar{\boldsymbol{\sigma}})\,.
\end{aligned}
\tag{8.4}
$$

For computing the effective stress increment $\Delta\bar{\boldsymbol{\sigma}}$ in (8.4), departing from the stress-strain relation in total form (8.2), the relation between the effective stress rate $\dot{\bar{\boldsymbol{\sigma}}}$ and the strain rate $\dot{\boldsymbol{\varepsilon}}$ is expressed as

$$
\dot{\bar{\boldsymbol{\sigma}}} = \mathbb{C} : (\dot{\boldsymbol{\varepsilon}} - \dot{\boldsymbol{\varepsilon}}^{\mathrm{p}} - \dot{\boldsymbol{\varepsilon}}^{\mathrm{ve}} - \dot{\boldsymbol{\varepsilon}}^{\mathrm{f}} - \dot{\boldsymbol{\varepsilon}}^{\mathrm{shr}})\,, \tag{8.5}
$$

which is converted to the incremental form by integrating over the interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$:

$$
\Delta\bar{\boldsymbol{\sigma}} = \mathbb{C} : (\Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}^{\mathrm{p}} - \Delta\boldsymbol{\varepsilon}^{\mathrm{ve}} - \Delta\boldsymbol{\varepsilon}^{\mathrm{f}} - \Delta\boldsymbol{\varepsilon}^{\mathrm{shr}})\,. \tag{8.6}
$$

Hence, for computing the nominal stress $\boldsymbol{\sigma}_{\mathrm{n+1}}$ within the stress update algorithm for a given interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$ and a given strain increment $\Delta\boldsymbol{\varepsilon}$, the increment of the effective stress $\bar{\boldsymbol{\sigma}}$ is computed based on the partial strain increments $\Delta\boldsymbol{\varepsilon}^{\mathrm{p}}$, $\Delta\boldsymbol{\varepsilon}^{\mathrm{ve}}$, $\Delta\boldsymbol{\varepsilon}^{\mathrm{f}}$ and $\Delta\boldsymbol{\varepsilon}^{\mathrm{shr}}$, and the updated damage parameter $\omega_{\mathrm{n+1}}$ is calculated. The involved individual steps are presented in the following.

It is useful to illustrate the effective stress-strain relation according to the SCDP model by means of an equivalent uniaxial rheological model, as depicted in Figure 8.1. While this uniaxial rheological model is a simplification of the actual structure of the three-dimensional SCDP model, it allows to present the basic

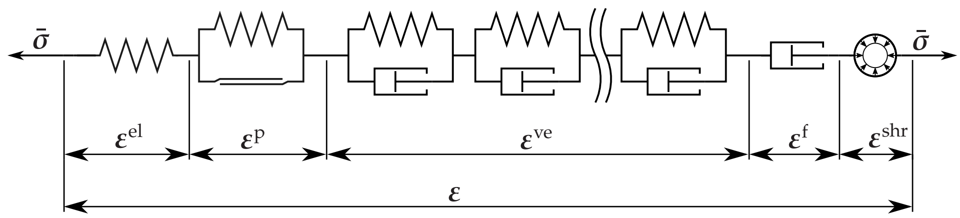

Figure 8.1: Uniaxial rheological model representing the relation between the effective stress $\bar{\boldsymbol{\sigma}}$ and the total strain $\boldsymbol{\varepsilon}$ according to the SCDP model, based on the decomposition of $\boldsymbol{\varepsilon}$ into the elastic strain $\boldsymbol{\varepsilon}^{\mathrm{el}}$, the plastic strain $\boldsymbol{\varepsilon}^{\mathrm{p}}$, the viscoelastic strain $\boldsymbol{\varepsilon}^{\mathrm{ve}}$, the flow strain $\boldsymbol{\varepsilon}^{\mathrm{f}}$ and the shrinkage strain $\boldsymbol{\varepsilon}^{\mathrm{shr}}$.

mechanisms of the model in a clear way. Therein,

- the elastic part is represented by a spring with time-independent stiffness,
- the plastic part by a hardening Saint-Venant unit,
- the viscoelastic part by an aging Kelvin chain,
- the flow part by a dashpot with time-dependent viscosity and
- the shrinkage part by a stress-independent, time-dependent share of the total strain.

The representation of the viscoelastic part in terms of an aging Kelvin chain corresponds to the numerical approximation of the viscoelastic formulation, as explained in the subsequent sections.

8.1.1 Shrinkage Strain Increment

The increment of the shrinkage strain is independent of the stress, and it is directly computed according to the time function of the Bažant-Panula model, defined in Publication I (26),

$$\Delta\boldsymbol{\varepsilon}^{\text{shr}} = \boldsymbol{\varepsilon}^{\text{shr}}(t_{\text{n+1}}) - \boldsymbol{\varepsilon}^{\text{shr}}(t_{\text{n}}) = \boldsymbol{I}\,\varepsilon^{\text{shr}}_{\infty}\,k_{\text{h}}\,(S(t+\Delta t, \tau_{\text{shr}}) - S(t, \tau_{\text{shr}}))\,. \tag{8.7}$$

8.1.2 Viscoelastic Strain Increment

For computing the viscoelastic strain increment, the evolution equation according to the modified solidification theory, defined in Publication I (15b) and extended in Publication V,

$$\dot{\boldsymbol{\varepsilon}}^{\text{ve}}(t) = \frac{F(\bar{\boldsymbol{\sigma}}(t))}{v_{\text{sc}}(t)} \int_{\bar{\boldsymbol{\sigma}}(0)}^{\bar{\boldsymbol{\sigma}}(t)} \frac{\mathrm{d}}{\mathrm{d}t}\,(\Phi(t-t'))\,\mathbb{C}_{\nu}^{-1} : \mathrm{d}\bar{\boldsymbol{\sigma}}(t') \tag{8.8}$$

with

$$\Phi(t-t') = q_2\,\ln(1+(t-t')^{0.1}) \tag{8.9}$$

is integrated over $[t_{\text{n}}, t_{\text{n+1}}]$.

However, since the integral expression in (8.8) is formulated in terms of the complete stress history, its direct numerical implementation for computing $\dot{\boldsymbol{\varepsilon}}^{\text{ve}}(t)$ is not reasonable, as the number of arithmetic operations and the required memory rapidly increase with increasing number of increments. Instead, for an efficient numerical implementation a differential expression is preferable, for which the complete history can be represented by a finite number of internal variables. To this end, the approach proposed by Bažant and Prasannan (1989b) and reformulated by Jirásek and Bažant (2002) is adopted, which is based on two steps, (i) and (ii):

(i) The compliance function $\Phi(t-t')$ in (8.8) is approximated by means of a Dirichlet series, consisting of a finite number of terms:

$$\Phi(t-t') \approx \sum_{i=1}^{M} \frac{1}{D_i}\left(1 - \exp\left(\frac{-(t-t')}{\tau_i}\right)\right). \tag{8.10}$$

Therein, M is the number of Dirichlet terms, D_i are the moduli, and τ_i are the retardation times of the Dirichlet series. The determination of D_i and τ_i is presented in Section 8.1.8.

Accordingly, the viscoelastic strain rate (8.8) is rewritten as

$$\dot{\varepsilon}^{\mathrm{ve}}(t) = \frac{F(\bar{\sigma}(t))}{v_{\mathrm{sc}}(t)} \sum_{i=1}^{M} \gamma_i(t) \tag{8.11}$$

with

$$\gamma_i(t) = \int_{\bar{\sigma}(0)}^{\bar{\sigma}(t)} \frac{1}{\tau_i\, D_i} \exp\left(\frac{-(t-t')}{\tau_i}\right) \mathbb{C}_\nu^{-1} : \mathrm{d}\bar{\sigma}(t')\,, \tag{8.12}$$

in which $\gamma_1(t), ..., \gamma_M(t)$ are the internal history variables of the Dirichlet series.

(ii) The aforementioned preferable differential expression for describing the evolution of the internal variables is obtained in terms of the solution of a first order differential equation. For constituting the first order differential equation, the first derivative of $\gamma_i(t)$ is computed by application of the Leibniz rule as

$$\dot{\gamma}_i(t) = -\frac{1}{\tau_i} \int_{\bar{\sigma}(0)}^{\bar{\sigma}(t)} \frac{1}{\tau_i\, D_i} \exp\left(\frac{-(t-t')}{\tau_i}\right) \mathbb{C}_\nu^{-1} : \mathrm{d}\bar{\sigma}(t') + \frac{1}{\tau_i\, D_i} \mathbb{C}_\nu^{-1} : \dot{\bar{\sigma}}(t)\,. \tag{8.13}$$

Combining (8.12) and (8.13) yields the first order differential equation

$$\frac{1}{\tau_i}\, \gamma_i(t) + \dot{\gamma}_i(t) = \frac{1}{\tau_i\, D_i} \mathbb{C}_\nu^{-1} : \dot{\bar{\sigma}}\,. \tag{8.14}$$

If the right hand side forms a time-independent expression, an analytical solution for (8.14) can be computed by application of the exponential algorithm (Zienkiewicz et al., 1968; Bažant, 1971). This can be exploited by assuming a constant stress rate within the interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$ given as $\dot{\bar{\sigma}}(t) = \Delta\bar{\sigma}/\Delta t$, and consequently, the analytical solution for the interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$ is obtained as

$$\gamma_i(t) = \mathbb{C}_\nu^{-1} : \frac{\Delta\bar{\sigma}}{\Delta t\, D_i} + \boldsymbol{K}_i \exp\left(\frac{-t}{\tau_i}\right). \tag{8.15}$$

Constant $\boldsymbol{K}_i$ is determined from the initial condition $\gamma_i(t_{\mathrm{n}}) = \gamma_{i_{\mathrm{n}}}$ within in the interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$:

$$\boldsymbol{K}_i = \left(\gamma_{i_{\mathrm{n}}} - \mathbb{C}_\nu^{-1} : \frac{\Delta\bar{\sigma}}{\Delta t\, D_i}\right) \exp\left(\frac{t_{\mathrm{n}}}{\tau_i}\right). \tag{8.16}$$

Substitution into (8.15) finally yields

$$\gamma_i(t) = \gamma_{i_{\mathrm{n}}} \exp\left(\frac{-(t-t_{\mathrm{n}})}{\tau_i}\right) + \mathbb{C}_\nu^{-1} : \frac{\Delta\bar{\sigma}}{\Delta t\, D_i} \left(1 - \exp\left(\frac{-(t-t_{\mathrm{n}})}{\tau_i}\right)\right), \tag{8.17}$$

which, in contrast to (8.12), is not formulated in terms of the complete stress history, and allows to describe the evolution of γ_i for each interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$

starting from the known state γ_{i_n}.

The viscoelastic strain increment is computed by integrating (8.11) over the interval $[t_\mathrm{n}, t_\mathrm{n+1}]$:

$$\Delta\boldsymbol{\varepsilon}^\mathrm{ve} = \int_{t_\mathrm{n}}^{t_\mathrm{n+1}} \frac{F(\bar{\boldsymbol{\sigma}}(t))}{v_\mathrm{sc}(t)} \sum_{i=1}^{M} \boldsymbol{\gamma}_i(t)\mathrm{d}t \,. \tag{8.18}$$

For evaluating the integral, the solidified volume is approximated by its value at midstep of the interval as $v_\mathrm{sc}(t_\mathrm{n} + \Delta t/2)$, and the nonlinear creep amplification function $F(\bar{\boldsymbol{\sigma}}(t))$ is approximated in terms of the effective stress at the beginning of the increment as $F(\bar{\boldsymbol{\sigma}}_\mathrm{n})$:

$$\Delta\boldsymbol{\varepsilon}^\mathrm{ve} \approx \frac{F(\bar{\boldsymbol{\sigma}}_\mathrm{n})}{v_\mathrm{sc}(t_\mathrm{n} + \Delta t/2)} \sum_{i=1}^{M} \int_{t_\mathrm{n}}^{t_\mathrm{n+1}} \boldsymbol{\gamma}_i(t)\mathrm{d}t \,. \tag{8.19}$$

Substitution of (8.17) into (8.19) and evaluation of the integral yields

$$\Delta\boldsymbol{\varepsilon}^\mathrm{ve} = \frac{F(\bar{\boldsymbol{\sigma}}_\mathrm{n})}{v_\mathrm{sc}(t_\mathrm{n} + \Delta t/2)} \sum_{i=1}^{M} \left(\mathbb{C}_\nu^{-1} : \frac{\Delta\bar{\boldsymbol{\sigma}}}{D_i} \, (1 - \lambda_i) + \boldsymbol{\gamma}_{i_\mathrm{n}} (1 - \beta_i) \right) , \tag{8.20}$$

with

$$\beta_i = \exp\left(\frac{-\Delta t}{\tau_i}\right) , \tag{8.21}$$

$$\lambda_i = \frac{\tau_i}{\Delta t} \, (1 - \beta_i) \,. \tag{8.22}$$

As apparent from (8.20), the increment of the viscoelastic strain $\Delta\boldsymbol{\varepsilon}^\mathrm{ve}$ is an inhomogeneous linear function of the effective stress increment $\Delta\bar{\boldsymbol{\sigma}}$. If a different rule for approximating the amplification function $F(\bar{\boldsymbol{\sigma}}(t))$ is employed, e.g., the midpoint rule as $F(\bar{\boldsymbol{\sigma}}(t_\mathrm{n}+\Delta t/2))$, (8.20) becomes nonlinear in the stress increment, and hence, an iterative solution procedure for computing the viscoelastic strain increment must be employed.

It can be shown that the approximation in terms of a Dirichlet series is equivalent to the representation of the viscoelastic part of the model by means of an aging Kelvin-chain, i.e., a serial coupling of a finite number of parallel spring-dashpot units with time-dependent stiffness and viscosity, as shown in Figure 8.1. To demonstrate equivalence, the stress-strain relation for a single Kelvin unit in one dimension with time-dependent properties according to Bažant (1979)

$$E(t)\, \dot{\varepsilon}(t) + \dot{\eta}(t)\, \dot{\varepsilon}(t) + \, \eta(t)\ddot{\varepsilon}(t) = \dot{\sigma}(t) \tag{8.23}$$

is considered, with $E(t)$ denoting the stiffness of the spring and $\eta(t)$ denoting the dashpot viscosity.

For the viscoelastic part of the SCDP model, the contribution of a single term of the Dirichlet series (8.11) to the viscoelastic strain rate is given as

$$\dot{\boldsymbol{\varepsilon}}_i^{\mathrm{ve}}(t) = \frac{F(\bar{\boldsymbol{\sigma}}(t))}{v_{\mathrm{sc}}(t)}\,\boldsymbol{\gamma}_i(t), \tag{8.24}$$

from which $\boldsymbol{\gamma}_i(t)$ and $\dot{\boldsymbol{\gamma}}_i(t)$ follow as

$$\boldsymbol{\gamma}_i(t) = \frac{v_{\mathrm{sc}}(t)}{F(\bar{\boldsymbol{\sigma}}(t))}\,\dot{\boldsymbol{\varepsilon}}_i^{\mathrm{ve}}(t), \tag{8.25}$$

$$\dot{\boldsymbol{\gamma}}_i(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{v_{\mathrm{sc}}(t)}{F(\bar{\boldsymbol{\sigma}}(t))}\right)\dot{\boldsymbol{\varepsilon}}_i^{\mathrm{ve}}(t) + \frac{v_{\mathrm{sc}}(t)}{F(\bar{\boldsymbol{\sigma}}(t))}\,\ddot{\boldsymbol{\varepsilon}}_i^{\mathrm{ve}}(t). \tag{8.26}$$

Subsequent substitution into (8.14) yields

$$\frac{1}{\tau_i}\,\frac{v_{\mathrm{sc}}(t)}{F(\bar{\boldsymbol{\sigma}}(t))}\,\dot{\boldsymbol{\varepsilon}}_i^{\mathrm{ve}}(t) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{v_{\mathrm{sc}}(t)}{F(\bar{\boldsymbol{\sigma}}(t))}\right)\dot{\boldsymbol{\varepsilon}}_i^{\mathrm{ve}}(t) + \frac{v_{\mathrm{sc}}(t)}{F(\bar{\boldsymbol{\sigma}}(t))}\,\ddot{\boldsymbol{\varepsilon}}_i^{\mathrm{ve}}(t) = \frac{1}{\tau_i\,D_i}\,\mathbb{C}_\nu^{-1} : \dot{\bar{\boldsymbol{\sigma}}}(t)\,, \tag{8.27}$$

which has a structure similar to (8.23).

8.1.3 Flow Strain Increment

For integrating the flow strain rate of the extended SCDP model defined in Publication V (15) as

$$\dot{\boldsymbol{\varepsilon}}^{\mathrm{f}}(t) = q_4\,\frac{F(\bar{\boldsymbol{\sigma}}(t))}{t^{e_{\mathrm{flow}}}}\,\mathbb{C}_\nu^{-1} : \bar{\boldsymbol{\sigma}}(t) \tag{8.28}$$

over the interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$, the numerical approximation

$$\Delta\boldsymbol{\varepsilon}^{\mathrm{f}} = \int_{t_{\mathrm{n}}}^{t_{\mathrm{n+1}}} q_4\,\frac{F(\bar{\boldsymbol{\sigma}}(t))}{t^{e_{\mathrm{flow}}}}\,\mathbb{C}_\nu^{-1} : \bar{\boldsymbol{\sigma}}(t)\mathrm{d}t \approx q_4\,\frac{F(\bar{\boldsymbol{\sigma}}_{\mathrm{n}})\,\Delta t}{(t_{\mathrm{n}} + \Delta t/2)^{e_{\mathrm{flow}}}}\,\mathbb{C}_\nu^{-1} : (\bar{\boldsymbol{\sigma}}_{\mathrm{n}} + \Delta\bar{\boldsymbol{\sigma}}/2) \tag{8.29}$$

is used. Similar as for the viscoelastic strain increment, the amplification function $F(\bar{\boldsymbol{\sigma}}(t))$ is evaluated at t_{n} as $F(\bar{\boldsymbol{\sigma}}_{\mathrm{n}})$, in order to preserve the inhomogeneous linear relationship between the flow strain increment $\Delta\boldsymbol{\varepsilon}^{\mathrm{f}}$ and the effective stress increment $\Delta\bar{\boldsymbol{\sigma}}$.

8.1.4 Plastic Strain Increment

For computing the increment of the plastic strain, the fully implicit return mapping algorithm is employed, as presented by Simo and Hughes (1998) and in particular by Valentini (2011) for the damage plasticity model for concrete. For the sake of brevity, the following description is limited to the application of the return mapping algorithm for the stress update of the SCDP model and the thereby resulting nonlinear algebraic equation system, while reference is made to (Valentini, 2011) for a detailed description of the computational treatment of the return mapping algorithm. In addition, issues related to the yield function and the plastic potential of the SCDP model, as well as a substepping scheme are discussed briefly in the following.

Following the return mapping algorithm, a trial effective stress $\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}$ assuming $\Delta\boldsymbol{\varepsilon}^{\mathrm{p}} = \mathbf{0}$, denoted as the elastic predictor, is computed to determine if the yield condition is violated, i.e., $f(\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}, \alpha_{\mathrm{p_n}}, t_{\mathrm{n+1}}) > 0$.

If $f(\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}, \alpha_{\mathrm{p_n}}, t_{\mathrm{n+1}}) \leq 0$, no plastic flow occurs, i.e., $\Delta\boldsymbol{\varepsilon}^{\mathrm{p}} = \mathbf{0}$, and accordingly the strain-like internal variables α_{p} and α_{d} do not change within the current interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$, i.e., $\alpha_{\mathrm{p_{n+1}}} = \alpha_{\mathrm{p_n}}$, $\alpha_{\mathrm{dn+1}} = \alpha_{\mathrm{dn}}$ and consequently, $\omega_{\mathrm{n+1}} = \omega_{\mathrm{n}}$. Otherwise, i.e., if $f(\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}, \alpha_{\mathrm{p_n}}, t_{\mathrm{n+1}}) > 0$, the return mapping algorithm is performed.

For computing the trial effective stress, departing from (8.6) the relation between the effective stress $\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}$ and the total strain increment $\Delta\boldsymbol{\varepsilon}$ is given as

$$\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} = \bar{\boldsymbol{\sigma}}_{\mathrm{n}} + \mathbb{C} : (\Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}^{\mathrm{shr}} - \Delta\boldsymbol{\varepsilon}^{\mathrm{ve}}(\Delta\bar{\boldsymbol{\sigma}}) - \Delta\boldsymbol{\varepsilon}^{\mathrm{f}}(\Delta\bar{\boldsymbol{\sigma}}) - \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}(\Delta\bar{\boldsymbol{\sigma}}))\,, \tag{8.30}$$

and accordingly, the effective trial stress is computed as

$$\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}} = \bar{\boldsymbol{\sigma}}_{\mathrm{n}} + \mathbb{C} : (\Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}^{\mathrm{shr}} - \Delta\boldsymbol{\varepsilon}^{\mathrm{ve}}(\Delta\bar{\boldsymbol{\sigma}}) - \Delta\boldsymbol{\varepsilon}^{\mathrm{f}}(\Delta\bar{\boldsymbol{\sigma}}))\,. \tag{8.31}$$

As indicated in (8.30) and (8.31), both the viscoelastic strain and the flow strain increment are dependent on the effective stress increment $\Delta\bar{\boldsymbol{\sigma}}$, which is not known a priori in case of plastic flow, and consequently, the structure of (8.30) differs from the standard problem of small strain plasticity.

However, the inhomogeneous linear relation between the increments of the viscoelastic strain and the flow strain on the one hand, and the increment of the effective stress on the other hand can be exploited by means of an effective stiffness tensor, leading to a computationally efficient algorithm. This is accomplished in two steps: (i) The effective stiffness tensor is derived, and (ii) (8.30) and (8.31) are reformulated in order to obtain the standard problem of small strain plasticity, expressed in terms of the effective stiffness tensor.

(i) For deducing the effective stiffness tensor, the effective stiffness $\tilde{E}$ for the interval $[t_{\mathrm{n}}, t_{\mathrm{n+1}}]$ is computed according to the approach presented by Jirásek and Bažant (2002): As apparent from (8.20) and (8.29), both the viscoelastic strain increment and the flow strain increment are composed of terms which are independent of the stress increment, and terms which are linear in the effective stress increment. Departing from this observation, a decomposition of the total strain increment is introduced as

$$\Delta\boldsymbol{\varepsilon} = \Delta\boldsymbol{\varepsilon}'' + \Delta\hat{\boldsymbol{\varepsilon}} + \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}, \tag{8.32}$$

with

$$\Delta\boldsymbol{\varepsilon}'' = \Delta\boldsymbol{\varepsilon}^{\mathrm{shr}} + \frac{F(\bar{\boldsymbol{\sigma}}_{\mathrm{n}})}{v_{\mathrm{sc}}(t_{\mathrm{n}} + \Delta t/2)} \sum_{i=1}^{M} \gamma_{i_{\mathrm{n}}} (1 - \beta_i) \; + \; q_4 \, \frac{F(\bar{\boldsymbol{\sigma}}_{\mathrm{n}})\, \Delta t}{(t_{\mathrm{n}} + \Delta t/2)^{e_{\mathrm{flow}}}} \, \mathbb{C}_{\nu}^{-1} : \bar{\boldsymbol{\sigma}}_{\mathrm{n}} \tag{8.33}$$

and

$$\Delta\hat{\boldsymbol{\varepsilon}} = \left(q_1 + \frac{F(\bar{\boldsymbol{\sigma}}_\mathrm{n})}{v_\mathrm{sc}(t_\mathrm{n} + \Delta t/2)} \sum_{i=1}^{M} \frac{1}{D_i} \, (1 - \lambda_i) + \frac{1}{2} q_4 \, \frac{F(\bar{\boldsymbol{\sigma}}_\mathrm{n})\,\Delta t}{(t_\mathrm{n} + \Delta t/2)^{e_\mathrm{flow}}} \right) \mathbb{C}_\nu^{-1} : \Delta\bar{\boldsymbol{\sigma}} \,, \tag{8.34}$$

based on a decomposition of the viscoelastic strain increment and the flow strain increment, and based on the definition of $\mathbb{C}$ presented in Publication I:

$$\mathbb{C}^{-1} = q_1 \, \mathbb{C}_\nu^{-1}. \tag{8.35}$$

The expression in parentheses in (8.34),

$$\frac{1}{\tilde{E}} = q_1 + \frac{F(\bar{\boldsymbol{\sigma}}_\mathrm{n})}{v_\mathrm{sc}(t_\mathrm{n} + \Delta t/2)} \sum_{i=1}^{M} \frac{1}{D_i} \, (1 - \lambda_i) + \frac{1}{2} \, q_4 \, \frac{F(\bar{\boldsymbol{\sigma}}_\mathrm{n})\,\Delta t}{(t_\mathrm{n} + \Delta t/2)^{e_\mathrm{flow}}} \,, \tag{8.36}$$

defines the effective stiffness $\tilde{E}$ for the interval $[t_\mathrm{n}, t_\mathrm{n+1}]$, from which the effective stiffness tensor $\tilde{\mathbb{C}}$ is computed as

$$\tilde{\mathbb{C}} = \mathbb{C}_\nu \, \tilde{E} \,. \tag{8.37}$$

(ii) Expressions (8.30) and (8.31) are reformulated in order to obtain a structure amenable to the standard return mapping algorithm. Departing from the stress-strain relation in (8.34), and substitution of the effective stiffness tensor (8.37) into (8.34) and considering the decomposition of the total strain (8.32) yields after rearrangement

$$\Delta\bar{\boldsymbol{\sigma}} = \tilde{\mathbb{C}} : \Delta\hat{\boldsymbol{\varepsilon}} = \tilde{\mathbb{C}} : (\Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}'' - \Delta\boldsymbol{\varepsilon}^\mathrm{p}) \tag{8.38}$$

and finally

$$\begin{aligned} \bar{\boldsymbol{\sigma}}_\mathrm{n+1} &= \bar{\boldsymbol{\sigma}}_\mathrm{n} + \tilde{\mathbb{C}} : (\Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}'' - \Delta\boldsymbol{\varepsilon}^\mathrm{p}) \\ &= \bar{\boldsymbol{\sigma}}_\mathrm{n} + \tilde{\mathbb{C}} : (\Delta\tilde{\boldsymbol{\varepsilon}} - \Delta\boldsymbol{\varepsilon}^\mathrm{p}) \end{aligned} \tag{8.39}$$

with

$$\Delta\tilde{\boldsymbol{\varepsilon}} = \Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}'' \,, \tag{8.40}$$

for which the identical structure to the standard problem of small strain plasticity is immediately apparent. Accordingly, the trial effective stress is computed as

$$\bar{\boldsymbol{\sigma}}^\mathrm{tr}_\mathrm{n+1} = \bar{\boldsymbol{\sigma}}_\mathrm{n} + \tilde{\mathbb{C}} : \Delta\tilde{\boldsymbol{\varepsilon}}, \tag{8.41}$$

and the return mapping algorithm is performed based on the effective stiffness tensor $\tilde{\mathbb{C}}$.

Further discussion is required on the yield function and the plastic potential function: For both the yield function and the plastic potential function the partial derivatives with respect to the effective stress are not defined uniquely for stress states located on the hydrostatic axis. For the yield surface, this issue is manifested by two vertices at its intersection with the hydrostatic axis. Consequently, the standard return mapping algorithm fails for trial stress states located in those domains of the effective stress space, for which the back projected stress state is located at one of those vertices. As a remedy, the approach described in (Hofstetter and Taylor, 1991; Valentini, 2011) is followed: A second cut-off yield function is employed, defining two plane yield surfaces orthogonal to the hydrostatic axis, and cutting the main yield function slightly beneath the vertices.

The cut-off yield function is given as

$$\begin{aligned} f_{\text{cut-off}}(\bar{\sigma}_{\mathrm{m}}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) = &\left(\frac{1 - q_{\mathrm{h}}(\alpha_{\mathrm{p}})}{f_{\mathrm{cu}}^2(t)}\right)^2 (\bar{\sigma}_{\mathrm{m}} + \operatorname{sgn}(\bar{\sigma}_{\mathrm{m}})\, \sigma_{\mathrm{m}}^{\text{shift}})^4 \\ &+ \frac{m_0\, q_{\mathrm{h}}^2(\alpha_{\mathrm{p}})}{f_{\mathrm{cu}}(t)} (\bar{\sigma}_{\mathrm{m}} + \operatorname{sgn}(\bar{\sigma}_{\mathrm{m}})\, \bar{\sigma}_{\mathrm{m}}^{\text{shift}}) - q_{\mathrm{h}}^2(\alpha_{\mathrm{p}}) \end{aligned} \tag{8.42}$$

with $q_{\mathrm{h}}(\alpha_{\mathrm{p}})$ denoting the stress-like internal hardening variable given as

$$q_{\mathrm{h}}(\alpha_{\mathrm{p}}) = \begin{cases} f_{\mathrm{cy}}/f_{\mathrm{cu}} + (1 - f_{\mathrm{cy}}/f_{\mathrm{cu}})\, \alpha_{\mathrm{p}}\, (\alpha_{\mathrm{p}}^2 - 3\,\alpha_{\mathrm{p}} + 3) & \text{if } \alpha_{\mathrm{p}} < 1\,, \\ 1 & \text{otherwise,} \end{cases} \tag{8.43}$$

and with $\bar{\sigma}_{\mathrm{m}}^{\text{shift}}$ denoting a numerical parameter for shifting the cut-off yield function a small amount beneath the vertices of the main yield surface, defined as $\bar{\sigma}_{\mathrm{m}}^{\text{shift}} = 10^{-4}$ MPa. For the cut-off yield surface, an associated plastic potential

$$g_{\text{cut-off}}(\bar{\sigma}_{\mathrm{m}}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) = f_{\text{cut-off}}(\bar{\sigma}_{\mathrm{m}}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) \tag{8.44}$$

is employed.

Consequently, the return mapping algorithm is formulated within the framework of multi-surface plasticity, based on Koiter's generalized flow rule (Koiter, 1953, 1964). The evolution equations for the plastic strain and the strain-like internal hardening variable α_{p} are given according to Publication I (5) and Publication I (6), and extended to account for the cut-off yield surface as

$$\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}} = \sum_{j \in I_{\text{act}}} \dot{\lambda}_j \frac{\partial g_j(\bar{\boldsymbol{\sigma}}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t)}{\partial \bar{\boldsymbol{\sigma}}}\,, \tag{8.45}$$

$$\dot{\alpha}_{\mathrm{p}} = \|\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}}\| \frac{1 + 3\, \frac{\bar{\rho}^2}{\bar{\rho}^2 + \vartheta_{\mathrm{h}}} \cos^2(1.5\,\theta)}{x_{\mathrm{h}}(\bar{\sigma}_{\mathrm{m}}, t)}\,, \tag{8.46}$$

with

$$I_{\text{act}} = \{ j \in \{\mathrm{p}, \text{cut-off}\} \,|\, f_j(\bar{\boldsymbol{\sigma}}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) = 0 \} \tag{8.47}$$

denoting the active set of yield functions, complemented by the Kuhn-Tucker conditions

$$\dot{\lambda}_j \geq 0, \quad f_j(\bar{\boldsymbol{\sigma}}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) \leq 0, \quad \dot{\lambda}_j\, f_j(\bar{\boldsymbol{\sigma}}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) = 0 \tag{8.48}$$

and the consistency condition

$$\dot{\lambda}_j\, \dot{f}_j(\bar{\boldsymbol{\sigma}}, q_{\mathrm{h}}(\alpha_{\mathrm{p}}), t) = 0\,. \tag{8.49}$$

Fully implicit integration of the rate equations yields the incremental form of (8.45) and (8.46) as

$$\Delta\boldsymbol{\varepsilon}^{\mathrm{p}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \alpha_{\mathrm{p_{n+1}}}, \Delta\lambda_j, t_{\mathrm{n+1}}) = \sum_{j\in I_{\mathrm{act}}} \Delta\lambda_i \, \frac{\partial g_i(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, q_{\mathrm{h}}(\alpha_{\mathrm{p_{n+1}}}), t_{\mathrm{n+1}})}{\partial \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}}, \tag{8.50}$$

$$\Delta\alpha_{\mathrm{p}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \alpha_{\mathrm{p_{n+1}}}, \Delta\lambda_j, t_{\mathrm{n+1}}) = \left\|\Delta\boldsymbol{\varepsilon}^{\mathrm{p}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \alpha_{\mathrm{p_{n+1}}}, \Delta\lambda_j)\right\| \, \frac{1 + 3\,\frac{\bar{\rho}^2_{\mathrm{n+1}}}{\bar{\rho}^2_{\mathrm{n+1}} + \vartheta_{\mathrm{h}}}\, \cos^2(1.5\,\theta)}{x_{\mathrm{h}}((\bar{\boldsymbol{\sigma}}_{\mathrm{m}})_{\mathrm{n+1}}, t_{\mathrm{n+1}})}, \tag{8.51}$$

expressed as nonlinear functions of the unknowns $\{\boldsymbol{\sigma}_{\mathrm{n+1}}, \alpha_{\mathrm{p_{n+1}}}, \Delta\lambda_j\}$.

Finally, based on those incremental evolution equations, the system of nonlinear algebraic equations of the return mapping algorithm is formulated as:

$$\begin{Bmatrix} \bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}} \\ \alpha_{\mathrm{p_n}} \\ \mathbf{0} \end{Bmatrix} = \begin{Bmatrix} \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} + \tilde{\mathbb{C}} : \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \alpha_{\mathrm{p_{n+1}}}, \Delta\lambda_j, t_{\mathrm{n+1}}) \\ \alpha_{\mathrm{p_{n+1}}} - \Delta\alpha_{\mathrm{p}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \alpha_{\mathrm{p_{n+1}}}, \Delta\lambda_j, t_{\mathrm{n+1}}) \\ \boldsymbol{f}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, q_{\mathrm{h}}(\alpha_{\mathrm{p_{n+1}}}), t_{\mathrm{n+1}}) \end{Bmatrix}, \tag{8.52}$$

in which $\boldsymbol{f}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, q_{\mathrm{h}}(\alpha_{\mathrm{p_{n+1}}}), t_{\mathrm{n+1}})$ is the vector of all active yield functions:

$$\boldsymbol{f}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, q_{\mathrm{h}}(\alpha_{\mathrm{p_{n+1}}}), t_{\mathrm{n+1}}) = \begin{cases} \left\{ f_{\mathrm{p}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, q_{\mathrm{h}}(\alpha_{\mathrm{p_{n+1}}}), t_{\mathrm{n+1}}) \right\} & \text{if } I_{\mathrm{act}} = \{f_{\mathrm{p}}\}, \\ \left\{ f_{\text{cut-off}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, q_{\mathrm{h}}(\alpha_{\mathrm{p_{n+1}}}), t_{\mathrm{n+1}}) \right\} & \text{if } I_{\mathrm{act}} = \{f_{\text{cut-off}}\}, \\ \begin{Bmatrix} f_{\mathrm{p}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, q_{\mathrm{h}}(\alpha_{\mathrm{p_{n+1}}}), t_{\mathrm{n+1}}) \\ f_{\text{cut-off}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, q_{\mathrm{h}}(\alpha_{\mathrm{p_{n+1}}}), t_{\mathrm{n+1}}) \end{Bmatrix} & \text{if } I_{\mathrm{act}} = \{f_{\mathrm{p}}, f_{\text{cut-off}}\}. \end{cases} \tag{8.53}$$

Within the stress update algorithm, the nonlinear system (8.52) is solved using the Newton-Raphson method.

For improving the robustness of the return mapping algorithm, the substepping scheme presented by Pérez-Foguet et al. (2001) for plasticity models and extended by Valentini (2011) to the damage plasticity model for concrete is employed: If the return mapping procedure fails, i.e., if the Newton-Raphson procedure does not converge, the trial strain increment $\Delta\tilde{\boldsymbol{\varepsilon}}$ is additively split into N subincrements $\Delta\tilde{\boldsymbol{\varepsilon}}_k$, defined by subincrement ratios α_k given as

$$\Delta\tilde{\boldsymbol{\varepsilon}} = \sum_{k=1}^{N} \Delta\tilde{\boldsymbol{\varepsilon}}_k = \sum_{k=1}^{N} \alpha_k \, \Delta\tilde{\boldsymbol{\varepsilon}}\,. \tag{8.54}$$

Following the substepping scheme, the return mapping algorithm is applied in a sequential manner for N subincrements. The stress at the end of each subincrement is given as

$$(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}})_k = (\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}})_{k-1} + \tilde{\mathbb{C}} : (\Delta\tilde{\boldsymbol{\varepsilon}}_k - \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}_k), \tag{8.55}$$

with $(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}})_0 = \bar{\boldsymbol{\sigma}}_{\mathrm{n}}$, and with the respective trial effective stress computed as

$$(\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}})_k = (\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}})_{k-1} + \tilde{\mathbb{C}} : \Delta\tilde{\boldsymbol{\varepsilon}}_k. \tag{8.56}$$

Consequently, the total plastic strain increment $\Delta\boldsymbol{\varepsilon}^{\mathrm{p}}$ is computed as the sum of the plastic subincrements as

$$\Delta\boldsymbol{\varepsilon}^{\mathrm{p}} = \sum_{k=1}^{N} \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}_k. \tag{8.57}$$

The total number of subincrements N is not known a priori, but follows from the adaptive scheme which becomes active if the return mapping fails. For the adaptive scheme, a cutback ratio of 10 is employed: If the return mapping algorithm fails, the current subincrement size is divided by a factor of 10, and the return mapping procedure is repeated with the reduced subincrement. As soon as a number of 5 successful subincrements are computed, the subincrement size is increased again by a factor of 1.5 in order to minimize the overall number of involved computational steps. If a specified minimum size of the subincrement is attained, i.e., if $\alpha_k \leq 1 \times 10^{-3}$, the complete stress update procedure for the present strain increment $\Delta\boldsymbol{\varepsilon}$ is abandoned, and the global displacement increment size must be decreased.

Basically, the adaptive substepping procedure can be combined with an integration error estimator as proposed by Fellin et al. (2009). However, for the SCDP model it was found that the accuracy of the standard fully implicit scheme is completely acceptable, provided that the global increment size is moderate, and it is concluded that the additional computational effort of the adaptive scheme is rather disadvantageous.

8.1.5 Damage Increment

With the increment of the plastic strain at hand, departing from the rate form evolution law for the damage-driving internal variable α_{d}, defined in Publication I (10), the increment $\Delta\alpha_{\mathrm{d}}$ is computed as

$$\Delta\alpha_{\mathrm{d}} = \begin{cases} 0 & \text{if } \alpha_{\mathrm{p}_{\mathrm{n+1}}} < 1, \\ \Delta\varepsilon^{\mathrm{p}}_{\mathrm{V}} / x_{\mathrm{s}}(\Delta\boldsymbol{\varepsilon}^{\mathrm{p}}) & \text{otherwise,} \end{cases} \tag{8.58}$$

and consequently, the damage parameter ω is updated as

$$\omega_{\mathrm{n+1}} = 1 - \exp\left(-\frac{\alpha_{\mathrm{d}_{\mathrm{n}}} + \Delta\alpha_{\mathrm{d}}}{\epsilon_{\mathrm{f}}}\right). \tag{8.59}$$

8.1.6 Algorithmic Tangent Stiffness

For preserving the quadratic rate of convergence within the global Newton-Raphson scheme, the finite element stiffness matrices are computed by means of the algorithmic tangent stiffness matrices. The algorithmic tangent stiffness describes the change of $\boldsymbol{\sigma}_{\mathrm{n+1}}$ with respect to the strain increment $\Delta\boldsymbol{\varepsilon}$, and it is computed by consistent linearization of the stress update algorithm:

$$\mathbb{A}_{\mathrm{n+1}} = \frac{\partial \boldsymbol{\sigma}_{\mathrm{n+1}}}{\partial \Delta\boldsymbol{\varepsilon}} = \frac{\partial \Delta\boldsymbol{\sigma}}{\partial \Delta\boldsymbol{\varepsilon}} \,. \tag{8.60}$$

Taking the stress-strain relation of the stress update algorithm (8.4), and deriving the latter with respect to $\Delta\boldsymbol{\varepsilon}$ yields

$$\frac{\partial \boldsymbol{\sigma}_{\mathrm{n+1}}}{\partial \Delta\boldsymbol{\varepsilon}} = \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \otimes \frac{\partial \omega_{\mathrm{n+1}}}{\partial \Delta\boldsymbol{\varepsilon}} + (1-\omega_{\mathrm{n+1}}) \frac{\partial \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}}{\partial \Delta\boldsymbol{\varepsilon}} \,. \tag{8.61}$$

Therein, the first term on the right hand side represents the change of the nominal stress with respect to the strain increment due to an increase of the damage, whereas the second term represents the change of the nominal stress due to plastic material behavior. Departing from (8.39), the latter is expressed as

$$\frac{\partial \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}}{\partial \Delta\boldsymbol{\varepsilon}} = \frac{\partial \Delta\bar{\boldsymbol{\sigma}}}{\partial \Delta\boldsymbol{\varepsilon}} = \tilde{\mathbb{C}} : \frac{\partial}{\partial \Delta\boldsymbol{\varepsilon}} \left(\Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}'' - \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}\right) = \tilde{\mathbb{C}} : \left(\mathbb{I} - \frac{\partial \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}}{\partial \Delta\boldsymbol{\varepsilon}}\right), \tag{8.62}$$

with $\mathbb{I}$ denoting the fourth order unit tensor. This expression represents the standard algorithmic elasto-plastic tangent stiffness

$$\tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}} = \tilde{\mathbb{C}} : \left(\mathbb{I} - \frac{\partial \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}}{\partial \Delta\boldsymbol{\varepsilon}}\right) \tag{8.63}$$

in the context of the implicit return mapping algorithm and in terms of the effective stiffness tensor $\tilde{\mathbb{C}}$. The computation of $\tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}}$ is performed directly within the return mapping algorithm, as described by Valentini (2011). The same holds for the presented substepping scheme, for which the algorithmic tangent can be computed efficiently exploiting a recursive scheme, as explained by Pérez-Foguet et al. (2001).

Expanding the first term on the right hand side in (8.61) by application of the chain rule and substitution of (8.63) yields

$$\begin{aligned} \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \otimes \frac{\partial \omega_{\mathrm{n+1}}}{\partial \Delta\boldsymbol{\varepsilon}} &= \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \otimes \frac{\partial \omega_{\mathrm{n+1}}}{\partial \alpha_{\mathrm{d_{n+1}}}} \frac{\partial \alpha_{\mathrm{d_{n+1}}}}{\partial \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}} : \frac{\partial \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}}{\partial \Delta\boldsymbol{\varepsilon}} \\ &= -\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \otimes \frac{\partial \omega_{\mathrm{n+1}}}{\partial \alpha_{\mathrm{d_{n+1}}}} \frac{\partial \alpha_{\mathrm{d_{n+1}}}}{\partial \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}} : (\tilde{\mathbb{C}}^{-1} : \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}} - \mathbb{I}) \,. \end{aligned} \tag{8.64}$$

Expression $\frac{\partial \alpha_{\mathrm{d_{n+1}}}}{\partial \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}}$ is obtained by consistent linearization of (8.58), and vanishes for $\alpha_{\mathrm{p_{n+1}}} < 1$. In (Valentini, 2011) a computationally efficient method for computing this linearization within the substepping scheme is presented.

Consequently, $\mathbb{A}_{n+1}$ can be rewritten as

$$\mathbb{A}_{n+1} = \bar{\boldsymbol{\sigma}}_{n+1} \otimes \frac{\partial \omega_{n+1}}{\partial \alpha_{d_{n+1}}} \frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \boldsymbol{\varepsilon}^{p}} : (\tilde{\mathbb{C}}^{-1} : \tilde{\mathbb{A}}^{ep}_{n+1} - \mathbb{I}) + (1 - \omega_{n+1})\, \tilde{\mathbb{A}}^{ep}_{n+1} \,. \tag{8.65}$$

In case of zero plastic flow, (8.65) simplifies to

$$\mathbb{A}_{n+1} = (1 - \omega_{n+1})\, \tilde{\mathbb{C}} \,, \tag{8.66}$$

and in case of plastic flow along with $\alpha_{p_{n+1}} < 1$, i.e., no softening material behavior, to

$$\mathbb{A}_{n+1} = \tilde{\mathbb{A}}^{ep}_{n+1} \,, \tag{8.67}$$

which resembles the standard elasto-plastic algorithmic stiffness in terms of the effective stiffness tensor $\tilde{\mathbb{C}}$.

8.1.7 Summary of the Stress Update Algorithm

After computing the nominal stress $\boldsymbol{\sigma}_{n+1}$ and the algorithmic tangent stiffness $\mathbb{A}_{n+1}$, the internal variables $\gamma_{1_n}, \dots, \gamma_{M_n}$ of the Dirichlet series are updated according to (8.17), and the updated stress $\boldsymbol{\sigma}_{n+1}$, the algorithmic stiffness $\mathbb{A}_{n+1}$, and the updated internal variables $\boldsymbol{\alpha}_{n+1}$ are passed back to the finite element framework. The presented stress update algorithm is summarized in Algorithm 1. For the sake of simplicity, the substepping scheme is omitted.

8.1.8 Determination of the Parameters for the Dirichlet Series

The compliance function $\Phi(t - t')$ of the viscoelastic part of the SCDP model is given as

$$\Phi(t - t') = q_2 \, \ln(1 + (t - t')^{0.1}) \,. \tag{8.68}$$

The approximation of $\Phi(t - t')$ in (8.10) by means of a Dirichlet series composed of M terms allows to describe the viscoelastic strain rate in terms of first order ordinary differential equations, corresponding to an aging Kelvin chain. To this end, the parameters of the Dirichlet series, i.e., the moduli D_i and the retardation times τ_i, are chosen in order to approximate the continuous retardation spectrum of the exact compliance function (8.68) by means of the discrete retardation spectrum of the Dirichlet series.

The parameters of the Dirichlet series are determined according to the approach proposed by Bažant and Xi (1995) and extended by Jirásek and Havlásek (2014). The basic idea behind that approach is that any compliance function can be computed from its continuous retardation spectrum $L(\tau)$ by means of transformation

$$\Phi(t) = \int_{\tau=0}^{\infty} L(\tau) \, \left(1 - \exp\left(-t/\tau\right)\right) \mathrm{d}(\ln \tau), \tag{8.69}$$

and that the continuous retardation spectrum $L(\tau)$ can be approximated by exploiting the Post-Widder inversion formula for Laplace transforms (Widder,

Algorithm 1 Stress update algorithm for the SCDP model

function Stress Update $(\Delta\boldsymbol{\varepsilon}, t_{\mathrm{n}}, t_{\mathrm{n+1}}, \boldsymbol{\sigma}_{\mathrm{n}}, \alpha_{\mathrm{p}_{\mathrm{n}}}, \alpha_{\mathrm{d}_{\mathrm{n}}}, \omega_{\mathrm{n}}, \gamma_{1_{\mathrm{n}}}, \ldots, \gamma_{M_{\mathrm{n}}})$
- $\bar{\boldsymbol{\sigma}}_{\mathrm{n}} \leftarrow \frac{1}{(1-\omega_{\mathrm{n}})}\boldsymbol{\sigma}_{\mathrm{n}}$ ▹ Compute effective stress at t_{n}
- $\Delta\boldsymbol{\varepsilon}^{\mathrm{shr}} \leftarrow \boldsymbol{\varepsilon}^{\mathrm{shr}}(t_{\mathrm{n}}, t_{\mathrm{n+1}})$ ▹ Shrinkage strain increment
- $\Delta\boldsymbol{\varepsilon}'' \leftarrow \Delta\boldsymbol{\varepsilon}''(\bar{\boldsymbol{\sigma}}_{\mathrm{n}}, \gamma_{1_{\mathrm{n}}}, \ldots, \gamma_{M_{\mathrm{n}}}, t_{\mathrm{n}}, t_{\mathrm{n+1}}, \Delta\boldsymbol{\varepsilon}^{\mathrm{shr}})$ ▹ Initial strain increment
- $\bar{E} \leftarrow \bar{E}(\bar{\boldsymbol{\sigma}}_{\mathrm{n}}, \gamma_{1_{\mathrm{n}}}, \ldots, \gamma_{M_{\mathrm{n}}}, t_{\mathrm{n}}, t_{\mathrm{n+1}},)$ ▹ Compute effective stiffness
- $\tilde{\mathbb{C}} \leftarrow \mathbb{C}_{\nu}\,\bar{E}$
- $\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}} \leftarrow \bar{\boldsymbol{\sigma}}_{\mathrm{n}} + \tilde{\mathbb{C}} : (\Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}'')$ ▹ Compute trial effective stress
- **if** $f(\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}, \alpha_{\mathrm{p}_{\mathrm{n}}}, t_{\mathrm{n+1}}) < 0$ **then** ▹ Yield condition not violated
  - $\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \leftarrow \bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}$
  - $\alpha_{\mathrm{p}_{\mathrm{n+1}}} \leftarrow \alpha_{\mathrm{p}_{\mathrm{n}}}$
  - $\alpha_{\mathrm{d}_{\mathrm{n+1}}} \leftarrow \alpha_{\mathrm{d}_{\mathrm{n}}}$
  - $\omega_{\mathrm{n+1}} \leftarrow \omega_{\mathrm{n}}$
  - $\mathbb{A}_{\mathrm{n+1}} \leftarrow (1 - \omega_{\mathrm{n+1}})\,\tilde{\mathbb{C}}$
- **else** ▹ Yield condition violated
  - $\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}, \alpha_{\mathrm{p}_{\mathrm{n+1}}}, \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}} \leftarrow$ **Return Mapping** $(\boldsymbol{\sigma}_{\mathrm{n}}, \bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}, \tilde{\mathbb{C}}, \alpha_{\mathrm{p}_{\mathrm{n}}}, t_{\mathrm{n+1}})$
  - $\alpha_{\mathrm{d}_{\mathrm{n+1}}} \leftarrow \alpha_{\mathrm{d}}(\alpha_{\mathrm{d}_{\mathrm{n}}}, \Delta\boldsymbol{\varepsilon}^{\mathrm{p}})$ ▹ Update damage driving variable
  - $\omega_{\mathrm{n+1}} \leftarrow \omega(\alpha_{\mathrm{d}_{\mathrm{n+1}}})$ ▹ Update damage parameter
  - $\mathbb{A}_{\mathrm{n+1}} \leftarrow \mathbb{A}_{\mathrm{n+1}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}, \alpha_{\mathrm{d}_{\mathrm{n+1}}}, \omega_{\mathrm{n+1}}, \tilde{\mathbb{C}}, \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}})$ ▹ Algorithmic tangent
- **end if**
- **for** $i = 1, \ldots, M$ **do**
  - $\gamma_{i_{\mathrm{n+1}}} \leftarrow \gamma_i(\gamma_{i_{\mathrm{n}}}, \Delta\bar{\boldsymbol{\sigma}}, t_{\mathrm{n}}, t_{\mathrm{n+1}})$ ▹ Update viscoelastic state variables
- **end for**
- $\boldsymbol{\sigma}_{\mathrm{n+1}} \leftarrow (1 - \omega_{\mathrm{n+1}})\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}$ ▹ Update nominal stress
- **return** $\boldsymbol{\sigma}_{\mathrm{n+1}}, \alpha_{\mathrm{p}_{\mathrm{n+1}}}, \alpha_{\mathrm{d}_{\mathrm{n+1}}}, \omega_{\mathrm{n+1}}, \gamma_{1_{\mathrm{n+1}}}, \ldots, \gamma_{M_{\mathrm{n+1}}}, \mathbb{A}_{\mathrm{n+1}}$

end function

1971), given as

$$L(\tau) \approx L_k(\tau) = \frac{(-k\,\tau)^k}{(k-1)!}\,\Phi^{(k)}(k\,\tau), \quad \text{for } k \in \{1, 2, \ldots\}. \tag{8.70}$$

Therein, k denotes the order of approximation and $\Phi^{(k)}$ the kth derivate of the compliance function. The Post-Widder inversion formula has the property

$$L(\tau) = \lim_{k\to\infty} L_k(\tau), \tag{8.71}$$

i.e., the approximation $L_k(\tau)$ converges to the exact continuous spectrum $L(\tau)$ with increasing order k.

For the discrete spectrum of the Dirichlet series, the retardation times τ_i are chosen such that they form a geometric progression with common ratio 10, i.e., $\tau_{i+1} = 10\,\tau_i$. The shortest retardation time τ_1 is chosen such that any duration of load of interest t in the numerical analyses is longer than τ_1, i.e., $\tau_1 \ll t$, whereas τ_M is chosen such that $\tau_M \gg t$.

For a chosen set of retardation times and a chosen order of approximation of the continuous spectrum $L(t)$ of the exact compliance function $\Phi(t)$ in (8.68), integral (8.69) is approximated as

$$\Phi(t) \approx \int_{\tau=0}^{\tau_1/\sqrt{10}} L_k \, \mathrm{d}(\ln(\tau)) \;+\; \ln(10) \sum_{i=1}^{M} L_k(\tau_i)\,(1 - \exp(-t/\tau_i)). \tag{8.72}$$

The second term on the right hand side is obtained by application of the midpoint rule with a step size of ln(10) for the interval $[\tau_1/\sqrt{10}, \tau_M]$, with each retardation time τ_i covering one order of magnitude of the time scale, defined by the interval $[\tau_i/\sqrt{10}, \tau_i \cdot \sqrt{10}]$ (Jirásek and Havlásek, 2014). This term corresponds directly to the presented approximation of the compliance function by means of a Dirichlet series, as a comparison of (8.72) with (8.10) reveals. The interval $[\tau_M \cdot \sqrt{10}, \infty]$ is neglected in the approximation. The first term on the right hand side requires further explanations: In (8.69), the expression $(1-\exp(-t/\tau))$ is very close to 1 for all $\tau \ll \tau_1$ and $t \gg \tau_1$. This is exploited in (8.72) by means of the approximation

$$\int_{\tau=0}^{\tau_1/\sqrt{10}} L_k\,(1 - \exp(-t/\tau))\,\mathrm{d}(\ln(\tau)) \approx \int_{\tau=0}^{\tau_1/\sqrt{10}} L_k \, \mathrm{d}(\ln(\tau)). \tag{8.73}$$

Accordingly, the approximation of $\Phi(t - t')$ in (8.10) can be rewritten as

$$\Phi(t - t') = \frac{1}{D_0} + \sum_{i=1}^{M} \frac{1}{D_i}\left(1 - \exp\left(\frac{-(t - t')}{\tau_i}\right)\right), \tag{8.74}$$

with the moduli expressed as

$$\frac{1}{D_0} = \int_{\tau=0}^{\tau_1/\sqrt{10}} L_k(\tau)\,\mathrm{d}(\ln(\tau))\,, \tag{8.75}$$

$$\frac{1}{D_i} = \ln(10)\,L_k(\tau_i) \quad \text{for } i = 1, 2, \ldots M\,. \tag{8.76}$$

As pointed out by Jirásek and Havlásek (2014), for practical applications the second order approximation $L_2(\tau)$ of the continuous retardation spectrum of $\Phi(t)$ is sufficiently accurate, which is computed by choosing $k = 2$ and substituting (8.68) into (8.70) as

$$\begin{aligned} L_2(\tau) &= \frac{(-2\,\tau)^2}{(2-1)!}\,\Phi^{(2)}(2\,\tau) \\ &= q_2\,\frac{0.1(2\tau)^{0.1}(0.9 + (2\tau)^{0.1})}{(1 + (2\tau)^{0.1})^2}\,. \end{aligned} \tag{8.77}$$

Based on the second order approximation, the moduli of the Dirichlet series are computed as

$$\frac{1}{D_0} = q_2 \left(\ln(1+\tilde{\tau}_0) - \frac{\tilde{\tau}_0}{10(1+\tilde{\tau}_0)} \right), \tag{8.78}$$

$$\frac{1}{D_i} = q_2 \ln(10) \frac{\tilde{\tau}_i(0.9+\tilde{\tau}_i)}{10(1+\tilde{\tau}_i)^2} \quad \text{for } i = 1, 2, \ldots, M\,, \tag{8.79}$$

with

$$\tilde{\tau}_0 = \left(\frac{2\tau_1}{\sqrt{10}} \right)^{0.1}, \tag{8.80}$$

$$\tilde{\tau}_i = (2\,\tau_i)^{0.1} \quad \text{for } i = 1, 2, \ldots, M\,. \tag{8.81}$$

To account for the additional time-independent term $\frac{1}{D_0}$ within the stress update algorithm of the SCDP model, it is incorporated into the expression for the effective stiffness (8.36):

$$\frac{1}{\tilde{E}} = q_1 + \frac{F(\bar{\sigma}_{\mathrm{n}})}{v_{\mathrm{sc}}(t_{\mathrm{n}} + \Delta t/2)} \left(\frac{1}{D_0} + \sum_{i=1}^{M} \frac{1}{D_i}\,(1-\lambda_i) \right) + \frac{1}{2}\,q_4\,\frac{F(\bar{\sigma}_{\mathrm{n}})\,\Delta t}{(t_{\mathrm{n}} + \Delta t/2)^{e_{\mathrm{flow}}}}\,. \tag{8.82}$$

In addition, Jirásek and Havlásek (2014) proposed two improvements in a heuristic manner for better agreement of the second order approximation with the exact compliance function: (i) After determining the moduli D_i, all retardation times are multiplied by a factor of 1.35, and (ii) the longest retardation time τ_M is additionally multiplied a factor of 1.2. A justification for those heuristic improvements despite the present convergence properties (8.71) of the Post-Widder formula is given also in (Jirásek and Havlásek, 2014).

8.2 Implementation of the Gradient-Enhanced Formulation

In the following sections, the derivation of the nonlinear system of algebraic equations for the implicit gradient-enhanced damage formulation is presented. To this end, the governing equations of the time-dependent, quasi-static problem are converted to the weak form, discretized in space employing the finite element method and subsequently discretized in time. The standard incremental-iterative procedure based on the Newton-Raphson scheme is employed for solving the IBVP. The here presented procedure follows the methods presented by de Borst, Crisfield, et al. (2012) and Belytschko et al. (2013). Finally, the stress update algorithm for the gradient-enhanced SCDP model is presented. It is considered as an extension of the respective algorithm of the *local* SCDP model, presented in Section 8.1.

8.2.1 Weak Form of the Governing Equations

The governing partial differential equations in the strong form are converted to the weak form by taking the integral over the complete domain Ω and applying test functions $\delta \boldsymbol{u}$ for the displacement field and $\delta\bar{\alpha}_\mathrm{d}$ for the field of the nonlocal damage driving internal strain-like variable, which is henceforth simply denoted as the *nonlocal* field. The framework is formulated within the regime of small strains and displacements, with the small strain tensor $\boldsymbol{\varepsilon}$ given as the symmetric part of the displacement gradient:

$$\boldsymbol{\varepsilon} = \nabla_\mathrm{sym}\boldsymbol{u} \,. \tag{8.83}$$

For the equilibrium equation (7.9), the weak form is given as

$$\int_\Omega \delta\boldsymbol{\varepsilon} : \boldsymbol{\sigma}\,\mathrm{d}V = \int_{\Gamma_t} \delta\boldsymbol{u}^T\bar{\boldsymbol{t}}\,\mathrm{d}S + \int_\Omega \delta\boldsymbol{u}^T\boldsymbol{f}\,\mathrm{d}V, \tag{8.84}$$

in which use of the Gauss theorem and the boundary conditions (7.10) is made. In a similar manner, departing from (7.8), conversion to the weak form and application of the Gauss theorem yields

$$\int_\Omega \delta\bar{\alpha}_\mathrm{d}\,\bar{\alpha}_\mathrm{d}\,\mathrm{d}V + \int_\Omega l^2\,\nabla\delta\bar{\alpha}_\mathrm{d}^T\,\nabla\bar{\alpha}_\mathrm{d}\,\mathrm{d}V - \int_\Omega \delta\bar{\alpha}_\mathrm{d}\,\alpha_\mathrm{d}\,\mathrm{d}V = 0, \tag{8.85}$$

in which use of the homogeneous Neumann boundary conditions (7.11) was made.

8.2.2 Spatial Discretization

The weak form (8.84) and (8.85) is spatially discretized by the finite element method following the Bubnov-Galerkin approach, i.e., using the same discretization scheme for trial and test functions. The displacement field and the nonlocal field are interpolated in terms of global interpolation operators $\mathbf{N}_\mathrm{u}$ for the displacement field and $\mathbf{N}_{\bar{\alpha}}$ for the nonlocal field:

$$\boldsymbol{u} = \mathbf{N}_\mathrm{u}\,\mathbf{q}_\mathrm{u}, \tag{8.86}$$
$$\bar{\alpha}_\mathrm{d} = \mathbf{N}_{\bar{\alpha}}\,\mathbf{q}_{\bar{\alpha}}, \tag{8.87}$$

in which $\mathbf{q}_\mathrm{u}$ and $\mathbf{q}_{\bar{\alpha}}$ denote the vectors of the nodal displacements and the nodal nonlocal values, respectively. Furthermore, the global strain-displacement operator $\mathbf{B}_\mathrm{u}$ for the displacement field and the global gradient operator $\mathbf{B}_{\bar{\alpha}}$ for the nonlocal field are introduced as:

$$\boldsymbol{\varepsilon} = \mathbf{B}_\mathrm{u}\,\mathbf{q}_\mathrm{u}, \tag{8.88}$$
$$\nabla\bar{\alpha}_\mathrm{d} = \mathbf{B}_{\bar{\alpha}}\,\mathbf{q}_{\bar{\alpha}}. \tag{8.89}$$

Departing from (8.84) and (8.85), the discretized weak form expressions are given as

$$\begin{aligned}\int_\Omega \delta\mathbf{q}_\mathrm{u}^T\,\mathbf{B}_\mathrm{u}^T\,\boldsymbol{\sigma}\,\mathrm{d}V &= \int_{\Gamma_t} \delta\mathbf{q}_\mathrm{u}^T\,\mathbf{N}_\mathrm{u}^T\,\bar{\boldsymbol{t}}\,\mathrm{d}S + \int_\Omega \delta\mathbf{q}_\mathrm{u}^T\mathbf{N}_\mathrm{u}^T\,\boldsymbol{f}\,\mathrm{d}V \\ \delta\mathbf{q}_\mathrm{u}^T\,\mathbf{f}_\mathrm{u}^\mathrm{int} &= \delta\mathbf{q}_\mathrm{u}^T\,\mathbf{f}_\mathrm{u}^\mathrm{ext}\end{aligned} \tag{8.90}$$

and

$$\begin{aligned}\int_\Omega \delta\mathbf{q}_{\bar\alpha}^T\,\mathbf{N}_{\bar\alpha}^T\,\mathbf{N}_{\bar\alpha}\mathbf{q}_{\bar\alpha}\,\mathrm{d}V + \int_\Omega l^2\,\delta\mathbf{q}_{\bar\alpha}^T\,\mathbf{B}_{\bar\alpha}^T\,\mathbf{B}_{\bar\alpha}\,\mathbf{q}_{\bar\alpha}\,\mathrm{d}V - \int_\Omega \delta\mathbf{q}_{\bar\alpha}^T\mathbf{N}_{\bar\alpha}^T\alpha_\mathrm{d}\,\mathrm{d}V &= 0 \\ \delta\mathbf{q}_{\bar\alpha}^T\,\mathbf{f}_{\bar\alpha}^\mathrm{int} &= 0,\end{aligned} \tag{8.91}$$

with $\mathbf{f}_\mathrm{u}^\mathrm{int}$ and $\mathbf{f}_\mathrm{u}^\mathrm{ext}$ denoting the standard internal and external nodal force vectors associated with displacement field, and $\mathbf{f}_{\bar\alpha}^\mathrm{int}$ the non-standard internal force vector associated with the nonlocal field:

$$\mathbf{f}_\mathrm{u}^\mathrm{int} = \int_\Omega \mathbf{B}_\mathrm{u}^T\,\boldsymbol{\sigma}\,\mathrm{d}V\,, \tag{8.92}$$

$$\mathbf{f}_\mathrm{u}^\mathrm{ext} = \int_{\Gamma_t} \mathbf{N}_\mathrm{u}^T\,\bar{\boldsymbol{t}}\,\mathrm{d}S + \int_\Omega \mathbf{N}_\mathrm{u}^T\,\boldsymbol{f}\,\mathrm{d}V\,, \tag{8.93}$$

$$\mathbf{f}_{\bar\alpha}^\mathrm{int} = \int_\Omega \mathbf{N}_{\bar\alpha}^T\,\mathbf{N}_{\bar\alpha}\,\mathbf{q}_{\bar\alpha}\,\mathrm{d}V + \int_\Omega l^2\,\mathbf{B}_{\bar\alpha}^T\,\mathbf{B}_{\bar\alpha}\,\mathbf{q}_{\bar\alpha}\,\mathrm{d}V - \int_\Omega \mathbf{N}_{\bar\alpha}^T\alpha_\mathrm{d}\,\mathrm{d}V\,. \tag{8.94}$$

The virtual work expressions (8.90) and (8.91) must hold for arbitrary, kinematically admissible values of $\delta\mathbf{q}_\mathrm{u}$ and $\delta\mathbf{q}_{\bar\alpha}$, which leads to the system of nonlinear algebraic equations

$$\begin{Bmatrix} \mathbf{f}_\mathrm{u}^\mathrm{int} - \mathbf{f}_\mathrm{u}^\mathrm{ext} \\ \mathbf{f}_{\bar\alpha}^\mathrm{int} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{0} \end{Bmatrix}\,. \tag{8.95}$$

Following the standard finite element procedure, the internal and external force vectors are obtained by application of the standard assembly procedure on the respective element load vectors.

8.2.3 Discretization in Time

The system of nonlinear algebraic equations (8.95) holds for any time t, and for any prescribed external load $\mathbf{f}_\mathrm{u}^\mathrm{ext}$. Due to the time- and path-dependent character of the considered IBVP, the solution is computed employing an incremental-iterative procedure. Following the standard incremental-iterative procedure for time-dependent quasi-static problems, a discretization in time is introduced, and the respective solution for each instant of time is computed iteratively in an incremental manner, starting in each increment from the computed solution of the preceding increment. For each increment, the system of nonlinear equations is formulated in terms of the unknown nodal increments $\Delta\mathbf{q}_\mathrm{u}$ and $\Delta\mathbf{q}_{\bar\alpha}$, and solved employing the Newton-Raphson scheme. In particular, considering an increment defined by the interval $[t_\mathrm{n}, t_\mathrm{n+1}]$ with $t_\mathrm{n+1} = t_\mathrm{n} + \Delta t$ and departing from the known state at t_n, the internal force vectors at the end of the interval expressed in terms of the increments $\Delta\mathbf{q}_\mathrm{u}$ and $\Delta\mathbf{q}_{\bar{\alpha}}$ are given as

$$\mathbf{f}^\mathrm{int}_\mathrm{u}(t_\mathrm{n+1}, \Delta\mathbf{q}_\mathrm{u}, \Delta\mathbf{q}_{\bar{\alpha}}) = \int_\Omega \mathbf{B}^T_\mathrm{u}\, \boldsymbol{\sigma}_\mathrm{n+1}\, \mathrm{d}V\,, \tag{8.96}$$

$$\mathbf{f}^\mathrm{int}_{\bar{\alpha}}(t_\mathrm{n+1}, \Delta\mathbf{q}_\mathrm{u}, \Delta\mathbf{q}_{\bar{\alpha}}) = \int_\Omega \mathbf{N}^T_{\bar{\alpha}}\, \mathbf{N}_{\bar{\alpha}}\, (\mathbf{q}_{\bar{\alpha}_\mathrm{n}} + \Delta\mathbf{q}_{\bar{\alpha}}) + l^2\, \mathbf{B}^T_{\bar{\alpha}}\, \mathbf{B}_{\bar{\alpha}}\, (\mathbf{q}_{\bar{\alpha}_\mathrm{n}} + \Delta\mathbf{q}_{\bar{\alpha}}) - \mathbf{N}^T_{\bar{\alpha}}\, \alpha_\mathrm{dn+1}\, \mathrm{d}V. \tag{8.97}$$

Therein, $\mathbf{q}_{\bar{\alpha}_\mathrm{n}}$ denotes the known nodal values of the nonlocal field at the beginning of the interval, and $\boldsymbol{\sigma}_\mathrm{n+1}$ and $\alpha_\mathrm{dn+1}$ denote the values of the stress and of the local damage driving variable the end of the interval $[t_\mathrm{n}, t_\mathrm{n+1}]$. They are computed within the extended stress update algorithm based on the constitutive relations, the prescribed strain increment $\Delta\boldsymbol{\varepsilon} = \mathbf{B}_\mathrm{u}\, \Delta\mathbf{q}_\mathrm{u}$ and the value of the nonlocal damage driving variable $\bar{\alpha}_\mathrm{dn+1}$, starting from the known stress $\boldsymbol{\sigma}_\mathrm{n}$, and the known set of internal variables $\boldsymbol{\alpha}_\mathrm{n}$ at the beginning of the increment. Accordingly, $\boldsymbol{\sigma}_\mathrm{n+1}$ is expressed as a function of those quantities:

$$\boldsymbol{\sigma}_\mathrm{n+1} = \boldsymbol{\sigma}(\Delta\boldsymbol{\varepsilon}, \bar{\alpha}_\mathrm{dn+1}, t_\mathrm{n}, t_\mathrm{n+1}, \boldsymbol{\sigma}_\mathrm{n}, \boldsymbol{\alpha}_\mathrm{n}), \tag{8.98}$$

and in a similar manner $\alpha_\mathrm{dn+1}$ is expressed as

$$\alpha_\mathrm{dn+1} = \alpha_\mathrm{d}(\Delta\boldsymbol{\varepsilon}, t_\mathrm{n}, t_\mathrm{n+1}, \boldsymbol{\sigma}_\mathrm{n}, \boldsymbol{\alpha}_\mathrm{n})\,, \tag{8.99}$$

which is however, independent of $\bar{\alpha}_\mathrm{dn+1}$.Due to the employed numerical integration schemes and the time- and path-dependent character of the considered problem, the accuracy of the solution crucially depends on the size of the increments, i.e., on the resolution of the time-discretization.

For solving iteratively the system of nonlinear equations for the unknown nodal increments $\Delta\mathbf{q}_\mathrm{u}$ and $\Delta\mathbf{q}_{\bar{\alpha}}$ by means of the Newton-Raphson method, the Jacobian of the nonlinear system is required. It is obtained by consistent linearization of (8.95) with respect to the nodal increments $\Delta\mathbf{q}_\mathrm{u}$ and $\Delta\mathbf{q}_{\bar{\alpha}}$, considering that the external forces are independent of the nodal values:

$$\begin{bmatrix} \dfrac{\partial \mathbf{f}^\mathrm{int}_\mathrm{u}}{\partial \Delta\mathbf{q}_\mathrm{u}} & \dfrac{\partial \mathbf{f}^\mathrm{int}_\mathrm{u}}{\partial \Delta\mathbf{q}_{\bar{\alpha}}} \\ \dfrac{\partial \mathbf{f}^\mathrm{int}_{\bar{\alpha}}}{\partial \Delta\mathbf{q}_\mathrm{u}} & \dfrac{\partial \mathbf{f}^\mathrm{int}_{\bar{\alpha}}}{\partial \Delta\mathbf{q}_{\bar{\alpha}}} \end{bmatrix} = \begin{bmatrix} \int_\Omega \mathbf{B}^T_\mathrm{u}\, \mathbb{A}_\mathrm{n+1} \mathbf{B}_\mathrm{u}\, \mathrm{d}V & \int_\Omega \mathbf{B}^T_\mathrm{u}\, \dfrac{\partial \boldsymbol{\sigma}_\mathrm{n+1}}{\partial \bar{\alpha}_\mathrm{dn+1}}\, \mathbf{N}_{\bar{\alpha}}\, \mathrm{d}V \\ -\int_\Omega \mathbf{N}^T_{\bar{\alpha}} \dfrac{\partial \alpha_\mathrm{dn+1}}{\partial \Delta\boldsymbol{\varepsilon}} \mathbf{B}_\mathrm{u}\, \mathrm{d}V & \int_\Omega \mathbf{N}^T_{\bar{\alpha}}\, \mathbf{N}_{\bar{\alpha}} + l^2\, \mathbf{B}^T_{\bar{\alpha}}\, \mathbf{B}_{\bar{\alpha}}\, \mathrm{d}V \end{bmatrix}. \tag{8.100}$$

Therein, $\mathbb{A}_\mathrm{n+1}$ is the algorithmic stiffness, and $\frac{\partial \sigma_\mathrm{n+1}}{\partial \bar{\alpha}_\mathrm{dn+1}}$ and $\frac{\partial \alpha_\mathrm{dn+1}}{\partial \Delta\boldsymbol{\varepsilon}}$ are tangents describing the coupling between the displacement and the nonlocal field, which are computed simultaneously to $\boldsymbol{\sigma}_\mathrm{n+1}$, $\alpha_\mathrm{dn+1}$ and $\mathbb{A}_\mathrm{n+1}$ within the extended stress update algorithm. It is apparent that the Jacobian is nonsymmetric. Similar to the internal and external force vectors, the Jacobian is obtained by assembling the respective element stiffness matrices.

8.2.4 Extended Stress Update Algorithm for the Gradient-Enhanced SCDP Model

The stress update algorithm of the *local* SCDP model presented in Section 8.1 is extended to the gradient-enhanced formulation. Accordingly, the nonlocal damage driving variable $\bar{\alpha}_{\mathrm{d}}$ is introduced as an additional input parameter, and the local damage driving variable $\alpha_{\mathrm{dn+1}}$, as well as algorithmic tangents $\frac{\partial \sigma_{\mathrm{n+1}}}{\partial \bar{\alpha}_{\mathrm{dn+1}}}$ and $\frac{\partial \alpha_{\mathrm{dn+1}}}{\partial \Delta \varepsilon}$, which are required for computing the Jacobian (8.100), are considered as additional output parameters. In contrast to the local algorithm, the computation of the updated damage parameter $\omega_{\mathrm{n+1}}$ is performed independent of any plastic flow, and it is formulated in terms of the weighted over-nonlocal parameter $\hat{\alpha}_{\mathrm{d}}$ as described in (7.6) and (7.7).

Algorithm 2 Stress update algorithm for the gradient-enhanced SCDP model

function STRESS UPDATE $(\Delta\boldsymbol{\varepsilon}, \bar{\alpha}_{\mathrm{dn+1}}, t_{\mathrm{n}}, t_{\mathrm{n+1}}, \boldsymbol{\sigma}_{\mathrm{n}}, \alpha_{\mathrm{p_n}}, \alpha_{\mathrm{dn}}, \omega_{\mathrm{n}}, \boldsymbol{\gamma}_{1_{\mathrm{n}}}, \ldots, \boldsymbol{\gamma}_{M_{\mathrm{n}}})$
    $\bar{\boldsymbol{\sigma}}_{\mathrm{n}} \leftarrow \frac{1}{(1-\omega_{\mathrm{n}})} \boldsymbol{\sigma}_{\mathrm{n}}$ ▷ Compute effective stress at t_{n}
    $\Delta\boldsymbol{\varepsilon}^{\mathrm{shr}} \leftarrow \boldsymbol{\varepsilon}^{\mathrm{shr}}(t_{\mathrm{n}}, t_{\mathrm{n+1}})$
    $\Delta\boldsymbol{\varepsilon}'' \leftarrow \Delta\boldsymbol{\varepsilon}''(\bar{\boldsymbol{\sigma}}_{\mathrm{n}}, \boldsymbol{\gamma}_{1_{\mathrm{n}}}, \ldots, \boldsymbol{\gamma}_{M_{\mathrm{n}}}, t_{\mathrm{n}}, t_{\mathrm{n+1}}, \Delta\boldsymbol{\varepsilon}^{\mathrm{shr}})$
    $\bar{E} \leftarrow \bar{E}(\bar{\boldsymbol{\sigma}}_{\mathrm{n}}, \boldsymbol{\gamma}_{1_{\mathrm{n}}}, \ldots, \boldsymbol{\gamma}_{M_{\mathrm{n}}}, t_{\mathrm{n}}, t_{\mathrm{n+1}},)$ ▷ Compute effective stiffness
    $\tilde{\mathbb{C}} \leftarrow \mathbb{C}_{\nu}\, \bar{E}$
    $\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}} \leftarrow \bar{\boldsymbol{\sigma}}_{\mathrm{n}} + \tilde{\mathbb{C}} : (\Delta\boldsymbol{\varepsilon} - \Delta\boldsymbol{\varepsilon}'')$ ▷ Compute trial effective stress
    if $f(\bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}, \alpha_{\mathrm{n}}, t_{\mathrm{n+1}}) < 0$ **then** ▷ Yield condition not violated
        $\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \leftarrow \bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}$
        $\alpha_{\mathrm{p_{n+1}}} \leftarrow \alpha_{\mathrm{p_n}}$
        $\alpha_{\mathrm{dn+1}} \leftarrow \alpha_{\mathrm{dn}}$
    else ▷ Yield condition violated
        $\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}, \alpha_{\mathrm{p_{n+1}}}, \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}} \leftarrow$ RETURN MAPPING $(\boldsymbol{\sigma}_{\mathrm{n}}, \bar{\boldsymbol{\sigma}}^{\mathrm{tr}}_{\mathrm{n+1}}, \tilde{\mathbb{C}}, \alpha_{\mathrm{p_n}}, t_{\mathrm{n+1}})$
        $\alpha_{\mathrm{dn+1}} \leftarrow \alpha_{\mathrm{d}}(\alpha_{\mathrm{dn}}, \Delta\boldsymbol{\varepsilon}^{\mathrm{p}})$
    end if
    $\hat{\alpha}_{\mathrm{dn+1}} = m\, \bar{\alpha}_{\mathrm{dn+1}} + (1-m)\, \alpha_{\mathrm{dn+1}}$ ▷ Update weighted variable
    $\omega_{\mathrm{n+1}} \leftarrow \omega(\hat{\alpha}_{\mathrm{dn+1}})$ ▷ Update damage parameter
    $\frac{\partial \boldsymbol{\sigma}_{\mathrm{n+1}}}{\partial \bar{\alpha}_{\mathrm{dn+1}}} \leftarrow \frac{\partial \boldsymbol{\sigma}_{\mathrm{n+1}}}{\partial \bar{\alpha}_{\mathrm{dn+1}}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \hat{\alpha}_{\mathrm{dn+1}}, \bar{\alpha}_{\mathrm{dn+1}})$ ▷ Algorithmic tangents
    $\frac{\partial \alpha_{\mathrm{dn+1}}}{\partial \Delta\boldsymbol{\varepsilon}} \leftarrow \frac{\partial \alpha_{\mathrm{dn+1}}}{\partial \Delta\boldsymbol{\varepsilon}}(\Delta\boldsymbol{\varepsilon}^{\mathrm{p}}, \alpha_{\mathrm{dn+1}})$
    $\mathbb{A}_{\mathrm{n+1}} \leftarrow \mathbb{A}_{\mathrm{n+1}}(\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}, \Delta\boldsymbol{\varepsilon}^{\mathrm{p}}, \hat{\alpha}_{\mathrm{dn+1}}, \alpha_{\mathrm{dn+1}}, \omega_{\mathrm{n+1}}, \tilde{\mathbb{C}}, \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}})$
    for $i = 1, \ldots, M$ **do**
        $\boldsymbol{\gamma}_{i_{\mathrm{n+1}}} \leftarrow \boldsymbol{\gamma}_i(\boldsymbol{\gamma}_{i_{\mathrm{n}}}, \Delta\bar{\boldsymbol{\sigma}}, t_{\mathrm{n}}, t_{\mathrm{n+1}})$ ▷ Update viscoelastic state variables
    end for
    $\boldsymbol{\sigma}_{\mathrm{n+1}} \leftarrow (1-\omega_{\mathrm{n+1}})\bar{\boldsymbol{\sigma}}_{\mathrm{n+1}}$ ▷ Update nominal stress
    return $\boldsymbol{\sigma}_{\mathrm{n+1}}, \alpha_{\mathrm{p_{n+1}}}, \alpha_{\mathrm{dn+1}}, \omega_{\mathrm{n+1}}, \boldsymbol{\gamma}_{1_{\mathrm{n+1}}}, \ldots, \boldsymbol{\gamma}_{M_{\mathrm{n+1}}}, \mathbb{A}_{\mathrm{n+1}}, \frac{\partial \boldsymbol{\sigma}_{\mathrm{n+1}}}{\partial \bar{\alpha}_{\mathrm{dn+1}}}, \frac{\partial \alpha_{\mathrm{dn+1}}}{\partial \Delta\boldsymbol{\varepsilon}}$
end function

Accordingly, the algorithmic tangent stiffness $\mathbb{A}_{\mathrm{n+1}}$ (8.65) is modified as

$$\begin{aligned}\mathbb{A}_{\mathrm{n+1}} &= \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \otimes \frac{\partial \omega_{\mathrm{n+1}}}{\partial \hat{\alpha}_{\mathrm{dn+1}}} \frac{\partial \hat{\alpha}_{\mathrm{dn+1}}}{\partial \alpha_{\mathrm{dn+1}}} \frac{\partial \alpha_{\mathrm{dn+1}}}{\partial \Delta \boldsymbol{\varepsilon}^{\mathrm{p}}} : (\tilde{\mathbb{C}}^{-1} : \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}} - \mathbb{I}) + (1 - \omega_{\mathrm{n+1}})\, \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}} \\ &= (1 - m)\, \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \otimes \frac{\partial \omega_{\mathrm{n+1}}}{\partial \hat{\alpha}_{\mathrm{dn+1}}} \frac{\partial \alpha_{\mathrm{dn+1}}}{\partial \Delta \boldsymbol{\varepsilon}^{\mathrm{p}}} : (\tilde{\mathbb{C}}^{-1} : \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}} - \mathbb{I}) + (1 - \omega_{\mathrm{n+1}})\, \tilde{\mathbb{A}}^{\mathrm{ep}}_{\mathrm{n+1}} \,.\end{aligned} \tag{8.101}$$

The algorithmic tangent $\frac{\partial \sigma_{\mathrm{n+1}}}{\partial \bar{\alpha}_{\mathrm{dn+1}}}$ is obtained as

$$\begin{aligned}\frac{\partial \boldsymbol{\sigma}_{\mathrm{n+1}}}{\partial \bar{\alpha}_{\mathrm{dn+1}}} &= \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \left(- \frac{\partial \omega_{\mathrm{n+1}}}{\partial \hat{\alpha}_{\mathrm{dn+1}}} \frac{\partial \hat{\alpha}_{\mathrm{dn+1}}}{\partial \bar{\alpha}_{\mathrm{dn+1}}} \right) \\ &= -m\, \bar{\boldsymbol{\sigma}}_{\mathrm{n+1}} \frac{\partial \omega_{\mathrm{n+1}}}{\partial \hat{\alpha}_{\mathrm{dn+1}}} \,.\end{aligned} \tag{8.102}$$

The extended, nonlocal algorithm is summarized in Algorithm 2, with the modified sections highlighted.

9 Summary and Conclusions

The present work addressed the complex, time-dependent material behavior of shotcrete both by experimental investigations and by numerical modeling. A new damage plasticity model, the SCDP model, representing time-dependent material properties, hardening and softening behavior, inelastic strains, stiffness degradation, shrinkage and nonlinear creep was presented. The SCDP model is based on three constitutive models for concrete, which are the damage plasticity model by Grassl and Jirásek (2006a), the solidification theory for concrete creep by Bažant and Prasannan (1989a), and the semi-empirical shrinkage model by Bažant and Panula (1978). For assessing the new model, two other advanced models for shotcrete were selected from the literature, i.e., the model by Meschke (1996) and the model by Schädlich and Schweiger (2014).

An evaluation of the three models, and in addition of a multi-field model by Gawin et al. (2006a) was presented in Publication I, *An Extended Damage Plasticity Model for Shotcrete: Formulation and Comparison with Other Shotcrete Models*. To this end, experimental tests from the literature were employed, in particular those presented by M. Müller (2001) and Huber (1991) for dry-mix shotcrete compositions, and the material models were calibrated and validated at material point level by means of numerical simulations of those tests.

Consequently, the SCDP model, the Meschke model, the Schädlich model, and in addition a linear-elastic reference model based on the approach by Pöttler (1990b), were assessed Publication II, *Influence of the Constitutive Model for Shotcrete on the Predicted Structural Behavior of the Shotcrete Shell of a Deep Tunnel*, by means of an initial boundary value problem. Based on a stretch of the Brenner Base Tunnel, a benchmark for studying the influence of the constitutive model for shotcrete on the predicted structural response of shotcrete shells in finite element simulations of deep tunnel advance was developed. It was concluded that substantially different results are predicted by the advanced shotcrete models compared to the simplified linear-elastic approach with a hypothetical modulus of elasticity. The presented results indicate a considerably better agreement of the results based on the advanced shotcrete models with geodetic and in-situ measurement results compared to the simplified approach based on linear elasticity. Furthermore, it was shown that creep and shrinkage have a considerable influence on the predicted mechanical behavior.

Motivated by the fact that the experimental test series on shotcrete available in the literature do not represent the material behavior of modern shotcrete compositions anymore, a new experimental program on shotcrete was designed and presented in Publication IV, *Time-Dependent Material Properties of Shotcrete: Experimental and Numerical Study*. The new program focused on the evolution of material stiffness and strength, shrinkage and creep. The design of the experimental program on creep and shrinkage took advantage of the experience gained during a preceding experimental program on concrete creep and shrinkage, presented Publication III, *Comprehensive Study of Concrete Creep, Shrinkage, and Water Content Evolution under Sealed and Drying Conditions*. For sampling the shotcrete specimens, two different techniques were employed: Drill cores from spray boxes and sprayed specimens, sampled directly from tubular molds. The use of sprayed specimens allows for a start of the lab tests at a very early age of the material, i.e., a few hours after mixing the composition, without any cutting or grinding of the susceptible young specimens. Based on the new experimental program, it was shown that the material properties of the investigated wet-mix composition are substantially different from those of the dry-mix compositions used in the past decades and documented in the literature. Furthermore, a calibration of the three investigated material models was presented.

In Publication V, *Comparative Investigation of Constitutive Models for Shotcrete Based on Numerical Simulations of Deep Tunnel Advance*, a further development of the SCDP model, aiming at a simplified calibration procedure and an improved agreement with the experimentally observed material behavior of shotcrete was presented. Furthermore, the improved SCDP model, as well as the models by Meschke and by Schädlich and Schweiger were applied within the benchmark finite element study presented in Publication II, employing the material parameters calibrated in Publication IV.

Finally, motivated by the deficiencies of the crack band approach for regularizing softening material behavior in finite element analyses, an advanced regularization scheme for the SCDP model was examined. To this end, the over-nonlocal implicit gradient-enhanced formulation proposed by Poh and Swaddiwudhipong (2009b) for the concrete model by Grassl and Jirásek (2006a) was adopted for the SCDP model. In Publication VI, *Evaluation of a Gradient Enhanced Damage Plasticity Model for Shotcrete*, the gradient-enhanced SCDP model was applied to a benchmark problem for mode I failure, and the good performance of the over-nonlocal implicit gradient-enhancement was proven. Finally, in Publication VII, *Study of the Regularization Scheme of an Advanced Rock Model*, an extended version of the damage plasticity model for rock mass employing the same gradient-enhanced formulation was applied to biaxial compression tests, and thereby it was shown that this approach is also suitable for mode II failure.

Based on the presented results within this thesis, an outlook and several proposals for future research activities are given. Concerning future experimental programs, following proposals are made:

- The presented technique to use sprayed specimens for experimental tests has proven to be a simple and effective alternative to drill cores. Since no preparation of the specimens in terms of cutting or grinding is necessary, experimental tests can be started at a very early age of the material. Hence, it is proposed to employ this technique for future experimental tests.

- Nonlinear creep of shotcrete shells is considered as a very important issue in tunneling: Due to the imposed deformations, high degrees of loading can arise, and consequently nonlinear viscoelastic material behavior occurs. However, an experimental study on this phenomenon of shotcrete is not yet available in the literature. Hence, it is proposed to investigate nonlinear creep of shotcrete by means of creep tests with different levels of the applied and sustained stress.
- For validating the deformations of shotcrete shells predicted in numerical simulations, detailed in-situ measurements of displacements and strains are required. Routine geodetic measurements performed during tunnel advance allow only for a very limited comparison and validation of the those deformations. More sophisticated techniques, e.g., based on optical fibers installed in the shotcrete shell, may be employed to gain more insight on the actual mechanical response during tunnel advance.

Regarding constitutive modeling of shotcrete in numerical simulations and in particular regarding the SCDP model, following recommendations and prospects are given:

- The SCDP model represents nonlinear creep behavior by means of the amplification function of the solidification theory by Bažant and Prasannan. However, due to the lack of experimental results on nonlinear creep of shotcrete, a validation of this formulation is still pending. Hence, for future development of the SCDP model, a validation of this amplification function based on new experimental tests is recommended.
- Due to the lack of experimental data, a time-independent softening modulus is currently assumed within the SCDP model, computed from the specific mode I fracture energy, the uniaxial tensile strength and the Young's modulus at the age of 28 days, and following the regularization scheme presented by Valentini (2011). However, this simplification remains to be validated based on experimental results, and it is proposed that a more sophisticated model for describing time-dependent softening material behavior is developed, accompanied by a new experimental program on the evolution of the specific mode I fracture energy. In this context, Lackner and Mang (2004) proposed a thermodynamic framework for considering the evolution of the specific mode I fracture energy dependent on the degree of hydration in numerical simulations.
- It was shown that the implicit gradient-enhancement of the SCDP model allows to describe softening material behavior in an objective manner with respect to the employed mesh in finite element simulations. The application to a two dimensional numerical study of deep tunnel advance complementing the one presented in Publication II is currently in preparation. Thereby the gradient-enhanced SCDP model presented in Publication VI as well as the gradient-enhanced damage plasticity model for rock mass by Unteregger (2015) presented in Publication VII will be employed.

- Within the present thesis, the performance of the SCDP model was demonstrated by means of two dimensional finite element simulations. The application within a three dimensional numerical study along with the damage plasticity model for rock mass by Unteregger (2015) is currently in preparation. In this context, a validation of the assumed stress release layout employed in Publication II to account for the time-dependent excavation process in two dimensional finite element simulations is recommended.

A follow up experimental program on shotcrete, including laboratory tests for determining the nonlinear creep behavior, as well as wedge splitting tests for determining the evolution of the specific mode I fracture energy is currently being prepared at the University of Innsbruck.

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